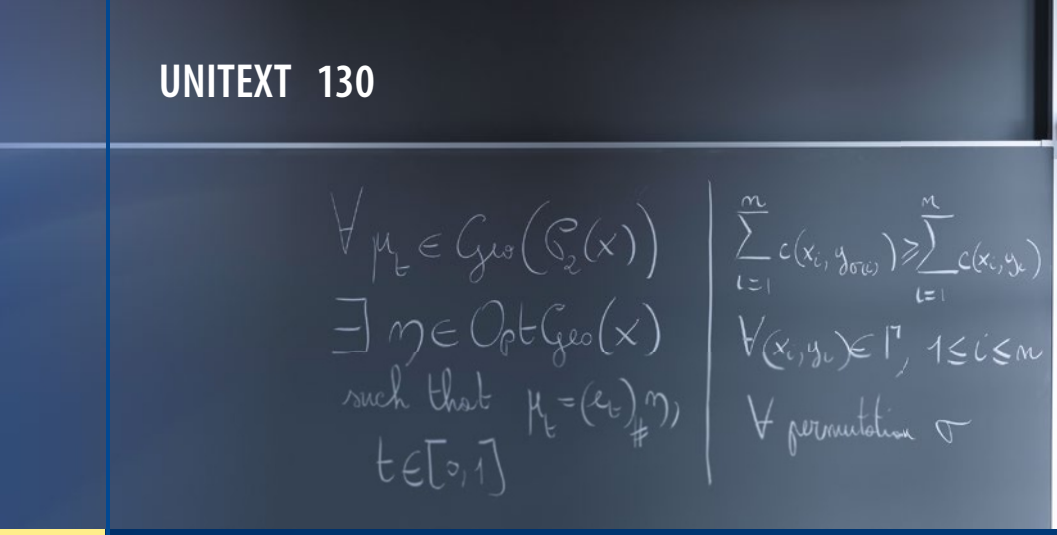
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Luigi Ambrosio · Elia Brué
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Lectures on Optimal Transport

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Lectures on Optimal Transport

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Preface

This textbook originated from the teaching experience of the first author at the Scuola Normale Superiore, where a course on optimal transport and its applications has been given many times during the last 20 years. Gradually, the content has been refined and updated, until the present form. On the basis of the past teaching experience, we have tried to make this book, as much as possible, suitable for a PhD course (even though in Pisa the course is attended also by senior undergraduate students), calling the single chapters "Lectures": ideally, with a few exceptions, each lecture can take 2 h, for a course of approximately 40 h. With respect to the already existing excellent texts devoted to the theory of optimal transport (above all [118], but see also [9, 12, 102, 103, 106, 117] and the forthcoming [61]), this book keeps the organization into lectures, based on the original course.

The theory of optimal transport has been blooming in the last 25 years at an enormous rate, with applications covering geometric and functional inequalities, partial differential equations, geometry of metric measure spaces, mathematical finance, computer graphics, and even, more recently, machine learning. Therefore, it is impossible to cover all this material in a single book, particularly in ours, which is designed for use in a PhD course. On the other hand, the topics and the tools have been chosen at a sufficiently general and advanced level, so that the student or scholar interested in a more specific theme will gain from the book the necessary background to explore it. In any case, after a large and detailed introduction to the classical theory (formulations of the problem, duality, necessary and sufficient optimality conditions, existence of optimal maps in Euclidean spaces as well as Riemannian manifolds), a more specific attention is devoted to applications in geometric and functional inequalities and in partial differential equations. With this aim, we treat extendedly the theory of gradient flows, which has an independent interest, and the metric, differentiable, and "Riemannian" structure of the space of probability measures. On the basis of Otto's calculus, these two topics can be combined showing how many basic PDEs of mathematical physics (and, above all, the heat equation) inherit a new gradient flow structure from optimal transport. The final chapter is devoted to the close links between Ricci curvature and heat equation in Riemannian manifolds: it hints at the most recent developments in metric measure

geometry, where these links have been studied in the much more general context of metric measure spaces.

The second and third coauthors already contributed in 2016 to a preliminary version of these notes, together with several other students attending the most recent edition of the course: G. E. Comi, G. Franz, F. Glaudo, A. Minne, L. Portinale, and D. Tewodrose. Their help has been fundamental to provide the initial draft on which we have been intensively working in the last months, expanding some proofs and adding a few more results and references.

Pisa, Italy — Luigi Ambrosio
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January 2021

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Lecture 1: Preliminary Notions and the Monge Problem



In this chapter we present basic notions and tools that will play a role in the rest of the book and we introduce the first formulation of the optimal transport problem, due to Monge.

1 Notation and Preliminary Results

We shall denote by $\mathbb{N} = \{0, 1, 2, \ldots\}$ the set of natural numbers and by $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ the sets of integer, rational and real numbers, respectively. We will denote by $\overline{\mathbb{R}}$ the extended real line. The characteristic function $\chi_E : X \to \{0, 1\}$ is defined by

$$\chi_E(x) := \begin{cases} 1 & \text{if } x \in E \\ 0 & \text{if } x \in X \setminus E. \end{cases}$$

We shall also use the notation 1_E for the $\{0, +\infty\}$-valued indicator function, namely

$$1_E(x) := \begin{cases} 0 & \text{if } x \in E \\ +\infty & \text{if } x \in X \setminus E. \end{cases}$$

The Lebesgue measure in $\mathbb{R}^n$ will be denoted by $\mathscr{L}^n$.

Given a metric space (X, d), we denote by $C(X)$ the vector space of continuous functions $f : X \to \mathbb{R}$ and by $C_b(X)$ the subspace of bounded continuous functions.

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Analogously, $\mathrm{Lip}(X)$ and $\mathrm{Lip}_b(X)$ denote the spaces of Lipschitz and bounded Lipschitz functions, with

$$\mathrm{Lip}(f) := \sup_{x \neq y} \frac{|f(x) - f(y)|}{|x - y|}$$

denoting the Lipschitz constant.

Given an open set $\Omega \subset \mathbb{R}^n$ and $k \in \mathbb{N}$ we denote by $C^k(\Omega)$ the set of functions with k continuous derivatives in Ω, while any element of $C^k(\overline{\Omega})$ is the restriction to $\overline{\Omega}$ of a function in $C^k(\mathbb{R}^n)$. The set $C^\infty(\Omega)$ (respectively $C^\infty(\overline{\Omega})$) is defined as the intersection of $C^k(\Omega)$ (respectively $C^k(\overline{\Omega})$) for $k \in \mathbb{N}$. The subscript c in $C_c(\Omega)$, $C_c^k(\Omega)$ and $C_c^\infty(\Omega)$ will stand for compact support.

In a metric space (X, d), $\mathscr{B}(X)$ denotes its Borel σ-algebra and $\mathscr{M}(X)$ the set of the σ-additive functions $\mu : \mathscr{B}(X) \to \mathbb{R}$. Furthermore we denote by

$$\mathscr{M}_+(X) := \big\{\mu \in \mathscr{M}(X) : \mu \geq 0\big\}, \qquad \mathscr{P}(X) = \big\{\mu \in \mathscr{M}_+(X) : \mu(X) = 1\big\}$$

the subsets of nonnegative and probability measures, respectively.

Two useful operations mapping measures to measures are introduced in the next definition.

Definition 1.1 (Total Variation and Restriction) Given $\mu \in \mathscr{M}(X)$, the total variation measure $|\mu|$ is the set function defined on $\mathscr{B}(X)$ by

$$|\mu|(B) := \sup\left\{\sum_{i \in \mathbb{N}} |\mu(B_i)| \, : \{B_i\}_{i \in \mathbb{N}} \text{ is a Borel partition of } B\right\}$$

and, for $E \in \mathscr{B}(X)$, the restriction $\mu \llcorner E$ of μ to E is defined by

$$\mu \llcorner E(B) := \mu(E \cap B) \qquad B \in \mathscr{B}(X).$$

Sometimes we write $\chi_E \mu$ to denote $\mu \llcorner E$.

It is not hard to prove (see for instance Theorem 1.6 in [10] for a detailed proof) that $|\mu| \in \mathscr{M}_+(X)$, so that $|\mu|$ is the smallest $\sigma \in \mathscr{M}_+(X)$ satisfying $|\mu(B)| \leq \sigma(B)$ for any Borel set B. Also, using $|\mu|$ we can define the positive and negative parts of $\mu \in \mathscr{M}(X)$ by

$$\mu^+ := \frac{|\mu| + \mu}{2}, \qquad \mu^- := \frac{|\mu| - \mu}{2},$$

so that $\mu^\pm \in \mathscr{M}_+(X)$ and $\mu = \mu^+ - \mu^-$.

Definition 1.2 (Support and Concentration) Given $\mu \in \mathscr{M}(X)$, its support is the closed set defined by

$$\operatorname{supp}\mu := \big\{x \in X : |\mu|(U) > 0 \text{ for all } U \ni x \text{ open}\big\}.$$

We say that μ is concentrated on $A \in \mathscr{B}(X)$ if $|\mu|(X \setminus A) = 0$.

More generally, if A is not a Borel set, we say that μ is concentrated on A if there exists $B \in \mathscr{B}(X)$ contained in A such that $\mu(X \setminus B) = 0$ (an equivalent formulation is that the outer measure of $X \setminus A$ is 0).

By definition $\mu \llcorner E = \chi_E \mu$ is concentrated on E and, if $\mu \in \mathscr{M}_+(X)$, it is the largest measure smaller than μ with this property.

It is easily seen that if (X, d) is separable, then μ is concentrated on $\operatorname{supp}\mu$, actually the smallest closed set where μ is concentrated. On the other hand, a measure μ can be concentrated on a set much smaller than the support, as the following simple example shows.

Example 1.3 The measure $\mu = \sum_1^\infty 2^{-n}\delta_{q_n}$ on $X = \mathbb{R}$, where $\{q_n\}_{n\in\mathbb{N}} = \mathbb{Q}$, is concentrated on $\mathbb{Q}$, whereas its support is the whole $\mathbb{R}$.

Now let us introduce the key notion of Polish space: as we will see, many measure-theoretic properties valid in compact or locally compact spaces extend to this class of spaces.

Definition 1.4 (Polish Space) A topological space (X, τ) is said to be Polish if there exists a distance d on X inducing τ such that (X, d) is complete and separable.

Let us point out that every open subset U of a Polish space is Polish. Indeed a distance $\bar{d}$ on U inducing the subspace topology and such that $(U, \bar{d})$ is complete can be defined as follows. First, whenever d induces the topology τ, one can easily verify that

$$\hat{d}(x, y) := \frac{d(x, y)}{1 + d(x, y)}$$

is a distance on X inducing the same topology and with the additional property $\hat{d} < 1$. Then, if we let

$$\bar{d}(x, y) := \hat{d}(x, y) + \left|\frac{1}{\hat{d}(x, X \setminus U)} - \frac{1}{\hat{d}(y, X \setminus U)}\right|,$$

where $\hat{d}(\cdot, X \setminus U) := \inf\{\hat{d}(\cdot, z) : z \in X \setminus U\}$, it is possible to check that:

(i) $\bar{d}$ is a distance on U;
(ii) $\bar{d}$ induces the subspace topology on U;
(iii) $(U, \bar{d})$ is complete.

Lemma 1.5 (Ulam) *If (X, τ) is a Polish space, for every $\mu \in \mathscr{M}_+(X)$ and for every $\epsilon > 0$ there exists $K \subset X$ compact such that $\mu(X \setminus K) < \epsilon$.*

Proof Let d be a distance satisfying the definition of Polish space for (X, τ) and let $D = \{x_i\}_{i \in \mathbb{N}}$ be a dense subset of X. Then, for every $k \geq 1$, we have that

$$\bigcup_{i=0}^{\infty} \overline{B}(x_i, 2^{-k}) = X.$$

Consequently, since μ is finite, we can choose $n(k) \in \mathbb{N}$ such that

$$\mu\left(X \setminus \bigcup_{i=0}^{n(k)} \overline{B}(x_i, 2^{-k})\right) < \frac{\epsilon}{2^k}$$

and construct the closed and totally bounded set

$$K = \bigcap_{k=1}^{\infty} \bigcup_{i=0}^{n(k)} \overline{B}(x_i, 2^{-k}).$$

By the completeness assumption, K is compact. We conclude by noticing that

$$\mu(X \setminus K) \leq \sum_{k=1}^{\infty} \mu\left(X \setminus \bigcup_{i=0}^{n(k)} \overline{B}(x_i, 2^{-k})\right) < \epsilon.$$

□

Recall that finite Borel measures μ in a metric space always satisfy

$$\mu(B) = \sup\{\mu(C) : C \subset B,\ C \text{ closed}\}.$$

In the Polish setting, thanks to Ulam's theorem, we can improve this inner regularity property by requiring the sets C to be compact.

Now we introduce a basic concept of the theory, the push forward operator between Borel probability measures in metric spaces.

Definition 1.6 (Push Forward Measure) Given a Borel function $f : X \to Y$, we define the push forward operator $f_\# : \mathscr{M}(X) \to \mathscr{M}(Y)$ by

$$f_\#\mu(B) := \mu(f^{-1}(B)) \qquad \forall B \in \mathscr{B}(Y)$$

and call $f_\#\mu$ push forward measure.

Notice that $f_\#(\mathscr{M}(X)) \subset \mathscr{M}(Y)$, $f_\#(\mathscr{P}(X)) \subset \mathscr{P}(Y)$ and that, for $g : Y \to Z$ Borel, it is simple to verify that $(g \circ f)_\# = g_\# \circ f_\#$.

This definition works also at the more general level of measure spaces and measurable maps (i.e. without assuming that the σ-algebras are the Borel ones) and occasionally it will be used also at this level. In the language of Probability, when μ is a probability measure, f can be thought as a Y-valued random variable and then $f_\#\mu$ is the law of f under μ. Since the class

$$\left\{B \in \mathscr{B}(Y) : \mu(f^{-1}(B)) = \nu(B)\right\}$$

is easily seen to be a σ-algebra, the condition $f_\#\mu = \nu$ needs only to be checked on a family of generators of the σ-algebra $\mathscr{B}(Y)$.

We now show how to compute integrals with respect to the push forward measure.

Proposition 1.7 (Change of Variables Formula) *For any Borel function $f : X \to Y$ and any Borel function $\phi : Y \to [0, \infty]$ one has*

$$\int_Y \phi \, \mathrm{d} f_\#\mu = \int_X (\phi \circ f) \, \mathrm{d}\mu. \tag{1.1}$$

It follows that $\psi : X \to \overline{\mathbb{R}}$ is $f_\#\mu$-integrable if and only if $\psi \circ f$ is μ-integrable.

Proof For characteristic functions $f = \chi_B$ the statement corresponds to the definition of push forward measure. By linearity of both sides with respect to ϕ, the statement still holds for nonnegative and simple Borel functions. In the general case there exists a nondecreasing sequence (ϕ_i) of simple functions monotonically converging to ϕ and the conclusion follows by the monotone convergence theorem.

The last part of the statement can be proved splitting ψ in positive and negative parts. □

The change of variables formula also suggests a useful criterion to check the condition $T_\#\mu = \nu$.

Proposition 1.8 *For $T : X \to Y$ Borel, one has $T_\#\mu = \nu$ if and only if*

$$\int_Y \phi \, \mathrm{d}\nu = \int_X (\phi \circ T) \, \mathrm{d}\mu \qquad \forall \phi \in C_b(Y). \tag{1.2}$$

Proof The condition (1.2) is obviously necessary. Its sufficiency can be obtained by noticing that characteristic functions of open sets are monotone nondecreasing limits of functions in $C_b(Y)$. □

It is useful to compare this "soft" change of variables formula with the one usually taught in calculus courses, requiring some regularity of the function f: the main point in Proposition 1.7 is that we do not look for a representation of $f_\#\mu$ in terms of another reference measure ν in Y. In the case $X = Y = \mathbb{R}$ and $\mu = \nu = \mathscr{L}^1$ this is well illustrated by the next example.

Example 1.9 If $X = Y = \mathbb{R}$ and $f \in C^1$ with $f' > 0$, then $f_\# \mathscr{L}^1 = \frac{1}{f' \circ f^{-1}} \mathscr{L}^1$. Indeed, if $E = (c, d)$ and $f^{-1}(E) = (a, b)$, the classical change of variables formula gives

$$\int_c^d \frac{1}{f' \circ f^{-1}(y)} \, \mathrm{d}y = \int_a^b \frac{f'(x)}{f'(x)} \, \mathrm{d}x = (b - a) = \mathscr{L}^1((a, b)).$$

Another nice link betwen the change of variables formula and push-forward is provided by the identity

$$v_\#(Dv) = \mathscr{L}^1 \llcorner [a, b], \tag{1.3}$$

valid for continuous functions of bounded variation $v : [0, 1] \to \mathbb{R}$ with $v(0) = a < b = v(1)$ (see Definition 6.3 for the general definition of function of bounded variation). Here Dv is the derivative in the sense of distributions, representable thanks to the BV property as a signed measure. In order to check it, assume first that v is continuously differentiable and notice that (by differentiation of both sides with respect to s)

$$\int_0^s \phi(v(t)) v'(t) \, \mathrm{d}t = \int_a^{v(s)} \phi(z) \, \mathrm{d}z \qquad \forall s \in [0, 1]$$

for any bounded continuous function ϕ. Setting $s = 1$ gives (1.3), since $Dv = v' \mathscr{L}^1$ in this case. The general case can be obtained by approximating v uniformly by C^1 functions v_h, using also the weak convergence of Dv_h to Dv.

2 Monge's Formulation of the Optimal Transport Problem

Monge's optimal transport problem is concerned with the problem of finding the way to carry a given mass distribution from a place to another in order to minimize a suitable transport cost. The modern formulation of the optimal transport problem has as data probability measures $\mu \in \mathscr{P}(X)$, $\nu \in \mathscr{P}(Y)$ and a Borel cost function $c(x, y) : X \times Y \to [0, \infty]$, representing the cost of shipping a unit of mass from x to y. Given these data, the problem is

$$\inf \left\{ \int_X c(x, T(x)) \, \mathrm{d}\mu(x) : \ T : X \to Y \text{ Borel}, \ T_\# \mu = \nu \right\}. \tag{M}$$

In Monge's original formulation [93], $X = Y$ were Euclidean spaces and $c(x, y) = |x - y|$ so that (under the assumption of proportionality between force and mass) the cost is proportional to the work done. We shall denote by $\mathscr{C}_\mu(T)$ the transport cost $\int_X c(x, T(x)) \, \mathrm{d}\mu(x)$, omitting μ when it is clear from the context.

Before developing the general theory, we illustrate two basic examples of optimal transport problems. Then we will discuss the technical problems related to the minimization (M).

Example 1.10 (Discrete Case) Let us consider

$$\mu = \sum_{i=1}^{N} \frac{1}{N} \delta_{x_i}, \qquad \nu = \sum_{j=1}^{N} \frac{1}{N} \delta_{y_i}$$

probability "empirical" measures on the discrete spaces $X = \{x_1, \ldots, x_N\}$ and $Y = \{y_1, \ldots, y_N\}$ respectively, with cardinality N.

A function $T : X \to Y$ is a bijection if and only if $T_{\#}\mu = \nu$, since

$$T_{\#}\mu = \sum_{i=1}^{N} \frac{1}{N} \delta_{T(x_i)}$$

and, for any choice of the cost c, existence of an optimal transport is obvious, since the class of admissible transports is a finite set.

Now, let us consider a less restrictive way to transport μ to ν. In particular, call π_{ij} the amount of mass sent from x_i to y_j. Since any y_j has to receive a fraction $\frac{1}{N}$ of mass, it must be $\sum_{i=1}^{N} \pi_{ij} = \frac{1}{N}$ for every $j = 1, \ldots, N$. Analogously, since any x_i has to distribute a fraction $\frac{1}{N}$ of mass, one has $\sum_{j=1}^{N} \pi_{ij} = \frac{1}{N}$ for every $i = 1, \ldots, N$. The set S of $N \times N$ matrices $\pi = \{\pi_{ij}\}_{i,\, j=1,\ldots,N}$, with $\pi_{ij} \geq 0$, respecting these conditions is called set of bi-stochastic matrices and it is easily seen to be a compact and convex set. Inside this set, we can single out the set $\tilde{S}$ corresponding to transport maps, namely, adopting in this circumstance the classical notation δ_{ij} for the Kronecker delta,

$$\tilde{S} = \left\{ \pi \in S : \text{there exists } T : X \to Y \text{ such that } \pi_{ij} = \frac{1}{N} \delta_{jT(i)} \right\}.$$

Accordingly, given a cost function $c : X \times Y \to [0, \infty]$, we can define

$$\mathscr{C}(T) := \sum_{i=1}^{N} \frac{1}{N} c(x_i, T(x_i)), \qquad \mathscr{C}(\pi) = \sum_{i,\, j=1}^{N} \frac{1}{N} c(x_i, y_j) \pi_{ij}, \tag{1.4}$$

the costs obtained by a transport map T or a bi-stochastic matrix π in S, so that $\mathscr{C}(T) = \mathscr{C}(\pi)$ whenever π is generated by the transport map T (notice that $\mathscr{C}(T) = \mathscr{C}_{\mu}(T)$). Consequently

$$\min_{T} \mathscr{C}(T) = \min_{\pi \in \tilde{S}} \mathscr{C}(\pi) \geq \min_{\pi \in S} \mathscr{C}(\pi). \tag{1.5}$$

However in this particular case we see that there is no gain from the larger freedom in the transportation process: indeed, we know from finite-dimensional convex analysis that S is the convex hull of its extremal points (namely points in S that cannot be written as a nontrivial convex combination of points in S) and, since $\pi \mapsto \mathscr{C}(\pi)$ is an affine function, the minimum in the right hand side of (1.5) is attained on extremal points. On the other hand, Birkhoff's theorem [21] (see also the introduction of [117] for a strategy of proof) states that extremal points of S correspond precisely to matrices in $\tilde{S}$. Though the proof of Birkhoff's theorem is not hard, the interested reader can also consult [27] for a proof of the equality in (1.5) not based on Birkhoff's theorem.

However, the extra degree of freedom is already crucial when the points x_i are not distinct or the weights are not equal to $1/N$: for instance if $\mu = \delta_0$ and $\nu = \frac{1}{2}(\delta_{-1} + \delta_1)$ simply there is no transport map!

The following theorem provides many properties of the optimal transport on the real line, at least under the assumption that μ has no atom. We shall only prove the existence of the monotone transport map, since uniqueness and optimality will follow by the general theory we are going to develop in the next chapters. In the sequel we shall use the notation

$$F_\mu(x) := \mu((-\infty, x]) \tag{1.6}$$

for the *cumulative distribution function* of a probability measure μ in $\mathbb{R}$.

Theorem 1.11 (Existence, Uniqueness and Optimality) *If $\mu,\ \nu \in \mathscr{P}(\mathbb{R})$ and μ has no atom, then there exists $T : \mathbb{R} \to [-\infty, \infty]$ nondecreasing pushing μ into ν and any other map S with these properties coincides with T on* $\operatorname{supp}\mu$*, with at most countably many exceptions.*

If $c(x, y) = \phi(|y - x|)$ with $\phi : [0, \infty) \to [0, \infty)$ convex and nondecreasing, and if $\mathscr{C}_\mu(T) < \infty$, then T is an optimal map. If ϕ is strictly convex, T is the unique optimal map.

Proof Notice that $F_\mu(x)$ is continuous, since μ has no atom, and $F_\nu(y)$ is right continuous. By looking at the detailed mass balance condition on halflines, one sees that it must be $F_\nu(T(x)) = F_\mu(x)$. If $\operatorname{supp}\nu = \mathbb{R}$, then F_ν is strictly increasing and the function $T := F_\nu^{-1} \circ F_\mu$ does the job.

In general, let us prove that

$$T(x) := \inf\{y \in \operatorname{supp}\nu : F_\nu(y) \geq F_\mu(x)\}$$

is an admissible transport map. Indeed, it is obvious that F is nondecreasing, so let us check the mass balance property on halflines $(-\infty, y]$, i.e. $F_\nu(y) = \mu(\{T \leq y\})$. Now, by monotonicity, $\{T \leq y\}$ contains $(-\infty, x)$ and is contained in $(-\infty, x]$ for some $x \in \mathbb{R}$. All these sets have the same μ-measure, and we need only to prove that $F_\nu(y) = F_\mu(x)$. The inclusion $(-\infty, x) \subset \{T \leq y\}$ gives that for any $x' < x$ and any $\epsilon > 0$ there exists $y_\epsilon < y + \epsilon$ with $F_\nu(y_\epsilon) \geq F_\mu(x')$, so that

$F_\nu(y+\epsilon) \geq F_\nu(y_\epsilon) \geq F_\mu(x')$. Letting $\epsilon \downarrow 0$ and $x' \uparrow x$ gives $F_\nu(y) \geq F_\mu(x)$. On the other hand, the inclusion $\{T \leq y\} \subset (-\infty, x]$ gives that $T(x') > y$ for any $x' > x$, hence $F_\nu(y) < F_\mu(x')$. Letting $x' \downarrow x$ gives $F_\nu(y) \leq F_\mu(x)$. □

Remark 1.12 (Push-Forward of the Cumulative Distribution Function) Another remarkable property linking F_μ to $\mathscr{L}^1$, when μ has no atom, is $(F_\mu)_\# \mu = \mathscr{L}^1 \llcorner (0,1)$. In addition, $F_\mu^{-1} \circ F_\mu(x) = x$ for μ-a.e. $x \in \mathbb{R}$, where

$$t \mapsto F_\mu^{-1}(t) := \sup\left\{x : F_\mu(x) \leq t\right\}$$

is the so-called pseudoinverse of the cumulative distribution function F_μ.

We will see that in higher dimensions, for the cost $c(x, y) = |x-y|^2$, the correct replacement of the monotonicity is given by the condition $\langle T(x) - T(y), x - y\rangle \geq 0$, together with the fact that T is a gradient, in a generalized sense (for general costs the monotonicity condition will be more implicit). Notice that we allowed for $\pm\infty$ valued T's because it may happen that the support of μ is bounded, while the support of ν is not. For instance, if $\sup \operatorname{supp} \nu = +\infty$, because of monotonicity, T is forced to be equal to $+\infty$ on $(\max \operatorname{supp} \mu, \infty)$. However the values of T on this interval really do not matter, because this interval is μ-negligible. Analogously, the case $\mu = \mathscr{L}^1 \llcorner [0,1]$ and $\nu = \frac{1}{2}(\delta_0 + \delta_1)$ shows that T needs not to be unique: $T \equiv 0$ on $(0, 1/2)$ and $T \equiv 1$ on $(1/2, 1)$, but the value of T at $1/2$ can be any number in $[0, 1]$ and, as we said, the values of T out of $[0, 1]$ are irrelevant.

The following classical example shows that uniqueness can fail, when $c(x, y) = \phi(|x-y|)$ and ϕ is not strictly convex.

Example 1.13 (Book Shifting) Given an integer $M \geq 2$, consider $\mu = \frac{1}{M}\mathscr{L}^1 \llcorner [0, M]$, $\nu = \frac{1}{M}\mathscr{L}^1 \llcorner [1, M+1]$ and the cost $c(x, y) = |y-x|^p$ with $0 < p < \infty$. Consider the admissible transport maps

$$T_1(x) := x + 1, \qquad T_2(x) := \begin{cases} x + M, & \text{if } 0 \leq x \leq 1 \\ x, & \text{otherwise} \end{cases}.$$

Then, Theorem 1.11 gives that T_1 is optimal for $p \geq 1$, while it can be proved that T_2 is optimal for $0 < p \leq 1$.

Let us just prove the optimality of both maps for $p = 1$: obviously $\mathscr{C}(T_1) = \mathscr{C}(T_2) = 1$, moreover for every admissible T we have

$$\begin{aligned} \mathscr{C}(T) &= \int_{\mathbb{R}} |T(x) - x| \, \mathrm{d}\mu(x) \geq \int_{\mathbb{R}} T(x) \, \mathrm{d}\mu(x) - \int_{\mathbb{R}} x \, \mathrm{d}\mu(x) \\ &= \int_{\mathbb{R}} y \, \mathrm{d}\nu(y) - \int_{\mathbb{R}} x \, \mathrm{d}\mu(x) = 1. \end{aligned}$$

For general cost functions $c(x, y)$, we will see that a strategy similar to the one of the previous example can be exploited, provided one builds appropriate functions $\phi(x)$, $\psi(y)$ with $\phi(x) + \psi(y) \le c(x, y)$.

In order to attack Monge's minimization problem it would be desirable to have a topology (or at least a notion of sequential convergence) making the class of admissible transport maps sequentially closed and compact, guaranteeing also the lower semicontinuity of $T \mapsto \mathscr{C}(T)$. Unfortunately no such topology exists, as the non existence example for transport maps Example 1.14 shows.

Let us give a simple proof of the fact that neither the strong L^p topology nor the weak one are suitable for our purposes. Indeed the class of transport maps is not compact in the strong L^p topology and it is not closed in the weak one.

Let us consider the case $Y = \mathbb{R}$ and assume that ν has finite p-th moment for some $p \ge 1$, namely $\int_{\mathbb{R}} |y|^p \, \mathrm{d}\nu(y) < \infty$. Then the set of admissible transport maps is contained in $L^p(X, \mu)$ (as a consequence of the change of variables formula) and it is a closed subset. Indeed, if T_i are admissible transport maps convergent in L^p to T, we can assume without loss of generality that $T_i \to T$ μ-a.e. in X and pass to the limit as $i \to \infty$ in

$$\int_X \phi(T_i(x)) \, \mathrm{d}\mu(x) = \int_{\mathbb{R}} \phi(y) \, \mathrm{d}\nu(y) \qquad \phi \in C_b(\mathbb{R})$$

to obtain that T satisfies the same identity. By Proposition 1.8 we obtain that $T_\# \mu = \nu$. By Fatou's lemma, a similar argument gives also the lower semicontinuity of $T \mapsto \mathscr{C}(T)$ with respect to the strong L^p convergence, if $c(x, \cdot)$ is lower semicontinuous for μ-a.e. x.

Unfortunately, in general the class of admissible transport maps is not closed in the weak topology of L^p, as the following simple example shows: consider $\mu = \mathscr{L}^1 \llcorner [0, 1]$, $\nu = \frac{1}{2}(\delta_1 + \delta_{-1})$ and let $f : \mathbb{R} \to \mathbb{R}$ be the 1-periodic function such that

$$f(x) = \begin{cases} 1 & \text{if } 0 \le x < \frac{1}{2} \\ -1 & \text{if } \frac{1}{2} \le x < 1. \end{cases}$$

The functions $T_h(x) = f(hx)$ are easily seen to be transport maps for all $h \in \mathbb{N} \setminus \{0\}$, but $T_h \overset{*}{\rightharpoonup} T := 0$ (and therefore weakly in all $L^p(\mu)$, $1 \le p < \infty$) as $h \to \infty$ and $T_\# \mu = \delta_0 \ne \nu$.

Finally, with the following example, we want to see that in general the infimum in Monge's formulation is not a minimum. It also shows, in view of the previous remarks, that no compactness property with respect to strong L^p topologies can be expected.

Example 1.14 Let us consider $\mu = \mathscr{H}^1 \llcorner \{0\} \times [0, 1] \in \mathscr{P}(\mathbb{R}^2)$, where $\mathscr{H}^1$ denotes the one-dimensional Hausdorff measure in $\mathbb{R}^2$. In more elementary terms, μ is uniformly distributed on the vertical segment $\{0\} \times [0, 1]$ and it can be represented as the push forward of $\mathscr{L}^1 \llcorner [0, 1]$ under the map $t \mapsto (0, t)$. Analogously, let $\nu = \frac{1}{2}\mathscr{H}^1 \llcorner \{-1\} \times [0, 1] + \frac{1}{2}\mathscr{H}^1 \llcorner \{1\} \times [0, 1]$ and take $|y - x|$ as cost.

It is easy to prove that 1 is the infimum of Monge's problem. The lower bound is trivial since $|x - y| \geq 1$ for all $x \in \operatorname{supp}\mu$ and $y \in \operatorname{supp}\nu$. The upper bound can be obtained dividing the segment $\{0\} \times [0, 1]$ in $2N$ equal pieces, the intervals $\{\pm 1\} \times [0, 1]$ in N equal pieces and mapping (linearly) the $(2i + 1)$-th piece of $\{0\} \times [0, 1]$ to the $(i+1)$-th piece of $\{-1\} \times [0, 1]$, the $(2i+2)$-th piece of $\{0\} \times [0, 1]$ to the $(i+1)$-th piece of $\{1\} \times [0, 1]$, $i = 0, \ldots, N-1$. Denoting by T_N the resulting map, it is easy to check that $\mathscr{C}(T_N) \leq 1 + O(1/N)$.

Let us prove that no optimal map T exists. If, by contradiction

$$\int_{\mathbb{R}^2} (|T(x) - x| - 1)\, \mathrm{d}\mu(x) = 0$$

the nonnegativity of the integrand gives that $|T(x) - x| = 1$ for μ-a.e. x. Hence, for $\mathscr{L}^1$-a.e. $t \in [0, 1]$ either $T((0, t)) = (1, t)$ or $T((0, t)) = (-1, t)$. Denoting by $A_\pm \subset [0, 1]$ the sets of points t where the two possibilities occur, one has

$$T_\# \mu = \mathscr{H}^1 \llcorner \{-1\} \times A_- + \mathscr{H}^1 \llcorner \{1\} \times A_+,$$

so that $T_\# \mu \neq \nu$, a contradiction.

Lecture 2: The Kantorovich Problem



1 Kantorovich's Formulation of the Optimal Transport Problem

We can now introduce Kantorovich's formulation of the optimal transport problem. It involves the concept of *transport plan* (also called coupling in the Probability literature) between probability measures. In the discrete setting of Example 1.10, transport plans correspond to bi-stochastic matrices.

Definition 2.1 (Transport Plans) Given $\mu \in \mathscr{P}(X)$ and $\nu \in \mathscr{P}(Y)$, define

$$\Gamma(\mu, \nu) := \{\pi \in \mathscr{P}(X \times Y) : \pi(A \times Y) = \mu(A),\ \pi(X \times B) = \nu(B) \quad \forall \text{ Borel } A,\ B\}.$$

Denoting by $p_X : X \times Y \to X$, $p_Y : X \times Y \to Y$ the coordinate projections, the property of being a transport plan is equivalent to

$$(p_X)_{\#}\pi = \mu, \qquad (p_Y)_{\#}\pi = \nu,$$

since $p_X^{-1}(A) = A \times X$ and $p_Y^{-1}(B) = X \times B$. The class of couplings is always not empty, since $\mu \times \nu$ is a coupling between μ and ν.

Similarly to transport maps, transport plans represent a way to carry mass from X to Y. In particular, $\pi(A \times B)$ is the mass initially in A sent to B (recall also that Borel probability measures in $X \times Y$ are uniquely determined by their value on Borel cartesian products).

Kantorovich's formulation of the optimal transport problem asks to find

$$\inf \left\{ \int_{X \times Y} c(x, y)\, \mathrm{d}\pi(x, y) : \pi \in \Gamma(\mu, \nu) \right\}. \tag{K}$$

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We will denote by

$$\mathscr{C}(\pi) := \int_{X\times Y} c(x, y)\, \mathrm{d}\pi(x, y)$$

the cost involved in (K) and by $\Gamma_o(\mu, \nu)$ the (possibly empty) set of transport plans $\pi \in \Gamma(\mu, \nu)$ which attain the minimum in (K).

2 Transport Plans Versus Transport Maps

Given a transport map T, we can define the transport plan $\pi_T := (\mathrm{id} \times T)_{\#}\mu$, where $\mathrm{id} \times T : X \to X \times Y$ is the map $x \mapsto (x, T(x))$. By the change of variables formula (1.1) we obtain immediately that

$$\mathscr{C}(\pi_T) = \int_X c(x, T(x))\, \mathrm{d}\mu(x) = \mathscr{C}(T).$$

Thus $\inf_{(M)} \geq \inf_{(K)}$, i.e. the infimum in Monge's formulation is greater than or equal to the infimum in Kantorovich's one. Obviously the inequality can be strict, simply because the class of transport plans can be empty due to atoms in the support of μ, for instance when μ is a Dirac mass and ν is not a Dirac mass. It is a classical result in Probability (see [98]) that, for Polish spaces, the presence of atoms is essentially the unique restriction, namely transport maps from μ to ν exist whenever μ has no atom. Still in the Polish setting, the following refinement [100], due to A. Pratelli, of this classical result does not play a role in the sequel, but it is worth being pointed out.

Theorem 2.2 (Pratelli) *If μ has no atom and $c : X \times Y \to [0, \infty)$ is continuous, then*

$$\inf_{(M)} = \min_{(K)},$$

where the infimum in Kantorovich's formulation is a minimum thanks to Theorem 2.10 below.

Transport plans reveal to be a more flexible tool, compared to transport maps. For instance, if we have transport maps T_1 and T_2 from μ to ν, then we can "combine" them to obtain

$$\pi = \frac{1}{2}(\mathrm{id} \times T_1)_{\#}\mu + \frac{1}{2}(\mathrm{id} \times T_2)_{\#}\mu \in \Gamma(\mu, \nu).$$

The simple construction above will play an important role in some proofs of uniqueness of transport maps. Also, given a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ and

measurable maps $S : \Omega \to X$, $T : \Omega \to Y$, it will be useful to rely on the implication

$$S_{\#}\mathbb{P} = \mu, \quad T_{\#}\mathbb{P} = \nu \quad \Longrightarrow \quad (S, T)_{\#}\mathbb{P} \in \Gamma(\mu, \nu), \tag{2.1}$$

which is a simple consequence of the associativity of the push-forward operator.

Now we would like to reverse the implication from transports to plans, providing a sufficient (and necessary) condition ensuring that $\pi = \pi_T$ for some T.

Theorem 2.3 *Let X, Y be Polish spaces. If $\Gamma \subset X \times Y$ is a π-measurable graph and π is concentrated on Γ, then there exists a Borel transport map T such that $\pi = (\mathrm{id} \times T)_{\#}\mu$.*

Proof Since π is concentrated on Γ there exists a Borel set $\Gamma_1 \subset \Gamma$ such that $\pi(X \times Y \setminus \Gamma_1) = 0$. Hence, thanks to the inner regularity of π, there exists a nondecreasing sequence (K_k) of compact subsets of Γ_1 such that $\pi(X \times Y \setminus \cup_k K_k) = 0$. Call $C_k = p_X(K_k)$ and define $T_k : C_k \to Y$ such that $(x, T_k(x)) \in \Gamma$ for every $x \in C_k$. Since the graph K_k of T_k is compact, T_k is continuous and furthermore $T_k|_{C_\ell} = T_\ell$ whenever $\ell \leq k$.
Since $\mu(X \setminus C_k) = \pi(p_X^{-1}(X \setminus C_k)) \leq \pi(X \times Y \setminus K_k)$, the set $X \setminus \cup_k C_k$ is μ-negligible. Thus choosing $y_0 \in Y$ and defining

$$T(x) = \begin{cases} T_k(x), & \text{if } x \in C_k \\ y_0, & \text{if } x \in X \setminus \bigcup_k C_k, \end{cases}$$

we obtain that T is Borel and well defined. To conclude the proof, it suffices to check that

$$(Id \times T)_{\#}\mu \llcorner C_k = \pi \llcorner K_k \qquad \forall k \in \mathbb{N}$$

and to let $k \to \infty$. To see this, it suffices to integrate a nonnegative Borel ϕ with respect to both measures to get

$$\int_{X \times Y} \phi \, \mathrm{d}(Id \times T)_{\#}\mu \llcorner C_k = \int_{C_k} \phi(x, T(x)) \, \mathrm{d}\mu(x)$$

and

$$\int_{X \times Y} \phi \, \mathrm{d}\pi \llcorner K_k = \int_{K_k} \phi(x, y) \, \mathrm{d}\pi(x, y) = \int_{C_k} \phi(x, T(x)) \, \mathrm{d}\mu(x),$$

where we used that $\chi_{K_k}(x, y)\phi(x, y) = \chi_{C_k}(x)\phi(x, T(x))$. □

We are now going to see that transport plans can still be seen as transport maps, understanding the map to be measure-valued. This interpretation can be seen using

the disintegration theorem (see for instance [110] for the proof in Polish spaces, [10] in locally compact and separable spaces), stated below.

Theorem 2.4 (Disintegration) *Let Z, X be Polish spaces, $\sigma \in \mathscr{M}_+(Z)$, $f : Z \to X$ a Borel function and set $\theta := f_\# \sigma \in \mathscr{M}_+(X)$. Then there exists a family $\{\sigma_x\}_{x\in X} \subset \mathscr{P}(Z)$ such that*

(i) $x \mapsto \sigma_x$ is Borel, i.e. $\sigma_x(A)$ is Borel for all $A \in \mathscr{B}(Z)$;
(ii) $\sigma = \int_X \sigma_x \, \mathrm{d}\theta$, i.e. $\sigma(A) = \int_X \sigma_x(A) \, \mathrm{d}\theta(x)$ for every $A \in \mathscr{B}(Z)$;
(iii) σ_x is concentrated on $f^{-1}(x)$ for θ-a.e. $x \in X$.

Any other family $\{\sigma'_x\}_{x\in X} \subset \mathscr{P}(Z)$ with these properties satisfies $\sigma'_x = \sigma_x$ for θ-a.e. $x \in X$.

In the language of Probability, σ_x are the conditional probabilities, namely $\sigma_x = \mathrm{E}[\sigma|\{f = x\}]$. Strictly speaking, the measurability of $x \mapsto \sigma_x$ we introduced should be called *weak* Borel property*, since it does not refer to a σ-algebra in the set $\mathscr{P}(Y)$. By standard measure-theoretic arguments, it implies the Borel property of $x \mapsto \int_Y \phi(x, y) \, \mathrm{d}\sigma_x(y)$ for any nonnegative Borel function ϕ in $X \times Y$.

We will often use the disintegration theorem when $Z = X \times Y$ and f is the projection on one of the factors, say $f = p_X$. In this case σ_x are supported on $f^{-1}(x) = \{x\} \times Y$ and can also be viewed as probability measures on Y. When we adopt this viewpoint, instead of the "superposition" notation $\sigma = \int_X \sigma_x \, \mathrm{d}\theta$ we shall use the "Fubini" notation $\sigma = \sigma_x \otimes \theta$, or even

$$\sigma(\mathrm{d}x, \mathrm{d}y) = \sigma_x(\mathrm{d}y)\theta(\mathrm{d}x),$$

when we want to emphasize the role of the variables in the process of integration. This notation is also motivated by the formula

$$\int_{X\times Y} \phi \, \mathrm{d}\sigma = \int_X \left(\int_Y \phi(x, y) \, \mathrm{d}\sigma_x(y) \right) \mathrm{d}\theta(x)$$

for integration with respect to σ.

It follows that any $\pi \in \Gamma(\mu, \nu)$ can be represented as a *measure-valued* transport map, namely the map $x \mapsto \pi_x$.

3 Advantages of Kantorovich's Formulation

Let us list a few consequences of Kantorovich's formulation.

(i) The class of transport plans is not empty, since $\mu \times \nu \in \Gamma(\mu, \nu)$, and convex. In addition $\pi \mapsto \mathscr{C}(\pi)$ is an affine map, so that Γ_o is convex as well. Notice that none of these properties holds at the level of transport maps.

(ii) The formulation in terms of transport plans is more symmetric, compared to the one in terms of transport maps. To see this, notice that a transport map from μ to ν does not induce a transport map from ν to μ, unless T is invertible. On the contrary, the switching map $(x, y) \mapsto (y, x)$ induces a mapping from $\Gamma(\mu, \nu)$ to $\Gamma(\nu, \mu)$, mapping optimal plans relative to c to optimal plans relative to $\tilde{c}(y, x) = c(x, y)$.

(iii) We will see that existence of minimizers is ensured under a very mild regularity condition on the cost.

(iv) Kantorovich's formulation allows many important extensions of the optimal transport problem, let us see three of them, that have been recently considered in the literature, also in combination.

(a) *Martingale optimal transport.* In the transportation problem we might add additional constraints, besides the obvious ones coming from the pairs (x, y) such that $c(x, y) = \infty$. For instance, in Mathematical Finance it is natural to add the *martingale constraint*: in the language of disintegrations, if $X = Y = \mathbb{R}^m$, this amounts to the condition that $\int_Y y \, d\pi_x(y) = x$ for μ-a.e. x. In Probability terms, this corresponds to the condition $\mathrm{E}[y|x] = x$. Then, for every martingale transport π from μ to ν, Jensen's inequality gives

$$\int_Y \phi(y) \, d\nu(y) = \int_{X\times Y} \phi(y) \, d\pi(y) = \int_X \left(\int_Y \phi(y) \, d\pi_x(y) \right) d\mu(x)$$
$$\geq \int_X \phi(x) \, d\mu(x) \tag{2.2}$$

for any convex and lower semicontinuous function $\phi : Y \to [0, \infty]$. A beautiful theorem of Strassen [109] ensures that the validity of (2.2) for any convex and lower semicontinuous function $\phi : Y \to [0, \infty]$ is also sufficient for the existence of a martingale transport.

(b) *Multi-marginal optimal transportation*. We can consider more than two factors, in this case we are considering the so-called multi-marginal transportation. Assume for simplicity that we have three factors X_1, X_2, X_3 and a cost function $c : X_1 \times X_2 \times X_3 \to [0, \infty]$. We might consider different marginal conditions: for instance, given $\mu_i \in \mathscr{P}(X_i)$, $i = 1, 2, 3$ we can minimize

$$\int_{X_1\times X_2\times X_3} c(x_1, x_2, x_3) \, d\pi(x_1, x_2, x_3)$$

among all $\pi \in \mathscr{P}(X_1 \times X_2 \times X_3)$ whose projections on the factors are μ_i. Particularly studied in the last years (in connection with the Kohn-Sham

theory [76]) is the case when $X_i = \mathbb{R}^3$, $\mu_i = \varrho \mathscr{L}^n$ (representing the electronic density) and the cost is the Coulomb cost:

$$\frac{1}{|x_1 - x_2|} + \frac{1}{|x_2 - x_3|} + \frac{1}{|x_3 - x_1|}.$$

Instead, we might consider $\pi^{12} \in \mathscr{P}(X_1 \times X_2)$ and $\pi^{23} \in \mathscr{P}(X_2 \times X_3)$ and look for all $\pi \in \mathscr{P}(X_1 \times X_2 \times X_3)$ whose projections on $X_1 \times X_2$ and $X_2 \times X_3$ are π^{12} and π^{23} respectively. A necessary condition for the existence of at least one π is that π^{12} and π^{23} should have the same marginal on X_2. We will see that Dudley's Lemma 8.4 shows that this condition is also sufficient.

(c) *Weak transport problem*. In this setting, the cost $\mathbf{c}$ is "nonlocal", namely it is defined on $X \times \mathscr{P}(Y)$, so that we minimize

$$\int_X \mathbf{c}(x, \pi_x)\, \mathrm{d}\mu(x)$$

among all $\pi = \pi_x \otimes \mu \in \Gamma(\mu, \nu)$. Notice that in the case $\mathbf{c}(x, \sigma) = \int_Y c(x, y)\, \mathrm{d}\sigma(y)$ we recover the standard Kantorovich version of the problem. We refer to [4, 69] for more details on this problem.

4 Existence of Optimal Plans

Let us prove existence of the minimum in the Kantorovich formulation for lower semicontinuous costs. In view of the application of the so-called direct method of Calculus of Variations, we need a topology (or at least a notion of sequential convergence) on $\mathscr{P}(X \times Y)$ ensuring at the same time lower semicontinuity of $\pi \mapsto \mathscr{C}(\pi)$ as well as compactness of $\Gamma(\mu, \nu)$.

Definition 2.5 (Weak Topology on $\mathscr{M}(Z)$) In a topological space (Z, τ), we consider the weak topology on $\mathscr{M}(Z)$, namely the topology induced by duality with $C_b(Z)$. Hence, a fundamental system of neighborhoods of $\mu \in \mathscr{M}(Z)$ is given by finite intersections of the sets

$$\left\{ \nu \in \mathscr{M}(Z) : \left| \int_Z \phi\, \mathrm{d}\nu - \int_Z \phi\, \mathrm{d}\mu \right| < \epsilon \right\}$$

on varying of $\phi \in C_b(Z)$ and $\epsilon > 0$.

If (Z, d) is a compact metric space, we know from Riesz's theorem that $\mathscr{M}(Z)$ is canonically isometric to the dual of $C(Z)$, if we endow $\mathscr{M}(Z)$ with the norm $\|\mu\| = |\mu|(Z)$. It follows that the weak topology is nothing but the weak* topology. Under this assumption, since $\mathscr{P}(Z)$ is a closed and bounded subset of $\mathscr{M}(Z)$, from the Banach-Alaoglu-Bourbaki theorem it follows that $\mathscr{P}(Z)$ is compact.

More generally it can be proved (see for instance Remark 5.1.1 in [12]) that, endowed with the weak topology, $\mathscr{P}(Z)$ is a Polish space whenever (Z, τ) is Polish. These are instances of a general phenomenon, we will see in Lectures 8 and 9 that many properties of the base space (as compactness, completeness,...) can be "lifted" from Z to $\mathscr{P}(Z)$.

We now show that the map $\pi \mapsto \mathscr{C}(\pi)$ is lower semicontinuous if the cost is lower semicontinuous, and that $\Gamma(\mu, \nu)$ is compact with respect to the weak topology on $\mathscr{P}(X \times Y)$. It will then follow that there exists the minimum in Kantorovich's formulation.

Theorem 2.6 (Lower Semicontinuity of $\mathscr{C}(\pi)$) *If $c : X \times Y \to [0, \infty]$ is lower semicontinuous, then $\pi \mapsto \mathscr{C}(\pi)$ is lower semicontinuous in $\mathscr{P}(X \times Y)$ with respect to the weak topology.*

Proof For every $k \in \mathbb{N}$, let us define

$$c_k(x, y) := \inf_{x' \in X,\ y' \in Y} \{c(x', y') \wedge k + k d_X(x, x') + k d_Y(y, y')\},$$

which obviously satisfy $0 \le c_k \le c_{k+1} \le c \wedge k$. In addition, since c_k is the pointwise infimum of a family (indexed by (x', y')) of equi-Lipschitz functions in $X \times Y$, one has $c_k \in \mathrm{Lip}_b(X \times Y)$.

The lower semicontinuity of c ensures that $c_k \uparrow c$ as $k \to \infty$ (notice that lower semicontinuity is also a necessary condition). Indeed, fixing $(x, y) \in X \times Y$ and assuming with no loss of generality that $\sup_k c_k(x, y)$ is finite, let for any $k \in \mathbb{N}^*$ points (x'_k, y'_k) with

$$c(x'_k, y'_k) \wedge k + k d_X(x, x'_k) + k d_Y(y, y'_k) \le c_k(x, y) + \frac{1}{k}.$$

From this inequality we immediately get that $x'_k \to x$ and $y'_k \to y$ as $k \to \infty$. In addition, from $c(x'_k, y'_k) \wedge k \le c_k(x, y) + 1/k$ and the lower semicontinuity of c it follows that $c(x, y) \le \sup_k c_k(x, y)$, as claimed.

Now, if π_i weakly converge to π in $\mathscr{P}(X \times Y)$, for any $k \in \mathbb{N}$ one has

$$\liminf_{i \to \infty} \mathscr{C}(\pi_i) \ge \liminf_{i \to \infty} \int_{X \times Y} c_k \, \mathrm{d}\pi_i = \int_{X \times Y} c_k \, \mathrm{d}\pi.$$

Finally, by taking the limit as $k \to \infty$, the monotone convergence theorem gives

$$\liminf_{i \to \infty} \mathscr{C}(\pi_i) \ge \sup_{k \in \mathbb{N}} \int_{X \times Y} c_k \, \mathrm{d}\pi = \mathscr{C}(\pi).$$

□

Remark 2.7 (Semicontinuity Properties with Respect to Weak Convergence) The monotonicity argument we just used proves that the evaluation on open sets is lower semicontinuous with respect to the weak convergence. The evaluation on closed sets instead is upper semicontinuous, namely

$$\liminf_{n\to\infty} \mu_n(A) \geq \mu(A), \qquad \limsup_{n\to\infty} \mu_n(C) \leq \mu(C)$$

whenever $A \subset X$ is open, $C \subset X$ is closed and $\mu_n \to \mu$ weakly in $\mathscr{P}(X)$. Also, we have proved that

$$\liminf_{n\to\infty} \int_X f \, \mathrm{d}\mu_n \geq \int_X f \, \mathrm{d}\mu$$

whenever $\mu_n \to \mu$ weakly and $f : X \to [0, \infty]$ is lower semicontinuous.

Now we state a basic compactness criterion, based on the uniform validity of Ulam's property stated in Lemma 1.5, and called *equi-tightness*.

Theorem 2.8 (Prokhorov) *Let (Z, d) be a Polish space and let $\mathcal{F} \subset \mathscr{M}_+(Z)$ with $\sup_{\mu\in\mathcal{F}} \mu(X) < \infty$. Then $\mathcal{F}$ is relatively compact with respect to the weak topology if and only if $\mathcal{F}$ is equi-tight, i.e. for every $\epsilon > 0$ there exists $K \subset Z$ compact such that $\mu(Z \setminus K) < \epsilon$ for every $\mu \in \mathcal{F}$.*

Proof Possibly normalizing the family $\mathcal{F}$ and using the compactness of closed intervals of $\mathbb{R}$ we can assume with no loss of generality that $\mathcal{F} \subset \mathscr{P}(Z)$.

Let us begin by proving that if $\mathcal{F}$ is equi-tight then it is relatively compact with respect to the weak topology. By the assumption we can find a nondecreasing family of compact sets K_k such that

$$\omega_k := \sup_{\mu\in\mathcal{F}} (Z \setminus K_k) \to 0 \quad \text{as } k \to \infty.$$

Since K_k are compact, given any sequence $(\mu_n) \subset \mathcal{F}$, a diagonal argument ensures the existence of a subsequence $n(p)$ such that $\mu_{n(p)} \llcorner K_k$, viewed as measures in $\mathscr{M}_+(K_k)$, weakly converge to some measure ν_k in $\mathscr{M}_+(K_k)$. Then, viewing now ν_k as a measure in $\mathscr{M}_+(Z)$ with support in K_k, $\mu_{n(p)} \llcorner K_k$ weakly converge to ν_k in $\mathscr{M}_+(Z)$ and in particular one has $\nu_k \leq \nu_{k+1}$. Moreover, since $1 - \omega_k \leq \mu_{n(p)} \llcorner K_k(Z) \leq 1$, we obtain that $1 - \omega_k \leq \nu_k(Z) \leq 1$, therefore

$$\nu(B) := \sup_{k\in\mathbb{N}} \nu_k(B) \quad \text{belongs to } \mathscr{P}(Z).$$

Notice that, because of monotonicity, ν is additive on disjoint sets and σ-subadditive, hence σ-additive.

Finally, for all $\phi \in C_b(Z)$ one has

$$\left|\int_Z \phi \,\mathrm{d}(\mu_{n(p)} - \nu)\right| \leq \left|\int_Z \phi \,\mathrm{d}\mu_{n(p)} - \int_Z \phi \,\mathrm{d}\mu_{n(p)} \llcorner K_k\right| + \left|\int_Z \phi \,\mathrm{d}\mu_{n(p)} \llcorner K_k - \int_Z \phi \,\mathrm{d}\nu_k\right| + \left|\int_Z \phi \,\mathrm{d}\nu_k - \int_Z \phi \,\mathrm{d}\nu\right|.$$

Consequently, taking $p \to \infty$, we have

$$\limsup_{p\to\infty} \left|\int_Z \phi \,\mathrm{d}(\mu_{n(p)} - \nu)\right| \leq 2 \sup|\phi|\, \omega_k,$$

which leads to the conclusion as $k \to \infty$.

Let us now prove the converse implication. Fix $\epsilon > 0$ and a dense and countable sequence $(x_i) \subset Z$. It is enough to prove that for any $j \in \mathbb{N}$ there exists $k_j \in \mathbb{N}$ such that

$$\mu(Z \setminus \bigcup_{i=1}^{k_j} B(x_i, 1/j)) \leq 2^{-j}\epsilon \quad \text{for any } \mu \in \mathcal{F}. \tag{2.3}$$

Indeed (2.3) implies that $\sup_{\mu\in\mathcal{F}} \mu(Z \setminus K) \leq \epsilon$ where

$$K := \bigcap_{j=1}^{\infty} \bigcup_{i=1}^{k_j} \overline{B}(x_i, 1/j)$$

is a compact set.

Let us now prove (2.3) arguing by contradiction. If the conclusion is false we can find a positive integer j_0 such that for any $k \in \mathbb{N}$ there exists $\mu_k \in \mathcal{F}$ satisfying

$$\mu_k\big(Z \setminus \bigcup_{i=0}^{k} B(x_i, \frac{1}{j_0})\big) > 2^{-j_0}\epsilon.$$

By compactness, we can find a subsequence $(\mu_{k(n)})$ converging weakly to $\mu \in \mathscr{M}_+(Z)$. From Remark 2.7 we deduce that for any $k \in \mathbb{N}$ it holds

$$\mu\big(Z \setminus \bigcup_{i=0}^{k} B(x_i, \frac{1}{j_0})\big) \geq \limsup_{n\to\infty} \mu_{k(n)}\big(Z \setminus \bigcup_{i=0}^{k} B(x_i, \frac{1}{j_0})\big) \geq 2^{-j_0}\epsilon,$$

that leads to a contradiction as $k \to \infty$. □

Corollary 2.9 (Compactness of $\Gamma(\mu, \nu)$) *Let X, Y be Polish spaces, $\mu \in \mathscr{P}(X)$, $\nu \in \mathscr{P}(Y)$. Then the set $\Gamma(\mu, \nu)$ is compact with respect to the weak topology.*

Proof Writing the marginal condition in the form

$$\int_X \phi \, \mathrm{d}\mu = \int_{X\times Y} \phi \, \mathrm{d}\pi \quad \forall \phi \in C_b(X), \qquad \int_Y \psi \, \mathrm{d}\nu = \int_{X\times Y} \psi \, \mathrm{d}\pi \quad \forall \psi \in C_b(Y) \tag{2.4}$$

it is immediately seen that $\Gamma(\mu, \nu)$ is closed. Thanks to Prokhorov theorem, it is sufficient to prove that $\Gamma(\mu, \nu) \subset \mathscr{P}(X \times Y)$ is equi-tight. Thanks to Ulam's lemma, for every $\epsilon > 0$ there exist $K \subset X$ and $\tilde{K} \subset Y$ compact sets such that $\mu(X \setminus K) < \epsilon/2$ and $\nu(Y \setminus \tilde{K}) < \epsilon/2$. Thus

$$\pi(X \times Y \setminus K \times \tilde{K}) \leq \pi\big((X \setminus K) \times Y\big) + \pi\big(X \times (Y \setminus \tilde{K})\big) < \epsilon$$

and this proves the equi-tightness. □

Theorem 2.10 *Let X, Y be Polish spaces and let $c : X \times Y \to [0, \infty]$ be lower semicontinuous. Then the minimum in* (K) *is attained.*

Proof It is a direct corollary of Theorem 2.6 and Corollary 2.9, thanks to the existence of the minimum of lower semicontinuous functions on compact sets. □

Lecture 3: The Kantorovich–Rubinstein Duality



This lecture is devoted to the proof of the most basic result of the theory of Optimal Transport, namely the Kantorovich–Rubinstein duality. We assume as usual that X and Y are Polish spaces with $\mu \in \mathscr{P}(X)$ and $\nu \in \mathscr{P}(Y)$.

Theorem 3.1 (Kantorovich–Rubinstein Duality) *If $c : X \times Y \to [0, \infty]$ is lower semicontinuous then*

$$\min_{(K)} = \sup_{(\phi,\psi)\in I_c} \left\{ \int_X \phi(x)\, \mathrm{d}\mu(x) + \int_Y \psi(y)\, \mathrm{d}\nu(y) \right\} \tag{3.1}$$

where

$$I_c := \left\{ (\phi, \psi) \in \mathrm{Lip}_b(X) \times \mathrm{Lip}_b(Y) : \phi(x) + \psi(y) \le c(x, y) \right\}.$$

For $(\phi, \psi) \in I_c$ we shall call *contact set* the subset of $X \times Y$ defined by

$$\Gamma := \{(x, y) : c(x, y) = \phi(x) + \psi(y)\}. \tag{3.2}$$

Remark 3.2 (A Suggestive Interpretation of the Duality Formula) Imagine that X is a set of bakeries and Y is a set of cafes, then the problem in the Kantorovich formulation corresponds to minimize costs of a consortium between bakeries and cafes. Now suppose that there is a transport company which buys at price $\phi(x)$ from the bakery $x \in X$ and sells at price $\psi(y)$ to the cafe $y \in Y$. To be competitive with the direct agreement between bakeries and cafes, it must hold $\psi(y) - \phi(x) \le c(x, y)$. Then the profit is $\int_Y \psi\, \mathrm{d}\nu - \int_X \phi\, \mathrm{d}\mu$, which corresponds with the dual formulation of Theorem 3.1 above, up to the change of sign of ϕ. The duality formula then says that if the transport company has lower transportation costs c', compared to the cost c of the consortium, then it is possible for it to design a set of prices (ϕ, ψ) with a positive gain $\int_{X\times Y} (\psi - \phi - c')\, \mathrm{d}\pi$.

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It is easy to see that $\min_{(\mathrm{K})} \geq \sup$. Indeed for every $(\phi, \psi) \in I_c$ and $\pi \in \Gamma(\mu, \nu)$ it holds

$$\int_{X\times Y} c(x, y)\, \mathrm{d}\pi(x, y) \geq \int_{X\times Y} (\phi(x) + \psi(y))\, \mathrm{d}\pi(x, y) = \int_X \phi\, \mathrm{d}\mu + \int_Y \psi\, \mathrm{d}\nu. \tag{3.3}$$

In general, the supremum of the dual formulation is not a maximum. To have more chances to achieve the maximum, one may try to maximize on pairs (ϕ, ψ) with $\phi \in L^1(X, \mu)$ and $\psi \in L^1(X, \nu)$, since the argument in (3.3) works also with this more general choice of pairs. We are going to see that, whenever a maximizing pair exists, one can derive a lot of information on the structure of the optimal transport problem because π-a.e. it holds $\phi(x) + \psi(y) = c(x, y)$, i.e. π is concentrated on the contact set.

A *formal* proof of the duality formula can be obtained by writing the marginal constraint as a maximization problem, passing from $\mathscr{P}(X \times Y)$ to $\mathscr{M}_+(X \times Y)$ and then formally exchanging supremum and infimum:

$$\begin{aligned}
&\min_{(\mathrm{K})} \\
&= \inf_{\pi\in\mathscr{M}_+(X\times Y)} \sup_{\phi\in C_b(X),\, \psi\in C_b(Y)} \int_{X\times Y} c\, \mathrm{d}\pi + \int_X \phi\, \mathrm{d}\mu - \int_{X\times Y} \phi\, \mathrm{d}\pi + \int_Y \psi\, \mathrm{d}\nu - \int_{X\times Y} \psi\, \mathrm{d}\pi \\
&= \sup_{\phi\in C_b(X),\, \psi\in C_b(Y)} \left[\inf_{\pi\in\mathscr{M}_+(X\times Y)} \int_{X\times Y} (c - \phi - \psi)\, \mathrm{d}\pi \right] + \int_X \phi\, \mathrm{d}\mu + \int_Y \psi\, \mathrm{d}\nu \\
&= \sup_{(\phi,\psi)\in I_c} \int_X \phi\, \mathrm{d}\mu + \int_Y \psi\, \mathrm{d}\nu,
\end{aligned} \tag{3.4}$$

because the infimum inside the square brackets is 0 if $(\phi, \psi) \in I_c$, $-\infty$ otherwise. Our goal is to make this argument rigorous, using the Fenchel–Rockafellar theorem, at least when X and Y are compact. But, since these minmax arguments are not very constructive, we are going to provide a constructive proof of the duality formula along the lines of [64, 105]. What the two proofs have in common is the use of Convex Analysis tools, the classical ones for the first proof and their adaptation to the "nonlinear" setting for the second proof. For this reason we devote the next section to the illustration of some classical Convex Analysis tools.

1 Convex Analysis Tools

Given a normed vector space X, we denote

$$\Gamma(X) := \{f : X \to (-\infty, \infty] : f \text{ is convex and lower semicontinuous, } f \not\equiv +\infty\}$$

and for all $f \in \Gamma(X)$ we denote by $\mathrm{Dom}(f)$ its finiteness domain

$$\mathrm{Dom}(f) := \{x \in X : f(x) < \infty\}, \tag{3.5}$$

which is obviously a convex set.

The Fenchel–Moreau theorem below (see Theorem 3.4), which essentially follows by the Hahn-Banach theorem, provides the characterization

$$\Gamma(X) = \left\{f : X \to (-\infty, \infty] : \ f \text{ is the supremum of affine functions, } \ f \not\equiv +\infty\right\}, \tag{3.6}$$

which will be the key for the extension to the nonlinear setting.

Let us now denote by X^* the dual space of X and by $\langle \cdot, \cdot \rangle$ the canonical duality map from $X^* \times X$ to $\mathbb{R}$.

Definition 3.3 (Conjugate Functions) Given $f : X \to (-\infty, \infty]$, we define

$$f^*(x^*) := \sup_{x \in X} \left\{\langle x^*, x\rangle - f(x)\right\} \qquad x^* \in X^*.$$

Analogously, given $g : X^* \to (-\infty, \infty]$, we define

$$g^*(x) := \sup_{x^* \in X^*} \left\{\langle x^*, x\rangle - g(x^*)\right\} \qquad x \in X.$$

Notice that if $f \not\equiv +\infty$ then f^* takes its values in $(-\infty, \infty]$. In addition, if f is bounded from below by an affine function, then $f^* \not\equiv +\infty$ (more precisely, $f^*(y_0) < \infty$ if $f(x) \geq \langle x, y_0\rangle + c$). Analogous properties hold for the operator $g \mapsto g^*$. According to these remarks, the $*$ operators map $\Gamma(X)$ in $\Gamma(X^*)$ and $\Gamma(X^*)$ in $\Gamma(X)$.

Notice also that, in principle, g^* could be even defined on a space larger than X, namely the dual of X^*, but this extension will not be useful for us.

The following result will be proved in the more general setting of Theorem 3.14, using the characterization (3.6) of $\Gamma(X)$.

Theorem 3.4 (Fenchel–Moreau) *For all $f \in \Gamma(X)$ one has $(f^*)^* = f$, namely*

$$f(x) = \sup_{x^* \in X^*} \left\{\langle x^*, x\rangle - f^*(x^*)\right\} \qquad \forall x \in X,$$

for all $f \in \Gamma(X)$.

In order to make rigorous the argument in (3.4), we shall use the following result (see for instance Theorem 1.12 of [26]).

Theorem 3.5 (Fenchel–Rockafellar) *Let X be a normed space and let $\Theta, \Sigma : X \to (-\infty, \infty]$ be convex and lower semicontinuous. If there exists $x_0 \in Dom(\Theta) \cap Dom(\Sigma)$ such that Θ is continuous in x_0, then*

$$\inf_{x \in X} \{\Theta(x) + \Sigma(x)\} = \sup_{x^* \in X^*} \{-\Theta^*(-x^*) - \Sigma^*(x^*)\}. \tag{3.7}$$

We can write the left hand side in (3.7) as $\inf_{x,y} \sup_{x^*} \Theta(x) + \Sigma(y) + \langle x^*, x - y\rangle$. On the other hand, by definition

$$-\Theta^*(-x^*) = \inf_{x \in X} \{\Theta(x) + \langle x^*, x\rangle\}, \qquad -\Sigma(x^*) = \inf_{y \in X} \{\Sigma(y) - \langle x^*, y\rangle\}.$$

Consequently (3.7) reduces to

$$\inf_{x,\, y \in X} \sup_{x^* \in X^*} \{\Theta(x) + \Sigma(y) + \langle x^*, x - y\rangle\} = \sup_{x^* \in X^*} \inf_{x,\, y \in X} \{\Theta(x) + \Sigma(y) + \langle x^*, x - y\rangle\},$$

showing once more the basic property of exchange of supremum and infimum.

Remark 3.6 More generally, the commutation property

$$\inf_u \sup_v L(u, v) = \sup_v \inf_u L(u, v)$$

holds, under suitable topological and finiteness assumptions, whenever L is convex with respect to u and concave with respect to v, see [55].

Definition 3.7 (Subdifferential) For all $f \in \Gamma(X)$ and all $x \in \{f < \infty\}$ the subdifferential $\partial f(x)$ is defined as

$$\partial f(x) := \{x^* \in X^* : f(y) \geq f(x) + \langle x^*, y - x\rangle \quad \forall y \in X\}.$$

Observe that $\partial f(x)$ is a convex and closed (possibly empty) subset of X^* and that $x^* \in \partial f(x)$ if and only if $f(x) + f^*(x^*) = \langle x^*, x\rangle$. Moreover the monotonicity of difference quotients gives

$$\{\nabla f(x)\} = \partial f(x) \qquad \text{at any differentiability point of } f, \tag{3.8}$$

where differentiability is understood in the Gateaux sense.
Let us recall that f is said to be differentiable at $x \in X$ in the Gateaux sense if $r \mapsto f(x + ry)$ is differentiable at $r = 0$ for any $y \in X$.

In the following we will consider the graph of the subdifferential, considered as a multivalued operator

$$\text{Graph}(\partial f) := \{(x, x^*) \in X \times X^* : x^* \in \partial f(x)\}.$$

Definition 3.8 (Cyclical Monotonicity) We say that $\Gamma \subset X \times X^*$ is cyclically monotone if

$$\sum_{i=1}^{N} \langle x_i^*, x_{\sigma(i)} - x_i \rangle \leq 0$$

for any integer $N \geq 1$, $(x_1, x_1^*), \ldots, (x_N, x_N^*)$ points in Γ, and any permutation σ.

The usual monotonicity condition corresponds to the choice $N = 2$, namely

$$\langle x_2^* - x_1^*, x_2 - x_1 \rangle \geq 0.$$

Equivalent definitions of cyclical monotonicity can be given using only cyclical permutations (which justifies the terminology we adopted, see Remark 3.11), or permutations in the dual variable:

$$\sum_{i=1}^{N} \langle x_{\sigma(i)}^* - x_i^*, x_i \rangle \leq 0.$$

It is immediate to check that $\mathrm{Graph}(\partial f)$ is cyclically monotone: it suffices to add the inequalities

$$f(x_{\sigma(i)}) - f(x_i) \geq \langle x_i^*, x_{\sigma(i)} - x_i \rangle \qquad i = 1, \ldots, N.$$

The following fundamental result shows the validity of the converse implication. It will be proved, setting $c(x, x^*) = -\langle x^*, x \rangle$, in the more general context of Theorem 3.18.

Theorem 3.9 (Rockafellar) *Any cyclically monotone set $\Gamma \subset X \times X^*$ is contained in* $\mathrm{Graph}(\partial f)$ *for some $f \in \Gamma(X)$.*

Notice that the property of being cyclically monotone is stable by inclusion. A more precise statement of Rockafellar's theorem is that the sets $\mathrm{Graph}(\partial f)$, $f \in \Gamma(X)$, are precisely the *maximal* cyclically monotone sets, but we will not need this property in the sequel.

2 Proof of Duality via Fenchel–Rockafellar

Here we follow the presentation in [117], assuming X and Y to be compact and $c : X \times Y \to [0, \infty]$ to be lower semicontinuous. We refer to [73] for the study of the duality problem without the lower semicontinuity assumption on the cost.

The compactness of $X \times Y$ ensures that $(C(X \times Y))^*$ is isometrically isomorphic to $\mathscr{M}(X \times Y)$. Now we can prove Theorem 3.1 applying Theorem 3.5 with

$$\Theta(u) = \begin{cases} 0, & \text{if } u \geq -c \\ +\infty, & \text{otherwise,} \end{cases} \qquad \Sigma(u) = \begin{cases} \int_X \phi \, \mathrm{d}\mu + \int_Y \psi \, \mathrm{d}\nu, & \text{if } u(x, y) = \phi(x) + \psi(y) \\ +\infty, & \text{otherwise.} \end{cases}$$

First of all, notice that Σ is well defined, since $\phi(x) + \psi(y) = u(x, y) = \tilde{\phi}(x) + \tilde{\psi}(y)$ ensures the existence of $c \in \mathbb{R}$ such that $\phi - \tilde{\phi} \equiv c \equiv \psi - \tilde{\psi}$. Notice also that the continuity of u forces the continuity of ϕ and ψ, when $u = \phi + \psi$. In addition, both Θ and Σ are finite at $u_0 \equiv 1$ and Θ is continuous at u_0. Now, the left hand side in

$$\inf_{u \in C(X \times Y)} \{\Theta(u) + \Sigma(u)\} = \sup_{\pi \in \mathscr{M}(X \times Y)} \{-\Theta^*(-\pi) - \Sigma^*(\pi)\}.$$

corresponds to

$$\inf \left\{ \int_X \phi \, \mathrm{d}\mu + \int_Y \psi \, \mathrm{d}\nu : \phi + \psi \geq -c \right\}$$

and therefore, after a change of sign, it coincides with the supremum in (3.1).

In order to analyze the right hand side, let us compute Θ^* and Σ^*. One has

$$\Theta^*(-\pi) = \sup_u \left\{ -\int_{X \times Y} u \, \mathrm{d}\pi - \Theta(u) \right\} = \sup_{u \geq -c} -\int_{X \times Y} u \, \mathrm{d}\pi.$$

Now, if $\pi^- \neq 0$ it is easily seen that there exists u nonnegative with $\int_{X \times Y} u \, \mathrm{d}\pi < 0$. Considering the functions tu with $t \gg 1$ we obtain that the supremum is $+\infty$. On the other hand, if $\pi^- = 0$ then the supremum is $\int c \, \mathrm{d}\pi$. It follows that

$$\Theta^*(-\pi) = \begin{cases} \int_{X \times Y} c \, \mathrm{d}\pi & \text{if } \pi \in \mathscr{M}_+(X \times Y) \\ \\ +\infty & \text{otherwise.} \end{cases}$$

Analogously

$$\Sigma^*(\pi) = \sup_u \left\{ \int_{X \times Y} u \, \mathrm{d}\pi - \Sigma(u) \right\} = 1_{\Gamma(\mu, \nu)}(\pi).$$

Putting together these ingredients we see that the right hand side corresponds, up to a change of sign, to the infimum in (K).

3 The Theory of c-Duality

We are going to adapt several concepts of Convex Analysis to the case of general spaces X, Y and costs c. We assume in this section that c is $\mathbb{R}$-valued and call *c-affine functions* the functions of the form $c(\cdot, y) + \alpha$, or of the form $c(x, \cdot) + \beta$, for some $x \in X$, $y \in Y$ and $\alpha, \beta \in \mathbb{R}$.

Definition 3.10 (c-Cyclically Monotonicity) We say that $\Gamma \subset X \times Y$ is c-cyclically monotone if

$$\sum_{i=1}^{N} c(x_i, y_{\sigma(i)}) \geq \sum_{i=1}^{N} c(x_i, y_i) \tag{3.9}$$

for every $N \geq 1$, σ permutation of $\{1, \ldots, N\}$ and $(x_i, y_i) \in \Gamma$ for $i = 1, \ldots, N$.

Occasionally, we might also consider the weaker c-monotonicity, corresponding to (3.9) with $N = 2$, namely

$$c(x, y') + c(x', y) \geq c(x, y) + c(x', y') \qquad \text{for all } (x, y) \in \Gamma,\ (x', y') \in \Gamma. \tag{3.10}$$

Remark 3.11 In the definition of c-cyclically monotone set it is enough to ask for the validity of (3.9) for cyclical permutations, i.e. for those $\sigma : \{1, \ldots, N\} \to \{1, \ldots, N\}$ such that $\sigma(i) = i + 1$ for any $i \in \{1, \ldots, N + 1\}$ and $\sigma(N) = 1$. Indeed any permutation can be obtained as composition of cycles acting on disjoint subsets of $\{1, \ldots, N\}$ and trivial cycles on fixed points.

Motivations for the definition above come, on the one hand from Convex Analysis and, on the other hand, from the study of the optimal transport problem in discrete setting. More specifically, the contact set Γ in (3.2) is obviously c-cyclically monotone, since it suffices to replace in the right hand side of (3.9) $c(x_i, y_i)$ with $\phi(x_i) + \psi(y_i)$ and, for the left hand side, use the inequality $c(x, y) \geq \phi(x) + \psi(y)$. Moreover, in the case $Y = X^*$ and $c(x, x^*) = -\langle x^*, x \rangle$, a set Γ is c-cyclically monotone if and only if it is cyclically monotone according to Definition 3.8. In addition, if X is a Hilbert space and we identify X and X^*, then the same equivalence holds for the cost $|x - y|^2$: it suffices to expand the squares in (3.9).

Definition 3.12 (c-Conjugate Function) Given $\phi : X \to [-\infty, \infty)$ we define the c-conjugate function ϕ^c by

$$\phi^c(y) := \inf_{x \in X} \{c(x, y) - \phi(x)\}.$$

Analogously, given $\psi : Y \to [-\infty, \infty)$, we define

$$\psi^c(x) := \inf_{y \in Y} \{c(x, y) - \psi(y)\}.$$

If $\phi \not\equiv -\infty$ (resp. $\psi \not\equiv -\infty$), then $\phi^c : Y \to [-\infty, \infty)$ (resp. $\psi^c : X \to [-\infty, \infty)$). Furthermore, if ϕ is bounded from above by a c-affine function $c(\cdot, y_0) + \alpha$, then $\phi^c(y_0) > -\infty$, so that $\phi^c \not\equiv -\infty$. An analogous statement holds for ψ^c.

Definition 3.13 (c-Concavity) A function $\phi : X \to [-\infty, \infty)$ is said to be c-concave if it is the infimum of a family of c-affine functions $c(\cdot, y)+\alpha$. Analogously, $\psi : Y \to [-\infty, \infty)$ is said to be c-concave if it is the infimum of a family of c-affine functions $c(x, \cdot) + \beta$.

Observe that a function $\phi : X \to [-\infty, \infty)$ such that $\phi \not\equiv -\infty$ is c-concave if and only if it is the c-conjugate of a function ψ. Indeed $\phi(x) = \inf_i c(x, y_i) + \alpha_i$ implies $\phi = \psi^c$ with $\psi : Y \to [-\infty, \infty)$ given by (notice that the possibility that $\psi(y) = +\infty$ for some y is ruled out by the assumption $\phi \not\equiv -\infty$)

$$\psi(y) := \begin{cases} -\inf\{\alpha_i : y_i = y\} & \text{if } y = y_i \text{ for some } i \\ \\ -\infty & \text{otherwise.} \end{cases}$$

Theorem 3.14 *Assume that $\phi : X \to [-\infty, \infty)$, with $\phi \not\equiv -\infty$. Then $\phi^{cc} \geq \phi$, with equality if and only if ϕ is c-concave.*

Proof Our assumption ensures that $\phi^c : Y \to [-\infty, \infty)$, so that the c-transform can be iterated. Observe that ϕ^c is the largest function ψ compatible with the constraint $\phi + \psi \leq c$. Analogously ψ^c is the largest function ϕ compatible with the constraint $\phi + \psi \leq c$. From this variational characterization (or by a direct verification) it follows that $\phi \leq \tilde{\phi}$ implies $\phi^c \geq \tilde{\phi}^c$ and that $\phi^{cc} \geq \phi$.

Considering the equality case, if ϕ is c-concave we can write $\phi = \psi^c$ and then conclude that $\phi^c = (\psi^c)^c \geq \psi$ and that $\phi^{cc} \leq \psi^c = \phi$. It follows that $\phi^{cc} = \phi$, since the converse inequality is always satisfied. On the other hand, if we assume that $\phi^{cc} = \phi$, then ϕ is the conjugate function of ϕ^c and we conclude that it is c-concave. □

Finally, still pursuing the analogy with the classical case, we can give the following definition.

Definition 3.15 (c-Subdifferential) Given $\phi : X \to [-\infty, \infty)$ and $x \in \{\phi > -\infty\}$ we define the c-superdifferential as

$$\partial^c \phi(x) := \left\{ y \in Y : c(x', y) - \phi(x') \text{ is minimal at } x' = x \right\}.$$

The analogy becomes even more clear if we write the minimality condition in the form

$$\phi(x') \leq \phi(x) - c(x, y) + c(x', y) \qquad \forall x' \in X.$$

Also, it is easy to prove that $\phi(x) + \phi^c(y) = c(x, y)$ if and only if $y \in \partial^c \phi(x)$, if and only if $x \in \partial \phi^c(y)$.

4 Proof of Duality and Dual Attainment for Bounded and Continuous Cost Functions

Returning now to the general Polish setting, we are going to provide the duality formula for bounded and continuous cost functions. In the proof we will need the following general tool from measure theory.

Lemma 3.16 *Let $X_1, \ldots, X_k$ be Polish spaces and $\gamma_1 \in \mathscr{P}(X_1), \ldots, \gamma_k \in \mathscr{P}(X_k)$. Then there exist a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ and measurable maps $T_i : \Omega \to X_i$, $i = 1, \ldots, k$, such that $(T_i)_\# \mathbb{P} = \gamma_i$ for every $i = 1, \ldots, k$.*

Its proof is standard, since one can take as Ω the product space $X_1 \times \cdots \times X_k$, as $\mathcal{F}$ the product σ-algebra, as $\mathbb{P}$ the product of the γ_i. With these choices the canonical projections T_i on the i-th coordinate map $\mathbb{P}$ to γ_i.

Theorem 3.17 (c-Cyclical Monotonicity of Supports of Optimal Plans) *Assume that $c : X \times Y \to [0, \infty)$ is continuous and that $\pi \in \Gamma_o(\mu, \nu)$ with $\int_{X \times Y} c \, d\pi < \infty$. Then $\operatorname{supp} \pi$ is c-cyclically monotone.*

Proof Assume by contradiction that the conclusion is false. Then there exist $N \geq 1$, a permutation σ and points $(x_1, y_1), \ldots, (x_N, y_N) \in \operatorname{supp} \pi$ such that

$$\sum_{i=1}^{N} c(x_i, y_{\sigma(i)}) < \sum_{i=1}^{N} c(x_i, y_i). \tag{3.11}$$

By the continuity of the cost function c there exist neighbourhoods $U_i \times V_i \ni (x_i, y_i)$ where inequality (3.11) continues to hold, replacing (x_i, y_i) with any choice of $(x_i', y_i') \in U_i \times V_i$. Now we want to construct a measure $\theta \in \mathscr{M}(X \times Y)$ such that:

(a) θ has null marginals on X and Y;
(b) $\theta^- \leq \pi$ (recall that θ^- denotes the negative part of the measure θ);
(c) $\int_{X \times Y} c \, d\theta < 0$.

If we are able to construct such a measure θ, then we get a contradiction since from (a) and (b) it follows that $\pi + \theta \in \Gamma(\mu, \nu)$ and from (c) we get

$$\int_{X\times Y} c\,\mathrm{d}(\pi+\theta) < \int_{X\times Y} c\,\mathrm{d}\pi = \min_{(\mathrm{K})}.$$

In order to construct such a measure θ, let us define $z_i := \pi(U_i \times V_i)$ for any $i = 1, \dots, N$. Observe that $z_i > 0$ since $(x_i, y_i) \in \operatorname{supp}\pi$. Then define

$$\pi_i := \frac{\pi \llcorner (U_i \times V_i)}{z_i}$$

and observe that $\pi_i \in \mathscr{P}(U_i \times V_i)$. Thanks to Lemma 3.16 we obtain the existence of a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ and of measurable maps $X_i \times Y_i : \Omega \to U_i \times V_i$ such that $\pi_i = (X_i \times Y_i)_{\#}\mathbb{P}$ for any $i = 1, \dots, N$. Now we define

$$\theta := \sum_{i=1}^{N} (X_i \times Y_{\sigma(i)})_{\#}\mathbb{P} - \sum_{i=1}^{N} (X_i \times Y_i)_{\#}\mathbb{P}.$$

It is obvious that $\theta \in \mathscr{M}(X \times Y)$ and it is immediate to check that θ satisfies (a). By the change of variables formula it follows that

$$\int_{X\times Y} c\,\mathrm{d}\theta = \int_{\Omega} \sum_{i=1}^{N} \left[c\left(X_i, Y_{\sigma(i)}\right) - c\left(X_i, Y_i\right) \right] \mathrm{d}\mathbb{P} < 0,$$

since the integrand is less than zero by construction, and we conclude that also (c) is satisfied. Observe that

$$\theta^- \le \sum_{i=1}^{N} \pi_i \le \frac{N}{\min_i z_i}\pi,$$

so, if we replace θ by $t\theta$, where $t > 0$ is sufficiently small, conditions (a) and (c) continue to hold and (b) is satisfied. The conclusion follows. □

The following result implies the classical one, Theorem 3.9, simply choosing $c(x, x^*) = -\langle x^*, x\rangle$. It can also be rephrased by saying that c-cyclically monotone sets are contained in contact sets, according to (3.2) (and we already observed that contact sets are c-cyclically monotone).

Theorem 3.18 (Generalized Rockafellar) *Assume that $c : X \times Y \to \mathbb{R}$ and that $\Gamma \subset X \times Y$ is c-cyclically monotone. Then there exists a c-concave function $\phi : X \to [-\infty, \infty)$, $\phi \not\equiv -\infty$, such that $\Gamma \subset \operatorname{Graph}(\partial^c \phi)$.*

Proof We fix $(x_0, y_0) \in \Gamma$ and we try to build ϕ in such a way that $\phi(x_0) = 0$. The condition $\Gamma \subset \mathrm{Graph}(\partial^c \phi)$ is equivalent to

$$\phi(x) \le c(x, y_*) - c(x_*, y_*) + \phi(x_*) \qquad \forall (x_*, y_*) \in \Gamma, \ \forall x \in X. \tag{3.12}$$

Now, if we choose $(x_1, y_1) \in \Gamma$, using (3.12) first with (x_1, y_1) and then with (x_0, y_0) (with $x = x_1$ in the second case) we get

$$\phi(x) \le c(x, y_1) - c(x_1, y_1) + \phi(x_1) \le c(x, y_1) - c(x_1, y_1) + c(x_1, y_0) - c(x_0, y_0).$$

Continuing in this way, we see that it must be

$$\phi(x) \le \inf \big\{ c(x, y_N) - c(x_N, y_N) + c(x_N, y_{N-1}) \\ - c(x_{N-1}, y_{N-1}) + \cdots + c(x_1, y_0) - c(x_0, y_0) \big\}, \tag{3.13}$$

where the infimum is made over all the possible choices of $N \ge 1$ and points

$$(x_1, y_1), \ldots, (x_N, y_N) \in \Gamma.$$

This suggests that ϕ should be defined as the infimum in the right hand side of (3.13). Since c is real valued, ϕ takes its values in $[-\infty, \infty)$ and is c-concave. Now, let us check that it satisfies (3.12) and that $\phi(x_0) = 0$. Indeed, if first we minimize keeping (x_N, y_N) fixed, then we minimize with respect to (x_N, y_N), we see that

$$\phi(x) = \inf_{(x_N, y_N)} c(x, y_N) - c(x_N, y_N) + \phi(x_N)$$

which implies (3.12), changing in an obvious way names to the variables.

Finally, the inequality $\phi(x_0) \ge 0$ follows by the c-cyclical monotonicity of Γ (notice that we are using here only cyclical permutations $x_i \mapsto x_{i+1}$ for $0 \le i < N$, $x_N \mapsto x_0$), while the inequality $\phi(x_0) \le 0$ can be obtained choosing $N = 1$ and $(x_1, y_1) = (x_0, y_0)$ in (3.13). □

Corollary 3.19 *The duality formula* (3.1) *holds for any cost function* $c \in \mathrm{Lip}_b(X \times Y)$. *In addition, in this case the supremum in the dual formulation is a maximum, attained at a pair* (ϕ, ϕ^c) *with* ϕ, ϕ^c *bounded and Lipschitz.*

Proof Consider an optimal transport plan $\pi \in \Gamma_o(\mu, \nu)$ (whose existence follows from Theorem 2.10). Then by Theorem 3.17 we get that $\Gamma := \mathrm{supp}\, \pi$ is c-cyclically monotone and by Theorem 3.18 we obtain the existence of a c-concave function ϕ such that $\phi + \phi^c = c$ on Γ and $\phi(x_0) = 0$. The function ϕ was defined by

$$\phi(x) := \inf \big\{ c(x, y_N) - c(x_N, y_N) + c(x_N, y_{N-1}) \\ - c(x_{N-1}, y_{N-1}) + \cdots + c(x_1, y_0) - c(x_0, y_0) \big\}, \tag{3.14}$$

and the very construction gives that ϕ is a Lipschitz function (being the infimum of a family of equi-Lipschitz functions), uniformly bounded from above (thanks to the boundedness of c). Now, $\phi^c(y) \le c(x_0, y) - \phi(x_0) \le \sup c$, so that also ϕ^c is uniformly bounded from above. Since

$$\phi(x) = \phi^{cc}(x) = \inf_y c(x, y) - \phi^c(y) \ge \inf c - \sup \phi^c$$

we obtain that ϕ is uniformly bounded from below. A similar argument turns the uniform bound from above on ϕ into a uniform from below on ϕ^c. It follows that both ϕ and ϕ^c are bounded Lipschitz functions.

Finally, since $c = \phi + \phi^c$ π-a.e. on Γ, we get

$$\mathscr{C}(\pi) = \int_{X\times Y} c \, \mathrm{d}\pi = \int_{X\times Y} (\phi + \phi^c) \, \mathrm{d}\pi = \int_X \phi \, \mathrm{d}\mu + \int_Y \phi^c \, \mathrm{d}\mu$$

and the supremum in (3.1) is attained at the pair $(\phi, \phi^c) \in \mathrm{Lip}_b(X) \times \mathrm{Lip}_b(Y)$. □

We refer to the very recent [15] for a generalization of the Rockafellar Theorem 3.18 allowing for cost functions c which assume the value $\pm\infty$.

Lecture 4: Necessary and Sufficient Optimality Conditions



In this lecture we first extend the duality formula to lower semicontinuous costs (by monotone approximation with bounded and Lipschitz costs), then we consider necessary and sufficient optimality conditions and attainment in the dual problem.

1 Duality and Necessary/Sufficient Optimality Conditions for Lower Semicontinuous Costs

Theorem 4.1 (Duality and Necessary Optimality Condition) *Assume that* $c : X \times Y \to [0, \infty]$ *is lower semicontinuous, then the duality formula* (3.1) *holds. Moreover, any* $\pi \in \Gamma_o(\mu, \nu)$ *such that* $\int_{X\times Y} c \, d\pi < \infty$ *is concentrated on a* c*-cyclically monotone* σ*-compact set.*

Proof Let us begin from the proof of duality. We already observed in the proof of Theorem 2.6 that c can be written as $\sup_k c_k$ where c_k are bounded Lipschitz functions monotonically converging to c. In particular we have that, for any $k \in \mathbb{N}$,

$$\begin{aligned}\min_{(K)_c} &\geq \sup_{\phi+\psi\leq c}\left(\int_X \phi \, d\mu + \int_Y \psi \, d\nu\right)\\ &\geq \sup_{\phi+\psi\leq c_k}\left(\int_X \phi \, d\mu + \int_Y \psi \, d\nu\right)\\ &= \min_{(K)_{c_k}},\end{aligned}$$

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where in the last line we applied Corollary 3.19 to the cost function c_k. To conclude that duality holds for the cost function c, it is now sufficient to prove that

$$\min_{(\mathrm{K})_{c_k}} \uparrow \min_{(\mathrm{K})_c} \qquad \text{as } k \text{ goes to infinity.}$$

To this aim for any $k \in \mathbb{N}$ take π_k optimal plan for the cost c_k. Observe that the class $\Gamma(\mu, \nu)$ is independent of k and compact with respect to weak convergence thanks to Corollary 2.9. Hence there exist a plan $\pi \in \Gamma(\mu, \nu)$ and a subsequence $(\pi_{k(\ell)})$ such that $\pi_{k(\ell)} \to \pi$ weakly as ℓ goes to infinity. Then

$$\begin{aligned}\lim_{k\to\infty} \min_{(\mathrm{K})_{c_k}} &= \lim_{\ell\to\infty} \int_{X\times Y} c_{k(\ell)}\, \mathrm{d}\pi_{k(\ell)} \\ &\geq \liminf_{\ell\to\infty} \int_{X\times Y} c_p\, \mathrm{d}\pi_{k(\ell)} = \int_{X\times Y} c_p\, \mathrm{d}\pi\end{aligned}$$

for any $p \in \mathbb{N}$. Then, letting p go to infinity, we obtain

$$\lim_{k\to\infty} \min_{(\mathrm{K})_{c_k}} \geq \int_{X\times Y} c\, \mathrm{d}\pi \geq \min_{(\mathrm{K})_c}.$$

Now, for any $k \in \mathbb{N}$ let us consider an optimal pair (ϕ_k, ψ_k) for the dual problem with cost c_k. We already observed that

$$\int_X \phi_k\, \mathrm{d}\mu + \int_Y \psi_k\, \mathrm{d}\nu = \min_{(\mathrm{K})_{c_k}} \uparrow \min_{(\mathrm{K})_c} = \int_{X\times Y} c\, \mathrm{d}\pi$$

as k goes to infinity. Since $\phi_k + \psi_k \leq c_k \leq c$ and $c \in L^1(\pi)$, we get that the sequence of nonnegative functions $(c - \phi_k - \psi_k)$ converges to 0 in $L^1(\pi)$. It follows that there exist a subsequence $(k(\ell))$ and a set Γ of full π-measure such that

$$\big(c - \phi_{k(\ell)} - \psi_{k(\ell)}\big) \to 0$$

pointwise on Γ as ℓ goes to infinity. Observe that, thanks to Ulam's Lemma 1.5, we can assume with no loss of generality that Γ is also σ-compact. Now, given $(x_1, y_1), \ldots, (x_N, y_N) \in \Gamma$ and a permutation σ, we have that

$$\begin{aligned}\sum_{i=1}^N c(x_i, y_{\sigma(i)}) &\geq \sum_{i=1}^N \phi_{k(\ell)}(x_i) + \sum_{i=1}^N \psi_{k(\ell)}(y_{\sigma(i)}) \\ &= \sum_{i=1}^N \phi_{k(\ell)}(x_i) + \sum_{i=1}^N \psi_{k(\ell)}(y_i).\end{aligned}$$

Letting ℓ go to infinity, we conclude that

$$\sum_{i=1}^{N} c(x_i, y_{\sigma(i)}) \geq \sum_{i=1}^{N} c(x_i, y_i).$$

□

Theorem 4.2 *Consider a lower semicontinuous cost function $c : X \times Y \to [0, \infty]$ and assume that there exist functions $a \in L^1(\mu)$, $b \in L^1(\nu)$ such that $c(x, y) \leq a(x) + b(y)$. Then:*

(i) $\pi \in \Gamma(\mu, \nu)$ is optimal if and only if it is concentrated on a c-cyclically monotone (σ-compact) set;

(ii) there exists a c-concave function $\phi : X \to [-\infty, \infty)$ such that $\phi \in L^1(\mu)$, $\phi^c \in L^1(\nu)$ and

$$\min_{(K)} = \int_X \phi \, \mathrm{d}\mu + \int_Y \phi^c \, \mathrm{d}\nu.$$

Proof The necessity of the c-cyclical monotonicity of the support for optimality is a consequence of Theorem 4.1 (notice also that our growth condition on c ensures that *any* admissible plan has finite cost).

In order to prove the sufficiency of c-cyclical monotonicity, let us consider $\pi \in \Gamma(\mu, \nu)$, with π concentrated on a c-cyclically monotone and σ-compact set Γ. Observe that Theorem 3.18 provides us with a c-concave function $\phi : X \to [-\infty, \infty)$, not identically equal to $-\infty$, such that $\phi + \phi^c = c$ on Γ. This property implies that $\phi > -\infty$ and $\phi^c > -\infty$ π-a.e. in $X \times Y$ and then, since $\pi \in \Gamma(\mu, \nu)$, $\phi > -\infty$ μ-a.e. in X and $\phi^c > -\infty$ ν-a.e. in Y.

Assume for a moment that $\phi \in L^1(\mu)$ and $\phi^c \in L^1(\nu)$. Then, for any $\tilde{\pi} \in \Gamma(\mu, \nu)$ we have

$$\begin{aligned} \int c \, \mathrm{d}\tilde{\pi} &\geq \int_{X \times Y} \left(\phi + \phi^c\right) \mathrm{d}\tilde{\pi} = \int_X \phi \, \mathrm{d}\mu + \int_Y \phi^c \, \mathrm{d}\nu \\ &= \int_{X \times Y} \left(\phi + \phi^c\right) \mathrm{d}\pi = \int_{X \times Y} c \, \mathrm{d}\pi. \end{aligned}$$

We are left with the verification of the integrability of the functions ϕ and ϕ^c. In this case, since we are not assuming that the cost is upper semicontinuous, the verification of the Borel property of the function ϕ in (3.14) is not straightforward, one has to use once more the approximation from below of the cost c by bounded Lipschitz functions and the σ-compactness of Γ, see Theorem 6.1.4 in [12] for details.

Now we have that π-almost everywhere $\phi^c(y) = c(x, y) - \phi(x)$, so that the function ϕ^c, seen as a function on $X \times Y$ independent of the first variable, is π-measurable. Since $(p_Y)_{\#}\pi = \nu$, it follows that ϕ^c is ν-measurable (now as a function

on Y). Now choose $y \in Y$ such that $b(y) < \infty$ and $\phi^c(y) > -\infty$. By integration of the inequality $\phi(x) \le c(x, y) - \phi^c(y)$ it follows that

$$\int_X \phi^+ \, \mathrm{d}\mu \le \int_X a \, \mathrm{d}\mu + b(y) - \phi^c(y) < \infty.$$

A symmetric argument (with an appropriate choice of $x \in X$ such that $a(x) < \infty$) gives $\int (\phi^c)^+ \, \mathrm{d}\nu < \infty$. Finally, by observing that the semi-integrability of ϕ and ϕ^c guarantees the validity of the identities

$$0 \le \int_{X\times Y} c \, \mathrm{d}\pi = \int_{X\times Y} (\phi + \phi^c) \, \mathrm{d}\pi = \int_X \phi \, \mathrm{d}\mu + \int_Y \phi^c \, \mathrm{d}\nu$$

we get that $\int_X \phi \, \mathrm{d}\mu > -\infty$ and $\int_Y \phi^c \, \mathrm{d}\nu > -\infty$ and the desired conclusion $\phi \in L^1(\mu)$ and $\phi^c \in L^1(\nu)$. □

The following definition will play a crucial role in the next sections. We are going to give it here in order to point out that the last result applies in this case.

Definition 4.3 (The Space $\mathscr{P}_p(X)$) Given a metric space (X, d) and $p \in [1, \infty)$, we define

$$\mathscr{P}_p(X) := \left\{ \mu \in \mathscr{P}(X) : \int_X d(x, x_0)^p \, \mathrm{d}\mu(x) < \infty \text{ for some (and hence for all) } x_0 \in X \right\}.$$

Observe that Theorem 4.2 applies in particular to the case when $X = Y = \mathbb{R}^n$, $c(x, y) = |x - y|^p$ and $\mu, \nu \in \mathscr{P}_p(\mathbb{R}^n)$. Indeed, it is sufficient to observe that $|x - y|^p \le 2^{p-1}(|x|^p + |y|^p)$ and that, by assumption, $\int_X |x|^p \, \mathrm{d}\mu + \int_Y |y|^p \, \mathrm{d}\nu < \infty$.

2 Remarks About Necessary and Sufficient Optimality Conditions

We proved that concentration of the support on a c-cyclically monotone set is a necessary optimality condition for the optimality of a transport plan π, if we assume that $\int c \, \mathrm{d}\pi < \infty$, and becomes a sufficient optimality condition if $c \le a + b$ with $a \in L^1(\mu)$ and $b \in L^1(\nu)$. W. Schachermeyer and J. Teichmann in [107] and then A. Pratelli in [101] proved that this condition is still sufficient for optimality if either c is lower semicontinuous and with values in $[0, \infty)$, or $c : X \times Y \to [0, \infty]$ is continuous. The following beautiful example (due to A. Pratelli, see [8]) shows that some finiteness/continuity assumption on the cost function is actually needed if we want to deduce optimality of a transport plan from the c-cyclical monotonicity of its support.

Example 4.4 Assume that $X = Y = [0, 1]$ with identification of the extreme points 0 and 1, $\mu = \nu = \mathscr{L}^1 \llcorner [0, 1]$ and fix $\alpha \in \mathbb{R} \setminus \mathbb{Q}$. We define the cost function c by the following expression

$$c(x, y) := \begin{cases} 1 & \text{if } y = x \\ 2 & \text{if } y = x + \alpha \pmod 1 \\ +\infty & \text{otherwise,} \end{cases}$$

where $a + b \pmod 1$ denotes the sum modulo integers.

Now we observe that the shift maps T_1 and T_2 defined respectively by $T_1(x) = x$ and $T_2(x) = x + \alpha \pmod 1$ are both admissible. One can easily check that T_1 is the unique optimal transport map and that $1 = \mathscr{C}_\mu(T_1) < \mathscr{C}_\mu(T_2) = 2$.

Let us prove that $\Gamma := \text{Graph}(T_2)$ is c-cyclically monotone. If this is not the case, there exist a minimal integer N and points $(x_1, y_1), \dots, (x_N, y_N)$ in Γ such that

$$c(x_1, y_1) + \cdots + c(x_N, y_N) > c(x_2, y_1) + \cdots + c(x_N, y_{N-1}) + c(x_1, y_N). \tag{4.1}$$

Since N is minimal, the points (x_i, y_i) are distinct. By the definition of the map T_2 we have that $y_i = x_i + \alpha \pmod 1$ for any $i = 1, \dots, N$, while the finiteness of the left hand side in (4.1) forces $y_i = x_{i+1}$ or $y_i = x_{i+1} + \alpha \pmod 1$ for any $i = 1, \dots, N$ (with $x_{N+1} = x_1$). If the second possibility occurs for some i, we conclude $x_i = x_{i+1}$ (and therefore $y_i = y_{i+1}$), contradicting the fact that the points are distinct. Therefore $x_{i+1} = x_i + \alpha \pmod 1$ for any $i = 1, \dots, N$ and we conclude $x_1 = x_1 + N\alpha \pmod 1$, leading to a contradiction thanks to the irrationality of α.

3 Remarks About c-Cyclical Monotonicity, c-Concavity and c-Transforms for Special Costs

In a very few special cases, c-cyclical monotonicity can be characterized. In some other cases, see [67] and Theorem 13.5 in [118], perturbative arguments can provide a sufficient condition for c-cyclical monotonicity but, as it is, c-cyclical monotonicity remains a pretty mysterious condition, not easily comparable to others.

4 Cost = distance^2

In the case of quadratic cost $c(x, y) = \frac{1}{2}|x - y|^2$ (in the Euclidean setting or more generally on a Hilbert space) one can prove that c-cyclical monotonicity is equivalent to the classical cyclical monotonicity and that a function ϕ is c-concave

if and only if $\phi(x) - \frac{1}{2}|x|^2$ is concave. This is a consequence of the following simple observation: if

$$\phi(x) = \inf_{i \in I} c(x, y_i) + \alpha_i,$$

then

$$\phi(x) - \frac{1}{2}|x|^2 = \inf_i \left\{ \frac{1}{2}|y_i|^2 + \alpha_i - \langle x, y_i \rangle \right\},$$

so that $\phi(\cdot) - \frac{1}{2}|\cdot|^2$ is a concave function (being the infimum of a family of affine functions). A similar argument gives

$$\frac{1}{2}|y|^2 - \phi^*(y) = \left(\frac{1}{2}|\cdot|^2 - \phi\right)^c(y). \tag{4.2}$$

Finally, this argument can be used (we will make this more explicit in the proof of Theorem 5.2, when dealing with differentiability points of ϕ) to show that

$$\partial\phi(x) = \partial^c\left(\frac{1}{2}|\cdot|^2 - \phi\right)(x). \tag{4.3}$$

5 Cost = Distance

As in Monge's original problem we can consider the case $X = Y$, with (X, d) metric space, and $c(x, y) = d(x, y)$. In this setting one can prove that a function ϕ is c-concave if and only if it is 1-Lipschitz. This follows from the fact that for any 1-Lipschitz function ϕ it holds

$$\phi(x) = \inf_{y \in Y} d(x, y) + \phi(y) \tag{4.4}$$

and, conversely, from the fact that any c-concave function is the infimum of a family of 1-Lipschitz functions, a property stable under infimum and supremum. In this case, for ϕ c-concave, the duality relation (4.4) gives immediately that $\phi^c = -\phi$.

6 Convex Costs on the Real Line

Assume that $X = Y = \mathbb{R}$ and that $c(x, y) = h(y - x)$, where $h : \mathbb{R} \to \mathbb{R}$ is a strictly convex function.

Proposition 4.5 *Assume that* $\Gamma \subset \mathbb{R}\times\mathbb{R}$ *is c-monotone according to* (3.10)*. Then* Γ *is a monotone graph, which means that, whenever* (x, y), $(x', y') \in \Gamma$ *and* $x < x'$, *one has* $y \leq y'$.

Proof Assume by contradiction that there exist x, x', y and y' such that (x, y), $(x', y') \in \Gamma$, $x < x'$ but $y > y'$. From c-monotonicity one deduces that

$$h(y - x) + h(y' - x') \leq h(y' - x) + h(y - x'). \tag{4.5}$$

Now we define $a := y - x$, $b := y' - x'$, $\delta := x' - x$ and observe that $a > b$ and that $\delta > 0$. Condition (4.5) can be rewritten as

$$h(a) + h(b) \leq h(b + \delta) + h(a - \delta).$$

Now it is sufficient to observe that $b + \delta = (1 - t)b + ta$ and $a - \delta = (1 - t)a + tb$ with $t = \frac{\delta}{a-b}$ and to notice that our hypothesis guarantees that $t \in (0, 1)$ to conclude that

$$\begin{aligned} h(a) + h(b) &\leq h(b + \delta) + h(a - \delta) \\ &< (1 - t)h(b) + th(a) + (1 - t)h(a) + th(b) \\ &= h(a) + h(b), \end{aligned}$$

reaching a contradiction. □

For a monotone graph, as in Proposition 4.5, the set of coordinates of the "vertical parts"

$$V := \{x \in \mathbb{R} : \#\{y : (x, y) \in \Gamma\} > 1\}$$

is at most countable, since the map associating to any $x \in V$ a rational number in the open interval between two points on Γ having the abscissa x is injective. Observe that Proposition 4.5 then gives that, on the "non-vertical part" $p_X(\Gamma) \setminus V$, the set Γ is the graph of a monotone nondecreasing map T.

The following corollary is an easy consequence of Proposition 4.5 and of the previous observation on the structure of a monotone graph. It completes the proof of Theorem 1.11, showing the optimality of the monotone rearrangement.

Corollary 4.6 *If* $c(x, y) = h(y - x)$ *with* h *strictly convex,* μ *has no atom and* $\min_{(K)} < \infty$, *then the monotone rearrangement is the unique optimal map.*

Let us just remark that to prove optimality for the monotone rearrangement for convex costs it is sufficient to approximate them by strictly convex ones, keeping the transport cost finite. By this procedure one gets optimality, but uniqueness may be lost, as the book shifting example shows.

Lecture 5: Existence of Optimal Maps and Applications



This lecture is mainly devoted to the study of the existence problem for optimal transport maps and its applications. In the last part we introduce the so-called Knothe map, which is not optimal in general but still useful for many purposes.

1 Existence of Optimal Transport Maps

Before giving the statement of the main result of this section let us list a few preliminary facts that will be needed in the sequel.

Theorem 5.1 (Rademacher) *Assume that $\Omega \subset \mathbb{R}^n$ is an open set and that $f : \Omega \to \mathbb{R}$ is locally Lipschitz. Then f is differentiable $\mathscr{L}^n$-almost everywhere in Ω.*

Assume now that $f : \mathbb{R}^n \to (-\infty, \infty]$, and let $\Omega = \mathrm{Dom}(f)$ be its finiteness domain, according to (3.5). The convexity of $\mathrm{Dom}(f)$ yields $\mathscr{L}^n(\partial\Omega) = 0$ (see for instance [80]).

In addition, by exploiting the very definition of convexity, it is not hard to prove that f is locally bounded in Ω. Thanks to the quantitative estimate

$$\mathrm{Lip}(f, B(x,r)) \le \frac{1}{R-r}\left(\sup_{B(x,R)} f - \inf_{B(x,R)} f\right) \quad \text{whenever } B(x,R) \Subset \Omega \text{ and } r \in (0,R)$$

(an easy consequence of the monotonicity of difference quotients) it follows that f is locally Lipschitz in Ω. It follows from this discussion and Theorem 5.1 that any lower semicontinuous convex function is differentiable $\mathscr{L}^n$-almost everywhere in the interior of its finiteness domain.

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The following result, due to Y. Brenier [23] and Knott-Smith [75], has a central role in the theory of Optimal Transport.

Theorem 5.2 (Brenier, Knott-Smith) *Assume that* $X = Y = \mathbb{R}^n$, $c(x,y) = \frac{1}{2}|x-y|^2$, $\mu, \nu \in \mathscr{P}_2(\mathbb{R}^n)$ *and that* $\mu \ll \mathscr{L}^n$. *Then*

(i) the problem (K) *has a unique solution* π. *In addition,* π *is induced by a transport map* T *(which is the unique solution of* (M)*) and* $T = \nabla\psi$, *where* $\psi : \mathbb{R}^n \to (-\infty, \infty]$ *is a lower semicontinuous convex function differentiable* μ*-almost everywhere;*

(ii) conversely, if ψ *is convex, lower semicontinuous, differentiable* μ*-almost everywhere with* $|\nabla\psi| \in L^2(\mu)$, *then* $T := \nabla\psi$ *is optimal from* μ *to* $\nu := T_\#\mu \in \mathscr{P}_2(\mathbb{R}^n)$;

(iii) if $\nu \ll \mathscr{L}^n$, *denoting by* $T^{\mu\to\nu}$ *(resp.* $T^{\nu\to\mu}$*) the unique optimal transport map between* μ *and* ν *(resp.* ν *and* μ*), we get that*

$$T^{\nu\to\mu} \circ T^{\mu\to\nu} = \mathrm{id} \quad \mu\text{-a.e. in } \mathbb{R}^n, \qquad T^{\mu\to\nu} \circ T^{\nu\to\mu} = \mathrm{id} \quad \nu\text{-a.e. in } \mathbb{R}^n. \tag{5.1}$$

Proof To begin we concentrate on the existence part in (i). First of all, by Theorem 2.10, we know that (K) has a solution π. Since $\mathscr{C}(\pi) \le \mathscr{C}(\mu\times\nu)$ and $\mathscr{C}(\mu\times\nu) < \infty$ by the finiteness of the second moments of μ and ν, Theorem 3.17 applies and we get that $\Gamma := \mathrm{supp}(\pi)$ is c-monotone. By Theorem 3.18 we obtain the existence of a c-concave function $\phi : \mathbb{R}^n \to (-\infty, \infty]$ such that $\Gamma \subset \mathrm{Graph}\,(\partial^c\phi) \subset \mathbb{R}^n_x \times \mathbb{R}^n_y$. In particular, ϕ is finite on $p_{\mathbb{R}^n_x}(\Gamma)$, which is a set of full μ-measure. Remember that $\psi := \frac{1}{2}|\cdot|^2 - \phi$ is a convex function, as we observed in Sect. 4 of Lecture 4. It follows that its domain $\mathrm{Dom}(\phi)$ is a convex set of full μ-measure and that, if we define Ω to be its interior part, Ω has full μ-measure as well (since the boundary of any convex set is $\mathscr{L}^n$-negligible and then μ-negligible by the absolute continuity assumption on μ). From the discussion preceding the statement of the theorem it follows that ϕ is differentiable μ-a.e. on Ω.

Now take $x \in \Omega$ where $\nabla\varphi(x)$ exists. We want to prove that there exists a unique y such that $(x,y) \in \mathrm{Graph}\big(\partial^c\phi\big)$. To this aim observe that the condition $(x,y) \in \mathrm{Graph}\big(\partial^c\phi\big)$ implies that $c(x',y) - \phi(x')$ is minimal at $x = x'$. By differentiation we obtain that $(x-y) - \nabla\phi(x) = 0$ and then

$$y = x - \nabla\phi(x) = \nabla\left(\frac{1}{2}|x|^2 - \phi(x)\right) = \nabla\psi(x).$$

It follows that π is concentrated on the graph of the function $\nabla\psi$ and then, by Theorem 2.3, we conclude that $\pi = (\mathrm{id}\times\nabla\psi)_\#\mu$.

Now we come to uniqueness. Assume that we have optimal transport plans π and π'. We already know that there exist Borel maps T and T' such that $\pi = (\mathrm{id}\times T)_\#\mu$ and $\pi' = (\mathrm{id}\times T')_\#\mu$. Observe that the plan $\pi'' := \frac{1}{2}(\pi+\pi')$ is still an optimal transport plan, by linearity of the cost function with respect to the plan. It follows

that also π'' is concentrated on the graph of a Borel function T''. But, since

$$\pi'' = \frac{1}{2}\big(\delta_{T(x)} + \delta_{T'(x)}\big) \otimes \mu,$$

this can happen if and only if $T' = T$ μ-a.e. in $\mathbb{R}^n$.

To prove (ii), observe that our assumptions guarantee that $\nu := T_\#\mu$ belongs to $\mathscr{P}_2(\mathbb{R}^n)$. Then Theorem 4.2 applies and to prove optimality for the map T we need only to check that

$$\Gamma := \{(x, \nabla\psi(x)) : \psi \text{ is differentiable at } x\}$$

is c-cyclically monotone. To prove this, we could apply (3.8) and (4.3) (see also Remark 5.3 for a genuinely nonlinear argument), but we prefer to repeat directly here the "adding/removing the squares" argument of Sect. 1 in Lecture 3. Choose differentiability points $x_1, \dots, x_N$ of ψ and a permutation σ. From the convexity of the function ψ we obtain that

$$\langle \nabla\psi(x_i), x_{\sigma(i)} - x_i\rangle \le \psi(x_{\sigma(i)}) - \psi(x_i)$$

for any $i = 1, \dots, N$. Adding these N inequalities we obtain that

$$\sum_{i=1}^{N} \langle \nabla\psi(x_i), x_{\sigma(i)} - x_i\rangle \le 0,$$

and we observe that the last condition is equivalent, after expanding the squares, to

$$\sum_{i=1}^{N} |\nabla\psi(x_i) - x_i|^2 \le \sum_{i=1}^{N} |\nabla\psi(x_i) - x_{\sigma(i)}|^2,$$

obtaining the desired c-cyclical monotonicity.

To prove (iii) let us introduce the map $i : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n \times \mathbb{R}^n$ defined by $i(x, y) := (y, x)$ and define $\tilde{c}(y, x) := c(x, y)$. Now it is easy to check that $i_\#$ sends $\Gamma(\mu, \nu)$ into $\Gamma(\nu, \mu)$ and $\Gamma_{o,c}(\mu, \nu)$ into $\Gamma_{o,\tilde{c}}(\nu, \mu)$. In our case, since $\tilde{c} = c$, denoting by T the unique optimal transport map between μ and ν and by S the unique optimal transport map between ν and μ, we deduce from the uniqueness of optimal transport plans that

$$(\mathrm{id} \times T)_\#\mu = (S \times \mathrm{id})_\#\nu.$$

It follows by the change of variables formula that for any nonnegative Borel function f one has

$$\int_{\mathbb{R}^n} f(x, T(x))\, \mathrm{d}\mu(x) = \int_{\mathbb{R}^n} f(S(y), y)\, \mathrm{d}\nu(y).$$

By choosing $f(x, y) = |y - T(x)|$ we obtain

$$0 = \int_{\mathbb{R}^n} |y - T(S(y))| \, \mathrm{d}\nu(y),$$

finally concluding that $T \circ S = \mathrm{id}$ ν-a.e. in $\mathbb{R}^n$. By a similar argument we get $S \circ T = \mathrm{id}$ μ-a.e. in $\mathbb{R}^n$. □

Remark 5.3 (Twist Condition and dist^p *Cost Functions)* Let us point out the key ingredients of the proof of Theorem 5.2. The first one is the differentiability μ-almost everywhere of ϕ, while the second one is the existence of a unique y solving the equation

$$\nabla_x c(x, y) = \nabla \phi(x).$$

At least locally, the second condition is ensured by the so-called *twist* condition

$$\det \nabla^2_{xy} c(x, y) \neq 0. \tag{5.2}$$

In general, however, the link between the gradient of the Kantorovich potential and the optimal map is not linear: for instance, in the case when $c(x, y) = |y - x|^p/p$, one obtains $|y - x|^{p-2}(x - y) = \nabla \phi(x)$ and, inverting this relation, this leads to the formula

$$T(x) = x - |\nabla \phi(x)|^{\frac{2-p}{p-1}} \nabla \phi(x). \tag{5.3}$$

In Sect. 4 of Lecture 4 we observed that, in the case of quadratic cost $c(x, y) = \frac{1}{2}|x - y|^2$, a function ϕ is c-concave if and only if $\phi(x) - \frac{1}{2}|x|^2$ is concave. Since $-\frac{1}{2}|x|^2$ is uniformly concave, this means that if $\|\phi\|_{C^2}$ is sufficiently small (more precisely smallness of the Hessian is sufficient) then ϕ is c-concave. Equivalently, small perturbations of the identity map are optimal. The picture is less understood in the case $p \neq 2$. In the following remarks, that have been suggested to us by F. Santambrogio, we consider sufficient and insufficient c-concavity conditions in the nonlinear case.

Remark 5.4 (c-Concavity for $c = \mathrm{dist}^p$ *with* $p > 2$*)* We claim that in the case of cost function $c(x, y) = \frac{1}{p}|x - y|^p$ for $p > 2$ there is no $k \in \mathbb{N}$ for which the condition $\|\phi\|_{C^k} < \epsilon$ guarantees that ϕ is c-concave.

To this aim it is sufficient to observe that, for any $\epsilon > 0$, $\phi(x) := \frac{1}{2}\epsilon|x|^2$ is not c-concave. Indeed, if ϕ is c-concave, then

$$\phi(x) = \phi^{cc}(x) = \min_y \{c(x, y) - \phi^c(y)\}$$

and it follows that, if y is a minimizer, then $y \in \partial^c \phi(x)$ and $\partial^c \phi(x)$ is not empty. On the other hand, if x is a differentiability point of ϕ, by differentiation we get that

$\partial^c\phi(x) = \{y\}$ is a singleton, with $y = T(x)$ given by (5.3). Since

$$\Gamma := \{(x, T(x)) : \phi \text{ is differentiable at } x\}$$

is contained in the contact set, Γ is c-cyclically monotone and then T is optimal. Now we can compute $T(x) = x - (\epsilon|x|)^{\frac{2-p}{p-1}}\epsilon x$. Therefore, if μ is the uniform measure concentrated on the sphere of radius $R = R(\epsilon)$ with $2R^{\frac{p-2}{p-1}} = \epsilon^{\frac{1}{p-1}}$, we obtain that $T(x) = -x$ μ-a.e. and therefore $T_\sharp\mu = \mu$. It follows that T is not optimal and then, thanks to the discussion above, ϕ is not c-concave.

Let us also point out that it holds $R(\epsilon) \to 0$ as $\epsilon \downarrow 0$ (thanks to the choice $p > 2$). Therefore the smallness of the C^k norm is seen to be not a sufficient c-concavity condition also restricting to fixed compact subsets of $\mathbb{R}^n$.

Remark 5.5 (c-Concavity for $c = \text{dist}^p$ with $1 < p < 2$) In the case of cost $c(x, y) = \frac{1}{p}|y - x|^p$ for $1 < p < 2$ the same example considered above shows that the smallness of any C^k norm is insufficient for c-concavity on the entire $\mathbb{R}^n$. The counterexample is no more valid when restricting to compact sets since $R(\epsilon)$ goes to infinity as $\epsilon \downarrow 0$ if $1 < p < 2$.

Let us see that on compact sets any function ϕ with sufficiently small $C^{1,1}$ norm is c-concave for $c(x, y) = |x - y|^p$ and $1 < p < 2$.

Indeed, under this assumption, for any convex and bounded set $K \subset \mathbb{R}^n$ the function $v \mapsto |v|^p$ is α-convex in K for some $\alpha > 0$. Therefore, if $\|\phi\|_{C^{1,1}}$ is sufficiently small, $x \mapsto \frac{1}{p}|x - y|^p - \phi(x)$ is convex in K too, since its gradient is monotone. It follows that if $y = x - |\nabla\phi(x)|^{\frac{2-p}{p-1}}\nabla\phi(x)$, then x is a minimizer in K of $\frac{1}{p}|y - \cdot|^p - \phi(\cdot)$. Hence $\phi(x) + \phi^c(y) = c(x, y)$ for any $x \in K$ and ϕ is c-concave in K, as we claimed.

2 A Digression About Monge's Problem

In the original formulation of the optimal transport problem Monge was interested in the case of cost $c(x, y) = |x - y|$ (in the Euclidean setting). By analogy with the case of quadratic cost we can consider the transport problem between measures $\mu, \nu \in \mathscr{P}_1(\mathbb{R}^n)$ with $\mu \ll \mathscr{L}^n$. Following the strategy of Theorem 5.2, and recalling that c-concave functions are nothing but 1-Lipschitz functions in this setting, one finds a 1-Lipschitz function ϕ such that $\text{supp}\,\pi \subset \partial^c\phi$, with $\pi \in \Gamma_o(\mu, \nu)$. By differentiating the minimality property of $c(\cdot, y) - \phi$ at x, we obtain

$$\frac{x - y}{|x - y|} = \nabla\phi(x) \qquad \text{and then} \qquad y = x - t\nabla\phi(x),\ t > 0$$

whenever $(x, y) \in \operatorname{supp}\pi$ and $y \neq x$. Hence, in this case the direction of transportation but not the transportation length is determined by the Kantorovich potential (in view of Example 1.13, this is not too surprising).

Let us define *transport ray* any segment $[x, y]$ in $\mathbb{R}^n$ whose endpoints x, y satisfy

$$\phi(x) - \phi(y) = |x - y|.$$

With this terminology, π is concentrated on endpoints of transport rays and the direction $y - x$ of the ray is opposite to $\nabla\phi(x)$, whenever ϕ is differentiable at x. It is not hard to prove that:

(i) any segment contained in a transport ray is itself a transport ray;
(ii) two transport rays with different directions cannot meet in the (relative) interior of one of them;
(iii) ϕ is differentiable in the interior of any transport ray;
(iv) if two transport rays with different directions meet at a common endpoint x, then ϕ is not differentiable at x.

Let us point out that, in particular, the properties (iii) and (iv) depend on the fact that the slope of ϕ is maximal along the ray.

All in all, invoking also Rademacher's theorem and (iv) to rule out the common endpoints, we have a decomposition of μ-almost all of $\mathbb{R}^n$ into a family of pairwise disjoint and (relatively) open transport rays.

The strategy pursued to solve the transport problem, first suggested by N. Sudakov in [112], is then to "factor" the n-dimensional problem into a family of one-dimensional problems along transport rays, prove that for these problems the hypotheses of Theorem 1.11 are fulfilled, and then construct the optimal transport map by gluing together a family of optimal maps for the one dimensional problems (obtained, for instance, with the monotone rearrangements). An essential ingredient of this strategy is the fact that the conditional measures μ_i obtained by disintegrating μ along the transport ray R_i are absolutely continuous with respect to $\mathscr{L}^1$ or, at least, that have no atom. The only fact that $\mu \ll \mathscr{L}^n$ and the interiors of transport rays are disjoint are sufficient to prove this property in dimension $n = 2$, but not enough in dimension $n > 2$, [11]. So, Sudakov's proof was not complete and, for the Euclidean distance, the first rigorous proof of the existence of an optimal map came in [58]. Later on, Sudakov's proof has been fixed in [5, 8, 115], by identifying an "extra" regularity condition of the decomposition of transport rays induced by ϕ. Finally, the case when $c(x, y) = \|x - y\|$, with $\|\cdot\|$ a norm in $\mathbb{R}^n$, is even more challenging, due to the potential lack of regularity/uniform convexity of the norm: the full solution came in [37, 38, 40].

3 Applications

We will illustrate two applications of Theorem 5.2. The first one is related to the polar factorization of maps. Let us begin by recalling the following result.

Theorem 5.6 (Helmholtz's Decomposition) *Let $\Omega \subset \mathbb{R}^n$ be an open bounded domain. Then any vector field $F \in L^2(\Omega;\mathbb{R}^n)$ can be written in a unique way as $F = \nabla\phi + G$, where $\phi \in H^1(\Omega)$ and $G \in L^2(\Omega,\mathbb{R}^n)$ is a divergence-free vector field in the distributional sense.*

To prove the existence part of Theorem 5.6 one starts by solving the following minimization problem

$$\min_{\psi\in H^1(\Omega)} \int_\Omega |F - \nabla\psi|^2 \,\mathrm{d}x, \tag{5.4}$$

which, by strict convexity, admits a unique solution $\nabla\phi$. Then, the Euler-Lagrange equation associated to the minimization problem (5.4), namely

$$\int_\Omega (F - \nabla\phi) g \,\mathrm{d}x = 0 \qquad \forall g \in C_c^\infty(\Omega),$$

corresponds precisely to the vector field $G := F - \nabla\phi$ being divergence-free in the distributional sense.

While Theorem 5.6 above provides us with an *additive* decomposition of vector fields, we are interested in a *multiplicative* decomposition. In this connection, let us remind that any linear mapping $L : \mathbb{R}^n \to \mathbb{R}^n$ can be factored as $L = S \circ O$, where $S : \mathbb{R}^n \to \mathbb{R}^n$ is self-adjoint, nonnegative and $O : \mathbb{R}^n \to \mathbb{R}^n$ is orthogonal (namely $O^* = O^{-1}$, O^* being the adjoint). This is known as "polar decomposition". The next result generalizes, to some extent, the polar decomposition to the nonlinear case.

Theorem 5.7 (Polar Decomposition for Vector Fields) *Let $D \subset \mathbb{R}^n$ be a bounded Borel set with $z := \mathscr{L}^n(D) > 0$. Define a probability measure $\mu \in \mathscr{P}_2(\mathbb{R}^n)$ by $\mu := z^{-1}\mathscr{L}^n \llcorner D$. Then any nondegenerate vector field $F \in L^2(D,\mathbb{R}^n)$ (i.e. such that $F_\#\mu \ll \mathscr{L}^n$) can be written in a unique way as $F = (\nabla\phi) \circ s$, where $\nabla\phi$ is the gradient of a convex function ϕ and $s : D \to D$ is μ-measure preserving (i.e. $s_\#\mu = \mu$).*

Proof To prove the existence of the decomposition define $\nu := F_\#\mu \ll \mathscr{L}^n$ and observe that $\nu \in \mathscr{P}_2(\mathbb{R}^n)$ since $F \in L^2(D;\mathbb{R}^n)$. If S is the unique solution of the optimal transport problem with quadratic cost from ν to μ provided by Theorem 5.2 we obtain that $s := S \circ F$ is μ-measure preserving. Now, if T is the unique solution for the optimal transport problem with quadratic cost from μ to ν then $T = \nabla\phi$ for some convex function ϕ and we get that $T \circ s = (T \circ S) \circ F = F$ (by the third conclusion in Theorem 5.2).

To prove uniqueness, one can observe that the preceding argument is basically reversible. Indeed, if we consider another decomposition $F = (\nabla\tilde{\phi}) \circ \tilde{s}$ we obtain that $\nu = F_{\#}\mu = (\nabla\tilde{\phi}) \circ \tilde{s}_{\#}\mu = (\nabla\tilde{\phi})_{\#}\mu$ and it follows from the second conclusion in Theorem 5.2 that $\nabla\phi = \nabla\tilde{\phi}$. By taking the left composition with S (namely the inverse of T) we obtain $s = \tilde{s}$. □

Remark 5.8 (L^2 Projection on μ-Measure Preserving Maps) Let $S(D)$ be the space of μ-measure preserving maps. It is easily seen that $S(D)$ is a closed subset of $L^2(D; \mathbb{R}^n)$. One can observe that the μ-measure preserving map s in the statement of Theorem 5.7 is the unique solution of the projection problem

$$\min_{r \in S(D)} \left\{ \int_D |F - r|^2 \, \mathrm{d}\mu \right\}$$

and the minimum value equals $\min_{(K)}$ with $c(x, y) = |x - y|^2$.

Indeed, with the same notation of the proof of the previous theorem, the fact that $(F, r)_{\#}\mu \in \Gamma(\nu, \mu)$ for any $r \in S(D)$ provides the lower bound, while the upper bound follows by the choice $r = s$, where s is given by the polar decomposition $F = T \circ s$:

$$\int_D |F - s|^2 \, \mathrm{d}\mu = \int_D |T - \mathrm{id}|^2 \, \mathrm{d}\mu = \min_{(K)}.$$

For uniqueness, one starts from the observation that $(F, r)_{\#}\mu \in \Gamma_o(\nu, \mu)$ whenever F is a minimizer, and then the uniqueness of transport plans gives $(F, r)_{\#}\mu = (\nabla\phi \times \mathrm{id})_{\#}\mu$. Since $F = T \circ s$, this yields $r = s$ arguing as in the proof of (iii) of Theorem 5.2, choosing the test function

$$f(x, y) = |y - S(x)|$$

and using (5.1). The existence of the closest point s is somehow surprising, since $S(D)$ is far from being convex, so that standard Hilbert space techniques do not apply: one can realize this extreme lack of convexity noticing that, according to (5.5), a smooth and injective map s is μ-measure preserving if and only if $|\det \nabla s| \equiv 1$ in D.

In this connection, notice that closed subsets of Hilbert spaces with the unique projection property are called Chebyshev sets, and obviously closed convex sets are Chebyshev; the validity of the converse implication is known to be true in finite-dimensional spaces and still open in infinite-dimensional ones, see [53] for more on this fascinating problem. If it were not for the non-degeneracy condition $F_{\#}\mu \ll \mathscr{L}^n$, which guarantees uniqueness of the projection, the nonconvex subset $S(D)$ of $L^2(\mu; \mathbb{R}^n)$ would be a counterexample.

4 Iterated Monotone Rearrangement

Let $\mu,\ \nu \in \mathscr{P}_2(\mathbb{R}^n)$ with $\mu \ll \mathscr{L}^n$. Brenier's theorem (Theorem 5.2) ensures the existence of an optimal map T between μ and ν, representable as the gradient of some convex function f. In general this map is difficult to compute, for instance if $\mu = \varrho\mathscr{L}^n$ and $\nu = \eta\mathscr{L}^n$, with $\varrho,\ \eta$ smooth and $\eta > 0$, we will see that finding an injective map $T = \nabla f$ pushing μ to ν is equivalent (if f is sufficiently smooth) to solving the Monge-Ampère equation

$$\det \nabla^2 f = \frac{\varrho}{\eta(\nabla f)}. \tag{5.5}$$

This is a highly nonlinear PDE, very difficult to solve in general, even numerically. On the other hand, we have seen that in the case $n = 1$ the picture is much easier and, recalling the notation $F_\mu(x) = \mu((-\infty, x])$, $F_\nu(y) = \nu((-\infty, y])$, we can directly compute T by $F_\nu^{-1} \circ F_\mu$, at least when μ has no atom and F_ν is strictly increasing.

It turns out that, considering an iterated monotone rearrangement, we can "constructively" build a transport map even in dimension higher than 1 (this idea is due to H. Knothe [74], and the Knothe map was known a long before the Brenier map). For simplicity, we describe this construction in the two-dimensional case, even though it can be done similarly in any dimension (see Remark 5.10). Taking $\mu,\ \nu \in \mathscr{P}(\mathbb{R}^2)$ and denoting $\mu_1 = (p_x)_\#\mu$ and $\nu_1 = (p_x)_\#\nu$, where (x, y) are the coordinates in $\mathbb{R}^2$ and p_x is the projection on the first variable, by the disintegration theorem (see Theorem 2.4) we have the following equalities:

$$\mu = \int_{\mathbb{R}} \mu_x \otimes \mu_1(x), \qquad \nu = \int_{\mathbb{R}} \nu_x \otimes \nu_1(x)$$

with $\mu_x,\ \nu_x \in \mathscr{P}(\mathbb{R}_y)$.

Proposition 5.9 *Assume that μ_1 has no atom and that μ_x has no atom for μ_1-a.e. $x \in \mathbb{R}$. Denote by S the monotone rearrangement from μ_1 to ν_1. Then the Knothe map*

$$T_K : (x, y) \to \left(S(x), T_{\mu_x}^{\nu_{S(x)}}(y)\right),$$

where $T_{\mu_x}^{\sigma}$ denotes the monotone rearrangement from μ_x to σ, satisfies $(T_K)_\#\mu = \nu$.

Proof For any nonnegative function $\psi : \mathbb{R}^2 \to \mathbb{R}$, we compute

$$\begin{aligned}\int_{\mathbb{R}^2} \psi(T_K(x,y))\,\mathrm{d}\mu(x,y) &= \int_{\mathbb{R}}\int_{\mathbb{R}} \psi(S(x), T_{\mu_x}^{\nu_{S(x)}}(y))\,\mathrm{d}\mu_x(y)\,\mathrm{d}\mu_1(x)\\ &= \int_{\mathbb{R}}\int_{\mathbb{R}} \psi(S(x), y')\,\mathrm{d}\nu_{S(x)}(y')\,\mathrm{d}\mu_1(x)\\ &= \int_{\mathbb{R}}\int_{\mathbb{R}} \psi(x', y')\,\mathrm{d}\nu_{x'}(y')\,\mathrm{d}\nu_1(x') = \int_{\mathbb{R}^2} \psi(x', y')\,\mathrm{d}\nu(x', y').\end{aligned}$$

□

Remark 5.10 Let us collect a few remarks about Knothe's map.

(i) In dimension $n = 3$, with coordinates (x, y, z), the construction goes as follows:

$$T_K(x,y,z) := \big(S(x,y), T_{\mu_{xy}}^{\nu_{S(x,y)}}(z)\big),$$

where this time $S(x, y)$ is the Knothe map between the projections of μ and ν on $\mathbb{R}^2_{x,y}$ and μ_{xy}, $\nu_{xy} \in \mathscr{P}(\mathbb{R}_z)$ are the corresponding disintegrations of μ and ν. In an analogous recursive way one can build the Knothe map in higher dimensions.

(ii) If $\mu = f\mathscr{L}^2$ and $\nu = g\mathscr{L}^2$, then $\mu_1 = \varrho\mathscr{L}^1$ with $\varrho(x) = \int_{\mathbb{R}} f(x,y)\,\mathrm{d}y$ and $\mu_x = \varrho_x\mathscr{L}^1$ with $\varrho_x(y) = f(x,y)/\varrho(x)$. Similar formulas holds for ν, making the practical computation of T_K very easy.

(iii) The regularity of T_K depends of course on the regularity of $S(x)$ and $T_{\mu_x}^{\nu_{S(x)}}(y)$. In particular T_K is smooth if the densities f and g are smooth, with $g > 0$.

(iv) By construction, the map ∇T has an upper triangular structure with nonnegative entries on the diagonal:

$$\det \nabla T = \prod_{i=1}^{n} \partial_i T_i \geq 0.$$

(v) In [39], G. Carlier, A. Galichon and F. Santambrogio presented a beautiful variational interpolation scheme between the Brenier and Knothe maps. In the two dimensional case the scheme involves the optimal transport maps T_ϵ relative to the costs

$$c_\epsilon\big((x,y),(x',y')\big) := \frac{1}{\epsilon}|x - x'|^2 + |y - y'|^2,$$

whose existence, since the twist condition (5.2) is obviously satisfied, can be proved arguing as in Brenier's theorem. As $\epsilon \to 0^+$, under the assumptions of Proposition 5.9, it can be proved that T_ϵ converge to T_K in $L^2(\mu;\mathbb{R}^2)$.

Lecture 6: A Proof of the Isoperimetric Inequality and Stability in Optimal Transport

1 Isoperimetric Inequality

Another striking application of the optimal transport theory is the proof of the isoperimetric inequality. In [92] M. Gromov gave a proof of this inequality based on Knothe's map [74] and, as we will see, essentially the same proof works with Brenier's map. This strategy has been pushed even further by A. Figalli, F. Maggi and A. Pratelli in [63] to treat the quantitative (and even the anisotropic) version of the inequality.

In the sequel we shall denote by σ^{n-1} the surface measure, elementarily defined on C^1 hypersurfaces via local parametrizations. For $k \geq 1$ integer, we also say that a set $E \subset \mathbb{R}^n$ is C^k-regular if, at any $x \in \partial E$, the set E is locally the subgraph of a C^k function, in a suitable system of coordinates.

Theorem 6.1 (Isoperimetric Inequality) *Let $E \subset \mathbb{R}^n$ be a C^1-regular bounded open set and let $B \subset \mathbb{R}^n$ be the ball with $\mathscr{L}^n(E) = \mathscr{L}^n(B)$. Then $\sigma^{n-1}(\partial E) \geq \sigma^{n-1}(\partial B)$.*

Proof For the moment we give a formal proof, assuming by scaling invariance that $\mathscr{L}^n(E) = \omega_n$. Since E is bounded, both measures

$$\mu := \frac{1}{\omega_n}\mathscr{L}^n \llcorner E, \qquad \nu := \frac{1}{\omega_n}\mathscr{L}^n \llcorner B$$

belong to $\mathscr{P}_2(\mathbb{R}^n)$. Let T be the Brenier map from μ to ν and assume that it is C^1-regular up to the boundary of E. Observe that (5.5) gives $|\det \nabla T(x)| = 1$ for all $x \in E$. Since $\nabla T = \nabla^2 \phi$ is a symmetric matrix, it is diagonalizable. Applying the inequality between arithmetic and geometric mean to the nonnegative eigenvalues of ∇T we obtain:

$$1 = \det(\nabla T)^{1/n} = (\lambda_1 \cdots \lambda_n)^{1/n} \leq \frac{1}{n}(\lambda_1 + \cdots + \lambda_n) = \frac{1}{n}\operatorname{div} T.$$

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Using the divergence theorem and the fact that T takes values in $\overline{B}(0,1)$ we have

$$\sigma^{n-1}(\partial B) = n\mathscr{L}^n(E) \le \int_E \operatorname{div} T \, \mathrm{d}x = \int_{\partial E} \langle T, \nu \rangle \, \mathrm{d}\sigma^{n-1} \le \sigma^{n-1}(\partial E) \tag{6.1}$$

and the statement follows. □

The previous formal argument can be made rigorous by using one of these strategies:

(i) since the target measure ν has a convex support and positive density inside, one may use deep regularity results for solutions of Monge-Ampère equations due to L.A. Caffarelli, [31–36];
(ii) try to understand to what extent the computation (6.1) can be replicated using only the information that T is the gradient of a convex map;
(iii) work with approximate transport maps, trying to retain the good properties of Brenier's map.

The strategy (i) is not feasible for us since, as we said, the regularity theory for solutions of Monge-Ampère equations is very deep. The most powerful strategy seems to be (ii), since combined with many (nontrivial) extra ideas led in [63] (see also the lecture notes [87]) even to the quantitative version $A(E) \le C(n)\sqrt{\delta(E)}$ of the isoperimetric inequality. Here $A(E)$ is the *spherical asymmetry index*

$$A(E) := \min\left\{ \frac{\mathscr{L}^n\big(E \Delta B\big)}{\mathscr{L}^n(E)} : B \text{ ball}, \ \mathscr{L}^n(B) = \mathscr{L}^n(E) \right\}$$

and $\delta(E)$ is the *isoperimetric deficit*

$$\delta(E) := \frac{\sigma^{n-1}(\partial E)}{n\omega_n^{1/n}|E|^{(n-1)/n}} - 1 \ge 0.$$

Notice that both $A(E)$ and $\delta(E)$ are scaling invariant, and that $A(E) \le C(n)\sqrt{\delta(E)}$ shows in particular that the only solution to the isoperimetric problem is the ball, a property hard to prove with strategy (iii).

In these notes we have chosen to use a combination of strategies (ii) and (iii), so let us study more in detail the regularity properties of gradients of convex functions, starting from the one-dimensional case.

For all $f : (a,b) \subset \mathbb{R} \to \mathbb{R}$ convex, the quantities

$$f'_+(x) = \lim_{h \to 0^+} \frac{f(x+h) - f(x)}{h}, \qquad f'_-(x) = \lim_{h \to 0^-} \frac{f(x+h) - f(x)}{h}$$

are well-defined for all $x \in (a,b)$, thanks to the monotonicity of difference quotients. These two functions are moreover nondecreasing and, since $f'_-(x) \le f'_+(x) \le f'_-(y)$ whenever $x < y$, it is easily seen that they differ only in a at most countable subset of (a,b).

Because of monotonicity, $f'_\pm$ are of locally bounded variation in (a,b), their *pointwise* derivatives f''_+, f''_- exist and coincide $\mathscr{L}^1$-a.e. in (a,b). In addition, their common *distributional* derivative $Df'_+ = Df'_-$ is a locally finite and nonnegative measure in (a,b) whose density with respect to $\mathscr{L}^1$ is $f''_\pm$ (see for instance [10, 104] for the proof of these elementary properties of monotone functions).

Example 6.2 Let us consider the convex function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} -x & \text{if } x \in (-\infty, 0], \\ \dfrac{x^2}{2} & \text{if } x \in (0, \infty). \end{cases}$$

Its right derivative $f'_+(x)$ is identically equal to -1 in $(-\infty, 0)$ and equal to x on $[0,\infty)$, so it has a jump discontinuity in 0 and therefore it cannot be "better than of bounded variation". More rigorously, we have that the distributional derivative Df' of f' is given by

$$Df' = \mathscr{L}^1 \llcorner (0,\infty) + \delta_0,$$

a measure with nonzero singular part.

In higher dimension the situation is similar. In a convex domain Ω in $\mathbb{R}^n$, a convex function $f : \Omega \to \mathbb{R}$ is locally Lipschitz, so that (as we have seen in the proof of Theorem 5.2), according to Rademacher's theorem its gradient ∇f exists $\mathscr{L}^n$-a.e. in Ω. Passing to second-order properties, one has $\nabla f \in BV_{\mathrm{loc}}(\Omega;\mathbb{R}^n)$, where BV_{loc} denotes the space of (vector valued) functions with *locally bounded variation*, whose definition is recalled below for the sake of completeness. We refer to [10] or [59] for a more detailed account about this topic.

Definition 6.3 Let $\Omega \subset \mathbb{R}^n$ be an open set and let $u \in L^1_{\mathrm{loc}}(\Omega)$ be a locally integrable function. We say that u has locally bounded variation (and write $g \in BV_{\mathrm{loc}}(\Omega)$) if its distributional derivative is representable with a locally finite measure, i.e. if

$$\int_\Omega u \operatorname{div} \varphi \, \mathrm{d}x = -\sum_{i=1}^n \int_\Omega \varphi_i \, \mathrm{d}D_i u, \quad \text{for any } \varphi \in C^1_c(\Omega,\mathbb{R}^n), \tag{6.2}$$

for some $\mathbb{R}^n$-valued measure $Du = (D_1 u, \dots, D_n u)$ with locally finite total variation on Ω.

If $g \in L^1_{\mathrm{loc}}(\Omega,\mathbb{R}^N)$, $N \geq 1$, then we say that g has locally bounded variation if $g_i \in BV_{\mathrm{loc}}(\Omega)$ for any $i = 1, \dots, N$.

Let us summarize in the next theorem all the properties we need for a rigorous proof of the isoperimetric inequality.

Theorem 6.4 *If $\Omega \subset \mathbb{R}^n$ is convex and $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ is a convex function, then $\nabla f \in BV_{\rm loc}(\Omega, \mathbb{R}^n)$. In addition*

(i) for $\mathscr{L}^n$-a.e. $x \in \Omega$ we have the second-order Taylor expansion

$$f(y) = f(x) + \langle \nabla f(x), y - x\rangle + \frac{1}{2}\langle \nabla^2 f(x)(y-x), (y-x)\rangle + o(|y-x|^2); \tag{6.3}$$

(ii) if $\mathcal{D} \subset \Omega$ is the set of differentiability points of f, $\nabla f : \mathcal{D} \to \mathbb{R}^n$ is differentiable $\mathscr{L}^n$-a.e. and its differential coincides $\mathscr{L}^n$-a.e. in Ω with the symmetric matrix $\nabla^2 f$ in (i);

(iii) the symmetric matrix $\nabla^2 f$ in (6.3) above is the density of $D(\nabla f) = D^2 f$ with respect to $\mathscr{L}^n$.

Let us remark that in (ii), even though $\mathcal{D}$ is not necessarily open, $\nabla^2 f(x)$ is identified by the standard property $\nabla f(y) - \nabla f(x) - \langle \nabla^2 f(x), y - x\rangle = o(|x-y|)$ as $D \ni y \to x$.

We will prove only the assertion $\nabla f \in BV_{\rm loc}(\Omega, \mathbb{R}^n)$. The proof of (i), known as Alexandrov theorem, can be found in [1, 59].

We start with two simple lemmas. Recall that a distribution T is nonnegative if $\langle T, \psi\rangle \geq 0$ for any nonnegative test function ψ.

Lemma 6.5 *Every nonnegative distribution T in an open set $\Omega \subset \mathbb{R}^n$ is representable by a nonnegative and locally finite measure μ in Ω, namely*

$$T(\psi) = \int_\Omega \psi \, \mathrm{d}\mu, \qquad \forall \psi \in C_c^\infty(\Omega).$$

Proof Let $A \Subset \Omega$ and fix $\chi \in C_c^\infty(\Omega)$ with $\chi \equiv 1$ in A. For all $\psi \in C_c^\infty(\Omega)$ with $\operatorname{supp}\psi \subset A$ we have

$$T(\psi) = T(\chi\psi) \leq T(\chi) \max|\psi|.$$

Using $-\psi$ instead of ψ we obtain that $|T(\psi)| \leq T(\chi)\max|\psi|$, and the statement follows by Riesz representation theorem. □

Lemma 6.6 *If $\Omega \subset \mathbb{R}^n$ is convex, $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ is a convex function and $v \in \mathbb{R}^n$, then the distribution*

$$\left\langle D_v\left(\frac{\partial f}{\partial v}\right), \psi \right\rangle := \int_\Omega f(x) \frac{\partial^2 \psi}{\partial^2 v}(x)\,\mathrm{d}x, \qquad \psi \in C_c^\infty(\Omega),$$

is nonnegative.

Proof Let us start by observing that, when $f \in C^2(\Omega)$, the distribution $D_v(\frac{\partial f}{\partial v})$ is representable by the second directional derivative of f with respect to v, which is obviously nonnegative.

In the general case we use an approximation argument. Let $\varrho_\epsilon(\cdot) = \epsilon^{-n}\varrho(\cdot/\epsilon)$ be standard mollifiers, with ϱ even and compactly supported in $B(0,1)$, and let $f_\epsilon := f * \varrho_\epsilon$, which are smooth and locally convex in the open set $\Omega_\epsilon = \{x \in \Omega : \operatorname{dist}(x, \Omega) > \epsilon\}$. For all $\psi \in C_c^\infty(\Omega)$, since $\operatorname{supp}\psi \subset \Omega_\epsilon$ for $\epsilon > 0$ small enough, we have

$$\int_\Omega f(x)\frac{\partial^2\psi}{\partial^2 v}(x)\,\mathrm{d}x = \lim_{\epsilon\to 0^+}\int_\Omega f_\epsilon(x)\frac{\partial^2\psi}{\partial^2 v}(x)\,\mathrm{d}x = \lim_{\epsilon\to 0^+}\int_\Omega \psi(x)\frac{\partial^2 f_\epsilon}{\partial^2 v}(x)\,\mathrm{d}x \geq 0.$$

□

By polarization we can write

$$D_i\left(\frac{\partial f}{\partial x_j}\right) = \frac{1}{4}\left(D_v\left(\frac{\partial f}{\partial v}\right) + D_w\left(\frac{\partial f}{\partial w}\right)\right),$$

for every $1 \leq i,\ j \leq n$, where $v = e_i + e_j$ and $w = e_i - e_j$. This, in combination with Lemmas 6.5 and 6.6 above, proves the BV property stated in Theorem 6.4.

We now go back to the proof of the isoperimetric inequality. Let T be the Brenier transport map between μ and ν. We know that, in general, there exists $f : \mathbb{R}^n \to (-\infty, \infty]$ convex, differentiable μ-almost everywhere, such that $\nabla f = T$. We will see later that, in this special case, we can represent T as ∇f, where $f : \mathbb{R}^n \to \mathbb{R}$ is a 1-Lipschitz and convex map. Taking the convolution of f with a standard mollifier ϱ_ϵ and setting $T_\epsilon = (\nabla f) * \varrho_\epsilon = T * \varrho_\epsilon$, we have enough regularity to make the same estimation (6.1) of the formal proof. Indeed, since f is 1-Lipschitz we have $|T_\epsilon| = |(\nabla f) * \varrho_\epsilon| \leq 1$ and then

$$\int_E \operatorname{div} T_\epsilon\,\mathrm{d}x = \int_{\partial E}\langle T_\epsilon, \nu\rangle\,\mathrm{d}\sigma^{n-1} \leq \sigma^{n-1}(\partial E).$$

The identity of measures $\operatorname{div} T_\epsilon \mathscr{L}^n = \operatorname{tr}(D^2 f) * \varrho_\epsilon$, together with the inequality $D^2 f \geq \nabla^2 f \mathscr{L}^n$ guaranteed by Theorem 6.4, gives the pointwise inequality $\operatorname{div} T_\epsilon \geq \operatorname{tr}(\nabla^2 f) * \varrho_\epsilon$. Now, if we are able to prove that the identity $\det(\nabla^2 f) = 1$ holds $\mathscr{L}^n$-almost everywhere in E (a detailed discussion about this property will be given in Lecture 7), we can apply the arithmetic-geometric mean inequality to the eigenvalues of $\nabla^2 f$ to conclude that the inequality $\operatorname{tr}(\nabla^2 f) \geq n$ holds $\mathscr{L}^n$-almost everywhere in E. Hence, we could pass to the limit as $\epsilon \to 0$ in the inequality

$$\int_E \operatorname{tr}(\nabla^2 f) * \varrho_\epsilon\,\mathrm{d}x \leq \sigma^{n-1}(\partial E)$$

to conclude that $n\mathscr{L}^n(E) \leq \sigma^{n-1}(\partial E)$.

Let us now clarify the construction of f. Since ν is supported in $\overline{B}_1$, we can apply the general theory with $X = \mathbb{R}^n$ and $Y = \overline{B}_1$ to get the expression

$$\phi(x) = \inf_{y \in \overline{B}_1} \left\{ \frac{1}{2}|x-y|^2 - \phi^c(y) \right\}, \qquad \forall x \in \mathbb{R}^n,$$

for the Kantorovich potential ϕ. Now, subtracting the squares, it is easily seen that the function $f(x) := \frac{|x|^2}{2} - \phi(x)$ is not only convex, but also 1-Lipschitz because it is the supremum, indexed by $y \in \overline{B}_1$, of a family of 1-Lipschitz functions.

Example 6.7 (Caffarelli's Example [35]) Fix $\alpha > 0$, let

$$B_1^+ = B_1 \cap \{x > 0\} \subset \mathbb{R}^2, \qquad B_1^- = B_1 \cap \{x < 0\} \subset \mathbb{R}^2$$

and let $S^+ = B_1^+ + (\alpha, 0)$, $S^- = B_1^- - (\alpha, 0)$. Let us define

$$\mu := \frac{1}{\pi} \mathscr{L}^2 \llcorner B_1, \qquad \nu := \frac{1}{\pi} \mathscr{L}^2 \llcorner (S^+ \cup S^-).$$

It is clear that the map

$$T(x, y) := (x, y) + \alpha \, \mathrm{sign}(x)(1, 0)$$

is optimal from μ to ν, if the cost is the square of the Euclidean distance: indeed, T is the gradient of the convex function $f(x, y) := \frac{1}{2}(x^2 + y^2) + \alpha|x|$ and therefore Theorem 5.2(ii) applies.

Obviously the optimal map T is discontinuous, since the support of the target measure ν is not connected. However, we can use Caffarelli's example to understand that the presence of discontinuities is not due only to topological reasons.

Let S_ϵ be a connected set close to $S^+ \cup S^-$ as in Fig. 1 below and consider the normalized Lebesgue measures ν_ϵ on S_ϵ. We will prove in Sect. 2 that the optimal transport maps T_ϵ from μ to ν_ϵ converge in $L^2(B_1; \mathbb{R}^2)$ to T and we claim that for $\epsilon \ll 1$ also T_ϵ is discontinuous.

Obviously the only convergence in $L^2(B_1; \mathbb{R}^2)$ of T_ϵ is not sufficient to prove the claim, and we need also to invoke the monotonicity of T_ϵ. Indeed, choose tangent tiny balls D_1, D_2 inside B_1, centered on the positive y axis (say with D_2 below D_1) and notice that for $\epsilon > 0$ small enough most of the points in $D_1 \cap \{x > 0\}$

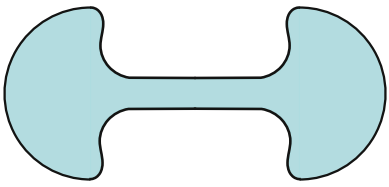


Fig. 1 Perturbation of the disconnected set $S^+ \cup S^-$

are sent almost horizontally to the right, and most of the points in $D_1 \cap \{x < 0\}$ are sent almost horizontally to the left. If T_ϵ is continuous, by continuity we can find $z \in D_1$ with $T_\epsilon(z) - z$ making an angle close to $\pi/4$ with the positive x axis. Moreover, there exists $\beta > 0$ such that for ϵ small enough one has $\mathrm{dist}(D_1, S_\epsilon) \geq \beta$, and then $|T_\epsilon(z) - z| \geq \beta$. Analogously, we can find $w \in D_2 \cap \{x < 0\}$ such that $T_\epsilon(w) - w$ is almost horizontal, in the direction of the negative x axis. With these choices, if the balls D_1, D_2 are sufficiently small it turns out that $w - z$ is almost vertical in the direction of the negative y axis, while $T_\epsilon(w) - T_\epsilon(z)$ has a positive second coordinate, say larger than $\beta\sqrt{2}/4$. Hence, since this coordinate is dominant in the computation of the scalar product, $\langle T_\epsilon(w) - T_\epsilon(z), w - z\rangle < 0$, contradicting the monotonicity of T_ϵ.

In Lecture 7 we shall say more about the regularity of the optimal transport map. This perturbation argument shows that, even when $\mathrm{supp}\,\nu$ is connected, we can only expect *partial* regularity of the transport map (namely, regularity out of a singular set), see [51, 62] for results in this direction.

2 Stability of Optimal Plans and Maps

In the investigation of the stability problem the strategy we adopted in the proof of Theorem 5.2 will be once more useful: first we prove the stability of optimal plans and then, under more restrictive assumptions on μ which ensure the existence of optimal maps, we see how this property can be rephrased in terms of stability of optimal maps.

Theorem 6.8 *Let $c : X \times Y \to [0, \infty)$ be a continuous cost function and let $(\mu_n) \subset \mathscr{P}(X)$, $(\nu_n) \subset \mathscr{P}(Y)$, weakly convergent to $\mu \in \mathscr{P}(X)$ and $\nu \in \mathscr{P}(Y)$ respectively. For any choice of $\pi_n \in \Gamma_o(\mu_n, \nu_n)$ with $\int_{X\times Y} c\,\mathrm{d}\pi_n < \infty$, it holds that:*

(i) the sequence (π_n) admits limit points in $\mathscr{P}(X \times Y)$ with respect to the weak convergence;
(ii) if $c(x, y) \leq a(x) + b(y)$ for some $a \in L^1(\mu)$, $b \in L^1(\nu)$, every limit point belongs to $\Gamma_o(\mu, \nu)$.

Proof The families $\{\mu_n\}$ and $\{\nu_n\}$ are equi-tight by Theorem 2.8 (notice that here we are using for the first time the implication in this direction) and, as we have seen in the proof of Corollary 2.9, this implies that the family $\{\pi_n\}$ is equi-tight as well. This proves the relative compactness in $\mathscr{P}(X \times Y)$ with respect to the weak convergence.

Thanks to Theorem 4.2(i), in order to conclude the proof it is sufficient to show that every limit point $\pi = \lim_k \pi_{n(k)}$ is supported in a c-cyclically monotone set.

Let us show first that for every $(x, y) \in \mathrm{supp}\,\pi$ there exist $(x_k, y_k) \in \mathrm{supp}\,\pi_{n(k)}$ convergent to (x, y). If the assertion were false we could find a neighborhood U of (x, y) that does not intersect $\mathrm{supp}\,\pi_{n_k}$ for infinitely many k, yielding a contradiction

because the evaluation on open sets is lower semicontinuous with respect to the weak convergence. Take now $\operatorname{supp}\pi \ni (x_i, y_i)$ for $i = 1, \dots, N$. By the previous argument, there exist $(x_i^k, y_i^k) \in \operatorname{supp}\pi_{n(k)}$ convergent to (x_i, y_i) as $k \to \infty$. Thanks to Theorem 3.17, $\operatorname{supp}\pi_{n(k)}$ are c-cyclically monotone, hence for any permutation σ we can pass to the limit as $k \to \infty$ in

$$\sum_{i=1}^{N} c(x_i^k, y_i^k) \le \sum_{i=1}^{N} c(x_i^k, y_{\sigma(i)}^k),$$

to obtain $\sum_i c(x_i, y_i) \le \sum_i c(x_i, y_{\sigma(i)})$. This proves the c-cyclical monotonicity of $\operatorname{supp}\pi$. □

Now we want to justify the statement made in the discussion of Caffarelli's example, namely that optimal maps are $L^2(\mu)$-stable with respect to perturbations of the target measure ν.

We start by recalling some useful probabilistic concepts, dealing with convergence of maps with values in a metric space, thought as random variables.

Definition 6.9 (Convergence in $\mathbb{P}$-Probability) Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let (X, d) be a metric space. If $f_h, f : \Omega \to X$ are measurable, we say that $f_h \to f$ in $\mathbb{P}$-*probability* if

$$\lim_{h\to\infty} \mathbb{P}(\{d(f_h, f) > \epsilon\}) = 0, \qquad \forall \epsilon > 0.$$

Here $\mathbb{P}(\{d(f_h, f) > \epsilon\})$ stands for $\mathbb{P}(\{x \in \Omega : d(f(x), f_h(x)) > \epsilon\})$. Convergence in $\mathbb{P}$-probability is also called by *convergence in measure*, in the measure-theoretic literature. It is easily seen that convergence in $\mathbb{P}$-probability is induced by the distance

$$d_{\mathbb{P}}(f, g) := \inf\{\epsilon > 0 : \mathbb{P}(\{d(f, g) > \epsilon\}) < \epsilon\}.$$

Lemma 6.10 *Let $(\Omega, \mathcal{F}, \mathbb{P})$, (X, d), f_h, f be as in Definition 6.9. Then $f_h \to f$ in $\mathbb{P}$-probability if and only if*

$$\lim_{h\to\infty} \int_\Omega 1 \wedge d(f_h, f)\, \mathrm{d}\mathbb{P} = 0.$$

Proof Assume that $f_h \to f$ in $\mathbb{P}$-probability. Then, since $1 \ge \mathbb{P}(\{d(f_h, f) > s\}) \to 0$ for all $s > 0$, we infer from Cavalieri's formula and the dominated convergence theorem that

$$\int_\Omega 1 \wedge d(f_h, f)\, \mathrm{d}\mathbb{P} = \int_0^1 \mathbb{P}(\{d(f_h, f) > s\})\, \mathrm{d}s \xrightarrow{h\to\infty} 0.$$

On the other hand, if $\epsilon \in (0,1)$ is fixed and $\lim_{h\to\infty} \int_\Omega 1 \wedge d(f_h, f)\, \mathrm{d}\mathbb{P} = 0$, we see from Markov's inequality that

$$\mathbb{P}(\{d(f_h, f) > \epsilon\}) \le \frac{1}{\epsilon} \int_\Omega 1 \wedge d(f_h, f)\, \mathrm{d}\mathbb{P} \xrightarrow{h\to\infty} 0.$$

□

Remark 6.11 (Relation with $\mathbb{P}$-a.e. and L^p Convergence) Thanks to the characterization provided in Lemma 6.10 and the dominated convergence theorem, convergence $\mathbb{P}$-a.e. implies convergence in $\mathbb{P}$-probability. On other hand, since convergence in $L^1(\mathbb{P})$ of $1 \wedge d(f_h, f)$ implies convergence $\mathbb{P}$-a.e. to 0 of a subsequence, we see also that convergence in $\mathbb{P}$-probability yields convergence $\mathbb{P}$-a.e., possibly passing to a subsequence. Loosely speaking, convergence in $\mathbb{P}$-probability provides a way to metrize $\mathbb{P}$-a.e. convergence.

Moreover, for $X = \mathbb{R}^n$, if $f_h \to f$ in $L^p((\Omega, \mathcal{F}, \mathbb{P}); \mathbb{R}^n)$ for some $p \ge 1$, then $f_h \to f$ in $\mathbb{P}$-probability, thanks to Markov's inequality. Conversely, if $p > 1$, $\sup_h \|f_h\|_{L^p((\Omega,\mathcal{F},\mathbb{P});\mathbb{R}^n)} < \infty$ and $f_h \to f$ in $\mathbb{P}$-probability, then $f_h \to f$ in $L^q((\Omega, \mathcal{F}, \mathbb{P}); \mathbb{R}^n)$ for all $q \in [1, p)$.

Now we can state and prove a beautiful and technically very useful relation between convergence in $\mathbb{P}$-probability (that, as we have just seen, is closely related to notions of *strong* L^p-convergence) and *weak* convergence of probability measures in the product space. Its origins go back to the theory of L.C. Young, see [119, 120] and the lecture notes [116].

Theorem 6.12 (Convergence of Maps Versus Convergence of Plans) *Let (X, d_X), (Y, d_Y) be Polish spaces, let $\mu \in \mathscr{P}(X)$ and let $f_h : X \to Y$ and $f : X \to Y$ be Borel functions. Then, as $h \to \infty$,*

$$f_h \to f \text{ in } \mu\text{-probability} \iff (\mathrm{id} \times f_h)_\# \mu \to (\mathrm{id} \times f)_\# \mu \quad \text{weakly}.$$

Proof Let $f_h \to f$ in μ-probability and assume by contradiction that the assertion does not hold. Then there exist a subsequence $(h(k))$, $\epsilon > 0$ and $\varphi \in C_b(X \times Y)$ such that

$$\left| \int_X \varphi(x, f_{h(k)}(x))\, \mathrm{d}\mu(x) - \int_X \varphi(x, f(x))\, \mathrm{d}\mu(x) \right| \ge \epsilon, \qquad \forall k. \tag{6.4}$$

By Remark 6.11, there exists a further subsequence $(f_{h(k(\ell))})$ converging μ-a.e. to f as $\ell \to \infty$. By the dominated convergence theorem,

$$\left| \int_X \varphi(x, f_{h(k(\ell))}(x))\, \mathrm{d}\mu(x) - \int_X \varphi(x, f(x))\, \mathrm{d}\mu(x) \right| \to 0, \qquad \ell \to \infty,$$

an immediate contradiction to (6.4).

To show the converse implication, let $\epsilon > 0$. Then, by Lusin's theorem, there exists a compact set $K_\epsilon \subset X$ such that $\mu(X \setminus K_\epsilon) < \epsilon$ and the restriction of f to K_ϵ is continuous. By Tietze's extension theorem (in the general version for functions with values in separable metric spaces due to R. Ellis [56]), there exists a continuous extension $\tilde{f} : X \to Y$ of f. Let $\varphi \in C_b(X \times Y)$ be defined by

$$\varphi(x, y) := 1 \wedge d_Y(y, \tilde{f}(x)).$$

By assumption,

$$\int_{X\times Y} \varphi \, \mathrm{d}(\mathrm{id} \times f_h)_\# \mu \to \int_{X\times Y} \varphi \, \mathrm{d}(\mathrm{id} \times f)_\# \mu, \qquad h \to \infty. \tag{6.5}$$

Furthermore, since $\tilde{f}|_{K_\epsilon} = f|_{K_\epsilon}$, one has

$$\begin{aligned}\int_{X\times Y} \varphi \, \mathrm{d}(\mathrm{id} \times f)_\# \mu &= \int_X \varphi(x, f(x)) \, \mathrm{d}\mu(x) = \int_X 1 \wedge d(f(x), \tilde{f}(x)) \, \mathrm{d}\mu(x) \\ &= \int_{X\setminus K_\epsilon} 1 \wedge d(f(x), \tilde{f}(x)) \, \mathrm{d}\mu(x) \le \mu(X \setminus K_\epsilon) < \epsilon.\end{aligned}$$

For h large enough, (6.5) shows that $\int_X 1 \wedge d(f_h, \tilde{f}) \, \mathrm{d}\mu < \epsilon$. Consequently,

$$\begin{aligned}\int_X 1 \wedge d(f_h, f) \, \mathrm{d}\mu &\le \int_X 1 \wedge (d(f_h, \tilde{f}) + d(f, \tilde{f})) \, \mathrm{d}\mu \\ &\le \int_X 1 \wedge d(f_h, \tilde{f}) \, \mathrm{d}\mu + \int_X 1 \wedge d(f, \tilde{f}) \, \mathrm{d}\mu < 2\epsilon.\end{aligned}$$

Since $\epsilon > 0$ is arbitrary we conclude by Lemma 6.10. □

The heuristic idea behind this proof is that the special structure of the limit measure (namely its concentration on a single graph) forces the absence of oscillations, and then the convergence in the strong sense of the maps. This is well illustrated by the next example, in which we see the structure of the limit measure when strong convergence does not occur.

Example 6.13 Let

$$f(x) := \begin{cases} +1 & 0 \le x < 1/2, \\ -1 & 1/2 \le x < 1, \end{cases}$$

and extend f to be 1-periodic on $\mathbb{R}$. Then $f_h(x) := f(hx) \rightharpoonup 0$ weakly star in $L^\infty(0, 1)$ as $h \to \infty$ (this could be seen for instance by viewing f_h as g_h', with $g_h : \mathbb{R} \to [0, 1/h]$ sawtooth functions, uniformly converging to 0) and it is easily seen that the convergence, in any space $L^p(0, 1)$, $1 \le p \le \infty$, is not strong. In

agreement with Theorem 6.12 we have

$$(\mathrm{id} \times f_h)_{\#} \mathscr{L}^1 \llcorner [0,1] \to \frac{1}{2} \left(\mathscr{L}^1 \llcorner [0,1] \times \delta_1 + \mathscr{L}^1 \llcorner [0,1] \times \delta_{-1} \right),$$

weakly, so that the limit measure is concentrated on the union of two distinct graphs.

Corollary 6.14 (Strong Stability of Optimal Maps) *Let $\mu \in \mathscr{P}_2(\mathbb{R}^n)$ with $\mu \ll \mathscr{L}^n$ and let $(\nu_h) \subset \mathscr{P}(\mathbb{R}^n)$, with $\cup_h \operatorname{supp} \nu_h$ bounded and $\nu_h \to \nu$ weakly. Then the optimal maps $T_\mu^{\nu_h}$ converge in $L^p(\mu; \mathbb{R}^n)$ to T_μ^ν for any $p \in [1, \infty)$.*

Proof Since $(T_\mu^{\nu_h})_{\#}\mu = \nu_h \to \nu = (T_\mu^\nu)_{\#}\mu$ weakly, we immediately infer by Theorem 6.8 that $(\mathrm{id} \times T_\mu^{\nu_h})_{\#}\mu$ weakly converge in $\mathscr{P}(\mathbb{R}^n \times \mathbb{R}^n)$ to $(\mathrm{id} \times T_\mu^\nu)_{\#}\mu$. Hence we can apply Theorem 6.12 with $f_h = T_\mu^{\nu_h}$ to obtain that $T_\mu^{\nu_h} \to T_\mu^\nu$ in μ-probability.

It remains to prove the uniform L^∞-boundedness of $(T_\mu^{\nu_h})$, since convergence in μ-probability and uniform L^∞-boundedness imply convergence in L^p for any $p \in [1, \infty)$, by Remark 6.11. This conclusion follows immediately by the boundedness of $\cup_h \operatorname{supp} \nu_h$, since this set contains the (essential) image of $T_\mu^{\nu_h}$ for any h. □

Lecture 7: The Monge-Ampére Equation and Optimal Transport on Riemannian Manifolds



In this section we translate the transport condition in pointwise terms by introducing the so-called Monge–Ampère equation. We then discuss different notions of weak solutions to this problem.

1 A General Change of Variables Formula

Consider measures $\mu, \nu \in \mathscr{P}_2(\mathbb{R}^n)$ absolutely continuous with respect to $\mathscr{L}^n$ and let ϱ and η be the respective probability densities, i.e. $\mu = \varrho\mathscr{L}^n$, $\nu = \eta\mathscr{L}^n$. Let T be the optimal map from μ to ν, representable thanks to Theorem 5.2 as $T = \nabla f$ for some convex function f, with μ concentrated on the interior Ω of the finiteness domain of f, so that ν is concentrated on $T(\Omega)$. Then, if f is of class C^2 and ∇f is injective on Ω, it is immediately seen that f solves in Ω the Monge–Ampère equation in the form

$$\eta(\nabla f)\det(\nabla^2 f) = \varrho \qquad \text{in } \Omega. \tag{7.1}$$

Indeed, by using that $T_\#\mu = \nu$ and the classical change of variable formula, we obtain

$$\int_\Omega \psi(T(x))\eta(T(x))\det\nabla T(x)\,\mathrm{d}x = \int_{T(\Omega)} \psi(y)\eta(y)\,\mathrm{d}y = \int_\Omega \psi(T(x))\varrho(x)\,\mathrm{d}x$$

for any nonnegative Borel function ψ. Choosing $\psi = \chi_{T(K)}$, with $K \subset \Omega$ compact, we obtain

$$\int_K \eta(T(x))\det\nabla T(x)\,\mathrm{d}x = \int_K \varrho(x)\,\mathrm{d}x$$

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and therefore, since K is arbitrary and $T = \nabla f$, one has that (7.1) holds, modulo a modification of ϱ in a $\mathscr{L}^n$-negligible set.

We would like to extend this argument to a weaker setting since, as we have seen, optimal maps in general do not have this regularity. The derivation of (5.5) in the more general setting completes also the proof, by regularization, of the isoperimetric inequality given in Sect. 1 of Lecture 6.

Let us consider a Borel set $D \subset \mathbb{R}^n$ and a continuous function $F : D \to \mathbb{R}^n$, define

$$D_0 := \big\{x \in D : \exists \nabla F(x)\big\}, \qquad \Sigma := \big\{x \in D_0 : \det(\nabla F(x)) = 0\big\}. \tag{7.2}$$

Here, as we have written after Theorem 6.4, even though D is not necessarily open, $\nabla F(x)$ is identified by the standard property $F(y)-F(x)-\langle \nabla F(x), y-x\rangle = o(|x-y|)$ as $y \in D \to x$. With this definition, a priori $\nabla F(x)$ is not uniquely determined, but it is easily seen that this happens if D has positive density at x (because, in this case, the tangent cone to D at x spans the whole space). Hence, since $\mathscr{L}^n$-a.e. point of a Borel set is a density 1 point, this ambiguity occurs only in a $\mathscr{L}^n$-negligible set, irrelevant for the meaning and the validity of the next formulas.

Theorem 7.1 (Area Formula for Lipschitz Functions, [10, 60]) *With the previous notation, assume that F is Lipschitz. Then $\mathscr{L}^n(F(\Sigma)) = 0$ and, setting*

$$N(D_0, y) := \#(D_0 \cap F^{-1}(y)),$$

one has

$$\int_{\mathbb{R}^n} N(D_0, y)\varphi(y)\,\mathrm{d}y = \int_{D_0} \varphi(F(x))|\det \nabla F(x)|\,\mathrm{d}x, \tag{7.3}$$

for any Borel function $\varphi : \mathbb{R}^n \to [0, \infty]$. More generally, one has

$$\int_{\mathbb{R}^n} \sum_{x \in D_0 \cap F^{-1}(y)} \psi(x)\,\mathrm{d}y = \int_{D_0} \psi(x)|\det \nabla F(x)|\,\mathrm{d}x, \tag{7.4}$$

for any Borel function $\psi : D_0 \to [0, \infty]$.

The first statement can be considered as a refinement of the fact that Lipschitz functions map $\mathscr{L}^n$-negligible sets into $\mathscr{L}^n$-negligible sets: $\mathscr{L}^n(F(\Sigma)) = 0$ even though it may happen that $\mathscr{L}^n(\Sigma) > 0$. Notice that in Theorem 7.1 above the set D_0 has full measure in D, since we can assume with no loss of generality that F is a globally defined Lipschitz function (this extension can for instance be achieved arguing componentwise), so that Theorem 5.1 applies. In particular, in the right hand sides of (7.3) and (7.4) we can safely replace D_0 with D. The same is true for the left hand sides, since $\mathscr{L}^n(F(D \setminus D_0)) = 0$. We stated Theorem 7.1 in the more

involved form using D_0 in place of D because, in this form, it holds without any global regularity assumption on F:

Corollary 7.2 *The properties* (7.3) *and* (7.4) *in Theorem 7.1, as well as* $\mathscr{L}^n(F(\Sigma)) = 0$*, hold for any continuous function* $F : D \subset \mathbb{R}^n \to \mathbb{R}^n$.

Proof The idea is to approximate F on D_0, in Lusin's sense, with Lipschitz functions F_k. Set

$$C_k = \big\{x \in D_0 : |\nabla F(x)| \le k,\ |F(x)| \le k,\\ |F(y) - F(x)| \le (1 + |\nabla F(x)|)|x - y| \ \ \forall y \in D \cap B_{1/k}(x)\big\}.$$

For $x \in D_0$, let $\delta > 0$ be chosen such that $|F(y)-F(x)-\langle \nabla F(x), y - x\rangle| \le |y-x|$ for $y \in D \cap B_\delta(x)$, and define $k := \max(1/\delta, |F(x)|, |\nabla F(x)|)$. Then

$$|F(y)-F(x)| \le |\langle \nabla F(x), y - x\rangle| + |x-y| \le (1+|\nabla F(x)|)|x-y|, \qquad y \in D \cap B_{1/k}(x),$$

hence $x \in C_k$. This proves that $C_k \uparrow D_0$. Furthermore, $F|_{C_k}$ is Lipschitz since, for $x \in C_k$, one has

$$|F(x) - F(y)| \le \begin{cases} (1+k)|y - x| & \text{if } |x - y| \le \frac{1}{k}, \\ 2k \le 2k^2|y - x| & \text{if } |x - y| > \frac{1}{k}. \end{cases}$$

Hence, from (7.3) and the remarks made after Theorem 7.1, we get

$$\int_{\mathbb{R}^n} N(C_k, y)\varphi(y)\,\mathrm{d}y = \int_{C_k} \varphi(F(x))|\det \nabla F(x)|\,\mathrm{d}x \tag{7.5}$$

for any Borel function $\varphi : \mathbb{R}^n \to [0, \infty]$. Now, to conclude the proof of (7.3) in the non Lipschitz case it is sufficient to invoke the monotone convergence theorem. The proof of (7.4) is similar. □

The following result is useful to compute explicitly the density of the push-forward of absolutely continuous measures, under low regularity assumptions on the transport map, with a characterization of the cases when the push-forward measure is absolutely continuous.

Theorem 7.3 *Let* $\mu = \varrho\mathscr{L}^n \in \mathscr{P}_2(\mathbb{R}^n)$, $\nu \in \mathscr{P}_2(\mathbb{R}^n)$ *and let* $T = \nabla f$ *be the optimal map between* μ *and* ν. *Let* Ω *be the interior of the finiteness domain of* f, *let* $D \subset \Omega$ *be the set of differentiability points of* ∇f, $D_0 := \{x \in D : \exists \nabla^2 f(x)\}$ *and let*

$$\Sigma := \left\{x \in D_0 :\ \det(\nabla^2 f(x)) = 0\right\}. \tag{7.6}$$

Then

(i) $T|_{D_0\setminus\Sigma}$ *is injective;*
(ii) $\nu \ll \mathscr{L}^n \Leftrightarrow \mu(\Sigma) = 0$*;*
(iii) If $\nu = \eta\mathscr{L}^n$*, then*

$$\eta(\nabla\phi)\det(\nabla^2 f) = \varrho \qquad \mathscr{L}^n\text{-a.e. in } \Omega. \tag{7.7}$$

Proof If h is convex on the real line, differentiable at 0 and at 1, with $h''(0) > 0$, then $h'(1) - h'(0) \geq h''(0) > 0$. For $h(t) := f(x + t(y - x))$ and $x \in D_0 \setminus \Sigma$, this implies

$$\langle \nabla f(y) - \nabla f(x), y - x\rangle \geq c|y - x|^2 \quad \text{with} \quad c := \min_{|\xi|=1} \langle \nabla^2 f(x)\xi, \xi\rangle > 0.$$

It follows that the restriction of ∇f to $D_0 \setminus \Sigma$ is injective.

In order to prove (ii) and (iii), recall that μ is concentrated on D, and that $D \setminus D_0$ is $\mathscr{L}^n$-negligible, so that μ is concentrated on D_0 as well. Assume $\mu(\Sigma) > 0$. Then

$$\nu(T(\Sigma)) = \mu(T^{-1}(T(\Sigma))) \geq \mu(\Sigma) > 0.$$

We know however from Corollary 7.2 that $T(\Sigma)$ is Lebesgue-negligible, hence ν is not absolutely continuous with respect to $\mathscr{L}^n$. On the other hand, if $\mu(\Sigma) = 0$, using (7.4) and denoting by S the restriction of T to $D_0 \setminus \Sigma$ we obtain

$$\begin{aligned}\nu(E) &= \int_{T^{-1}(E)} \varrho\,\mathrm{d}x = \int_{T^{-1}(E)\cap D_0\setminus\Sigma} \varrho\,\mathrm{d}x \\ &= \int_{T^{-1}(E)\cap D_0\setminus\Sigma} \frac{\varrho}{\det\nabla T}\det\nabla T\,\mathrm{d}x = \int_{E\cap T(D_0\setminus\Sigma)} \frac{\varrho(S^{-1}(y))}{\det\nabla T(S^{-1}(y))}\,\mathrm{d}y = 0\end{aligned}$$

for any $\mathscr{L}^n$-negligible Borel set E, so that $\nu \ll \mathscr{L}^n$.

To prove (iii), we can use the injectivity of T on $D_0 \setminus \Sigma$ as well as the fact that μ is concentrated on $D_0 \setminus \Sigma$, arguing as at the beginning of this section for the derivation of (7.1), under more classical assumptions. □

Remark 7.4 A closer look at the proof of (ii) shows that the Radon-Nikodym decomposition $\nu = \nu^a + \nu^s$ of ν, with $\nu^a \ll \mathscr{L}^n$, $\nu^s \perp \mathscr{L}^n$, is given by $\nu^a = T_\#(\mu \llcorner D_0 \setminus \Sigma)$, $\nu^s = T_\#(\mu \llcorner \Sigma)$.

2 The Monge–Ampère Equation

Returning to the Monge–Ampère equation $\det\nabla^2 f = h$, solutions f in the $\mathscr{L}^n$-a.e. sense can be irregular, even discontinuous, as illustrated by the optimal transport with potential $\phi(x, y) = \alpha|x| + \frac{1}{2}(x^2 + y^2)$ of Caffarelli's Example 6.7. One can

therefore ask for a stronger notion of solution to this second-order PDE, leading to better regularity.

In this section we assume that Ω is a convex open set in $\mathbb{R}^n$ and that $f : \Omega \to \mathbb{R}$ is convex (and we may apply freely the results of Sect. 1 in Lecture 3 thinking that $f \equiv +\infty$ in $\mathbb{R}^n \setminus \Omega$). Let us recall the subdifferential $\partial f(x)$ at $x \in \Omega$ is defined by

$$\partial f(x) := \{p \in \mathbb{R}^n : f(y) \geq f(x) + \langle p, y - x\rangle \ \ \forall y \in \Omega\}.$$

Then, we define

$$\partial f(B) := \bigcup_{x \in B} \partial f(x) \qquad B \in \mathscr{B}(\Omega).$$

Definition 7.5 (Monge–Ampère Measure and Alexandrov Solutions) For $B \in \mathscr{B}(\Omega)$, we define the Monge–Ampère measure by $MA_f(B) := \mathscr{L}^n(\partial f(B))$. In addition, given a nonnegative Borel measure μ, a function $f : \Omega \to \mathbb{R}$ is said to solve the Monge–Ampère equation $\det \nabla^2 f = \mu$ in the *Alexandrov* (or *maximal monotone*) sense if

$$MA_f = \mu.$$

If so, f is called an *Alexandrov solution*.

The following proposition shows that the definition of Monge–Ampère measure is well posed. In particular also the notion of Alexandrov solution is well posed, and we prove its consistency with the classical setting.

Proposition 7.6 *The set function $\mathscr{B}(\Omega) \ni B \mapsto \mathscr{L}^n(\partial f(B))$ is a nonnegative Borel measure which coincides with $\det \nabla^2 f \mathscr{L}^n$ when $f \in C^2(\Omega)$.*

Proof We claim that $\mathscr{L}^n(\partial f(B) \cap \partial f(B')) = 0$ whenever $B \cap B' = \emptyset$. Indeed, we know that $y \in \partial f(x)$ if and only if $x \in \partial f^*(y)$, so that

$$y \in \partial f(x) \cap \partial f(x') \quad \Rightarrow \quad x,\, x' \in \partial f^*(y).$$

So, this implication with $x \in B$ and $x' \in B'$ shows that $\partial f(B) \cap \partial f(B')$ is contained in the set of points where f^* is finite, but not differentiable, which is Lebesgue negligible.

Now, let us define:

$$\mathcal{L} := \left\{B \in \mathscr{B}(\Omega) : \partial f(B) \text{ is } \mathscr{L}^n\text{-measurable}\right\}.$$

The class $\mathcal{L}$ is trivially closed under countable unions and contains generators of Borel sets: indeed, if $B \subset \Omega$ is compact, then it is easily seen that $\partial f(B)$ is compact as well (and then $\mathscr{L}^n$-measurable). Hence, in order to prove that $\mathcal{L}$ is a σ-algebra, and then that $\mathcal{L} = \mathscr{B}(\Omega)$, it is sufficient to prove stability under complement. It is

easy to check that

$$\begin{aligned}\partial f(\Omega \setminus B) &= (\partial f(\Omega \setminus B) \setminus \partial f(B)) \cup (\partial f(\Omega \setminus B) \cap \partial f(B)) \\ &= (\partial f(\Omega) \setminus \partial f(B)) \cup (\partial f(\Omega \setminus B) \cap \partial f(B)) .\end{aligned}$$

Since $\partial f(\Omega)$ is σ-compact, and then Borel, the stability follows from the claim made at the beginning of the proof. This proves that MA_f is well defined and its σ-additivity follows from exactly the same arguments.

Finally, if $f \in C^1(\Omega)$, then $MA_f(B) = \mathscr{L}^n(\nabla f(B))$. If, in addition, $f \in C^2(\Omega)$, the change of variable formula immediately gives

$$\mathscr{L}^n(\nabla f(B)) = \int_B \det \nabla^2 f \, dx, \qquad \forall B \in \mathscr{B}(\Omega),$$

since the argument at the beginning of the proof shows that $\#((\nabla f)^{-1}(y)) \le 1$ for $\mathscr{L}^n$-a.e. $y \in \mathbb{R}^n$. □

In Caffarelli's Example 6.7 above, $f(x, y) = \alpha|x| + \frac{1}{2}(x^2 + y^2)$, so that

$$\partial f(x, y) = \begin{cases} \{(x + \alpha \operatorname{sgn} x, y)\} & \text{for } x \neq 0, \\ -[\alpha, \alpha] \times \{y\} & \text{for } x = 0. \end{cases} \tag{7.8}$$

Hence

$$MA_f = \mathscr{L}^2 + 2\alpha \mathscr{H}^1 \llcorner \{x = 0\}.$$

Some regularity properties of Alexandrov solutions will be presented in Theorem 7.9. For instance, if $MA_f = \mu$ with $\mu = h\mathscr{L}^n \llcorner \Omega$, then f can be shown to be $C^{1,\alpha}(\Omega)$ for all $\alpha \in (0, 1)$, if $1/C \le h \le C$ for some $C > 0$.

Arguing as in the proof of Theorem 7.3 (namely, using Corollary 7.2), one can prove the following result:

Theorem 7.7 *$MA_f \ge \det \nabla^2 f \mathscr{L}^n$, with equality if ∇f is Lipschitz in Ω. More precisely, defining $D \subset \Omega$ as the set of differentiability points of f and D_0 as in (7.2), one has*

$$\int_B \det \nabla^2 f \, dx = \mathscr{L}^n\big(\{\nabla f(x) : x \in B \cap D_0\}\big) \qquad \forall B \in \mathscr{B}(\Omega).$$

In the previous identity, the parts of B contained in Σ give null contribution to both sides as a consequence of Theorem 7.1. Caffarelli's example shows that the inequality $MA_f \ge \det \nabla^2 f \mathscr{L}^n$ can be strict. Another classical example in this direction is given by the primitive of the Cantor-Vitali function:

Example 7.8 (Cantor-Vitali Function) The Cantor-Vitali function $v : [0, 1] \to [0, 1]$ is a continuous nondecreasing function, locally constant in the complement of

the middle third Cantor set C, with $v(0) = 0$ and $v(1) = 1$. Setting $\gamma = \ln 2/\ln 3$, by self-similarity arguments (see for instance [71]) it can be proved that

$$Dv = \frac{1}{\mathcal{H}^\gamma(C)}\mathcal{H}^\gamma \llcorner C.$$

The primitive function $f(x) := \int_0^x v(y)\,\mathrm{d}y$ is convex, and from the identity $v_\#(Dv) = \mathscr{L}^1 \llcorner [0,1]$ guaranteed by (1.3) it follows that

$$\mathscr{L}^1(v(E)) = Dv\big(v^{-1}(v(E))\big) = Dv(E) \qquad \forall E \in \mathscr{B}([0,1]),$$

since $v^{-1}(v(E)) \setminus E$ is Dv-negligible (it is contained in the union of the intervals $[1/3, 2/3]$, $[1/9, 2/9]$, $[7/9, 8/9]$, etc.). Hence, the Monge-Ampère measure MA_f coincides with $\mu := Dv$. While $v_\#(Dv) = \mathscr{L}^1 \llcorner [0,1]$, it is not hard to prove that

$$v_\#\left(\mathscr{L}^1 \llcorner [0,1]\right) = \frac{1}{3}\delta_{\frac{1}{2}} + \frac{1}{9}\left(\delta_{\frac{1}{4}} + \delta_{\frac{3}{4}}\right) + \frac{1}{27}\left(\delta_{\frac{1}{8}} + \delta_{\frac{3}{8}} + \delta_{\frac{5}{8}} + \delta_{\frac{7}{8}}\right) + \cdots.$$

Now let us see an important result by Caffarelli [33], with more recent contributions on Sobolev regularity of the transport map in [49, 52, 108].

Theorem 7.9 *Let Ω, $V \subset \mathbb{R}^n$ be convex, open sets and let ϱ, η be probability densities in Ω and V, respectively, with $\eta > 0$ in V and $\mathscr{L}^n(V) < \infty$. Then:*

(i) the optimal map $T = \nabla f$ between $\mu := \varrho\mathscr{L}^n \llcorner \Omega$ and $\nu := \eta\mathscr{L}^n \llcorner V$ solves the Monge-Ampère equation (5.5) not only in the "$\mathscr{L}^n$-a.e. sense", but also in the Alexandrov sense, namely:

$$MA_f = \frac{\varrho}{\eta \circ \nabla f}\mathscr{L}^n \llcorner \Omega;$$

(ii) if f solves $MA_f = g\mathscr{L}^n$ with $\tau \le g \le \tau^{-1}$ $\mathscr{L}^n$-a.e. in Ω for some $\tau > 0$, then $f \in C^{1,\alpha}_{\mathrm{loc}}(\Omega)$, for all $\alpha < 1$ and $f \in W^{2,p}_{\mathrm{loc}}(\Omega)$ for some $p = p(n,\tau) > 1$.

Proof We prove only the first part, since the proof of the second part goes beyond the scope of these lectures. We claim that $\partial f(x) \subset \overline{V}$ for all $x \in \Omega$. If so, one has

$$MA_f(\Omega) = \mathscr{L}^n(\partial f(\Omega)) \le \mathscr{L}^n(\overline{V}) = \mathscr{L}^n(V)$$

because of the convexity of V. On the other hand, since $\eta > 0$ in V and f is a $\mathscr{L}^n$-a.e. solution to the Monge–Ampère equation, we have:

$$\int_\Omega \det\nabla^2 f\,\mathrm{d}x = \int_\Omega \frac{\varrho}{\eta\circ\nabla f}\,\mathrm{d}x = \int_\Omega \frac{1}{\eta\circ\nabla f(x)}\,\mathrm{d}\mu(x) = \int_V \frac{1}{\eta(y)}\,\mathrm{d}\nu(y) = \mathscr{L}^n(V).$$

These two inequalities yield $MA_f(\Omega) \leq \int_\Omega \det \nabla^2 f \, \mathrm{d}x$. But, since Theorem 7.7 grants the inequality between measures $MA_f \geq \det \nabla^2 f \mathscr{L}^n$, the converse inequality between their total masses can hold only if the two measures are equal.

In order to obtain the claim we prove, for convex functions, that $\nabla f \in V$ $\mathscr{L}^n$-a.e. in Ω implies $\partial f(\Omega) \subset \overline{V}$. In our case, the property that $\nabla f \in V$ $\mathscr{L}^n$-a.e. in Ω is ensured by the fact that $(\nabla f)_\# \mu = \nu$. Assume also that f is uniformly convex (the reduction to this case can be achieved considering the functions $f(x) + \delta |x|^2$, assuming also Ω to be bounded, and letting $\delta \to 0$).

Let now $x \in \Omega$, $p \in \partial f(x)$ and $\overline{B}_r(x) \subset \Omega$. Consider $f_\epsilon := f * \varrho_\epsilon$, where ϱ_ϵ are standard Friedrichs mollifiers and notice that the convexity of V and the representation

$$\nabla f_\epsilon(x) = (\nabla f) * \varrho_\epsilon(x) = \int_{\mathbb{R}^n} \nabla f(x - y) \varrho_\epsilon(y) \, \mathrm{d}y$$

ensure that $\nabla f_\epsilon(\overline{B}_r(x)) \subset \overline{V}$ for $\epsilon > 0$ small enough. If

$$h_\epsilon(x') := f_\epsilon(x') - \langle p, x' \rangle, \qquad h(x') := f(x') - \langle p, x' \rangle,$$

it is immediately seen that h_ϵ converges to h as $\epsilon \to 0$ uniformly, and that (because of uniform convexity), x is the unique minimizer of h. So, if x_ϵ minimizes $h_\epsilon|_{\overline{B}_r(x)}$, one has $x_\epsilon \to x$ as $\epsilon \to 0$ (since limit points of minimizers have to be minimizers). For $\epsilon > 0$ small enough we have $x_\epsilon \in B_r(x)$ and then $\nabla h_\epsilon(x_\epsilon) = 0$ gives

$$p = \nabla f_\epsilon(x_\epsilon) \in \overline{V}.$$

□

3 Optimal Transport on Riemannian Manifolds

Let $(M, \mathbf{g})$ be a smooth, n-dimensional compact Riemannian manifold, without boundary (namely, a closed manifold). We denote by d_M and vol_M the two canonical objects induced by $\mathbf{g}$, namely the Riemannian distance[1]

$$d_M^2(x, y) := \min \left\{ \int_0^1 \mathbf{g}_{\gamma(t)}(\gamma'(t), \gamma'(t)) \, \mathrm{d}t : \gamma(0) = x, \ \gamma(1) = y, \ \gamma \in AC([0, 1]; M) \right\}$$

[1] See Sect. 1 in Lecture 9 for the definition of the class AC of absolutely continuous curves, $\mathbb{R}^n$-valued and even with values in metric spaces. In the manifold case the definition can be initially given working in local coordinates, thus giving also a meaning to the speed γ'.

obtained by minimizing the action induced by $\mathbf{g}$ and the volume measure $\mathrm{vol}_M \in \mathscr{M}_+(M)$. Besides its definition $\sqrt{\det(\mathbf{g}_x)_{ij}}\mathscr{L}^n$ in local coordinates, thanks to the area formula the measure vol_M can also be identified with the n-dimensional Hausdorff measure induced by d_M. Since the infimum in the definition of d_M is attained, (M, d_M) is a geodesic space, namely for any pair of points $x, y \in M$ there exists an action-minimizing absolutely continuous curve $\gamma : [0,1] \to M$ with $\gamma(0) = x$ and $\gamma(1) = y$. In particular γ has constant speed and it is length minimizing. Furthermore, being (in local coordinates) a weak solution to the geodesic system of ODE $\gamma_i'' + \sum_{j,k} \Gamma^i_{jk}\gamma_j'\gamma_k' = 0$, thanks to the smoothness of the Christoffel numbers Γ^i_{jk} it follows that γ is smooth and representable as

$$\gamma(t) = \exp_x(t\xi) \qquad \text{with } \xi = \gamma'(0) \in T_xM \text{ satisfying } \sqrt{\mathbf{g}_x(\xi,\xi)} = d_M(x,y). \tag{7.9}$$

Therefore the exponential map $\exp_x : T_xM \to M$ is onto (Hopf-Rinow theorem) for any $x \in M$. See [54, 99] for the proof of these basic facts of Riemannian Geometry.

In this setting, R. McCann established in [90] an analogous of Brenier's Theorem 5.2. McCann's proof is a beautiful application of techniques coming from non-smooth analysis to an apparently smooth situation. Here the lack of smoothness comes from the fact that, in general, $d^2_M(\cdot, y)$ is differentiable only sufficiently near to y: think for instance to the case when $M = \mathbf{S}^n$ and x is antipodal to y, so that $d_M(\cdot, y)$ has a singularity of the form $\pi - |z|$, in local coordinates z around x.

Theorem 7.10 *Let $\mu, \nu \in \mathscr{P}(M)$ with $\mu \ll \mathrm{vol}_M$ and $c = \frac{1}{2}d^2_M$. Then there exists a unique $\pi \in \Gamma_o(\mu,\nu)$, and π is induced by a map T. Moreover, there exists a c-concave and Lipschitz function $\phi : M \to \mathbb{R}$ such that*

$$T(x) = \exp_x(-\nabla\phi(x)) \qquad \textit{for } \mu\textit{-a.e. } x \in M.$$

To realize that this is a generalization of Brenier's Theorem 5.2 we notice that in the Euclidean setting we had

$$f(x) = \frac{|x|^2}{2} - \phi(x) \quad \text{with} \quad T(x) = \nabla f(x),$$

namely $T(x) = x - \nabla\phi(x) = \exp_x(-\nabla\phi(x))$, since geodesics in $\mathbb{R}^n$ are lines. The strategy of the proof of Theorem 7.10 is the same. First of all, thanks to the compactness of (M, d_M), one obtains a Lipschitz and c-concave Kantorovich potential $\phi : M \to \mathbb{R}$ whose graph of the subdifferential $\partial^c\phi$ contains the support of a given optimal plan π. Then, one needs to prove that, for μ-a.e. x, the relation $y \in \partial^c\phi(x)$ determines y uniquely, where we recall that

$$\partial^c\phi(x) = \left\{ y \in M : c(x', y) - \phi(x') \text{ is minimal at } x' = x \right\}.$$

Definition 7.11 $\psi : M \to \mathbb{R}$ is differentiable at x if there exists $w \in T_x M$ such that

$$\psi(\exp_x v) - \psi(x) - \mathbf{g}_x(w, v) = o\big(\sqrt{\mathbf{g}_x(v, v)}\big) \qquad v \in T_x M.$$

The vector w is uniquely determined by this condition, and denoted by $\nabla\psi(x)$.

This corresponds to the usual definition in local coordinates, so that Rademacher Theorem 5.1 applies and, since $\mu \ll \mathrm{vol}_M$, the set of non differentiability points of ϕ is μ-negligible. Hence, in the sequel we can assume that x is a differentiability point of ϕ.

Unfortunately, as we explained above, there is no reason why the function $c(\cdot, y)$ should be differentiable at x, and this requires a new argument. To do that, we need the concept of Fréchet sub/superdifferential.

Definition 7.12 We denote by $\partial_F^- \psi(x)$ the Fréchet subdifferential of ψ at x, defined as follows:

$$w \in \partial_F^- \psi(x) \quad \Leftrightarrow \quad \liminf_{v \to 0,\, v \in T_x M} \frac{\psi(\exp_x v) - \psi(x) - \mathbf{g}_x(w, v)}{\sqrt{\langle v, v\rangle_x}} \geq 0.$$

Analogously, $\partial_F^+ \psi(x)$ stands for the Fréchet superdifferential of ψ in x, defined by:

$$w \in \partial_F^+ \psi(x) \quad \Leftrightarrow \quad \limsup_{v \to 0,\, v \in T_x M} \frac{\psi(\exp_x v) - \psi(x) - \mathbf{g}_x(w, v)}{\sqrt{\langle v, v\rangle_x}} \leq 0.$$

Notice that if ψ is differentiable in x, then both $\partial_F^+ \psi(x)$ and $\partial_F^- \psi(x)$ are the singletons containing $\nabla\psi(x)$. Conversely, since $L_1 \leq L_2$ implies $L_1 = L_2$ when L_1 and L_2 are linear operators, whenever both $\partial_F^+ \psi(x)$ and $\partial_F^- \psi(x)$ are not empty, ψ is differentiable at x (and then, they are singletons). In the sequel we will also use the chain rule for superdifferentials

$$\partial_F^+ \left(\frac{1}{2} d_M^2(\cdot, y)\right)(x) = d_M(x, y)\, \partial_F^+ d_M(\cdot, y)(x). \tag{7.10}$$

As we already mentioned, given $x, y \in M$ there exists a constant speed and length minimizing $\gamma : [0, 1] \to M$ with $\gamma(0) = x$, $\gamma(1) = y$, representable as in (7.9). In the next theorem we prove a basic superdifferentiability property of the distance function, see [88] for much more on the cut locus and its relations with first and second order derivatives of the distance function.

Theorem 7.13 (Properties of $\frac{1}{2} d_M^2(\cdot, y)$) *The function $x \mapsto \frac{1}{2} d_M^2(x, y)$ is differentiable at $x = y$, with null gradient. In general, it is everywhere superdifferentiable, more precisely if $y = \exp_x(\xi)$ with ξ as in (7.9), one has*

$$-\xi \in \partial_F^+ \left(\frac{1}{2} d_M^2(\cdot, y)\right)(x).$$

Proof We simplify a bit the original proof in [90], avoiding more advanced tools of Riemannian theory. We shall only use the expansion (see also Exercise 14 in Chapter 6 of [99])

$$\|v\| + \|w\| \le r \quad \Rightarrow \quad \left| d_M^2(\exp_x v, \exp_x w) - \|v - w\|^2 \right| \le \omega(r)(\|v\|^2 + \|w\|^2) \tag{7.11}$$

for some modulus of continuity ω depending only on $\mathbf{g}$, where $\|\cdot\|$ is the Hilbert norm induced by $\mathbf{g}_x$ on $T_x M$. Its proof can be achieved working in local coordinates, so that $d_M^2(x, y)$ can be computed by minimizing $\int_0^1 \sum_{i,j} \mathbf{g}_{ij}(\gamma(t))\gamma_i'(t)\gamma_j'(t)\,\mathrm{d}t$: one has to replace $\mathbf{g}_{ij}(\gamma(t))$ with $\mathbf{g}_{ij}(x)$ and the exponential curves with straight lines, using the continuity of $\mathbf{g}$ and the uniform C^2 smoothness of the exponential map.

Without loss of generality, we can assume $y \neq x$. Let $\xi' = \xi/d_M(x, y)$, a unit vector, and notice that by the chain rule (7.10) it suffices to show that $-\xi' \in \partial_F^+ d_M(\cdot, y)(x)$. Assume by contradiction that, for some $\epsilon > 0$ and $(v_i) \subset T_x M$, one has

$$d_M(\exp_x v_i, y) - d_M(x, y) \ge -\mathbf{g}_x(v_i, \xi') + \epsilon t_i \tag{7.12}$$

with $t_i := \|v_i\| \to 0$. We can assume with no loss of generality that $v_i = t_i w_i$ with $w_i \to w \in T_x M$ unit vector. We choose points $y_i = \exp_x(L t_i \xi')$, so that $d_M(x, y_i) = L t_i$ and $d_M(y_i, y) = d_M(x, y) - L t_i$, and prove that for $L > L(\epsilon)$ sufficiently large we have a contradiction. From the triangle inequality we get

$$d_M(\exp_x v_i, y) \le d_M(\exp_x v_i, y_i) + d_M(y_i, y) = d_M(\exp_x v_i, y_i) + d_M(x, y) - d_M(x, y_i),$$

so that (7.12) gives

$$d_M(\exp_x v_i, y_i) - d_M(x, y_i) \ge -\mathbf{g}_x(v_i, \xi') + \epsilon t_i.$$

Using the definition of y_i, the expansion (7.11) gives

$$\|v_i - L t_i \xi'\| - L t_i \ge -t_i \mathbf{g}_x(w_i, \xi') + \frac{\epsilon}{2} t_i$$

for i large enough, so that dividing by $L t_i$ and taking limits gives

$$\|\frac{w}{L} - \xi'\| \ge 1 - \mathbf{g}_x(\frac{w}{L}, \xi') + \frac{\epsilon}{2L}.$$

On the other hand, the Hilbertian expansion $\|\alpha - \xi'\| = 1 - \mathbf{g}_x(\alpha, \xi') + O(\|\alpha\|^2)$ with $\alpha = w/L$ provides a contradiction, if $L \gg 1$. □

Continuing our presentation of the proof of Theorem 7.10, we notice that the minimality of $c(\cdot, y) - \phi$ at x and the differentiability of ϕ at x immediately give (Taylor expanding ϕ) that $c(\cdot, y)$ is *subdifferentiable* at x, more precisely

$$\nabla\phi(x) \in \partial_F^- \left(\frac{1}{2} d_M^2(\cdot, y)\right)(x).$$

If we combine this information with the one provided by Theorem 7.13 we obtain that $\xi = -\nabla\phi(x)$, so that $y = \exp_x(-\nabla\phi(x))$ is uniquely determined by x, via the gradient of ϕ. As in the proof of Theorem 5.2, this shows that there exists a unique optimal plan and that this optimal plan is induced by the map $T(x) = \exp_x(-\nabla\phi(x))$.

Remark 7.14 (Optimal Map and Cut Locus) A byproduct of this proof is the fact that not only $T(x)$, but also the vector ξ such that $\exp_x \xi = T(x)$ is uniquely determined μ-a.e., since $\xi = -\nabla\phi(x) = -\nabla\frac{1}{2}d^2(\cdot, y)$ with $y = T(x)$! This means that the optimal transport problem essentially avoids "antipodal" pairs (x, y) having at least two constant speed and length minimizing geodesics joining them, and $\exp_x(-t\nabla\phi(x))$ is not in the cut locus of x for all $t \in [0, 1)$. As a matter of fact, the regularity theory of optimal transport maps on manifolds, developed under rather stringent curvature conditions (including the case of spheres), requires quantitative estimates of these phenomena, see for instance [50, pg. 44–47].

A particular case is when $M = \mathbb{T}^n$ is the n-dimensional torus. In this case we can lift probability measures in M to $\mathbb{Z}^n$-periodic measures in $\mathbb{R}^n$ having unit mass in $[0, 1)^n$, and McCann's theorem reads as follows:

Theorem 7.15 (Cordero-Erausquin, [41]) *Let μ, ν be $\mathbb{Z}^n$-periodic measures such that $\mu([0, 1)^n) = 1 = \nu([0, 1)^n)$ and $\mu \ll \mathscr{L}^n$. Then there exists a convex function $f : \mathbb{R}^n \to \mathbb{R}$ such that $f(x) - \frac{1}{2}|x|^2$ is $\mathbb{Z}^n$-periodic and $(\nabla f)_\# \mu = \nu$. Viewing $\nabla f(x) - x$ as a vector field $-\nabla\phi(x)$ on $\mathbb{T}^n$, one has that $\exp_x(-\nabla\phi(x))$ is the optimal transport map of Theorem 7.10.*

Lecture 8: The Metric Side of Optimal Transport



In the first seven lectures we have introduced the optimal transport problem, starting from the classical Monge formulation, moving then to the general theory and to some applications. Now our goal is to show how the optimal transport problem can be used to endow $\mathscr{P}_2(X)$ with a natural metric structure. We will see how many metric (and even differential) properties can be "lifted" from X to $\mathscr{P}_2(X)$, as separability, compactness, completeness, geodesic, nonbranching, lower bounds on sectional curvature.

As usual, we will always assume at least that (X, τ) is Polish, so that our basic measure-theoretic tools (Prokhorov Theorem 2.8 and the disintegration Theorem 2.4) will be applicable.

1 The Distance W_2 in $\mathscr{P}_2(X)$

Let us recall the definition of $\mathscr{P}_2(X)$:

$$\mathscr{P}_2(X) := \left\{ \mu \in \mathscr{P}(X) : \int_X d^2(x, x_0)\, \mathrm{d}\mu(x) < \infty \ \text{ for some (and thus for all) } x_0 \in X \right\}.$$

Definition 8.1 (Wasserstein Distance in $\mathscr{P}_2(X)$) We define:

$$W_2^2(\mu, \nu) := \min \left\{ \int_{X\times X} d^2(x, y)\, \mathrm{d}\pi(x, y) : \pi \in \Gamma(\mu, \nu) \right\}.$$

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Remark 8.2 (General Powers $1 \le p \le \infty$) For $p \in [1, \infty)$, all the metric results of this section extend, with really minor modifications, to the case of the space $\mathscr{P}_p(X)$ of Definition 4.3, endowed with the distance

$$W_p^p(\mu, \nu) := \min \left\{ \int_{X \times X} d^p(x, y) \, \mathrm{d}\pi(x, y) : \pi \in \Gamma(\mu, \nu) \right\}.$$

Actually, viewing all W_p as potentially infinite distances in $\mathscr{P}(X)$ (the so-called *extended* distances), one can even define

$$W_\infty(\mu, \nu) := \inf \left\{ \| d(x, y) \|_{L^\infty(\pi)} : \pi \in \Gamma(\mu, \nu) \right\}$$

and prove that $W_p \uparrow W_\infty$ in $\mathscr{P}(X) \times \mathscr{P}(X)$ as $p \to \infty$.

We have decided to focus on the space $\mathscr{P}_2(X)$ because this space is more relevant for the analysis, not only of the metric, but also of the differential properties of the space of probability measures, see Lecture 16.

Theorem 8.3 *$(\mathscr{P}_2(X), W_2)$ is a metric space and the embedding $X \ni x \mapsto \delta_x \in \mathscr{P}_2(X)$ is isometric.*

Proof Let us verify the defining properties of the distance.

The quantity W_2 is finite, since the choice $\pi = \mu \times \nu \in \Gamma(\mu, \nu)$ and the inequality $d^2(x, y) \le 2(d^2(x, x_0) + d^2(y, x_0))$ give

$$W_2^2(\mu, \nu) \le 2 \int_X d^2(x, x_0) \, \mathrm{d}\mu(x) + 2 \int_X d^2(y, x_0) \, \mathrm{d}\nu(y) < \infty.$$

Moreover, W_2 is symmetric, by the symmetry of the distance d and the invariance properties already used in the proof of Theorem 5.2(iii).

Choosing $\pi = (\mathrm{id} \times \mathrm{id})_\# \mu$ gives $W_2(\mu, \mu)$=0. Conversely, if $W_2(\mu, \nu) = 0$ and $\pi \in \Gamma_\mathrm{o}(\mu, \nu)$, from $x = y$ for π-a.e. (x, y) we obtain

$$\int_X f(x) \, \mathrm{d}\mu(x) = \int_{X \times X} f(x) \, \mathrm{d}\pi(x, y) = \int_{X \times X} f(y) \, \mathrm{d}\pi(x, y) = \int_X f(y) \, \mathrm{d}\nu(y)$$

for any bounded Borel function f, therefore $\mu = \nu$.

It remains to prove the triangle inequality $W_2(\mu, \sigma) \le W_2(\mu, \nu) + W_2(\nu, \sigma)$. We shall prove it first, also to clarify the analogy between W_2 and L^2, assuming that both μ and ν have no atom. Under this assumption, Theorem 2.2 gives

$$W_2^2(\mu, \nu) = \inf \left\{ \| d(\mathrm{id}, T) \|_{L^2(\mu)}^2 : T_\# \mu = \nu \right\}$$

and

$$W_2^2(\nu, \sigma) = \inf \left\{ \| d(\mathrm{id}, S) \|_{L^2(\mu)}^2 : S_\# \nu = \sigma \right\}.$$

Then, using the change of variable formula, we get:

$$W_2(\mu,\sigma) \le \|d(\mathrm{id},(S\circ T))\|_{L^2(\mu)} \le \|d(\mathrm{id},T)\|_{L^2(\mu)} + \|d(T,S\circ T)\|_{L^2(\mu)}$$
$$= \|d(\mathrm{id},T)\|_{L^2(\mu)} + \|d(\mathrm{id},S)\|_{L^2(\nu)}$$

and, by taking the infimum with respect to S and T, we conclude.

The general case can be achieved either by approximating μ and ν by atomless measures (but this might be not possible in general, think to the case when the space (X,d) is discrete, or it has an isolated point where some mass might be concentrated), or by Dudley's Lemma 8.4 below. Thinking of μ as a measure in $\mathscr{P}(X_1)$, of ν as a measure in $\mathscr{P}(X_2)$ and of σ as a measure in $\mathscr{P}(X_3)$, the lemma provides, for any $\pi^{12} \in \mathscr{P}(X_1\times X_2)$ optimal plan from μ to ν, and $\pi^{23} \in \mathscr{P}(X_2\times X_3)$ optimal plan from ν to σ, $\pi \in \mathscr{P}(X_1\times X_2\times X_3)$ whose projections on the first two factors (resp. last two factors) is π^{12} (resp. π^{23}). In particular, since the first marginal of π is μ and the third marginal of π is σ, one has

$$W_2(\mu,\sigma) \le \|d(x_1,x_3)\|_{L^2(\pi)} \le \|d(x_1,x_2)\|_{L^2(\pi)} + \|d(x_2,x_3)\|_{L^2(\pi)}$$
$$= \|d(x_1,x_2)\|_{L^2(\pi^{12})} + \|d(x_2,x_3)\|_{L^2(\pi^{23})} = W_2(\mu,\nu) + W_2(\nu,\sigma).$$

□

The following lemma, that we already used in the proof of the triangle inequality, provides a useful, but not canonical, way to build a "composition" of transport plans.

Lemma 8.4 (Dudley) *Let (X_1,μ_1), (X_2,μ_2), (X_3,μ_3) be Polish spaces, $\pi^{12}\in\Gamma(\mu_1,\mu_2)$ and $\pi^{23}\in\Gamma(\mu_2,\mu_3)$. Then there exists $\pi\in\mathscr{P}(X_1\times X_2\times X_3)$ such that*

$$p^{1,2}_\#(\pi) = \pi^{12} \quad \text{and} \quad p^{2,3}_\#(\pi) = \pi^{23},$$

where $p^{1,2}(x_1,x_2,x_3) = (x_1,x_2)$ and $p^{2,3}(x_1,x_2,x_3) = (x_2,x_3)$.

Proof Using the disintegration Theorem 2.4, we can write

$$\pi^{12}(\mathrm{d}x_1,\mathrm{d}x_2) = \pi^{12}_{x_2}(\mathrm{d}x_1)\mu_2(\mathrm{d}x_2), \qquad \pi^{23}(\mathrm{d}x_2,\mathrm{d}x_3) = \pi^{23}_{x_2}(\mathrm{d}x_3)\mu_2(\mathrm{d}x_2).$$

Setting

$$\pi := \pi^{12}_{x_2}\times\pi^{23}_{x_2}(\mathrm{d}x_1,\mathrm{d}x_3)\mu_2(\mathrm{d}x_2) \in \mathscr{P}(X_1\times X_2\times X_3),$$

it is easy to check that π has all the required properties. □

Example 8.5 (Comparison of W_2 with the L^2 Distance of the Densities) Let consider a measure $\mu = \varrho\mathscr{L}^n$ with $\operatorname{supp}\varrho \subset \overline{B}_1$ and the shifted measures $\mu_h = \varrho_h\mathscr{L}^n$, with $\varrho_h(\cdot) = \varrho(\cdot+h)$. If $|h|>1$, then $\operatorname{supp}\varrho\cap\operatorname{supp}\varrho_h = \emptyset$ and therefore $\|\varrho_h-\varrho\|^2_{L^2(\mathbb{R})} = 2\|\varrho\|^2_{L^2(\mathbb{R}^n)}$.

Since the translations are optimal maps, $W_2(\mu_h, \mu) = |h|$, therefore $W_2(\mu_h, \mu) \gg \|\varrho_h - \varrho\|_{L^2(\mathbb{R}^n)}$ for $|h|$ large. On the other hand, if we look for small values of h, recall that a classical characterization of Sobolev spaces gives $\|\varrho_h - \varrho\|_{L^2(\mathbb{R})} = O(h)$ if and only if $\varrho \in H^{1,2}(\mathbb{R}^n)$ and, if this is the case,

$$\|\varrho_h - \varrho\|_{L^2(\mathbb{R}^n)} \sim |h| \left(\int_{\mathbb{R}^n} \left| \frac{\partial \varrho}{\partial h} \right|^2 \mathrm{d}x \right)^{1/2}.$$

This shows that there is no general inequality between the Wasserstein distance of absolutely continuous measures and the L^2 distance of their densities. In Sect. 1 of Lecture 18 we will see, instead, a closer relation with the H^{-1} norm.

2 Completeness of $(\mathscr{P}_2(X), W_2)$

In order to prove that also the metric completeness can be lifted from X to $\mathscr{P}_2(X)$ we represent a sequence of measures as marginals of a single measure in an infinite product. The first step of this construction is provided by the proposition below.

Proposition 8.6 (Iterated Dudley's Lemma) *Given $N \geq 3$, (X_n, d_n) Polish, $\mu_n \in \mathscr{P}(X_n)$, $1 \leq n \leq N$ and $\theta_n \in \Gamma(\mu_{n-1}, \mu_n)$, $2 \leq n \leq N$, there exist $\pi_n \in \mathscr{P}(X_1 \times \cdots \times X_n)$, $1 \leq n \leq N$, such that (with an obvious meaning of the notation for projections):*

(i) $p^{1,\dots,n-1}_{\#}\pi_n = \pi_{n-1}$ *for* $2 \leq n \leq N$;
(ii) $p^{i}_{\#}\pi_n = \mu_i$ *for* $1 \leq i \leq n \leq N$;
(iii) $p^{i-1,i}_{\#}\pi_n = \theta^i$ *for* $2 \leq i \leq n \leq N$.[1]

In the case $N = \infty$, (i), (ii), (iii) hold with the understanding that the inequalities $n \leq N$ are strict.

Proof For $N = 3$ we simply apply Lemma 8.4. In general, we repeatedly apply Dudley's Lemma 8.4 with

$$X_1 \times \cdots \times X_n = Z_1 \times Z_2 \times Z_3, \quad Z_1 = (X_1 \times \cdots \cdots X_{n-2}), \quad Z_2 = X_{n-1}, \quad Z_3 = X_n$$

and with $\pi_{n-1} \in \mathscr{P}(Z_1 \times Z_2)$ and $\theta_n \in \Gamma(\mu_{n-1}, \mu_n) \subset \mathscr{P}(Z_2 \times Z_3)$, to obtain π_n. □

[1] Observe that (ii) is a consequence of (iii), nevertheless we prefer to point it out explicitly in the statement.

It is useful to include the case $n = N$ even when $N = \infty$, in Proposition 8.6. In this case, because of property (i), the family $\{\pi_n\}_{n\geq 1}$ given by the proposition is consistent. Therefore, according to Kolmogorov's theorem (see Theorem 7.7.1 in [22]), one has

$$\text{there exists a unique } \pi_\infty \in \mathscr{P}(\mathbb{X}) \text{ such that } \quad p^{1,\dots,n}_{\#}\pi_\infty = \pi_n \ \ \forall n \geq 1,$$

where $\mathbb{X} = \prod_{i=1}^{\infty} X_i$, endowed with the product $\mathscr{B}_\infty$ of the Borel σ-algebras. This allows to extend (ii) to the case $1 \leq i < n = \infty$ and (iii) to the case $2 \leq i < n = \infty$.

Given a measure space $(\Omega, \mathcal{F}, \mathbb{P})$, with $\mathbb{P}$ finite, in the proof of the next theorem we shall use the space

$$L^p(\Omega, \mathcal{F}, \mathbb{P}, X) := \left\{ f : \Omega \to X : f \text{ is } (\mathcal{F} - \mathscr{B}(X)) - \text{measurable}, \int_\Omega d^p(f, z_0)\, \mathrm{d}\mathbb{P} < \infty \right\}$$

(whose definition is obviously independent of $z_0 \in X$), endowed with the distance

$$\varrho(f, g) := \left(\int_\Omega d^p(f, g)\, \mathrm{d}\mathbb{P} \right)^{1/p}.$$

The notation here is a little misleading, since obviously this is not a vector space in general. Nevertheless, many properties of L^p spaces are retained. We will only need its completeness, when the target space (X, d) is complete.

Theorem 8.7 (Completeness) *If (X, d) is a complete metric space, $(\mathscr{P}_2(X), W_2)$ is complete as well.*

Proof Let (μ_n) be a Cauchy sequence with respect to the W_2 distance. Without loss of generality, we can assume $\sum_n W_2(\mu_n, \mu_{n+1}) < \infty$ (if not, we extract a subsequence with this property, and the limit of that subsequence, because of the Cauchy property, is the limit of the full sequence).

Applying Proposition 8.6 to μ_n and $X_n = X$ for all n, we can realize μ_n and $\theta^n \in \Gamma_o(\mu_n, \mu_{n+1})$ as the marginals of $\pi_\infty \in \mathscr{P}(\mathbb{X})$, i.e. $\mu_n = (p_n)_{\#}\pi_\infty$ and $\theta^n = (p_n, p_{n+1})_{\#}\pi_\infty$. Now, the key observation is that

$$p_n : \mathbb{X} \to \ X_n \sim X \ \text{ is Cauchy in } L^2(\mathbb{X}, \mathscr{B}_\infty, \pi_\infty, X).$$

Indeed, one has

$$\int_{\mathbb{X}} d^2(p_n, p_{n+1})\, \mathrm{d}\pi_\infty = \int_{\mathbb{X}} d^2(x_n, x_{n+1})\, \mathrm{d}(p_n, p_{n+1})_{\#}\pi_\infty = W_2^2(\mu_n, \mu_{n+1})$$

because of Proposition 8.6(iii). Thanks to the completeness of $L^2(\mathbb{X}, \mathscr{B}_\infty, \pi_\infty, X)$, (p_n) converges in $L^2(\mathbb{X}, \mathscr{B}_\infty, \pi_\infty, X)$. Calling p_∞ its limit and defining

$$\mu_\infty := (p_\infty)_{\#}\pi_\infty,$$

we claim that $\mu_n \xrightarrow{W_2} \mu_\infty$. Taking (2.1) into account, the claim follows easily by

$$W_2^2(\mu_n, \mu_\infty) \le \int_{\mathbb{X}} d^2(p_n, p_\infty)\, \mathrm{d}\pi_\infty \to 0.$$

□

3 Characterization of Convergence in $(\mathscr{P}_2(X), W_2)$ and Applications

In this section we characterize the convergence in $\mathscr{P}_2(X)$ in terms of weak convergence plus convergence of quadratic moments, relative to some, and then to all, points $x_0 \in X$.

Theorem 8.8 *Let $(\mu_n) \subset \mathscr{P}_2(X)$ and $\mu \in \mathscr{P}_2(X)$. Then $\mu_n \xrightarrow{W_2} \mu$ implies that μ_n weakly converge to μ and*

$$\int_X d^2(x_0, x)\, \mathrm{d}\mu_n(x) \to \int_X d^2(x_0, x)\, \mathrm{d}\mu(x) \quad \text{for all } x_0 \in X. \tag{8.1}$$

Conversely, if $\mu_n \to \mu$ weakly and the convergence of moments occurs for some $x_0 \in X$, then $\mu_n \xrightarrow{W_2} \mu$.

The following corollary is an immediate consequence of the compactness of $\mathscr{P}(X) = \mathscr{P}_2(X)$ with respect to the weak topology, when (X, d) is compact.

Corollary 8.9 *If (X, d) is compact, then $(\mathscr{P}(X), W_2)$ is compact.*

Remark 8.10 Recall that a metric space (X, d) is said to be *proper* if bounded closed sets are compact. Surprisingly, in general the properness of (X, d) does not imply the properness of $(\mathscr{P}_2(X), W_2)$, not even in the case $X = \mathbb{R}$. Indeed, for $\epsilon > 0$ given, let us consider the measures

$$\mu_n := \left(1 - \frac{1}{n}\right)\delta_0 + \frac{1}{n}\delta_{c_n}$$

where $c_n \to \infty$, with $c_n^2/n \ge \epsilon^2$. Then $\mu_n \to \delta_0$ weakly, but $W_2^2(\mu_n, \delta_0) = c_n^2/n \ge \epsilon^2$. As a consequence, no subsequence of (μ_n) can be convergent with respect to W_2.

Another simple consequence of the characterization of convergence with respect to W_2 is the following.

Corollary 8.11 *If (X, d) is Polish, then $(\mathscr{P}_2(X), W_2)$ is separable.*

Proof Let $D \subset X$ be a countable dense set and consider the countable set of probability measures given by finite sums $\sum_i q_i \delta_{x_i}$ with $x_i \in D$ and $q_i \in \mathbb{Q}_+$.

Its closure with respect to W_2 obviously contains the collection $\mathcal{Z}$ of probability measures representable as countable sums $\sum_i t_i \delta_{x_i}$ with $t_i \in \mathbb{R}_+$ and $x_i \in X$. Now, it is easy to check that any measure $\mu \in \mathscr{P}_2(X)$ with bounded support belongs to the closure of $\mathcal{Z}$ with respect to W_2. Indeed, if $\operatorname{supp}\mu \subset B_R(x_0)$, for any integer $h \geq 1$ we may partition $B_R(x_0)$ in countably many disjoint Borel pieces $A_{i,h}$ with diameter smaller than $1/h$ and define

$$\mu_h := \sum_i t_{i,h} \delta_{x_{i,h}} \qquad \text{with } x_{i,h} \in A_{i,h} \text{ and } t_{i,h} = \mu(A_{i,h}).$$

Since

$$\left| \int_{A_{i,h}} f \,\mathrm{d}\mu - t_{i,h} \int_{A_{i,h}} f \,\mathrm{d}\delta_{x_{i,h}} \right| = \left| \int_{A_{i,h}} f \,\mathrm{d}\mu - \mu(A_{i,h}) f(x_{i,h}) \right| \leq h^{-1} \mu(A_{i,h}) \operatorname{Lip}(f)$$

for all $f \in \operatorname{Lip}_b(X)$, we get

$$\left| \int_X f \,\mathrm{d}\mu_t - \int_X f \,\mathrm{d}\mu_h \right| \leq h^{-1} \operatorname{Lip}(f).$$

Hence, from Lemma 8.12 below we obtain that $\mu_h \to \mu$ weakly as $h \to \infty$, and since the supports are uniformly bounded, convergence occurs also with respect to W_2.

To conclude, it is sufficient to notice that any $\mu \in \mathscr{P}_2(X)$ can be approximated in W_2-distance by measures with bounded support: for instance, given $x_0 \in \operatorname{supp}\mu$, the measures defined by

$$\mu_R := \frac{1}{\mu(B_R(x_0))} \mu \llcorner B_R(x_0)$$

converge to μ as $R \to \infty$. □

We need two lemmas before the proof of Theorem 8.8.

Lemma 8.12 *Let $(\mu_n) \subset \mathscr{P}(X)$ and $\mu \in \mathscr{P}(X)$. Then*

$$\mu_n \to \mu \quad \textit{weakly} \qquad \Longleftrightarrow \qquad \int_X f \,\mathrm{d}\mu_n \to \int_X f \,\mathrm{d}\mu \qquad \forall f \in \operatorname{Lip}_b(X).$$

Proof One implication is obvious. In order to prove the other one, notice that for all $f \in C_b(X)$, there exist sequences (f_k), $(g_k) \subset \operatorname{Lip}_b(X)$ such that $f_k \uparrow f$ and $g_k \downarrow f$, thanks to the simple construction in the proof of Theorem 2.6. Then

$$\liminf_{n\to\infty} \int_X f \,\mathrm{d}\mu_n \geq \liminf_{n\to\infty} \int_X f_k \,\mathrm{d}\mu_n = \int_X f_k \,\mathrm{d}\mu \qquad \forall k,$$

so that

$$\liminf_{n\to\infty} \int_X f \, \mathrm{d}\mu_n \geq \int_X f \, \mathrm{d}\mu.$$

Analogously, using g_k, we obtain:

$$\limsup_{n\to\infty} \int_X f \, \mathrm{d}\mu_n \leq \int_X f \, \mathrm{d}\mu.$$

□

Lemma 8.13 (Weak Convergence Criterion) *Let $(\mu_n) \subset \mathscr{P}(X)$ and $\mu \in \mathscr{P}(X)$. Then $\mu_n \to \mu$ weakly if and only if*

$$\liminf_{n\to\infty} \mu_n(A) \geq \mu(A) \qquad \text{for any open set } A \subset X. \tag{8.2}$$

Equivalently, $\mu_n \to \mu$ weakly if and only if $\limsup_{n\to\infty} \mu_n(C) \leq \mu(C)$ for any closed set $C \subset X$.

Proof The implication from weak convergence to upper/lower semicontinuity of the evaluation of open/closed sets can be achieved by monotone approximation, using the construction of Theorem 2.6, as we already noticed in Remark 2.7.

The converse implication can be obtained noticing that, first of all, $\mu_n(X) \to \mu(X)$ implies $\limsup_n \mu_n(C) \leq \mu(C)$ for any closed set $C \subset X$. Then, Cavalieri's formula gives

$$\int_X f \, \mathrm{d}\mu_n = \int_0^{\sup f} \mu_n(\{f > t\}) \, \mathrm{d}t \to \int_0^{\sup f} \mu(\{f > t\}) \, \mathrm{d}t = \int_X f \, \mathrm{d}\mu,$$

for any $f \in C_b(X)$ nonnegative. Here we used the dominated convergence theorem, together with the fact that upper/lower semicontinuity of the evaluation on closed/open sets imply $\mu_n(\{f > t\}) \to \mu(\{f > t\})$ for any $t > 0$ such that $\mu(\{f = t\}) = 0$ (and the set of exceptional t's is at most countable). Then, weak convergence follows splitting $f \in C_b(X)$ in positive and negative part. □

Proof of Theorem 8.8 Assume that $\mu_n \xrightarrow{W_2} \mu$. Since $\int_X d^2(x_0, x) \, \mathrm{d}\nu(x) = W_2^2(\delta_{x_0}, \nu)$, convergence of all moments is obvious and we need only to check that μ_n weakly converge to μ. Thanks to Lemma 8.12 it suffices to test convergence against $f \in \mathrm{Lip_b}(X)$. If $\Sigma_n \in \Gamma_o(\mu_n, \mu)$, then

$$\limsup_{n\to\infty} \left| \int_X f \, \mathrm{d}\mu_n - \int_X f \, \mathrm{d}\mu \right| \leq \mathrm{Lip}(f) \limsup_{n\to\infty} \int_{X\times X} d \, \mathrm{d}\Sigma_n \leq \mathrm{Lip}(f) \limsup_{n\to\infty} W_2(\mu_n, \mu) = 0.$$

Assume now that $\mu_n \to \mu$ weakly, with convergence of moments relative to some $x_0 \in X$. We first prove convergence with respect to W_2 under the assumption

that (X, d) is compact. For some fixed $z \in X$, let us consider the following set of functions:

$$Z := \{f \in \mathrm{Lip}(X) : \mathrm{Lip}(f) \le 1,\ \ f(z) = 0\}.$$

Any function $f \in Z$ obviously satisfies $\max|f| \le \mathrm{diam}(X, d)$, hence Ascoli-Arzelà theorem provides the compactness of Z, viewed as a subset of $C(X)$.

Now the linear operators $L_n(f) := \int_X f \,\mathrm{d}\mu_n$ converge pointwise to $L(f) := \int_X f \,\mathrm{d}\mu$ in $C(X)$. Being uniformly bounded, they converge uniformly in Z. Since translation invariance gives

$$\sup_{\mathrm{Lip}(f)\le 1} \int_X f \,\mathrm{d}(\mu_n - \mu) = \sup_{f\in Z} \int_X f \,\mathrm{d}(\mu_n - \mu) \to 0, \tag{8.3}$$

from Kantorovich duality we deduce the existence of $\pi_n \in \Gamma(\mu_n, \mu)$, optimal for the distance cost, satisfying

$$\lim_{n\to\infty} \int_{X\times X} d \,\mathrm{d}\pi_n = 0.$$

Since d is bounded, this provides also convergence of μ_n to μ with respect to W_2, and completes the proof in the compact case.

In order to deal with the general case, define

$$\sigma_n := \frac{1}{Z_n}\left(1 + d^2(x_0, \cdot)\right)\mu_n \in \mathscr{P}(X), \qquad \sigma := \frac{1}{Z}\left(1 + d^2(x_0, \cdot)\right)\mu \in \mathscr{P}(X),$$

where Z_n and Z are the normalization constants. By the moment assumption, we have the convergence of Z_n to Z and this, together with

$$\liminf_{n\to\infty} \int_A d^2(x, x_0) \,\mathrm{d}\mu_n(x) \ge \int_A d^2(x, x_0) \,\mathrm{d}\mu(x) \qquad \text{for any open set } A \subset X$$

guaranteed by Remark 2.7, shows that $\liminf_n \sigma_n(A) \ge \sigma(A)$ for any open set $A \subset X$. Consequently, applying Lemma 8.13, we get $\sigma_n \to \sigma$ weakly.

Now we can apply Prokhorov Theorem 2.8, finding a nondecreasing family $\{K_k\}$ of compact sets such that $x_0 \in K_1$ and

$$\lim_{k\to\infty} \sup_n \sigma_n(X \setminus K_k) = 0.$$

Since

$$Z_n \sigma_n(X \setminus K_k) \ge \int_{X\setminus K_k} d^2(x_0, \cdot) \,\mathrm{d}\mu_n$$

we obtain

$$\lim_{k\to\infty} \sup_n \int_{X\setminus K_k} d^2(x_0,\cdot)\,\mathrm{d}\mu_n = 0. \tag{8.4}$$

If we define $\mu_{n,k} := \mu_n \llcorner K_k + (1-\mu_n(K_k))\delta_{x_0} \in \mathscr{P}(K_k)$, thanks to a diagonal argument and the discussion of the compact case, we can find a subsequence $(n(p))$ such that $\mu_{n(p),k}$ converges with respect to W_2 for any k (notice also that W_2 convergence transfers immediately from the space K_k to the larger space X, viewing $\mu_{n,k}$ as measures in X). Since the transport plans

$$\pi_{n,k} := (\mathrm{id}\times\mathrm{id})_\#\mu_n \llcorner K_k + (\mathrm{id}\times x_0)_\#\mu_n \llcorner (X\setminus K_k)$$

provide the uniform estimate

$$W_2^2(\mu_n,\mu_{n,k}) \le \int_{X\setminus K_k} d^2(x_0,x)\,\mathrm{d}\mu_n(x) \tag{8.5}$$

from (8.4) and (8.5) we obtain that $\mu_{n(p)}$ is Cauchy with respect to W_2. The convergence of $\mu_{n(p)}$ to μ follows by the completeness of $(\mathscr{P}_2(X), W_2)$ and the first implication that we already proved. □

Lecture 9: Analysis on Metric Spaces and the Dynamic Formulation of Optimal Transport



In this section we introduce basic notions and tools of analysis on metric spaces. We begin by defining the property of absolute continuity for curves $\gamma : [a, b] \to X$, with (X, d) a metric space, and we prove structural properties of this family of curves. In particular we show the existence of the so-called *metric derivative*, a key differential concept for analysis in metric spaces. We then introduce the *action of a curve* and the notion of *geodesic*. We finally discuss a dynamic formulation of the optimal transport problem employing all the previously introduced tools.

1 Absolutely Continuous Curves and Their Metric Derivative

We recall that for $X = \mathbb{R}$ endowed with the Euclidean distance at least two equivalent definitions of absolute continuity are available:

(i) $f : [a, b] \to \mathbb{R}$ is absolutely continuous if there exists $g \in L^1(a, b)$ such that

$$f(y) - f(x) = \int_x^y g(t)\,\mathrm{d}t \qquad \forall x,\ y \in (a, b).$$

By Lebesgue theorem, for $\mathscr{L}^1$-a.e. $x \in (a, b)$ the function f is differentiable at x and its derivative coincides with $g(x)$ (in particular, g is unique as an L^1 function).

(ii) $f : [a, b] \to \mathbb{R}$ is absolutely continuous if for any $\epsilon > 0$ there exists $\delta > 0$ such that for any disjoint collection $\{(a_i, b_i)\}_i$ of open intervals contained in $[a, b]$ with $\sum_i (b_i - a_i) < \delta$ we have

$$\sum_i |f(b_i) - f(a_i)| < \epsilon.$$

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The second definition, due to Vitali, is clearly implied by the first one, because the absolute continuity property of the integral of summable functions

$$\forall \epsilon > 0\ \exists \delta > 0 \quad \text{such that} \quad A \in \mathscr{B}([a,b]),\ \mathscr{L}^1(A) < \delta \quad \Longrightarrow \quad \int_A |g|\,\mathrm{d}x < \epsilon$$

can be applied with $A = \cup_i (a_i, b_i)$. See for instance Chapter 7 of [104] for a proof of the implication from (ii) to (i).

Definition 9.1 (Metric Absolute Continuity) We say that $\gamma : [a,b] \to X$ is an absolutely continuous curve, and we write $\gamma \in AC([a,b]; X)$, if there exists $g \in L^1(a,b)$ such that

$$d(\gamma(y), \gamma(x)) \le \int_x^y g(t)\,\mathrm{d}t \qquad \forall a \le x \le y \le b. \tag{9.1}$$

If $g \in L^p(a,b)$ for some $p \in (1,\infty]$, we also write $\gamma \in AC^p([a,b]; X)$.

In light of the equivalence between (i) and (ii), it is easy to see that this definition is consistent with the classical case, since it is clearly weaker than (i) and stronger than (ii). Notice also that the absolute continuity property of the integral ensures that any absolutely continuous curve is uniformly continuous. The Cantor-Vitali function, being nonconstant but with null derivative $\mathscr{L}^1$-a.e., provides a counterexample to the converse implication.

Theorem 9.2 (Metric Derivative) *For any $\gamma \in AC([a,b]; X)$ the limit*

$$\lim_{h\to 0} \frac{d(\gamma(t), \gamma(t+h))}{|h|} =: |\gamma'|(t)$$

exists for $\mathscr{L}^1$-a.e. $t \in (a,b)$. In addition, $|\gamma'|$ is the minimal g that we can choose, up to $\mathscr{L}^1$-negligible sets, in Definition 9.1. It is called metric derivative of γ.

Proof Without loss of generality, we can assume X to be compact: indeed, by the continuity of γ we can replace X with the image $\gamma([a,b])$.

Let $\{z_i\}_{i\in\mathbb{N}}$ be countable and dense in X and set $f_i(t) := d(\gamma(t), z_i)$. Now, f_i are absolutely continuous, since

$$|f_i(y) - f_i(x)| \le d(\gamma(y), \gamma(x)) \le \int_x^y g(t)\,\mathrm{d}t \qquad \forall\ a \le x \le y \le b,$$

by (9.1). Since f_i are real valued, we know that $f_i'(t)$ exists for $\mathscr{L}^1$-a.e. $t \in (a,b)$. In addition, the previous inequality in combination with Lebesgue theorem gives that $|f_i'| \le g$ $\mathscr{L}^1$-a.e. in (a,b).

We set $m(t) := \sup_i |f_i'(t)|$ and notice that the inequality $m \le g$ $\mathscr{L}^1$-a.e. in (a,b) guarantees the integrability of m. For any $t \in (a,b)$ such that $f_i'(t)$ exists for any i one has

$$\liminf_{h\to 0} \frac{d(\gamma(t+h),\gamma(t))}{|h|} \ge \liminf_{h\to 0} \frac{|f_i(t+h)-f_i(t)|}{|h|} = |f_i'(t)|$$

for any i, that implies

$$\liminf_{h\to 0} \frac{d(\gamma(t+h),\gamma(t))}{|h|} \ge m(t). \tag{9.2}$$

We now consider the upper estimate on the difference quotients, considering positive increments h (the case $h<0$ is analogous). Since

$$|f_i(t+h)-f_i(t)| \le \int_t^{t+h} |f_i'(s)|\,\mathrm{d}s \le \int_t^{t+h} m(s)\,\mathrm{d}s$$

and

$$\sup_i |f_i(t+h)-f_i(t)| = \sup_i |d(\gamma(t+h),z_i)-d(\gamma(t),z_i)| = d(\gamma(t+h),\gamma(t)),$$

(since we can take z_i arbitrarily close to $\gamma(t)$), it follows that

$$d(\gamma(t+h),\gamma(t)) \le \int_t^{t+h} m(s)\,\mathrm{d}s. \tag{9.3}$$

If we take t to be a Lebesgue point of m, by (9.3) we obtain

$$\limsup_{h\to 0^+} \frac{d(\gamma(t+h),\gamma(t))}{h} \le \lim_{h\to 0^+} \frac{1}{h}\int_t^{t+h} m(s)\,\mathrm{d}s = m(t).$$

This, combined with the analogous estimate for negative increments and (9.2), gives

$$\lim_{h\to 0} \frac{d(\gamma(t+h),\gamma(t))}{|h|} = m(t) \qquad \text{for } \mathscr{L}^1\text{-a.e. } t \in (a,b),$$

so that m provides the metric derivative $|\gamma'|(t)$. Finally, by construction we have $|\gamma'| = m \le g$ for any admissible g in Definition 9.1 and (9.3) gives that $|\gamma'|$ is admissible in Definition 9.1. □

Definition 9.3 (Length) Given a curve $\gamma \in AC([a,b];X)$, we define its length as $\ell(\gamma) := \int_a^b |\gamma'|(t)\,\mathrm{d}t$. It is then clear that $\ell(\gamma) \ge d(\gamma(a),\gamma(b))$ by (9.1) and Theorem 9.2.

Remark 9.4 (Length and Polygonals) Another traditional definition of length involves the supremum of the lengths of inscribed polygonals

$$\ell'(\gamma) := \sup\left\{\sum_{i=0}^{n-1} d(\gamma(t_i),\gamma(t_{i+1})) : a = t_0 < t_1 < \cdots < t_{n-1} < t_n = b\right\} \in [0,\infty]$$

and it makes sense even for discontinuous curves. It is obvious from (9.1) with $g = |\gamma'|$ that $d(\gamma(a),\gamma(b)) \le \ell'(\gamma) \le \ell(\gamma)$ on $AC([a,b];X)$. On the other hand, adding always the intermediate point s to the polygonals involved in the definition of $\ell'(\gamma|_{[a,t]})$, it is easily seen that

$$\ell'(\gamma|_{[a,s]}) + \ell'(\gamma|_{[s,t]}) = \ell'(\gamma|_{[a,t]}) \qquad \forall\, a < s < t \le b.$$

This yields $\ell'(\gamma|_{[a,t]}) - \ell'(\gamma|_{[a,s]}) \le \int_s^t |\gamma'|(r)\,\mathrm{d}r$, so that $s \mapsto \ell'(\gamma|_{[a,s]})$ is absolutely continuous in $[a,b]$. Now, the inequality

$$\ell'(\gamma|_{[a,s+h]}) \ge \ell'(\gamma|_{[a,s]}) + d(\gamma(s),\gamma(s+h)) \qquad \forall\, a < s < s+h \le b$$

guarantees that its derivative is $\mathscr{L}^1$-a.e. larger than $|\gamma'|$. By integration in (a,b) it follows that $\ell'(\gamma) \ge \ell(\gamma)$.

Definition 9.5 (Reparameterization) We say that $\tilde\gamma : [c,d] \to X$ is a reparameterization of $\gamma : [a,b] \to X$ if there exists a strictly increasing and onto map $\varphi : [c,d] \to [a,b]$ such that $\tilde\gamma = \gamma \circ \varphi$.

It is not difficult to see that the length ℓ' is invariant under reparameterization, in particular this applies also with ℓ to the class of absolutely continuous curves. As in the classical case, if φ is sufficiently regular, this can also be proved with the differential definition of length, since $|(\gamma\circ\varphi)'| = (|\gamma'|\circ\varphi)\varphi'$.

We will say that an absolutely continuous curve γ has *constant speed* if $|\gamma'|$ is (equivalent to) a constant. Of course, constant speed curves are also Lipschitz.

Proposition 9.6 (Constant Speed Reparameterizations) *For any $\gamma \in AC([a,b];X)$ there exists $\tilde\gamma \in AC([0,1];X)$ with constant speed, equal to $\ell(\gamma)$, such that $\gamma([a,b]) = \tilde\gamma([0,1])$, $\gamma(a) = \tilde\gamma(0)$ and $\gamma(b) = \tilde\gamma(1)$.*

Proof We prove the result under the additional assumption that $t \mapsto \ell(\gamma|_{[a,t]})$ is strictly increasing in $[a,b]$. In the general case the reparameterization procedure is slightly more complex, see for instance Theorem 4.2.1 of [7] for details.

Setting $L(t) := \ell(\gamma|_{[a,t]}) : [a,b] \to [0,\ell(\gamma)]$, from (9.1) we obtain that

$$d\big(\gamma(L^{-1}(v)),\gamma(L^{-1}(u))\big) \le \int_{L^{-1}(u)}^{L^{-1}(v)} |\gamma'|\,\mathrm{d}t = v - u \qquad 0 \le u \le v \le \ell(\gamma),$$

so that $\gamma \circ L^{-1} : [0, \ell(\gamma)] \to X$ is 1-Lipschitz. On the other hand, from

$$\ell(\gamma) = \ell(\gamma \circ L^{-1}) = \int_0^{\ell(\gamma)} |(\gamma \circ L^{-1})'| \, \mathrm{d}t$$

we obtain that $|(\gamma \circ L^{-1})'| = 1$ $\mathscr{L}^1$-a.e. in $(0, \ell(\gamma))$. By a linear rescaling, we obtain $\tilde{\gamma}$ out of $\gamma \circ L^{-1}$. □

2 Geodesics and Action

In the metric theory, the word geodesic refers always to a length-minimizing property, since a priori no geodesic ODE, as in the Riemannian theory, is available. Recall also that Riemannian geodesics (now understood as solutions to the ODE system) have constant speed and are length minimizing only locally, not necessarily globally.

Definition 9.7 (Geodesic) We say that $\gamma \in AC([a, b]; X)$ is a geodesic if

$$\ell(\gamma) = d(\gamma(a), \gamma(b)).$$

We actually need to require only that $\ell(\gamma) \le d(\gamma(a), \gamma(b))$, since the converse inequality is always true. The following elementary lemma shows that restrictions of geodesics are still geodesics.

Lemma 9.8 (Restriction of Geodesics) *If $\gamma \in AC([a, b]; X)$ is a geodesic, then $\gamma|_{[x,y]}$ is still a geodesic for any $a \le x \le y \le b$.*

Proof We notice that

$$\begin{aligned} d(\gamma(a), \gamma(b)) &\le d(\gamma(a), \gamma(x)) + d(\gamma(x), \gamma(y)) + d(\gamma(y), \gamma(b)) \\ &\le \int_a^x |\gamma'|(s) \, \mathrm{d}s + \int_x^y |\gamma'|(s) \, \mathrm{d}s + \int_y^b |\gamma'|(s) \, \mathrm{d}s \\ &= \int_a^b |\gamma'|(s) \, \mathrm{d}s = d(\gamma(a), \gamma(b)). \end{aligned}$$

Since the extreme sides are equal, all intermediate inequalities have to be equalities. □

Definition 9.9 (The Space Geo(X)) We define Geo(X) as the space of constant speed geodesics on $[0, 1]$. That is, geodesic curves γ with constant metric derivative, then equal to $\ell(\gamma) = d(\gamma(0), \gamma(1))$.

Notice that $\mathrm{Geo}(X)$ is a closed subset of $C([0,1];X)$. Indeed, $\gamma \in \mathrm{Geo}(X)$ if and only if

$$d(\gamma(t),\gamma(s)) = |t-s| d(\gamma(0),\gamma(1)) \qquad \text{for all } s,\ t \in [0,1]. \tag{9.4}$$

One might ask which properties can be lifted from X to $C([0,1];X)$. It is well known that this happens for completeness and for separability.

We define the evaluation maps $e_t : C([0,1];X) \to X$, $t \in [0,1]$, as

$$e_t : C([0,1];X) \to X, \qquad e_t(\gamma) := \gamma(t). \tag{9.5}$$

Let us introduce the following important notion.

Definition 9.10 (Action of a Curve) The (possibly infinite) action $\mathcal{A}_2(\gamma)$ of $\gamma \in AC([0,1];X)$ is defined as

$$\mathcal{A}_2(\gamma) = \int_0^1 |\gamma'|^2(t)\,\mathrm{d}t.$$

Note that Hölder's inequality easily implies

$$\mathcal{A}_2(\gamma) \geq \ell^2(\gamma) \geq d^2(\gamma(0),\gamma(1)). \tag{9.6}$$

In addition, for any $x,\ y \in [0,1]$ with $x \leq y$ we have

$$d(\gamma(x),\gamma(y)) \leq \int_x^y |\gamma'|(t)\,\mathrm{d}t \leq |x-y|^{\frac{1}{2}} (\mathcal{A}_2(\gamma))^{\frac{1}{2}}, \tag{9.7}$$

which yields that absolutely continuous curves with finite action are (1/2)-Hölder continuous.

As in the Riemannian context the space $\mathrm{Geo}(X)$ can be characterized by a variational principle involving not the length, but the action.

Lemma 9.11 *For $\gamma \in AC([0,1];X)$, $\gamma \in \mathrm{Geo}(X)$ if and only if $\mathcal{A}_2(\gamma) = d^2(\gamma(0),\gamma(1))$.*

Proof By definition any $\gamma \in \mathrm{Geo}(X)$ is length minimizing and it has constant speed equal to $d(\gamma(0),\gamma(1))$, so that its action is the square of the distance between the endpoints.

The converse implication, from $\mathcal{A}_2(\gamma) = d^2(\gamma(0),\gamma(1))$ to the geodesic property, is immediate as (9.6) implies $\ell(\gamma) = d(\gamma(0),\gamma(1))$ and the equality in Hölder's inequality implies that $|\gamma'|$ has to be (equivalent to a) constant. □

It is possible to extend the functional $\mathcal{A}_2$ to the whole $C([0,1];X)$ by setting

$$\mathcal{A}_2(\gamma) := \begin{cases} \int_0^1 |\gamma'|^2(t)\,\mathrm{d}t & \text{if } \gamma \in AC([0,1];X), \\ +\infty & \text{if } \gamma \in C([0,1];X) \setminus AC([0,1];X). \end{cases} \tag{9.8}$$

In this way, $\mathcal{A}_2 : C([0,1];X) \to [0,\infty]$ is lower semicontinuous, as we will prove later on in Theorem 10.2.

3 Dynamic Reformulation of the Optimal Transport Problem

With the action functional we can give, at least in geodesic spaces, another equivalent formulation of the optimal transport problem for the cost function $c(x,y) = d^2(x,y)$. In this "dynamic" formulation, crucial for the many new developments of the theory, we care not only of the initial/final destination, but also of the paths (namely the curves in $\mathrm{Geo}(X)$) used to move the mass in between.

For $\mu,\ \nu \in \mathscr{P}_2(X)$, consider

$$\min\left\{ \int_{C([0,1];X)} \mathcal{A}_2(\gamma)\,\mathrm{d}\eta(\gamma) : \eta \in \mathscr{P}(C([0,1];X)),\ (e_0)_\#\eta = \mu,\ (e_1)_\#\eta = \nu \right\}. \tag{Dyn}$$

Since trajectories are sought, instead of pairs of initial and final points, we call any admissible η in (Dyn) a *dynamic transport plan* and any optimal η will be called *optimal geodesic plan*. We denote by $\mathrm{OptGeo}(\mu,\nu)$ the collection of optimal geodesic plans from μ to ν. We remark that our terminology is justified by the fact that $\eta \in \mathscr{P}(C([0,1];X))$ is a dynamic transport plan from μ to ν if and only if $(e_0,e_1)_\#\eta$ is an admissible transport plan from μ to ν, and that, in geodesic spaces, any $\eta \in \mathrm{OptGeo}(\mu,\nu)$ is supported on $\mathrm{Geo}(X)$ (see Theorem 9.13).

Definition 9.12 (Geodesic Metric Space) A metric space (X,d) is geodesic if for all $x,\ y \in X$ there exists $\gamma \in \mathrm{Geo}(X)$ with $\gamma(0) = x$ and $\gamma(1) = y$.

It is well known that any compact Riemannian manifold is a geodesic metric space. A weaker definition, that will not be used in these lectures, is the notion of *length* space, where we require only that $\inf\{\ell(\gamma) : \gamma(0) = x,\ \gamma(1) = y\}$ equals $d(x,y)$.

Theorem 9.13 *If (X,d) is a Polish and geodesic metric space, then*

$$\min_{\text{(Dyn)}} = \min_{\text{(K)}}.$$

In addition, η is optimal if and only if η is supported on $\mathrm{Geo}(X)$ *and* $(e_0,e_1)_\#\eta \in \Gamma_o(\mu,\nu)$.

Proof If η is admissible for (Dyn), then

$$\int_{C([0,1];X)} \mathcal{A}_2(\gamma)\,\mathrm{d}\eta(\gamma) \geq \int_{C([0,1];X)} d^2(\gamma(0),\gamma(1))\,\mathrm{d}\eta(\gamma) \tag{9.9}$$
$$= \int_{X\times X} d^2(x,y)\,\mathrm{d}(e_0,e_1)_\#\eta(x,y) \geq \min_{(K)}.$$

In order to prove the converse inequality, we start from an optimal plan π in Kantorovich's problem, and for any $(x,y) \in X\times X$ we choose $\gamma_{x,y} \in \mathrm{Geo}(X)$ such that $\gamma_{x,y}(0) = x$ and $\gamma_{x,y}(1) = y$. We can thus define the map[1]

$$\Gamma : X\times X \to \mathrm{Geo}(X),\ \ \Gamma(x,y) = \gamma_{x,y}$$

and the measure $\eta := \Gamma_\#\pi \in \mathscr{P}(C([0,1];X))$ which, by construction, is supported in $\mathrm{Geo}(X)$. Hence, we have

$$\min_{(K)} = \int_{X\times X} d^2(x,y)\,\mathrm{d}\pi(x,y) = \int_{\mathrm{Geo}(X)} d^2(\gamma(0),\gamma(1))\,\mathrm{d}\eta(\gamma) = \int_{\mathrm{Geo}(X)} \mathcal{A}_2(\gamma)\,\mathrm{d}\eta(\gamma),$$

by Lemma 9.11.

Note finally that η is optimal if and only if the inequalities in (9.9) are all equalities, which happens if and only if η is supported in $\mathrm{Geo}(X)$ and $(e_0,e_1)_\#\eta \in \Gamma_o(\mu,\nu)$. □

[1] We are tacitly assuming that there is a π-measurable way to select the geodesics γ_{xy}. For simplicity we do not discuss in detail this technical point, based on the fact that (9.4) grants the closedness of the graph in $X\times X\times \mathrm{Geo}(X)$ of the multi-valued map $(x,y)\mapsto\gamma_{xy}$ (see [22] for more details about measurable selection theorems, based on Suslin's theory of analytic sets). When the geodesic γ_{xy} is unique, it can even be proved that the map $(x,y)\mapsto\gamma_{xy}$ is Borel (and even continuous in proper spaces (X,d)). In this particular case, a deep measure-theoretic result shows that closedness of the graph implies the Borel property (see for instance Lemma 6.7.1 in [22]).

Lecture 10: Wasserstein Geodesics, Nonbranching and Curvature

1 Lower Semicontinuity of the Action $\mathcal{A}_2$

Let us now come to the proof of the lower semicontinuity of the action, defined as in (9.8). The proof could be achieved with more elementary tools, but we prefer to use a general lemma that will play a role also in the sequel.

Lemma 10.1 (Joint Lower Semicontinuity) *Let (X, d) be a Polish space and $p \in (1, \infty)$. Let $\mu_n, \mu \in \mathscr{M}_+(X)$ with $\mu_n \to \mu$ in the duality with $C_b(X)$ and let $f_n \in L^p(X, \mu_n)$, with $\sup_n \|f_n\|_{L^p(X,\mu_n)} < \infty$. Then there exist $f \in L^p(X, \mu)$ and a subsequence $n(k)$ such that $f_{n(k)}\mu_{n(k)} \to f\mu$ in the duality with $C_b(X)$ and*

$$\|f\|_{L^p(\mu)} \le \liminf_{k\to\infty} \|f_{n(k)}\|_{L^p(\mu_{n(k)})}.$$

This result can be rephrased as the *joint* lower semicontinuity with respect to the weak convergence of the functional

$$F(\sigma, \mu) := \begin{cases} \displaystyle\int_X \left|\frac{\sigma}{\mu}\right|^p \mathrm{d}\mu & \text{if } |\sigma| \ll \mu, \\ +\infty & \text{otherwise.} \end{cases}$$

Proof Splitting f_n in positive and negative parts and using the superadditivity of the lim inf we can assume with no loss of generality that $f_n \ge 0$. We also assume, modulo the extraction of a subsequence, that $\|f_n\|_{L^p(X,\mu_n)}$ has limit L as $n \to \infty$.

By the Hölder inequality, with $p' = p/(p-1)$ we have

$$f_n\mu_n(A) \le \left(\int_X f_n^p \,\mathrm{d}\mu_n\right)^{\frac{1}{p}} (\mu_n(A))^{1/p'}$$

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for any Borel set $A \subset X$. Therefore the measures $f_n\mu_n$ are equi-tight. Thanks to Prokhorov Theorem 2.8, possibly passing to a subsequence, there exists $\sigma \in \mathscr{M}_+(X)$ such that $f_n\mu_n \to \sigma$ in duality with $C_b(X)$.

Now, if $g \in C_b(X)$, from $\int_X f_n g \,\mathrm{d}\mu_n \to \int_X g \,\mathrm{d}\sigma$ and

$$\left| \int_X f_n g \,\mathrm{d}\mu_n \right| \le \left(\int_X f_n^p \,\mathrm{d}\mu_n \right)^{\frac{1}{p}} \left(\int_X |g|^{p'} \,\mathrm{d}\mu_n \right)^{\frac{1}{p'}},$$

we get

$$\left| \int_X g \,\mathrm{d}\sigma \right| \le L \left(\int_X |g|^{p'} \,\mathrm{d}\mu \right)^{\frac{1}{p'}} \qquad \forall g \in C_b(X).$$

It follows that $g \mapsto \int_X g \,\mathrm{d}\sigma$ is a linear continuous functional with respect to the $L^{p'}(X,\mu)$ norm, that by the density of $C_b(X)$ can be extended to a linear continuous functional on the whole $L^{p'}(X,\mu)$. Then, by Riesz theorem there exists $f \in L^p(X,\mu)$ with $\|f\|_{L^p(X,\mu)} \le L$ such that $\sigma = f\mu$. □

Theorem 10.2 *If for $\gamma \in C([0,1]; X)$ we set*

$$\mathcal{A}_2{}^*(\gamma) := \sup \left\{ \sum_{i=1}^{n-1} \frac{d^2(\gamma(t_{i+1}), \gamma(t_i))}{t_{i+1} - t_i} : 0 \le t_1 < \cdots < t_n \le 1 \right\},$$

we have $\mathcal{A}_2(\gamma) = \mathcal{A}_2{}^*(\gamma)$. In particular, $\mathcal{A}_2$ is lower semicontinuous, being representable as the supremum of a family of continuous functions.

Proof Let us first show that $\mathcal{A}_2{}^* \le \mathcal{A}_2$. Without loss of generality let us assume that $\gamma \in AC([0,1]; X)$. Then

$$d^2(\gamma(t_{i+1}), \gamma(t_i)) \le \left(\int_{t_i}^{t_{i+1}} |\gamma'|(s) \,\mathrm{d}s \right)^2 \le (t_{i+1} - t_i) \int_{t_i}^{t_{i+1}} |\gamma'|^2(s) \,\mathrm{d}s,$$

so that the inequality $\mathcal{A}_2{}^*(\gamma) \le \mathcal{A}_2(\gamma)$ follows by dividing both sides by $(t_{i+1} - t_i)$ and summing the resulting inequalities.

To show the converse one, assume that $\mathcal{A}_2{}^*(\gamma) < \infty$, since otherwise there is nothing to prove. Let us consider the equally spaced partition $t_{i,n} = i/n$, so that we have

$$\sum_{i=0}^{n-1} n \, d^2(\gamma(t_{i+1,n}), \gamma(t_{i,n})) \le \mathcal{A}_2{}^*(\gamma) < \infty. \tag{10.1}$$

To pass to the limit with respect to n we use Lemma 10.1. We set

$$\mu_n := \frac{1}{n}\sum_{i=0}^{n-1}\delta_{t_{i,n}} \to \mathscr{L}^1 \llcorner [0,1] =: \mu$$

and we take functions $f_n \in L^2((0,1),\mu_n)$ satisfying $f_n(t_{i,n}) = n\, d(\gamma(t_{i+1,n}), \gamma(t_{i,n}))$, since only such values are relevant. By (10.1), we have $\int_0^1 f_n^2 \,\mathrm{d}\mu_n \le \mathcal{A}_2{}^*(\gamma)$. Thus, possibly passing to a subsequence, by Lemma 10.1 we can assume that there exists a measure σ such that $\sigma_n := f_n\mu_n \to \sigma$ (in duality with respect to $C_b([0,1])$) and $\sigma = f\mu$, with $f \in L^2((0,1),\mu)$ and $\|f\|_{L^2(\mu)} \le \mathcal{A}_2{}^*(\gamma)$.

Hence, for any $i < j$ we have

$$d(\gamma(t_{i,n}), \gamma(t_{j,n})) \le \sum_{k=i}^{j-1} d(\gamma(t_{k+1,n}), \gamma(t_{k,n})) = \sigma_n\big([t_{i,n}, t_{j,n}]\big).$$

In particular, if we take $i_n := [sn]$, $j_n := [tn]$, for $s < t$, then $i_n/n \to s$ and $j_n/n \to t$. Thus, we pass to the limit and we obtain

$$d(\gamma(s), \gamma(t)) \le \sigma([s,t]) = \int_s^t f(r)\,\mathrm{d}r,$$

which implies that $\gamma \in AC([0,1]; X)$ and $|\gamma'| \le f$, so that

$$\mathcal{A}_2(\gamma) = \int_0^1 |\gamma'|^2(t)\,\mathrm{d}t \le \|f\|^2_{L^2(0,1)} \le \mathcal{A}_2{}^*(\gamma).$$

□

2 Compactness Criterion for Curves and Random Curves

We present now a basic compactness criterion for the probability measures η involved in (Dyn). Its proof involves this simple variant of the classical Ascoli-Arzelá compactness theorem.

Theorem 10.3 (Ascoli-Arzelá for Curves in Metric Spaces) *A family $\mathcal{F} \subset C([0,1]; X)$ is relatively compact in $C([0,1]; X)$ if*

(i) $\sup_{\gamma\in\mathcal{F}} \mathcal{A}_2(\gamma) < \infty$ *(which immediately implies that $\mathcal{F} \subset AC([0,1]; X)$ and that, because of (9.7), $\mathcal{F}$ is equicontinuous);*

(ii) there exists a dense set $D \subset [0,1]$ such that for all $t \in D$ the set $\{\gamma(t) : \gamma \in \mathcal{F}\}$ is relatively compact in X.

Proof We just sketch the argument since it is an immediate adaptation of the classical proof of Ascoli-Arzelá theorem.

Let $(\gamma_n) \subset \mathcal{F}$ be a sequence and assume with no loss of generality that D is countable. Since we know that for all $t \in D$ the set $\{\gamma_n(t)\}$ is relatively compact in X, by a diagonal argument we can find a subsequence $n(p)$ such that

$$\gamma(t) := \lim_{p\to\infty} \gamma_{n(p)}(t) \text{ exists for all } t \in D.$$

By equicontinuity, $\gamma : D \to X$ is uniformly continuous, hence it has a unique continuous extension to the whole $[0, 1]$. Finally, using once more the equicontinuity, pointwise convergence on D and density of D imply uniform convergence on $[0, 1]$. □

Theorem 10.4 (Random Ascoli–Arzelá) *Let $\mathcal{F} \subset \mathscr{P}(C([0, 1]; X))$ be satisfying:*

(i) $\sup_{\eta\in\mathcal{F}} \int_{C([0,1];X)} \mathcal{A}_2(\gamma)\, \mathrm{d}\eta(\gamma) < \infty$*;*

(ii) there exists a dense set $D \subset [0, 1]$ such that the family $\{(e_t)_\#\eta : \eta \in \mathcal{F}\}$ is tight in $\mathscr{P}(X)$ for all $t \in D$.

Then $\mathcal{F}$ is tight in $\mathscr{P}(C([0, 1]; X))$.

Proof We can assume with no loss of generality that D is countable. Let us fix $\epsilon > 0$, set $D = \{t_n\}_{n\geq 1}$ and call S the supremum in (i). Setting

$$\Gamma_0 := \{\gamma \in C([0, 1]; X) : \mathcal{A}_2(\gamma) \leq M\},$$

by the Markov inequality one has

$$\sup_{\eta\in\mathcal{F}} \eta(C([0, 1]; X) \setminus \Gamma_0) \leq \frac{1}{M} \sup_{\eta\in\mathcal{F}} \int_{C([0,1];X)} \mathcal{A}_2(\gamma)\, \mathrm{d}\eta(\gamma) = \frac{S}{M} < \frac{\epsilon}{2}$$

for M sufficiently large. By Prokhorov Theorem 2.8 and assumption (ii) we can find compact sets $K_n \subset X$ such that, setting

$$\Gamma_n := \{\gamma \in C([0, 1]; X) : \gamma(t_n) \in K_n\} \qquad n \geq 1,$$

one has

$$\eta(C([0, 1]; X) \setminus \Gamma_n) = (e_{t_n})_\#\eta(X \setminus K_n) < \frac{\epsilon}{2^{n+1}} \qquad \forall n \geq 1.$$

Setting $\Gamma := \cap_{i\geq 0}\Gamma_i$, it follows that $\eta(C([0, 1]; X) \setminus \Gamma) < \epsilon$ for all $\eta \in \mathcal{F}$. The set Γ is compact in $C([0, 1]; X)$ by Theorem 10.3 above. □

3 Lifting of Geodesics from X to $\mathscr{P}_2(X)$

Definition 10.5 (Lifting of a Curve in $\mathscr{P}(X)$) Given a curve $\mu_t : [0,1] \to \mathscr{P}(X)$, we say that $\eta \in \mathscr{P}(C([0,1];X))$ is a lifting of μ_t if $(e_t)_\# \eta = \mu_t$ for all $t \in [0,1]$.

Notice that a necessary condition for the existence of the lifting is the continuity of the curve $t \mapsto \mu_t$ with respect to the weak topology, as all curves $(e_t)_\#\eta$ with $\eta \in \mathscr{P}(C([0,1];X))$ have this property. When we restrict ourselves to curves $\mu_t \in \mathrm{Geo}(\mathscr{P}_2(X))$ and liftings $\eta \in \mathrm{OptGeo}(\mu_0, \mu_1)$, Lemma 9.11 provides the "commutation" property

$$\int_0^1 \mathcal{A}_2(\mu_t)\,\mathrm{d}t = \int_{\mathrm{Geo}(X)} \mathcal{A}_2(\gamma)\,\mathrm{d}\eta(\gamma). \tag{10.2}$$

Theorem 10.6 (Stability of the Geodesic Property) *If (X,d) is a geodesic metric space, then $(\mathscr{P}_2(X), W_2)$ is geodesic as well. More precisely, any $\eta \in \mathrm{OptGeo}(\mu,\nu)$ induces a constant speed geodesic $\mu_t = (e_t)_\#\eta$ in $\mathscr{P}_2(X)$ from μ to ν. Conversely, for any $\mu_t \in \mathrm{Geo}(\mathscr{P}_2(X))$ there exists a (possibly nonunique) lifting $\eta \in \mathrm{OptGeo}(\mu_0,\mu_1)$.*

Proof The proof of the first statement is elementary: since $(e_s, e_t)_\#\eta$ is an admissible plan from μ_s to μ_t, one has

$$\begin{aligned} W_2^2(\mu_s,\mu_t) &\le \int_{\mathrm{Geo}(X)} d^2(\gamma(s),\gamma(t))\,\mathrm{d}\eta(\gamma) \\ &= (s-t)^2 \int_{\mathrm{Geo}(X)} \mathcal{A}_2(\gamma)\,\mathrm{d}\eta(\gamma) = (s-t)^2 W_2^2(\mu,\nu) \end{aligned}$$

and we already observed that the $\le$ inequality with all pairs of intermediate times is sufficient to provide both the constant speed and the geodesic properties. This argument proves also the inequality $\le$ in (10.2) for any curve μ_t generated by $\eta \in \mathrm{OptGeo}(\mu_0,\mu_1)$ (even in the stronger form of a pointwise upper bound on the metric derivative of μ_t).

In order to prove the converse statement and inequality, let $\mu_t \in \mathrm{Geo}(\mathscr{P}_2(X))$ be given and let $n \ge 1$ integer. Let $\Sigma_i \in \Gamma_o(\mu_{i/n}, \mu_{(i+1)/n})$, $0 \le i \le n-1$. By applying repeatedly Dudley's Lemma 8.4 (see also Proposition 8.6) we can find $\Theta_n \in \mathscr{P}(X^{n+1})$ satisfying

$$\pi^{i,i+1}_\# \Theta_n = \Sigma_i\,, \qquad 0 \le i \le n-1. \tag{10.3}$$

Now we interpolate the random collection of $n+1$ points given by Θ_n, to build a probability measure η_n in $C([0,1];X)$ with these two properties:

(i) it is concentrated on curves γ having constant speed, equal to $nd(\gamma(i/n), \gamma((i+1)/n))$, in all intervals $[i/n, (i+1)/n]$;
(ii) $(e_t)_\#\eta_n = \mu_t$ for all $t = i/n$, $0 \le i \le n$.

More specifically, let $\Gamma_i : X \times X \to \mathrm{Geo}(X)$ be Σ_i-measurable geodesic selection maps, already mentioned in the proof of Theorem 9.13, and define

$$\Gamma^n : X^{n+1} \to C([0,1];X)$$

by saying that $\Gamma^n(x_1,\dots,x_{n+1})$ restricted to $[i/n,(i+1)/n]$ is $\Gamma_i(x_i,x_{i+1})(n(t-i/n))$. Then, η_n is precisely the image of Θ_n under Γ^n.

Using properties (i) and (ii) we compute:

$$\begin{aligned}\int_{C([0,1];X)} \mathcal{A}_2 \,\mathrm{d}\eta_n &= n\sum_{i=0}^{n-1}\int d^2(\gamma(\frac{i}{n}),\gamma(\frac{i+1}{n}))\,\mathrm{d}\eta_n(\gamma) = n\sum_{i=0}^{n-1}\int_{X^{n+1}} d^2(x_i,x_{i+1})\,\mathrm{d}\Theta_n(x)\\ &= n\sum_{i=0}^{n-1}\int_{X^2} d^2\,\mathrm{d}\Sigma_i = n\sum_{i=0}^{n-1} W_2^2(\mu_{i/n},\mu_{(i+1)/n}) = W_2^2(\mu_0,\mu_1). \qquad (10.4)\end{aligned}$$

Now we wish to pass to the limit as $n\to\infty$. To this aim, let us check that the family $\mathcal{F} = \{\eta_n\}_{n\ge1}$ satisfies the assumptions of Theorem 10.4. Obviously (i) holds, and to check (ii) it is sufficient to prove, thanks to Prokhorov Theorem 2.8, that $(e_t)_\#\eta_n$ is convergent to μ_t for all $t\in[0,1]$. One has, indeed,

$$\lim_{n\to\infty}\int_X \phi\,\mathrm{d}(e_t)_\#\eta_n = \lim_{n\to\infty}\int_X \phi\,\mathrm{d}(e_{[nt]/n})_\#\eta_n = \lim_{n\to\infty}\int_X \phi\,\mathrm{d}\mu_{[nt]/n} = \int_X \phi\,\mathrm{d}\mu_t \qquad (10.5)$$

for any $\phi\in \mathrm{Lip}_b(X)$ and we already noticed in Lemma 8.12 that, in the class of probability measures, it suffices to test the weak convergence on the class of bounded and Lipschitz test functions. The first equality in (10.5) is justified by

$$\begin{aligned}&\left|\int_{C([0,1];X)}\phi(\gamma(t))\,\mathrm{d}\eta_n(\gamma) - \int_{C([0,1];X)}\phi(\gamma(\frac{[nt]}{n}))\,\mathrm{d}\eta_n(\gamma)\right|\\ &\le \mathrm{Lip}(\phi)\sqrt{t-\frac{[nt]}{n}}\int\sqrt{\mathcal{A}_2(\gamma)}\,\mathrm{d}\eta_n(\gamma)\\ &\le \mathrm{Lip}(\phi)\sqrt{t-\frac{[nt]}{n}}\,W_2(\mu_0,\mu_1).\end{aligned}$$

We can then apply Theorem 10.4 to obtain a weakly convergent subsequence $(\eta_{n(p)})$, whose limit we denote by η. The lower semicontinuity of the action, together with (10.4), gives $\int\mathcal{A}_2\,\mathrm{d}\eta \le W_2^2(\mu_0,\mu_1)$. It remains to prove that

$(e_t)_\#\eta = \mu_t$ for all $t \in [0,1]$. This can be obtained directly from (10.5) and from the weak convergence of $\eta_{n(p)}$ to η. □

Remark 10.7 Actually it can be proved that any $\mu_t \in \mathrm{Geo}(\mathscr{P}_2(X))$ has a lifting $\eta \in \mathrm{OptGeo}(\mu_0, \mu_1)$ independently of the geodesic assumption on (X, d). The strategy, pursued in [85], is to interpolate using piecewise constant curves instead of piecewise geodesic ones. This approach requires different tightness criteria (since the curves become absolutely continuous only in the limit) and it has the advantage of avoiding the use of the geodesic selection map Γ.

Remark 10.8 (Lifting of Curves μ_t in AC^2) The proof of the existence of a "good" lifting η satisfying (10.2) works smoothly also for any curve $\mu_t \in AC^2([0,1], \mathscr{P}_2(X))$. Indeed the only step of the proof in which we used the geodesic property is (10.4), but there we can instead exploit the following inequality

$$n \sum_{i=0}^{n-1} W_2^2(\mu_{i/n}, \mu_{(i+1)/n}) \le \int_0^1 |\dot{\mu}_t|^2 \, \mathrm{d}t$$

that is an easy consequence of Cauchy-Schwarz. A byproduct of the proof is that the chosen lifting η satisfies, by construction, the inequality $\ge$ in (10.2). The converse inequality (this one valid for *all* liftings η concentrated on $AC^2([0,1]; X)$) can be proved arguing as in the geodesic case, starting from

$$W_2^2(\mu_s, \mu_t) \le \int_{C([0,1];X)} d^2(\gamma(s), \gamma(t)) \, \mathrm{d}\eta(\gamma) \le (s-t) \int_{C([0,1];X)} \int_s^t |\gamma'|^2(r) \, \mathrm{d}r \, \mathrm{d}\eta(\gamma).$$

We address for a moment the problem of uniqueness of dynamic transport plans and, as a consequence, of geodesics in $\mathscr{P}_2(X)$ under suitable assumptions.

Proposition 10.9 *Let (X, d) be a geodesic metric space. Fix $\mu_0, \mu_1 \in \mathscr{P}_2(X)$ and assume that there exists a unique $\Sigma \in \Gamma_o(\mu_0, \mu_1)$ and that Σ-a.e. $(x, y) \in X \times X$ are joined by a unique constant speed geodesic $\Gamma(x, y)$. Then there exist a unique $\eta \in \mathrm{OptGeo}(\mu_0, \mu_1)$ and a unique $\mu_t \in \mathrm{Geo}(\mathscr{P}_2(X))$ connecting μ_0 and μ_1, given by $(e_t)_\#\eta$.*

Proof Let $\eta \in \mathrm{OptGeo}(\mu_0, \mu_1)$ be an optimal geodesic plan, whose existence follows by Theorem 9.13. By the same result we also know that $(e_0, e_1)_\#\eta \in \Gamma_o(\mu_0, \mu_1)$ and that η is supported on $\mathrm{Geo}(X)$. By assumption $\Gamma_o(\mu_0, \mu_1) = \{\Sigma\}$ and we get $(e_0, e_1)_\#\eta = \Sigma$. On the other hand, since η is supported on $\mathrm{Geo}(X)$ one has $\eta_{xy} = \delta_{\Gamma(x,y)}$ for Σ-a.e. (x, y), where η_{xy} is the disintegration of η induced by Σ. The Borel property of the map $(x, y) \mapsto \eta_{xy}$ ensured by the disintegration Theorem 2.4 gives also that $(x, y) \mapsto \Gamma(x, y)$ is Σ-measurable. Also, it follows that $\eta = \Gamma_\#\Sigma$, thus the optimal geodesic plan is unique. Now, we know from Theorem 10.6 that any $\mu_t \in \mathrm{Geo}(\mathscr{P}_2(X))$ admits a lifting $\eta \in \mathrm{OptGeo}(\mu_0 \mu_1)$. Since $\mu_t = (e_t)_\#\eta$ and η is unique we obtain uniqueness also at the level of geodesics in $\mathscr{P}_2(X)$. □

Corollary 10.10 (Uniqueness of Geodesics in $\mathscr{P}_2(\mathbb{R}^n)$) *Assume that $\mu_0, \mu_1 \in \mathscr{P}_2(\mathbb{R}^n)$ and $\mu_0 \ll \mathscr{L}^n$. Then there exist a unique constant speed geodesic $\mu_t \in \mathrm{Geo}(\mathscr{P}_2(\mathbb{R}^n))$ joining μ_0 and μ_1 and a unique $\eta \in \mathrm{OptGeo}(\mu_0, \mu_1)$. In addition, μ_t is given by the formula*

$$\mu_t := (T_t)_{\#}\mu_0,$$

where $T = \nabla\phi$ is the optimal map from μ_0 to μ_1 and T_t is the interpolated transport map $T_t := (1-t)\mathrm{id} + tT$.

Proof The conclusion follows from Theorem 5.2 (which gives the uniqueness of the optimal transport plan $\Sigma = (\mathrm{id} \times T)_{\#}\mu_0$) and Proposition 10.9, together with the explicit expression $\gamma_{xy}(t) = (1-t)x + ty$ of constant speed geodesics in $\mathbb{R}^n$. □

An analogous uniqueness result holds in the Riemannian setting of Theorem 7.10, under the assumption that μ_0 is absolutely continuous with respect to the volume measure. In this case the interpolated transport maps are provided by the formula

$$T_t(x) := \exp_x(-t\nabla\phi(x)).$$

The following beautiful counterexample has been suggested by Filippo Santambrogio.

Example 10.11 (Non Uniqueness of the Lifting) Let $\mathbb{R}^2$ be endowed with the ℓ_∞ distance and with the equivalence relation $x \sim y$ if $x - y \in \mathbb{Z}^2$. We define $\mathbb{T}^2_\infty$ to be the quotient space (the two-dimensional torus), endowed with the quotient distance. In the sequel, for the sake of simplicity we do not write explicitly the equivalence class, since all concepts will be invariant. Then, all curves

$$\gamma_{y,a}(t) := (\frac{t}{2}, y + at) \mod \mathbb{Z}^2 \qquad 0 \le t \le 1$$

in $\mathbb{T}^2_\infty$ are constant speed geodesics, with speed $1/2$, as soon as $|a| \le 1/2$. It is easily seen that all the dynamic transport plans

$$\sigma_a := G^a_{\#}\mathscr{L}^1 \quad \text{where} \quad G^a(y) := \gamma_{y,a}(\cdot)$$

induce the same geodesic, with speed $1/2$, from the uniform length measure μ_0 on the interval $\{0\} \times [0, 1]$ to the uniform length measure μ_1 on the interval $\{\frac{1}{2}\} \times [0, 1]$. Namely, μ_t is given by the uniform length measure on the interval $\{\frac{t}{2}\} \times [0, 1]$.

By an averaging procedure, the same nonuniqueness example can also be produced when the initial and final measure are absolutely continuous with respect to the Lebesgue measure.

Definition 10.12 (Nonbranching Metric Space) A metric space (X, d) is said to be nonbranching if, for all $s \in (0, 1]$, the restriction map

$$\mathrm{restr}_{[0,s]} : \mathrm{Geo}(X) \to C([0, s]; X) \qquad \mathrm{restr}_{[0,s]}(\gamma) := \gamma|_{[0,s]}$$

is injective.

An equivalent definition is based on the injectivity of all the maps (e_0, e_s) with $s \in (0, 1)$. Indeed, assume that (X, d) is nonbranching and that γ_1, γ_2 satisfy $\gamma_1(0) = \gamma_2(0)$ and $\gamma_1(s) = \gamma_2(s)$ for some $s \in (0, 1)$. The new curve

$$\tilde{\gamma}(t) := \begin{cases} \gamma_1(t) & \text{if } t \in [0, s], \\ \\ \gamma_2(t) & \text{if } t \in (s, 1] \end{cases}$$

is still a constant speed geodesic and the nonbranching property gives $\gamma_1 \equiv \tilde{\gamma}$ on $[0, 1]$, whence $\gamma_1 \equiv \gamma_2$ on $[s, 1]$. Since constant speed geodesics are stable under time reversal, the nonbranching property can now be applied to the curves $\gamma_1(1-t)$, $\gamma_2(1-t)$, which coincide in $[0, 1-s]$, to get $\gamma_1 \equiv \gamma_2$.

Notice that the strength of the nonbranching assumption depends on the richness of constant speed geodesics: trivially satisfied if $\mathrm{Geo}(X)$ contains only constant curves, very meaningful when (X, d) is geodesic.

Example 10.13

(i) Any Riemannian manifold $(M, \mathbf{g})$ with $\mathbf{g}$ of class C^2 is nonbranching. This is due to the fact that geodesics solve in local coordinates the second order system of ODE's

$$\gamma_i'' = \sum_{j,k} \Gamma^i_{jk}(\gamma)\gamma_j'\gamma_k',$$

whose coefficients Γ^i_{jk}, namely the Christoffel symbols, depend on the first derivative of $\mathbf{g}$ (see [54, 99]). Hence, the Cauchy-Lipschitz theorem provides uniqueness for any given initial position and velocity.

(ii) The normed space $(\mathbb{R}^n, \|\cdot\|_\infty)$, viewed as a metric space, is highly branching. For instance, while there is a unique $\gamma \in \mathrm{Geo}(\mathbb{R}^2, \|\cdot\|_\infty)$ joining $(0, 0)$ to $(1, 1)$, all curves

$$\gamma(t) = (t, a(t)) \qquad \text{with } a(0) = 0,\ a(1) = 1,\ |a'| \le 1 \text{ in } [0, 1]$$

join $(0, 0)$ to $(1, 1)$ and belong to $\mathrm{Geo}(\mathbb{R}^2, \|\cdot\|_\infty)$.

In the following theorem we prove that $(\mathscr{P}_2(X), W_2)$ is nonbranching whenever (X, d) is nonbranching. Interestingly, the statement holds independently of the geodesic assumption.

Theorem 10.14 (Stability of the Nonbranching Property) *If (X, d) is a nonbranching metric space, then $(\mathscr{P}_2(X), W_2)$ is nonbranching as well. In addition, any $\mu_t \in \mathrm{Geo}(\mathscr{P}_2(X))$ has a unique lifting $\eta \in \mathrm{OptGeo}(\mu_0, \mu_1)$.*

Proof We prove the result under the additional assumption that (X, d) is geodesic. To avoid it, one should use Remark 10.7, which provides existence of liftings of constant speed geodesics in $\mathscr{P}_2(X)$ independently of the geodesic assumption on the base space (X, d).

Given an interval $I \subset [0, 1]$, we denote by $\eta_I \in \mathscr{P}(C(I; X))$ the push-forward of $\eta \in \mathscr{P}(C([0, 1]; X))$ under the operator restr_I, the restriction to I.

Let μ_t, $\tilde{\mu}_t$ be constant speed geodesics coinciding on the interval $[0, s]$, with $s \in (0, 1)$; the goal is to show that they coincide in $[0, 1]$. We denote by η and $\tilde{\eta}$ respective liftings, whose existence is ensured by Theorem 10.6.

We shall work with the space $C([0, s]; X)$, whose typical element will be denoted γ_s, and with the space $C([s, 1]; X)$, whose typical element will be denoted $\tilde{\gamma}_s$. By a "dynamic Dudley lemma", we are going to concatenate curves in the two spaces with $\gamma_s(s) = x = \tilde{\gamma}_s(s)$. Let us then denote

$$\varrho = \eta_{[0,s]} \in \mathscr{P}(C([0, s]; X)), \qquad \tilde{\varrho} = \tilde{\eta}_{[s,1]} \in \mathscr{P}(C([s, 1]; X)).$$

We claim that $\tilde{\varrho}$ is uniquely determined by ϱ, more precisely

$$\tilde{\varrho} = (\mathrm{restr}_{[s,1]} \circ \mathrm{restr}^{-1}_{[0,s]})_{\#}\varrho. \tag{10.6}$$

Since the marginal of ϱ at time s coincides with μ_s, the marginal of $\tilde{\varrho}$ at time s, we can disintegrate both to get $\varrho = \varrho_x \otimes \mu_s$, $\tilde{\varrho} = \tilde{\varrho}_x \otimes \mu_s$, with $\varrho_x \in \mathscr{P}(C([0, s]; X))$ concentrated on curves γ_s with $\gamma_s(s) = x$, $\tilde{\varrho}_x \in \mathscr{P}(C([s, 1]; X))$ concentrated on curves $\tilde{\gamma}_s$ with $\tilde{\gamma}_s(s) = x$.

Define now $\theta_x \in \mathscr{P}(C([0, 1]; X))$ as the image of $\varrho_x \times \tilde{\varrho}_x$ under the concatenation map, and $\theta := \int \theta_x \, \mathrm{d}\mu_s(x)$. By construction θ is a dynamic transport plan from $\mu_0 = \tilde{\mu}_0$ to $\tilde{\mu}_1$. We have then

$$\begin{aligned}
\|d(\gamma(0), \gamma(1))\|^2_{L^2(\theta)} &\le \|\mathcal{A}_2(\gamma)\|_{L^1(\theta)} = \| \int_0^s |\gamma'|^2 \, \mathrm{d}t \|_{L^1(\theta)} + \| \int_s^1 |\gamma'|^2 \, \mathrm{d}t \|_{L^1(\theta)} \\
&= \int_X \int \int_0^s |\gamma_s'|^2 \, \mathrm{d}t \, \mathrm{d}\varrho_x(\gamma_s) d\mu_s(x) + \int_X \int \int_s^1 |\tilde{\gamma}_s'|^2 \, \mathrm{d}t \, \mathrm{d}\tilde{\varrho}_x(\tilde{\gamma}_s) d\mu_s(x) \\
&= \int \int_0^s |\gamma'|^2 \, \mathrm{d}t \, \mathrm{d}\eta(\gamma) + \int \int_s^1 |\gamma'|^2 \, \mathrm{d}t \, \mathrm{d}\tilde{\eta}(\gamma) \\
&= s \int_{X \times X} d^2(\gamma(0), \gamma(1)) \, \mathrm{d}\eta(\gamma) + (1 - s) \int_{X \times X} d^2(\gamma(0), \gamma(1)) \, \mathrm{d}\tilde{\eta}(\gamma) \\
&= s W_2^2(\mu_0, \mu_1) + (1 - s) W_2^2(\tilde{\mu}_0, \tilde{\mu}_1) = W_2^2(\mu_0, \tilde{\mu}_1).
\end{aligned}$$

In the last equality we used the equality $W_2(\mu_0, \mu_1) = W_2(\tilde{\mu}_0, \tilde{\mu}_1)$, which follows by the fact that both geodesics have an initial segment in common, and therefore the

same speed. Since all inequalities in this chain have to be equalities, it follows that θ is supported on $\mathrm{Geo}(X)$, so that θ_x is supported on $\mathrm{Geo}(X)$ for μ_s-a.e. $x \in X$. Therefore the constant speed geodesic

$$\gamma(t) := \begin{cases} \gamma_s(t) & \text{if } t \in [0, s] \\ \tilde{\gamma}_s(t) & \text{if } t \in [s, 1] \end{cases}$$

is the unique extension of γ_s for ϱ_x-a.e. γ_s, for μ_s-a.e. x. It follows that, still for μ_s-a.e. x, $\tilde{\varrho}_x$ is uniquely determined by ϱ_x as in (10.6). By integration with respect to x we obtain that $\tilde{\varrho}$ is uniquely determined by ϱ as in (10.6).

We can now conclude the proof. Indeed, since $\tilde{\varrho}$ is uniquely determined by ϱ the same is for $\tilde{\eta}_{[s,1]}$, so that $\eta_{[s,1]} = \tilde{\eta}_{[s,1]}$. By taking the marginals we obtain that $\mu_t \equiv \tilde{\mu}_t$ in $[s, 1]$. Since both η and $\tilde{\eta}$ are concentrated on continuous curves, we obtain that $\eta = \tilde{\eta}$ by taking the limit as $s \downarrow 0$. This shows also the uniqueness of the lifting. □

In order to motivate the next definition, we start from the Hilbertian identity

$$|z - x_t|^2 = (1-t)|z - x_0|^2 + t|z - x_1|^2 - t(1-t)|x_1 - x_0|^2 \tag{10.7}$$

with $x_t := (1-t)x_0 + tx_1$. The identity can be reformulated by saying that $t \mapsto |z - x_t|^2 - t^2|x_1 - x_0|^2$ is an affine function in $[0, 1]$ and, written in this way, its proof is trivial, since

$$|z - x_t|^2 = |t(x_1 - x_0) - (z - x_0)|^2.$$

More generally, A.D. Alexandrov [3] noticed that one can ask concavity/convexity of $t \mapsto d^2(\gamma(t), z) - t^2 d^2(\gamma(0), \gamma(1))$ to encode lower/upper bounds on the sectional curvature on Riemannian manifolds (see also Chapter 11 in [99]). The advantage of the metric formulation is that it makes sense also on nonsmooth spaces, where the classical definition of sectional curvature cannot be given. The disadvantage is that this procedure provides only lower/upper bounds. We refer to [2, 28, 29, 70] as basic references for the theory of metric spaces with nonnegative and nonpositive sectional curvature.

Definition 10.15 (PC and NPC Spaces) A metric space (X, d) is said to be positively curved (PC in short) if

$$d^2(z, \gamma(t)) \geq (1-t)d^2(z, \gamma(0)) + td^2(z, \gamma(1)) - t(1-t)d^2(\gamma(0), \gamma(1)) \qquad \forall t \in [0, 1]$$

for all $\gamma \in \mathrm{Geo}(X)$ and $z \in X$.

Analogously, (X, d) is said to be negatively curved (NPC in short) if

$$d^2(z, \gamma(t)) \le (1-t)d^2(z, \gamma(0)) + td^2(z, \gamma(1)) - t(1-t)d^2(\gamma(0), \gamma(1)) \qquad \forall t \in [0, 1]$$

for all $\gamma \in \mathrm{Geo}(X)$ and $z \in X$.

Remark 10.16 Thanks to the Hilbertian identity (10.7), Hilbert spaces are of course both PC and NPC.

Theorem 10.17 *If (X, d) is PC, then $(\mathscr{P}_2(X), W_2)$ is PC as well.*

Proof Let $t \in [0, 1]$, $\mu_s \in \mathrm{Geo}(\mathscr{P}_2(X))$ and $\nu \in \mathscr{P}_2(X)$. Let $\eta \in \mathrm{OptGeo}(\mu_0, \mu_1)$ be a lifting of μ_s and let $\Sigma \in \Gamma_o(\mu_t, \nu)$. By a "Dudley-type" construction we are going to build $\Theta \in \mathscr{P}(\mathrm{Geo}(X) \times X)$ having η as first marginal and such that Σ is the image of Θ under the map $(\gamma, x) \mapsto (e_t(\gamma), x)$ (in particular the second marginal of Θ is ν).

Indeed, since μ_t is both the first marginal of Σ and the marginal of η under the evaluation map e_t, we can disintegrate $\Sigma = \Sigma_x \otimes \mu_t$ and $\eta = \eta_x \otimes \mu_t$ (with η_x concentrated on curves γ with $e_t(\gamma) = x$) to define, by superposition,

$$\Theta = \int_X (\eta_x \times \Sigma_x)\, \mathrm{d}\mu_t(x) \in \mathscr{P}(\mathrm{Geo}(X) \times X).$$

It is immediate to check that Θ has all the required properties.

Since $(e_0, x)_\# \Theta \in \Gamma(\mu_0, \nu)$, $(e_1, x)_\# \Theta \in \Gamma(\mu_1, \nu)$ and $(e_0, e_1)_\# \Theta = (e_0, e_1)_\# \eta$ is an optimal plan from μ_0 to μ_1 we have then

$$\begin{aligned} W_2^2(\mu_t, \nu) &= \int_{X\times X} d^2(x, y)\, \mathrm{d}\Sigma(x, y) = \int_{\mathrm{Geo}(X)\times X} d^2(e_t(\gamma), x)\, \mathrm{d}\Theta(\gamma, x) \\ &\ge \int_{\mathrm{Geo}(X)\times X} (1-t)d^2(e_0(\gamma), x) + td^2(e_1(\gamma), x) - t(1-t)d^2(e_0(\gamma), e_1(\gamma))\, \mathrm{d}\Theta(\gamma, x) \\ &\ge (1-t)W_2^2(\mu_0, \nu) + tW_2^2(\mu_1, \nu) - t(1-t)W_2^2(\mu_0, \mu_1). \end{aligned}$$

□

In Example 10.19 below, borrowed from [12], we show that the NPC property is not stable in the passage from X to $\mathscr{P}_2(X)$, even when $X = \mathbb{R}^2$. The case $X = \mathbb{R}$ is special, because of the following reason:

Theorem 10.18 (Hilbert Structure of $\mathscr{P}_2(\mathbb{R})$) *The map*

$$\mu \mapsto F_\mu^{-1}(t) := \sup\left\{x : F_\mu(x) \le t\right\},$$

where F_μ is the cumulative distribution function in (1.6), establishes an isometry between $\mathscr{P}_2(\mathbb{R})$ and the Hilbert space $L^2(0, 1)$.

Proof Let $\mu,\ \nu \in \mathscr{P}_2(\mathbb{R})$ and assume for simplicity that μ has no atom, and that F_ν is strictly increasing. Then Remark 1.12 gives that $F_\#\mu = \mathscr{L}^1 \llcorner (0,1)$ and $F_\mu^{-1} \circ F_\mu = \mathrm{id}$ μ-a.e., while Theorem 1.11 gives that $F_\nu^{-1} \circ F_\mu$ is the unique optimal transport map from μ to ν. It follows that

$$W_2^2(\mu,\nu) = \int_{\mathbb{R}} |F_\nu^{-1}\circ F_\mu - \mathrm{id}|^2 \,\mathrm{d}\mu = \int_{\mathbb{R}} |F_\nu^{-1}\circ F_\mu - F_\mu^{-1}\circ F_\mu|^2 \,\mathrm{d}\mu = \int_0^1 |F_\nu^{-1} - F_\mu^{-1}|^2 \,\mathrm{d}t.$$

□

Example 10.19 Let us set

$$\mu^1 := \frac{1}{2}\left(\delta_{(1,1)} + \delta_{(5,3)}\right), \qquad \mu^2 := \frac{1}{2}\left(\delta_{(-1,1)} + \delta_{(-5,3)}\right).$$

Then, coupling $(1,1)$ to $(-5,3)$ and $(5,3)$ to $(-1,1)$ it is immediate to check that $W_2^2(\mu^1,\mu^2) = 40$, hence the unique constant speed geodesic joining μ^1 to μ^2 is given by

$$\mu_t := \frac{1}{2}\left(\delta_{(1-6t,1+2t)} + \delta_{(5-6t,3-2t)}\right) \quad t \in [0,1]$$

and $\mu_{1/2} = \frac{1}{2}\left(\delta_{(-2,2)} + \delta_{(2,2)}\right)$. Now, setting $\mu^3 := \frac{1}{2}\left(\delta_{(0,0)} + \delta_{(0,-4)}\right)$, we easily compute $W_2^2(\mu^1,\mu^3) = 30$ and $W_2^2(\mu^2,\mu^3) = 30$. It turns out that

$$24 = W_2^2(\mu_{1/2},\mu^3) > \frac{30}{2} + \frac{30}{2} - \frac{40}{4},$$

so that the NPC property is violated.

Lecture 11: Gradient Flows: An Introduction



In this and in the next lectures we aim at a general introduction to the theory of gradient flows. We fix a Hilbert space H with scalar product $\langle \cdot, \cdot \rangle$ and associated norm $|\cdot|$, while $f : H \to (-\infty, \infty]$ will be a function whose finiteness domain $\{f < \infty\}$ will be denoted by $\mathrm{Dom}(f)$.

In the classical case when $H = \mathbb{R}^n$ and f is real-valued and everywhere differentiable, a gradient flow of f starting from $\bar{x} \in H$ is the solution of the ordinary differential equation

$$\begin{cases} x'(t) = b(x(t)) \quad t \geq 0 \\ x(0) = \bar{x}, \end{cases} \tag{11.1}$$

where the driving vector field b is $-\nabla f$. If $f \in C^{1,1}(H)$, existence and uniqueness of the solution follow from the Cauchy-Lipschitz theorem (this holds even in the infinite-dimensional case, understanding for instance ∇f as the Gateaux directional derivative).

On the other hand, the assumption $f \in C^{1,1}(H)$ is too strong for many infinite-dimensional applications to parabolic PDE's, viewed as infinite-dimensional ODE's. We will see, however, that convexity and lower semicontinuity of f are sufficient to provide a good existence and uniqueness theory. Properly understanding (11.1) with $b = -\nabla f$ when differentiability may fail, we will then study:

(i) the existence and uniqueness of solutions;
(ii) the rate of convergence as $t \to \infty$ to an equilibrium;

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(iii) the energy dissipation rate, namely the quantity $\frac{\mathrm{d}}{\mathrm{d}t} f(x(t))$. Whenever f is differentiable at $x(t)$ and $x'(t)$ exists, one has

$$\frac{\mathrm{d}}{\mathrm{d}t} f(x(t)) = \langle \nabla f(x(t)), x'(t) \rangle = -|\nabla f|^2(x(t))$$

and we will extend these energy dissipation identities also to the case of convex functions, often not differentiable in the classical sense.

1 λ-Convex Functions

Rather than settle for convexity, we will work in the broader context of λ-convexity.

Definition 11.1 (λ-Convexity) Given $\lambda \in \mathbb{R}$, we say that $f : H \to (-\infty, \infty]$ is λ-convex if $f - \frac{\lambda}{2}|\cdot|^2$ is convex.

It is easily seen that λ-convexity with $\lambda > 0$ is a stronger notion of convexity (actually a quantitative version of strict convexity), while λ-convexity with $\lambda < 0$ is a weaker notion of convexity.

If $H = \mathbb{R}^n$ and f is twice differentiable, f is λ-convex if and only if $\nabla^2 f \geq \lambda I$. The equivalence still persists if the differentiability assumption is dropped, understanding $\nabla^2 f \geq \lambda I$ in the sense of distributions, i.e.

$$\int_{\mathbb{R}^n} f \frac{\partial^2 \phi}{\partial^2 \xi} \, \mathrm{d}x \geq \lambda |\xi|^2 \int_{\mathbb{R}^n} \phi \, \mathrm{d}x$$

for any $\xi \in \mathbb{R}^n$ and $\phi \in C_c^\infty(\mathbb{R}^n)$ nonnegative.

Notice also that λ-convex functions f have the property that their negative part has at most quadratic growth in the case $\lambda < 0$ and at most linear growth in the case $\lambda = 0$ while, in the case $\lambda > 0$, f is uniformly bounded from below.

We can now extend to the case of λ-convex functions the notion of subdifferential introduced in Definition 3.7.

Definition 11.2 (λ-Subdifferential) The λ-subdifferential $\partial_\lambda f(x)$ of f at $x \in \mathrm{Dom}(f)$ is the set

$$\partial_\lambda f(x) := \left\{ p \in H : f(y) \geq f(x) + \langle p, y - x \rangle + \frac{\lambda}{2}|y - x|^2 \quad \forall y \in H \right\}.$$

We denote by $D(\partial_\lambda f) \subset \mathrm{Dom}(f)$ the set of all $x \in H$ such that $\partial_\lambda f(x) \neq \emptyset$.

Notice that $\partial_\lambda f(x)$ is a closed convex set and that $\partial_0 f(x) = \partial f(x)$. Since λ-convexity implies λ′-convexity for any $\lambda' < \lambda$, it is convenient to adopt a definition of subdifferential independent of the parameter λ that, when applied to λ-convex

functions, coincides with $\partial_\lambda f$. To this aim, recall that the monotonicity of difference quotients of convex functions gives the equivalence

$$p \in \partial f(x) \quad \Longleftrightarrow \quad \liminf_{t\to 0^+} \frac{f(x+tv)-f(x)}{t} \geq \langle p, v\rangle \quad \forall v \in H.$$

By applying this equivalence to $f - \lambda|\cdot|^2/2$ we obtain the following result.

Proposition 11.3 *For $x \in Dom(f)$, denoting*

$$\partial_G f(x) := \left\{ p \in H : \liminf_{t\to 0^+} \frac{f(x+tv)-f(x)}{t} \geq \langle v, p\rangle \quad \forall v \in H \right\}$$

the Gateaux subdifferential of f at x, one has $\partial_\lambda f(x) \subset \partial_G f(x)$, with equality if f is λ-convex.

2 Differentiability of Absolutely Continuous Curves

Before giving the definition of gradient flows, let us prove a basic differentiability property of absolutely continuous curves in Hilbert spaces.

Proposition 11.4 (Differentiability of Absolutely Continuous Curves) *Let I be an open interval of $\mathbb{R}$. Then any $x \in AC(I;H)$ is differentiable $\mathscr{L}^1$-a.e. in I, $|x'| \in L^1(I)$ and the fundamental theorem of calculus*

$$x(t) - x(s) = \int_s^t x'(r)\,\mathrm{d}r \qquad \forall s,\ t \in I$$

holds. In particular, the norm of $x'(t)$ coincides for $\mathscr{L}^1$-a.e. $t \in I$ with the metric derivative of the curve $x(t)$.

Proof Let $m(t)$ be the metric derivative of the curve $x(t)$, whose existence is ensured by Theorem 9.2, and let Y be the vector space spanned by the vectors $x(s)$, $s \in I$. Thanks to the continuity of x, Y is separable and we denote by $(e_n)_{n\geq 1}$ an orthonormal basis of $\overline{Y}$ and by $x_n = \langle x, e_n\rangle$ the components of x in this basis. The absolute continuity of x yields the absolute continuity of x_n and then the existence of the derivative x_n' on a subset A_n with full measure in I. Now, if we consider the difference quotients

$$\frac{x(t+h)-x(t)}{h} \tag{11.2}$$

at a Lebesgue point t of m, we see that they are bounded as $h \to 0$, thanks to the inequalities (for $h > 0$)

$$\frac{|x(t+h) - x(t)|}{h} \le \frac{1}{h} \int_t^{t+h} m(r)\, \mathrm{d}r, \qquad \frac{|x(t-h) - x(t)|}{h} \le \frac{1}{h} \int_{t-h}^{t} m(r)\, \mathrm{d}r. \tag{11.3}$$

In addition, if we assume also that $t \in \cap_n A_n$, we have also

$$\langle \frac{x(t+h) - x(t)}{h}, e_n \rangle = \frac{x_n(t+h) - x_n(t)}{h} \to x_n'(t) \qquad \forall n \ge 1.$$

From these remarks it follows that at any such point t the difference quotients have a unique limit point in $\overline{Y}$, with respect to the weak topology (since the scalar products of any limit point with e_n are given by $x_n'(t)$), so that

$$\frac{x(t+h) - x(t)}{h} \quad \xrightarrow{h \to 0} \quad x'(t) \quad \text{with } x'(t) := \sum_{n \ge 1} x_n'(t) e_n. \tag{11.4}$$

By orthogonal decomposition, since the difference quotients belong to Y, the weak convergence occurs also in H.

Passing to the limit as $h \to 0$ in (11.3), the lower semicontinuity of the norm with respect to the weak convergence gives $|x'(t)| \le m(t)$, so that $|x'| \in L^1(I)$. Now we improve the convergence from weak to strong using the following elementary criterion for strong convergence in Hilbert spaces (and, more generally, in uniformly convex Banach spaces): whenever w_h weakly converge to w in H and $\limsup_h |w_h| \le |w|$, one has $|w_h - w|^2 \to 0$ (its proof simply comes by expanding the squares). We use this criterion with w_h as in (11.2), $w = x'(t)$, assuming also that t is a Lebesgue point of $|x'|$ and considering for simplicity only the case $h > 0$.

By applying the fundamental theorem of calculus componentwise, we get

$$\langle w_h, v \rangle = \frac{1}{h} \sum_{i=1}^{n} a_i \int_t^{t+h} x_i'(r)\, \mathrm{d}r = \frac{1}{h} \int_t^{t+h} \langle x'(r), v \rangle\, \mathrm{d}r \tag{11.5}$$

for any vector v of the form $\sum_{i=1}^n a_i e_i$, so that by the density of this class of vectors we get $|w_h| \le \frac{1}{h} \int_t^{t+h} |x'(r)|\, \mathrm{d}r$. The conclusion comes from the fact that t is a Lebesgue point of $|x'|$.

Finally, the validity of the fundamental theorem of calculus follows again by (11.5), by a density argument. □

3 Gradient Flows

We can now give our basic definition of gradient flow.

Definition 11.5 (Gradient Flow) We say that $x : (0, \infty) \to \mathrm{Dom}(f)$ is a gradient flow of f if $x \in AC_{\mathrm{loc}}((0, \infty); H)$ and

$$x'(t) \in -\partial_G f(x(t)) \text{ for } \mathscr{L}^1\text{-a.e. } t \in (0, \infty).$$

We say that x starts from $\bar{x} \in H$ if $\lim_{t \to 0} x(t) = \bar{x}$.

Note that a necessary condition for the existence of a gradient flow starting from $\bar{x}$ is $\bar{x} \in \overline{\mathrm{Dom}(f)}$. A basic example of gradient flow (see Lecture 13 for more examples) is the heat equation, when treated as an infinite-dimensional ODE and from the functional-analytic point of view.

Example 11.6 (The Heat Equation in $\mathbb{R}^n$ and the Heat Flow) We define $H = L^2(\mathbb{R}^n)$ and $\mathbf{D} : H \to [0, \infty]$ by

$$\mathbf{D}(u) := \begin{cases} \frac{1}{2} \int_{\mathbb{R}^n} |\nabla u|^2 \, \mathrm{d}x & \text{if } u \in H^1(\mathbb{R}^n); \\ \\ +\infty & \text{otherwise.} \end{cases}$$

Then, it is immediately seen that $\mathbf{D}$ is convex, lower semicontinuous, with a dense domain. Therefore the theory of gradient flows (see in particular Theorem 11.7 below) applies and provides a contraction semigroup defined on the whole space H. It corresponds precisely to the heat flow

$$\frac{\mathrm{d}}{\mathrm{d}t} u = \Delta u, \tag{11.6}$$

where the left hand side is understood as the derivative of the H-valued map $t \mapsto u(t, \cdot)$, while the right hand side is understood as the distributional Laplacian of $u(t, \cdot)$, for $\mathscr{L}^1$-almost every t. In order to justify this statement, it is sufficient to prove the equivalence

$$\partial \mathbf{D}(u) \neq \emptyset \qquad \Longleftrightarrow \qquad \Delta u \in L^2(\mathbb{R}^n)$$

with the additional property $\partial \mathbf{D}(u) = \{-\Delta u\}$ whenever the subdifferential is not empty. We prove the implication $\Rightarrow$, the proof of the converse implication being similar.

Let $u \in H^1(\mathbb{R}^n)$, $\xi \in \partial \mathbf{D}(u)$, $v \in C_c^\infty(\mathbb{R}^n)$ and $\epsilon \in \mathbb{R} \setminus \{0\}$. From the inequality

$$\mathbf{D}(u + \epsilon v) \geq \mathbf{D}(u) + \epsilon \int_{\mathbb{R}^n} \xi v \, \mathrm{d}x$$

we get

$$O(\epsilon^2) + \epsilon \int_{\mathbb{R}^n} \langle \nabla u, \nabla v \rangle \, \mathrm{d}x \geq \epsilon \int_{\mathbb{R}^n} \xi v \, \mathrm{d}x.$$

Therefore, dividing by ϵ and taking into account the fact that ϵ can be positive and negative we get

$$\int_{\mathbb{R}^n} \langle \nabla u, \nabla v \rangle \, \mathrm{d}x = \int_{\mathbb{R}^n} \xi v \, \mathrm{d}x \qquad \forall v \in C_c^\infty(\mathbb{R}^n),$$

which means that $\Delta u = -\xi \in L^2(\mathbb{R}^n)$ in the sense of distributions.

Let us discuss now the existence of gradient flows. If A is a closed, densely defined linear operator acting on a Banach space X and satisfying the resolvent condition

$$\|(\lambda I - A)^{-n}\| \leq \frac{M}{(\lambda - \omega)^n} \qquad \forall \lambda > \omega, \ \forall n \geq 1$$

for some $\omega \in \mathbb{R}$, $M \geq 0$, then the Hille-Yosida theorem provides the existence of a continuous semigroup $(T_t)_{t>0}$ of linear operators in X, satisfying

$$\begin{cases} \dfrac{\mathrm{d}}{\mathrm{d}t} T_t u = A(T_t u) \\ \\ T_0 u = u. \end{cases} \tag{11.7}$$

Considering (11.1), the Hille-Yosida theorem applies only if f is a quadratic function (as in Example 11.6), with $D(A) = \{x : \partial f(x) \neq \emptyset\}$ and $\partial f(x) = \{Ax\}$ (and $M = 1$, $\omega = 0$). As we are working with convex functions, we need a more general and nonlinear statement.

Theorem 11.7 (Brézis-Komura) *Assume that f is λ-convex for some $\lambda \in \mathbb{R}$ and lower semicontinuous. For every $\bar{x} \in \overline{Dom(f)}$, there exists a unique gradient flow $x(t) = S_t \bar{x}$ starting from $\bar{x}$. The family of operators $S_t : \overline{Dom(f)} \to Dom(f)$, $t > 0$, satisfies the semigroup property $S_{t+s} = S_t \circ S_s$ and the contractivity property*

$$|S_t \bar{x} - S_t \bar{y}| \leq e^{-\lambda t} |\bar{x} - \bar{y}| \qquad \forall x, \ y \in \overline{Dom(f)}. \tag{11.8}$$

Proof We just prove the contractivity part, and as a consequence also the uniqueness, assuming for simplicity $\lambda = 0$. The construction of a solution is postponed to the next lectures. Assume that x and $\tilde{x}$ are gradient flows starting respectively from $\bar{x}$ and $\tilde{\bar{x}}$. Using the monotonicity inequality

$$\langle p - q, y - z \rangle \geq 0$$

with $y = x(t)$, $z = \tilde{x}(t)$, $p = -x'(t)$, $q = -\tilde{x}'(t)$, at any differentiability point t of x and $\tilde{x}$ we get

$$\frac{d}{dt}|x(t) - \tilde{x}(t)|^2 = 2\langle x'(t) - \tilde{x}'(t), x(t) - \tilde{x}(t)\rangle \leq 0.$$

It follows that $|x(t) - \tilde{x}(t)| \leq |x(s) - \tilde{x}(s)|$ for $0 < s \leq t$, and taking the limit as $s \to 0$ we obtain (11.8). □

In view of Definition 11.5, it is also convenient to set $S_0 x = x$ for all $x \in \overline{\mathrm{Dom}(f)}$.

Definition 11.8 (Gradient, or Minimal Selection) Whenever $\partial_G f(x)$ is a nonempty closed and convex set, we denote by $\nabla f(x)$ the element with minimal norm in $\partial_G f(x)$.

Let us give a list of properties of gradient flows. Remarkably, according to property (iii), the differential inclusion of Definition 11.5 yields the property that $-x'(t)$ is for $\mathscr{L}^1$-almost every t the minimal selection in $\partial_G f(x)$.

Proposition 11.9 *Let $S_t\bar{x} = x(t)$ be the gradient flow of the λ-convex and lower semicontinuous function f given by Theorem 11.7 above. The following properties hold:*

(i) the metric derivative from the right

$$|x'_+(t)| = \lim_{h \to 0^+} \frac{|x(t+h) - x(t)|}{h}$$

exists for any $t > 0$ and $t \mapsto e^{\lambda t}|x'_+(t)|$ is nonincreasing on $(0, \infty)$;

(ii) the function $t \mapsto f(x(t))$ is nonincreasing, locally Lipschitz on $(0, \infty)$ and

$$\frac{\mathrm{d}}{\mathrm{d}t} f(x(t)) = \langle \nabla f(x(t)), x'(t)\rangle = -|\nabla f|^2(x(t)) \qquad \text{for } \mathscr{L}^1\text{-a.e. } t \in (0, \infty); \tag{11.9}$$

(iii) $x'(t) = -\nabla f(x(t))$ for $\mathscr{L}^1$-a.e. $t \in (0, \infty)$;

(iv) if $\lambda = 0$, for all $t > 0$ one has

$$f(x(t)) \leq \inf_{v \in \mathrm{Dom}(f)} \left(f(v) + \frac{1}{2t}|\bar{x} - v|^2 \right) \tag{11.10}$$

and, denoting by $D(\partial f)$ the set of points x where $\partial f(x)$ is not empty, one has

$$|\nabla f|^2(x(t)) \leq \inf_{v \in D(\partial f)} \left(|\nabla f|^2(v) + \frac{1}{t^2}|\bar{x} - v|^2 \right); \tag{11.11}$$

(v) in the general λ-convex case, for all $t > 0$ one has

$$f(x(t)) \le \inf_{v \in \mathrm{Dom}(f)} \left(f(v) + \frac{\lambda}{2(e^{\lambda t} - 1)} |\bar{x} - v|^2 \right)$$

and

$$|\nabla f|^2(x(t)) \le \inf_{v \in D(\partial_\lambda f)} \left(|\nabla f|^2(v) + \frac{1}{t^2} |\bar{x} - v|^2 - \frac{\lambda}{2t} |x(t) - v|^2 - \frac{\lambda}{t^2} \int_0^t |x(s) - v|^2 ds \right).$$

Remark 11.10 (Refinements) A more refined analysis, that we will not pursue here (see [25]) yields that the right derivative exists and provides the minimal selection at any $t \in (0, \infty)$. In addition, also the energy dissipation identity (11.9) can be written in a pointwise form using the right derivative.

We prove these properties using the so-called EVI formulation of gradient flows and a "metric" characterization of $|\nabla f|$.

Definition 11.11 (EVI Gradient Flow) A curve $x \in AC_{\mathrm{loc}}((0, \infty); H)$ is called an EVI_λ solution if for any $y \in \mathrm{Dom}(f)$ one has

$$\frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{1}{2} |x(t) - y|^2 \right) + \frac{\lambda}{2} |x(t) - y|^2 + f(x(t)) \le f(y) \quad \text{for } \mathscr{L}^1\text{-a.e. } t \in (0, \infty). \tag{11.12}$$

We say that $x(t)$ starts from $\bar{x}$ if $\lim_{t \to 0^+} x(t) = \bar{x}$.

The acronym EVI stands for Evolution Variational Inequality. The advantage of this formulation is that, since involves only the energy f and the distance $d(x, y) = |x - y|$, it can in principle be applied also to evolution problems in general metric spaces (see [12, 45] for much more on this subject). Notice that the exceptional set of times in (11.12) a priori depends on y (see Lemma 11.13 below for a pointwise version of EVI), and that (11.12) also encodes the fact that the curve $x(t)$ takes its values in $\mathrm{Dom}(f)$, whenever $\mathrm{Dom}(f) \neq \emptyset$.

The next two lemmas will be useful to prove the implication from EVI to gradient flow, the first one in the case of separable Hilbert spaces, the second one in the general case.

Lemma 11.12 (Density in Energy) *For any function $f : H \to (-\infty, \infty]$ there exists a countable set $D \subset H$ dense in energy, meaning that for any $y \in Dom(f)$ there exists a sequence $(y_n) \subset D$ such that $y_n \to y$ and $f(y_n) \to f(y)$.*

Proof The result follows immediately from the separability of the graph of f on $\mathrm{Dom}(f)$, viewed as a subset of the separable space $H \times \mathbb{R}$ (endowed with the product Hilbertian distance), however we prefer to give a direct proof. For any integer $n \geq 1$, split $\mathrm{Dom}(f)$ into the disjoint "strips"

$$\mathrm{Dom}(f) := \bigcup_{p \in \mathbb{Z}} S_{n,p} \qquad \text{with} \qquad S_{n,p} = \{x \in \mathrm{Dom}(f) : 2^{-n} p \leq f(x) < 2^{-n}(p+1)\}.$$

Choose a countable dense subsets $D_{n,p}$ of $S_{n,p}$ and define the countable set $D := \cup_{n,p} D_{n,p}$.

For any $y \in \mathrm{Dom}(f)$ and n fixed, there exists a unique $p \in \mathbb{Z}$ such that $y \in S_{n,p}$ and, by the density of $D_{n,p}$, $y_n \in D_{n,p} \subset D$ with $|y - y_n| \leq \frac{1}{n}$. Then, as both y and y_n are in the same strip $S_{n,p}$, one has $|f(y) - f(y_n)| \leq 2^{-n}$, so that (y_n) is the desired sequence. □

Lemma 11.13 (EVI **Contractivity**) *Let $f : H \to (-\infty, \infty]$ be lower semicontinuous and such that $-\min\{f; 0\}$ is bounded on bounded sets. For any $\lambda \in \mathbb{R}$ and $\bar{x} \in \overline{Dom(f)}$ there exists at most one* EVI_λ *solution $x(t)$ starting from $\bar{x}$. Moreover, the* EVI_λ *solutions satisfy the same contractivity property of* (11.8) *in Theorem 11.7.*

Proof Again, we consider only the case $\lambda = 0$ and we assume for simplicity that f is bounded from below. Given an EVI_0 solution $x(t)$ and $y \in \mathrm{Dom}(f)$, by integration between t and $t + h$ one has

$$\frac{1}{2h}\big(|x(t+h) - y|^2 - |x(t) - y|^2\big) + \frac{1}{h}\int_t^{t+h} f(x(r))\,\mathrm{d}r \leq f(y) \qquad \forall h > 0.$$

By the lower semicontinuity of f and Fatou's lemma (here we use that -min$\{f; 0\}$ is bounded on bounded sets), we obtain the pointwise differential inequality

$$\frac{1}{2}\frac{\mathrm{d}^+}{\mathrm{d}t}|x(t) - y|^2 \leq f(y) - f(x(t)) \qquad \forall t \in (0, \infty), \tag{11.13}$$

where $\frac{\mathrm{d}^+}{\mathrm{d}t}$ is the lim sup of the difference quotients from the right. Now, given another EVI_0 solution $\tilde{x}(t)$, apply the fine Leibniz rule

$$\begin{aligned}\lim_{h \to 0^+} \frac{\delta(t+h, t+h) - \delta(t,t)}{h} &\leq \limsup_{h \to 0^+} \frac{\delta(t+h, t) - \delta(t,t)}{h} \\ &\quad + \limsup_{h \to 0^+} \frac{\delta(t, t+h) - \delta(t,t)}{h},\end{aligned} \tag{11.14}$$

valid for $\mathscr{L}^1$-a.e. $t \in (0, \infty)$, to $\delta(s,t) := \frac{1}{2}|x(s) - \tilde{x}(t)|^2$ to obtain that

$$\delta(t,t)' \leq f(x(t)) - f(y(t)) + f(y(t)) - f(x(t)) = 0 \qquad \text{for } \mathscr{L}^1\text{-a.e. } t \in (0, \infty).$$

It follows that $|x(t) - \tilde{x}(t)|$ is nonincreasing. The proof of (11.14) can be obtained immediately from the Hilbertian identity

$$\begin{aligned}|x(t+h) - \tilde{x}(t+h)|^2 - |x(t) - \tilde{x}(t)|^2 = &\big[|x(t+h) - \tilde{x}(t)|^2 - |x(t) - \tilde{x}(t)|^2\big] \\ &+ \big[|\tilde{x}(t+h) - x(t)|^2 - |\tilde{x}(t) - x(t)|^2\big] \\ &- 2\langle x(t+h) - x(t), \tilde{x}(t+h) - \tilde{x}(t)\rangle.\end{aligned}$$

See also Lemma 4.3.4 of [12] for the proof of contractivity of EVI in the general metric setting, that provides a variant of (11.14) without appealing to the Hilbertian identity and using also the inequality $\frac{\mathrm{d}^-}{\mathrm{d}t} d^2(x(t), y) \le 2f(y) - 2f(x(t))$, where $\frac{\mathrm{d}^-}{\mathrm{d}t}$ is the lim sup of the difference quotients from the left. □

Remark 11.14 (Kruzkhov Doubling of Variables) The scheme of putting $y = \tilde{x}(t)$ in the differential inequality $\frac{\mathrm{d}^+}{\mathrm{d}} |x(t) - y|^2 \le 2f(y) - 2f(x(t))$ and $y = x(t)$ in the differential inequality $\frac{\mathrm{d}^+}{\mathrm{d}} |\tilde{x}(t) - y|^2 \le 2f(y) - 2f(\tilde{x}(t))$, adding the resulting inequalities to get from the Leibniz rule the monotonicity of $|x(t) - \tilde{x}(t)|$, is inspired by Kruzkhov's uniqueness theorem of entropy solutions to scalar conservation laws, see [57, 78] for more details. In this setting, as in Lemma 4.3.4 of [12], the Leibniz rule can be proved understanding the derivative in the sense of distributions.

Theorem 11.15 (Equivalence of EVI_λ and Gradient Flow) *For a λ-convex and lower semicontinuous function f with $\mathrm{Dom}(f) \neq \emptyset$, a locally absolutely continuous curve x is a gradient flow if and only if it is an EVI_λ solution.*

Proof If x is a gradient flow, it is easy to show that x satisfies (11.12) by noticing that

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}|x(t) - y|^2\right) = \langle x'(t), x(t) - y\rangle = \langle -x'(t), y - x(t)\rangle \tag{11.15}$$

and that $-x'(t) \in \partial_G f(x(t)) = \partial_\lambda f(x(t))$ implies

$$\langle -x'(t), y - x(t)\rangle \le -\frac{\lambda}{2}|x(t) - y|^2 - f(x(t)) + f(y) \qquad \forall y \in H.$$

We prove the converse implication under the additional assumption that H is separable. Assume that x is an EVI_λ solution and let D be the countable subset provided by Lemma 11.12. Thanks to (11.15), one has

$$\langle -x'(t), y - x(t)\rangle + \frac{\lambda}{2}|x(t) - y|^2 + f(x(t)) \le f(y) \qquad \forall y \in D \tag{11.16}$$

for a set L of times t with $\mathscr{L}^1((0,\infty)\setminus L)=0$. The density in energy of D then provides the property

$$\langle -x'(t),\, y-x(t)\rangle+\frac{\lambda}{2}|x(t)-y|^2+f(x(t))\le f(y)\qquad \forall y\in \mathrm{Dom}(f), \tag{11.17}$$

so that $-x'(t)\in \partial_\lambda f(x(t))$ for all $t\in(0,\infty)\setminus L$.

Let us present a different argument for the non-separable case under the additional assumption $\lambda=0$. Arguing as in the first part of Lemma 11.13 we deduce the pointwise formulation (11.13) of EVI_0. Next we observe that the differential inequality (11.17) holds at any differentiability point of the curve $x(t)$. This implies the sought conclusion. □

Let us come now to the metric characterization of $|\nabla f(x)|$, where we recall that $\nabla f(x)$ was defined as the element with minimal norm of $\partial_G f(x)$. For λ-convex functions, it is based on the notion of *descending slope* $|\nabla^- f|$.

Definition 11.16 (Descending Slope) For $f: H\to(-\infty,\infty]$ and $x\in \mathrm{Dom}(f)$, the descending slope of f at x, denoted by $|\nabla^- f|(x)$, is defined by

$$|\nabla^- f|(x)=\limsup_{y\to x}\frac{(f(y)-f(x))^-}{|y-x|}.$$

If f is convex, the monotonicity of difference quotients gives

$$\frac{f(x)-f(y)}{|x-y|}\le\frac{f(x)-f(x_t)}{|x-x_t|}\quad\text{with}\quad x_t:=x+t(y-x),\ t\in(0,1]$$

for all $y\neq x$, and then

$$|\nabla^- f|(x)=\sup_{y\neq x}\frac{(f(y)-f(x))^-}{|y-x|}. \tag{11.18}$$

In the general λ-convex case, a similar argument gives

$$|\nabla^- f|(x)=\sup_{y\neq x}\left[\frac{f(y)-f(x)}{|y-x|}-\frac{\lambda}{2}|x-y|\right]^-. \tag{11.19}$$

Theorem 11.17 *For any λ-convex function f and any $x\in \mathrm{Dom}(f)$ one has that $\partial_G f(x)\neq\emptyset$ if and only if $|\nabla^- f|(x)<\infty$ and, in this case, it holds*

$$|\nabla f(x)|=|\nabla^- f|(x),$$

where $\nabla f(x)$ is the element with minimal norm in $\partial_G f(x)$.

In addition, assume that a continuous curve $\gamma : [0, 1] \to H$ *is differentiable at* 0 *with* $\gamma(0) = x$, $|\gamma'_+(0)| = |\nabla^- f|(x)$, *and that* $-(f \circ \gamma)'_+(0) = |\nabla^- f|^2(x)$. *Then* $-\gamma'_+(0) \in \partial_G f(x)$ *and therefore* $-\gamma'_+(0) = \nabla f(x)$.

Proof For simplicity we give the proof only when $\lambda = 0$ (in the general case, the quadratic perturbations proportional to λ do not affect the difference quotients at x). If $p \in \partial f(x)$ one has $f(x) - f(y) \le \langle p, x - y\rangle \le |p||x - y|$, so that $|\nabla^- f|(x) \le |p|$. By minimizing with respect to p we obtain the inequality $|\nabla^- f|(x) \le |\nabla f(x)|$.

Conversely, we need to prove that $\partial f(x) \neq \emptyset$ and that $|\nabla^- f|(x) \ge |\nabla f(x)|$ whenever $L = |\nabla^- f|(x)$ is finite. Without any loss of generality, we can assume $x = 0$ and $f(0) = 0$, so that (11.18) gives $L = \sup_{y\neq 0} f^-(y)/|y|$. Geometrically, the inequality $f(y) \ge -L|y|$ states that the epigraph of f, $\mathrm{epi}(f)$, is above the cone $C := \{(y, -L|y|) : y \in H\} \subset H \times \mathbb{R}$. In particular, the epigraph of f (which is convex) and the open and convex set $O := \{(y, t) : y \in H,\ t < -L|y|\} \subset H \times \mathbb{R}$ are separated. By the Hahn-Banach theorem (see Theorem 1.6 in [26]), there exists a hyperplane separating $\mathrm{epi}(f)$ and O, i.e. $p \in H$ such that

$$f(y) \ge \langle p, y\rangle \ge t \qquad \forall y \in H \ \text{ and } \ \forall t < -L|y|.$$

Hence

$$f(y) \ge \langle p, y\rangle \ge -L|y| \qquad \forall y \in H.$$

The first inequality states that $p \in \partial f(0)$ and the second one implies that $|p| \le L$, whence $|\nabla f(0)| \le |p| \le |\nabla^- f|(0)$.

In the proof of the second statement we can assume with no loss of generality that $x = 0$, $f(x) = 0$ and that $|\nabla^- f|(x) = 1$. Set $\xi = \gamma'_+(0)$ and assume by contradiction that $f(y) < -\langle y, \xi\rangle$ for some $y \in H$. Let α sufficiently close to 1 such that $f(y) \le -\alpha\langle y, \xi\rangle$, $(\alpha - 1)\langle y, \xi\rangle > 0$ and set

$$z_\tau(t) := \frac{1}{2}(\gamma(t) + t\tau y) \qquad \tau > 0.$$

We will prove that by computing the slope along the curve $z_\tau(t)$, for τ small enough we have a contradiction. Indeed, convexity gives

$$f(z_\tau(t)) \le \frac{1}{2}(f(\gamma(t)) + f(t\tau y)) \le \frac{1}{2}\big(-t - t\tau\alpha\langle y, \xi\rangle\big) + o(t),$$

while $|\nabla^- f|(0) = 1$ and $|\gamma'_+(0)| = 1$ give

$$f(z_\tau(t)) \ge -|z_\tau(t)| = -\frac{t}{2}\sqrt{1 + \tau^2|y|^2 + 2\tau\langle y, \xi\rangle} + o(t).$$

It follows that

$$1 + \tau\alpha\langle y, \xi\rangle \le \sqrt{1 + \tau^2|y|^2 + 2\tau\langle y, \xi\rangle}$$

which, by expansion with respect to τ, contradicts our choice of α. □

We are now in a position to prove the properties of gradient flows collected in Proposition 11.9. Notice that all proofs use the EVI formulation, rather than the standard gradient flow formulation.

Proof For simplicity, we assume $\lambda = 0$ and $f \ge 0$ (in the general case the proofs are very similar, only slightly longer, see [12]).

Metric Derivative from the Right Uniqueness gives the identity $S_s x(t) = x(s + t)$ for all $x \in \mathrm{Dom}(f)$ and all $s,\ t \ge 0$. Hence, contractivity yields

$$|x(s + t + h) - x(s + t)| = |S_s(x(t + h)) - S_s(x(t))| \le |x(t + h) - x(t)|$$

and this immediately implies the monotonicity of the metric differential quotients and the existence of the metric derivative from the right. By taking limits as $h \to 0$ we get that $|x'_+|$ is nonincreasing in $(0, \infty)$.

Local Lipschitz Properties By monotonicity and local integrability we obtain that $|x'| \in L^\infty(\epsilon, \infty)$ for all $\epsilon > 0$, whence the local Lipschitz property follows. In addition, denoting by L_ϵ the L^∞ norm of $|x'|$ in (ϵ, ∞), from EVI_0 we get

$$f(x(t)) - f(y) \le -\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}|x(t) - y|^2\right) \le |x'(t)||x(t) - y| \le L_\epsilon|x(t) - y| \qquad \forall y \in \mathrm{Dom}(f)$$

for $\mathscr{L}^1$-a.e. $t \in (\epsilon, \infty)$. This inequality actually holds for any $t \in [\epsilon, \infty)$, by the lower semicontinuity of f. Then, if we choose $y = x(s)$ for some $s \in [\epsilon, \infty)$, we get

$$f(x(t)) - f(x(s)) \le L_\epsilon|x(t) - x(s)| \le L_\epsilon^2|s - t|.$$

Exchanging the roles of s and t, this proves that also $f \circ x$ is locally Lipschitz on $(0, \infty)$.

Energy Dissipation Let us prove that

$$(f \circ x)'(t) \le -|x'(t)|^2 \qquad \text{for } \mathscr{L}^1\text{-a.e. } t \in (0, \infty). \tag{11.20}$$

Take $t > 0$ where both $f \circ x$ and x are differentiable, $h > 0$ and integrate EVI_0 between t and $t + h$:

$$\int_t^{t+h} (f(x(s)) - f(y))\,\mathrm{d}s \le \frac{1}{2}|x(t) - y|^2 - \frac{1}{2}|x(t+h) - y|^2 \qquad \forall y \in H.$$

Take now $y = x(t)$. As

$$\begin{aligned}\int_t^{t+h} (f(x(s)) - f(x(t)))\,\mathrm{d}s &= \int_t^{t+h} \left[(s-t)\frac{f(x(s)) - f(x(t))}{s-t}\right]\mathrm{d}s \\ &\underset{h\to 0}{=} \int_t^{t+h} \left[(s-t)\left((f\circ x)'(t) + \circ(1)\right)\right]\mathrm{d}s \\ &\underset{h\to 0}{=} \frac{h^2}{2}(f\circ x)'(t) + o(h^2)\end{aligned}$$

and

$$-\frac{1}{2}|x(t+h) - x(t)|^2 = -\frac{1}{2}h^2|x'(t)^2| + o(h^2)$$

we get $\frac{h^2}{2}(f\circ x)'(t) + o(h^2) \le -\frac{1}{2}h^2|x'(t)|^2 + o(h^2)$, whence (11.20) follows by dividing both sides by h^2 as $h \to 0$.

Note also that, as we already proved, $f(x(t)) - f(y) \le |x'(t)||x(t) - y|$ for any $y \in H$, hence

$$|\nabla^- f|(x(t)) \le |x'(t)| \qquad \text{for } \mathscr{L}^1\text{-a.e. } t \in (0,\infty). \tag{11.21}$$

Moreover, one has

$$-(f\circ x)'(t) \le |\nabla^- f|(x(t))|x'(t)| \qquad \text{for } \mathscr{L}^1\text{-a.e. } t \in (0,\infty) \tag{11.22}$$

thanks to the general one-sided chain rule

$$\frac{g(x(t)) - g(x(t+h))}{h} = \frac{g(x(t)) - g(x(t+h))}{|x(t) - x(t+h)|}\,\frac{|x(t+h) - x(t)|}{h} \qquad \forall h > 0,$$

which gives

$$\limsup_{h\to 0^+} \frac{g(x(t)) - g(x(t+h))}{h} \le |\nabla^- g|(x(t))|x'(t)|.$$

Then (11.20)–(11.22) imply

$$|x'(t)|^2 \le -(f\circ x)'(t) \le |\nabla^- f|(x(t))|x'(t)| \le |x'(t)|^2 \quad \text{for } \mathscr{L}^1\text{-a.e. } t \in (0,\infty),$$

so that all inequalities are actually equalities. Then $|x'(t)| = |\nabla^- f|(x(t))$ and from Theorem 11.17 it follows that $-x'(t) = \nabla f(x(t))$.

Regularization Estimates (11.10), (11.11) Take any $v \in \mathrm{Dom}(f)$. For $s < t$, the EVI_0 property and the fact that $f \circ x$ is decreasing imply that

$$\frac{\mathrm{d}}{\mathrm{d}s}\left(\frac{1}{2}|x(s)-v|^2\right) \le f(v) - f(x(t)).$$

An integration with respect to s between 0 and t gives

$$\frac{1}{2}|x(t)-v|^2 - \frac{1}{2}|\bar{x}-v|^2 \le t(f(v)-f(x(t))).$$

Dividing both sides by t and discarding the term $\frac{1}{2}|x(t)-v|^2$, (11.10) follows.

Since we assumed the starting point $\bar{x}$ to belong to the closure of $\mathrm{Dom}(f)$, the energy f at this point may be infinite. An important consequence of (11.10) is the control of the behavior of $f \circ x$ in a neighborhood of the singularity:

$$\lim_{t\to 0^+} tf(x(t)) = 0. \tag{11.23}$$

Indeed, (11.10) implies that for any $v \in \mathrm{Dom}(f)$ and $t > 0$,

$$tf(x(t)) \le tf(v) + \frac{1}{2}|v-\bar{x}|^2.$$

Taking the upper limit as t goes to 0 and then letting $v \to \bar{x}$ we get the claimed control. Let us use this fact to prove the energy dissipation rate estimate (11.11). By the monotonicity of $|\nabla f(x(t))| = |x'(t)|$ we get

$$\begin{aligned}
\frac{t^2}{2}|\nabla f(x(t))|^2 &= \int_0^t s|\nabla f(x(t))|^2\,\mathrm{d}s \le \int_0^t s|\nabla f(x(s))|^2\,\mathrm{d}s \\
&= -\int_0^t s(f\circ x)'(s)\,\mathrm{d}s = -\lim_{\epsilon\to 0}\int_\epsilon^t s(f\circ x)'(s)\,\mathrm{d}s \\
&= \lim_{\epsilon\to 0}\left(\int_\epsilon^t f(x(s))\,\mathrm{d}s - tf(x(t)) + \epsilon f(x(\epsilon))\right) \\
&= -tf(x(t)) + \lim_{\epsilon\to 0}\int_\epsilon^t f(x(s))\,\mathrm{d}s,
\end{aligned}$$

using (11.23). Note that we had to introduce a small $\epsilon > 0$ to perform the integration by parts, because of a potential non-integrability of $f \circ x$ in a neighborhood of 0. To go on, we use EVI_0. Take any $v \in \mathrm{Dom}(f)$. Then

$$\begin{aligned}\int_\epsilon^t f(x(s))\,\mathrm{d}s &= \int_\epsilon^t (f(x(s)) - f(v))\,\mathrm{d}s + (t-\epsilon) f(v) \\ &\leq -\frac{1}{2}\int_\epsilon^t \frac{\mathrm{d}}{\mathrm{d}s}|x(s) - v|^2\,\mathrm{d}s + (t-\epsilon) f(v) \\ &\leq -\frac{1}{2}|x(t) - v|^2 + \frac{1}{2}|x(\epsilon) - v|^2 + (t-\epsilon) f(v)\end{aligned}$$

As $\epsilon \to 0$, one gets

$$\frac{t^2}{2}|\nabla f(x(t))|^2 \leq t(f(v) - f(x(t))) - \frac{1}{2}|x(t) - v|^2 + \frac{1}{2}|\bar{x} - v|^2. \tag{11.24}$$

Using first the definition of descending slope and then Young's inequality we get

$$t(f(v) - f(x(t))) \leq t|\nabla^- f|(v)|x(t) - v| \leq \frac{t^2}{2}|\nabla f(v)|^2 + \frac{1}{2}|x(t) - v|^2,$$

that leads, in combination with (11.24), to the result.

□

Lecture 12: Gradient Flows: The Brézis-Komura Theorem



In this lecture, we will prove the Brézis-Komura Theorem 11.7. We start with a brief presentation of the classical proof, in the more general context of the theory of maximal monotone operators, based on the so-called Yosida regularization. Then, we describe in detail another strategy to prove the result, based on the so-called implicit Euler scheme.

1 Maximal Monotone Operators

The main reference for this part is Chapter 7 of Brézis' book [26] (see also the paper [77]).

Definition 12.1 An operator $A : D(A) \subset H \to H$ (possibly nonlinear) is said to be maximal monotone if

(i) A is monotone, meaning that for any $x, y \in D(A)$ one has $\langle A(x) - A(y), x - y\rangle \geq 0$;
(ii) $A + I$ is onto.

Note that both conditions (i) and (ii) make sense if the operator is multivalued. An important example of monotone multivalued operator is the subdifferential ∂f of a convex and lower semicontinuous function $f : H \to (-\infty, \infty]$. Indeed, notice that the surjectivity of $\partial f + I$ can simply be obtained by minimizing the strictly convex functional

$$x \mapsto \Phi(x) := f(x) + \frac{1}{2}|x - y|^2$$

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whose unique minimizer x satisfies $\xi + x = y$ for some $\xi \in \partial f(x)$ (see also (12.3) below). Notice that the existence of x is guaranteed by the compactness with respect to the weak convergence of sublevel sets of Φ, and by its lower semicontinuity with respect to the weak convergence.

The classical proof of Theorem 11.7 makes use of the so-called Yosida regularization.

Definition 12.2 (Yosida Regularization) Define for any $\epsilon > 0$ the Yosida regularization of parameter ϵ of A:

$$A_\epsilon := \frac{I - (I + \epsilon A)^{-1}}{\epsilon}.$$

Note also that A_ϵ is $\frac{2}{\epsilon}$-Lipschitz, and it satisfies

$$A_\epsilon \in A \circ (I + \epsilon A)^{-1}.$$

Let us consider the family of problems

$$\begin{cases} x_\epsilon'(t) = -A_\epsilon(x(t)), \\ x_\epsilon(0) = \bar{x}. \end{cases}$$

By the Cauchy-Lipschitz theorem, each of these problems admits a unique solution x_ϵ. Then, the result follows from local uniform Cauchy-type estimates on $(x_\epsilon)_{\epsilon>0}$ that allow to pass to the limit and to obtain a solution $x = \lim_\epsilon x_\epsilon$. In the particular case of cyclically monotone operators, that correspond to "gradient" operators according to the Rockafellar Theorem 3.9, a different strategy is possible, that makes the variational structure of the gradient flow more transparent and hints to important applications, much beyond the Hilbertian setting of these notes. Here are the steps of our strategy:

(i) reduction to the case of an initial condition $\bar{x}$ with finite energy;
(ii) application of the implicit Euler scheme to build a family of discrete solutions, indexed by the time step $\tau > 0$;
(iii) passage to the limit as $\tau \to 0^+$.

2 The Implicit Euler Scheme

Assume that step (i) has been done, and therefore that $f(\bar{x})$ is finite. For $\tau > 0$, let us define inductively a sequence (y_i) such that $y_0 = \bar{x}$ and, for any $i \geq 0$, y_{i+1} minimizes the function

$$y \mapsto f(y) + \frac{1}{2\tau}|y - y_i|^2.$$

Note that the sequence $(f(y_i))$ is nonincreasing. More precisely choosing $y = y_i$ as competitor one obtains

$$f(y_{i+1}) + \frac{1}{2\tau}|y_{i+1} - y_i|^2 \leq f(y_i). \tag{12.1}$$

Interpolation between the pairs (y_i, y_{i+1}) then provides a curve $x_\tau : [0, \infty) \to H$, that we call a discrete solution. There are several possible interpolations:

(i) the piecewise constant left-continuous one,

$$\hat{x}_\tau(t) = \begin{cases} \overline{x} & \text{if } t = 0; \\ y_i & \text{if } i \geq 1 \text{ and } t \in ((i-1)\tau, i\tau]; \end{cases} \tag{12.2}$$

(ii) the piecewise affine one,

$$x_\tau(t) = (1-s)y_{i-1} + sy_i$$

for any $i \geq 1$ and $t \in [(i-1)\tau, i\tau]$, where $s = \frac{t-(i-1)\tau}{\tau}$;

(iii) the variational one introduced by De Giorgi, denoted $\tilde{x}_\tau$ that we shall introduce in the next lecture.

The procedure for the construction of the sequence of points (y_i) is called implicit Euler scheme. Let us explain why "implicit". By definition of y_{i+1} as a minimizer of $y \mapsto f(y) + \frac{1}{2\tau}|y - y_i|^2$, for any $v \neq 0$ and $\epsilon > 0$,

$$f(y_{i+1} + \epsilon v) + \frac{1}{2\tau}|y_{i+1} + \epsilon v - y_i|^2 \geq f(y_{i+1}) + \frac{1}{2\tau}|y_{i+1} - y_i|^2$$

and so

$$\frac{f(y_{i+1} + \epsilon v) - f(y_{i+1})}{\epsilon} \geq \frac{1}{2\tau\epsilon}\left(|y_{i+1} - y_i|^2 - |y_{i+1} - y_i + \epsilon v|^2\right),$$

whence

$$\frac{f(y_{i+1} + \epsilon v) - f(y_{i+1})}{\epsilon} \geq \left\langle v, -\frac{y_{i+1} - y_i}{\tau} \right\rangle + O(\epsilon),$$

so that the definition of Gateaux subdifferential $\partial_G f$ and its coincidence with ∂f for convex functions, give

$$\frac{y_{i+1} - y_i}{\tau} \in -\partial f(y_{i+1}). \tag{12.3}$$

An explicit scheme should be such that $y_{i+1} - y_i \in -\tau \partial f(y_i)$ so that y_{i+1} would be defined explicitly using the subdifferential at y_i. Instead (12.3) involves the subdifferential at y_{i+1}, which is unknown: that is why this scheme is called implicit.

Even if an explicit scheme seems easier, the implicit scheme is more useful from a theoretical point of view. Indeed, if

$$y_{i+1} + \tau \partial f(y_{i+1}) = y_i,$$

we can write $y_{i+1} = (I + \tau \partial f)^{-1}(y_i)$ and then iterate to get

$$y_i = (I + \tau \partial f)^{-i}(\bar{x}).$$

As $(I + \tau \partial f)^{-1}$ is 1-Lipschitz, $(I + \tau \partial f)^{-i}$ is also 1-Lipschitz, so that this scheme has an intrinsic stability property. On the contrary (think for instance to the case when the operator is the Laplacian!) the iteration of $(I + \tau \partial f)$ is not equally stable.

Let us finally observe that the implicit scheme can be interpreted as the explicit scheme applied to the Yosida regularization of ∂f with parameter τ (cf. Definition 12.2):

$$(\partial f)_\tau := \frac{I - (I + \tau \partial f)^{-1}}{\tau}.$$

Indeed it holds

$$y_{i+1} - y_i = (I + \tau \partial f)^{-1}(y_i) - y_i = -\tau \frac{(I - (I + \tau \partial f)^{-1})(y_i)}{\tau} = -\tau (\partial f)_\tau (y_i).$$

3 Reduction to Initial Conditions with Finite Energy

We assume for simplicity that $\lambda = 0$, but the argument is similar in the general case $\lambda \in \mathbb{R}$. Let us assume that the solution $(S_t(\bar{x}))_{t>0}$ is defined for every initial condition $\bar{x}$ in the domain of f, and let us explain how we can build a solution $(S_t(\bar{x}))_{t>0}$ for any $\bar{x}$ in the closure of $\mathrm{Dom}(f)$.

Take a sequence $(\bar{x}_n) \subset \mathrm{Dom}(f)$ converging to $\bar{x} \in \overline{\mathrm{Dom}(f)}$. By contractivity (11.8), for any $t > 0$ the sequence $(S_t(\bar{x}_n))$ is Cauchy, then it converges to some element $x(t)$. To conclude, we need to prove that:

(i) $\lim_{t \to 0^+} x(t) = \bar{x}$;
(ii) x is a gradient flow.

As for any $t > 0$ and $n,\ p \in \mathbb{N}$ one has

$$|S_t(\bar{x}_{n+p}) - S_t(\bar{x}_n)| \le |\bar{x}_{n+p} - \bar{x}_n|,$$

letting $p \to \infty$ gives

$$|x(t) - S_t(\bar{x}_n)| \le |\bar{x} - \bar{x}_n|.$$

It follows that

$$\begin{aligned} |x(t) - \bar{x}| &\le |x(t) - S_t(\bar{x}_n)| + |S_t(\bar{x}_n) - \bar{x}_n| + |\bar{x}_n - \bar{x}| \\ &\le 2|\bar{x}_n - \bar{x}| + |S_t(\bar{x}_n) - \bar{x}_n| \end{aligned} \tag{12.4}$$

for all n. Taking the upper limit in (12.4) leads to

$$\limsup_{t \to 0^+} |x(t) - \bar{x}| \le 2|\bar{x}_n - \bar{x}|$$

and, since n is arbitrary, the proof of (i) is achieved.

Let us start by proving (ii) under the additional assumption that $x_n(t)$ are locally equi-Lipschitz in $(0, \infty)$, so that $x(t)$ is locally equi-Lipschitz as well. Writing $x_n(t) := S_t(\bar{x}_n)$ for simplicity, as $t \mapsto \bar{x}_n(t)$ is an EVI_0 solution for all $y \in \mathrm{Dom}(f)$ we have

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}|x_n(t) - y|^2\right) + f(x_n(t)) \le f(y) \tag{12.5}$$

both $\mathscr{L}^1$-a.e. in $(0, \infty)$ and in the sense of distributions. Since f is lower semicontinuous and

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}|x_n(t) - y|^2\right) \to \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}|x(t) - y|^2\right)$$

in the sense of distributions, we can pass to the limit in (12.5) to obtain the EVI_0 property of $x(t)$.

Let us now prove that $x_n(t)$ are locally equi-Lipschitz in $(0, \infty)$. Take $\epsilon > 0$. As each x_n is a gradient flow, by Proposition 11.9(iii), $|x_n'(t)| = |\nabla f|(x_n(t))$ for $\mathscr{L}^1$-a.e. $t > 0$. Then for $\mathscr{L}^1$-a.e. $t > \epsilon$, using Proposition 11.9(iv), for any v such that $\partial f(v) \neq \emptyset$ we can estimate

$$\begin{aligned} |x_n'(t)|^2 = |\nabla f|^2(x_n(t)) &\le |\nabla f|^2(v) + \frac{1}{t^2}|v - \bar{x}_n|^2 \\ &\le |\nabla f|^2(v) + \frac{1}{\epsilon^2}|v - \bar{x}_n|^2. \end{aligned}$$

This proves that the sequence (x_n') is uniformly bounded in $L^\infty(\epsilon, \infty)$.

4 Discrete EVI

Let us come back to the sequence (y_i) provided by the implicit Euler scheme in Sect. 2, namely minimizing recursively $f(y) + \frac{1}{2\tau}|y - y_i|^2$, with $y_0 = \bar{x}$. Use the piecewise constant interpolation (12.2) to define the curve $\hat{x}_\tau$ that we will also denote from time to time by $(S_\tau(\bar{x}, t))_{t>0}$. Let us start with an observation. If x is a solution of EVI and (t_i) is the sequence of times defined by $t_i = i\tau$ for any $i \in \mathbb{N}$, the derivative

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}|x(t) - v|^2\right)$$

can be approximated as τ goes to 0 by the rate of change

$$\frac{1}{\tau}\left(\frac{1}{2}|x(t_{i+1}) - v|^2 - \frac{1}{2}|x(t_i) - v|^2\right).$$

From this observation, we can define a discrete version of EVI, called EVI^τ and defined below, which turns out to be satisfied by the sequence $(y_i)_{i\in\mathbb{N}}$. Note that this property is very similar to the definition of y_{i+1} as a minimizer ($f(y_{i+1}) + \frac{1}{2\tau}|y_{i+1} - y_i|^2 \le f(v) + \frac{1}{2\tau}|v - y_i|^2$ for all $v \in H$) but it differs by y_i replaced by v in the left-hand side.

Proposition 12.3 *The sequence* (y_i) *satisfies the property* EVI^τ*, namely:*

$$\frac{|y_{i+1} - v|^2 - |y_i - v|^2}{2\tau} + f(y_{i+1}) \le f(v) \qquad \forall v \in H, \quad \forall i \in \mathbb{N}. \tag{12.6}$$

Proof Take $v \in \mathrm{Dom}(f)$ and $i \ge 1$. Define for every $t \in (0, 1)$, $\gamma(t) = (1 - t)y_{i+1} + tv$. By definition of y_{i+1}, for any $t \in (0, 1)$, $f(y_{i+1}) + \frac{1}{2\tau}|y_{i+1} - y_i|^2 \le f(\gamma(t)) + \frac{1}{2\tau}|\gamma(t) - y_i|^2$. By convexity, $f(\gamma(t)) \le (1-t)f(y_{i+1}) + tf(v)$. Besides, as pointed out in Remark 10.16, we can use the Hilbertian identity

$$\frac{1}{2\tau}|\gamma(t) - y_i|^2 = \frac{1-t}{2\tau}|y_{i+1} - y_i|^2 + \frac{t}{2\tau}|v - y_i|^2 - \frac{t(1-t)}{2\tau}|y_{i+1} - v|^2, \tag{12.7}$$

to get

$$\begin{aligned} f(y_{i+1}) + \frac{1}{2\tau}|y_{i+1} - y_i|^2 \le{}& (1-t)f(y_{i+1}) + tf(v) \\ &+ \frac{1-t}{2\tau}|y_{i+1} - y_i|^2 + \frac{t}{2\tau}|v - y_i|^2 - \frac{t(1-t)}{2\tau}|y_{i+1} - v|^2. \end{aligned}$$

Now we can remove $(1-t)f(y_{i+1}) + \frac{1-t}{2\tau}|y_{i+1}-y_i|^2$ from both sides of the inequality and divide by $t > 0$. As $f(y_{i+1}) \le f(y_{i+1}) + \frac{1}{2\tau}|y_{i+1}-y_i|^2$, one obtains

$$f(y_{i+1}) \le f(v) + \frac{1}{2\tau}|v-y_i|^2 - \frac{1-t}{2\tau}|y_{i+1}-v|^2$$

whence EVI^τ follows as $t \to 0^+$. □

Remark 12.4 (Convex Functions in NPC Spaces) The proof of the EVI^τ property works, basically with no modifications, for convex functions in non positively curved spaces (see Definition 10.15). Indeed, we only used the inequality $\le$ in (12.7). In this more general context, convexity (see also Definition 14.9) has to be understood as convexity along constant speed geodesics. By the techniques described in the sequel, the EVI^τ property then leads to existence of EVI gradient flows also in this much more general context.

Let us conclude by studying what happens when $\tau \to 0^+$, for simplicity we assume that f is convex and $f \ge 0$ (the general statement requires only minor modifications).

Theorem 12.5 *Let $f : H \to [0,\infty]$ be a convex and lower semicontinuous function and let $\bar{x} \in Dom(f)$. Then:*

(i) for any $t \ge 0$ the family $(S_\tau(\bar{x},t))_{\tau>0}$ is Cauchy as $\tau \to 0$, so that its limit $S(\bar{x},t)$ exists;
(ii) the curve $S(\bar{x},\cdot)$ is the gradient flow starting from $\bar{x}$;
(iii) $|S_\tau(\bar{x},t) - S(\bar{x},t)| \le C\sqrt{\tau f(\bar{x})}$, with $C = 2(\sqrt{2}+1)$, for any $\tau > 0$ and $t \ge 0$.

Particularly relevant for the applications is property (iii), since it provides an estimate of the gap between the continuous and the discrete solution depending only on the energy of the initial datum $\bar{x}$. To prove Theorem 12.5, we need two preliminary lemmas valid for any $\tau > 0$.

Lemma 12.6 *For any $t \in \mathbb{N}\tau$ one has*

$$|S_\tau(\bar{x},t) - S_{\tau/2}(\bar{x},t)|^2 \le 2\tau f(\bar{x}).$$

Proof The key point in this proof is the following estimate, involving the position at the solely time τ of gradient flows starting from two possibly different points $\bar{x}$ and $\bar{y}$ in H:

$$|S_{\tau/2}(\bar{x},\tau) - S_\tau(\bar{y},\tau)|^2 - |\bar{x}-\bar{y}|^2 \le 2\tau(f(\bar{x}) - f(S_{\tau/2}(\bar{x},\tau))). \tag{12.8}$$

Let us prove it. The sequence $(S_{\tau/2}(\bar{x}, i\tau/2))$ satisfies $\mathrm{EVI}^{\tau/2}$. The first time steps with $i = 0$ and $i = 1$ give

$$|S_{\tau/2}(\bar{x}, \tau/2) - v|^2 - |\bar{x} - v|^2 \leq \tau(f(v) - f(S_{\tau/2}(\bar{x}, \tau/2))) \qquad \forall v \in H, \tag{12.9}$$

$$|S_{\tau/2}(\bar{x}, \tau) - v|^2 - |S_{\tau/2}(\bar{x}, \tau/2) - v|^2 \leq \tau(f(v) - f(S_{\tau/2}(\bar{x}, \tau))) \qquad \forall v \in H. \tag{12.10}$$

The sum of (12.9) and (12.10), taking the monotonicity of $(f(y_i))$ into account, gives:

$$|S_{\tau/2}(\bar{x}, \tau) - v|^2 - |\bar{x} - v|^2 \leq 2\tau(f(v) - f(S_{\tau/2}(\bar{x}, \tau))) \qquad \forall v \in H. \tag{12.11}$$

The sequence $(S_\tau(\bar{y}, i\tau))$ satisfies EVI^τ. The first step with $i = 0$ gives:

$$|S_\tau(\bar{y}, \tau) - v|^2 - |\bar{y} - v|^2 \leq 2\tau(f(v) - f(S_\tau(\bar{y}, \tau))) \qquad \forall v \in H. \tag{12.12}$$

Take $v = S_\tau(\overline{y}, \tau)$ in (12.11) and $v = \bar{x}$ in (12.12), then add the two resulting inequalities to get (12.8).

We are now in position to conclude the proof. As $S_{\tau/2}(S_{\tau/2}(\bar{x}, \tau), \tau) = S_{\tau/2}(\bar{x}, 2\tau)$ and $S_\tau(S_\tau(\bar{y}, \tau), \tau) = S_\tau(\bar{y}, 2\tau)$, an iterative argument shows that for any $n \geq 1$,

$$\begin{aligned} &|S_{\tau/2}(\bar{x}, n\tau) - S_\tau(\bar{y}, n\tau)|^2 - |S_{\tau/2}(\bar{x}, (n-1)\tau) - S_\tau(\bar{y}, (n-1)\tau)|^2 \\ &\leq 2\tau(f(S_{\tau/2}(\bar{x}, (n-1)\tau)) - f(S_{\tau/2}(\bar{x}, n\tau))) . \end{aligned}$$

This leads, via a telescopic sum, to

$$|S_{\tau/2}(\bar{x}, n\tau) - S_\tau(\bar{y}, n\tau)|^2 - |\bar{x} - \bar{y}|^2 \leq 2\tau(f(\bar{x}) - f(S_{\tau/2}(\bar{x}, n\tau))).$$

It follows that $|S_{\tau/2}(\bar{x}, n\tau) - S_\tau(\bar{y}, n\tau)|^2 - |\bar{x} - \bar{y}|^2 \leq 2\tau f(\bar{x})$ as we have assumed $f \geq 0$. Taking $\bar{y} = \bar{x}$ we get the result. □

The second lemma is an estimate providing a sort of equi-$C^{0,1/2}$-Hölder continuity (say up to the scale τ) of the family of functions $S_\tau(\bar{x}, \cdot)$.

Lemma 12.7 *For all $s,\ t \geq 0$ with $s \leq t$ one has*

$$|S_\tau(\bar{x}, t) - S_\tau(\bar{x}, s)| \leq \sqrt{2f(\bar{x})}(t - s + \tau)^{1/2}.$$

Proof We already observed in (12.1) that $|y_{i+1} - y_i|^2 \le 2\tau(f(y_i) - f(y_{i+1}))$. Summing over $i \in \mathbb{N}$ these inequalities, as f is nonnegative we get $\sum_0^\infty |y_{i+1} - y_i|^2 \le 2\tau f(y_0) = 2\tau f(\bar{x})$. From that we deduce that for any integers $k < \ell$

$$\begin{aligned} |y_k - y_\ell| &\le \sum_{i=k}^{\ell-1} |y_{i+1} - y_i| = \sum_{i=k}^{\ell-1} \frac{|y_{i+1} - y_i|}{\sqrt{\tau}} \sqrt{\tau} \\ &\le \left(\sum_{i=k}^{\ell-1} \frac{|y_{i+1} - y_i|^2}{\tau} \right)^{1/2} (\tau|\ell - k|)^{1/2} \\ &\le \sqrt{2f(\bar{x})} \sqrt{\tau(\ell - k)}. \end{aligned}$$

Take $s, t > 0$. Let ℓ and k be the largest positive integers such that $\tau(\ell - 1) < s$ and $\tau(k - 1) < t$ respectively. As $S_\tau(\bar{x}, \cdot)$ has been defined via the piecewise constant interpolation, $S_\tau(\bar{x}, s) = y_\ell$ and $S_\tau(\bar{x}, t) = y_k$. The result follows from the observation $\tau k - \tau\ell \le t - s + \tau$. □

Let us prove statements (i) and (iii) of Theorem 12.5.

Proof of Theorem 12.5(i, iii) Fix $\tau_0 > 0$ and write $\tau_k := \tau_0/2^k$ for any $k \in \mathbb{N}$ and $x_k(t) := S_{\tau_k}(\bar{x}, t)$. Lemma 12.6 implies that $(x_k(t))$ is Cauchy for any $t \in \cup_n \tau_n \mathbb{N}$: indeed, for $t \in \tau_n \mathbb{N}$ and $n \le \ell < m$ one has

$$|S_{\tau_\ell}(\bar{x}, t) - S_{\tau_m}(\bar{x}, t)| \le \sum_{k=\ell}^{m-1} |S_{\tau_k}(\bar{x}, t) - S_{\tau_{k+1}}(\bar{x}, t)| \le \sqrt{2f(\bar{x})} \sum_{k=\ell}^{m-1} \tau_k^{1/2}. \tag{12.13}$$

Then $(x_k(t))$ converges to $x(t) \in H$. Taking $\ell = 0$ and letting $m \to \infty$ in (12.13), we get

$$|S_{\tau_0}(\bar{x}, t) - x(t)| \le C\sqrt{\tau_0 f(\bar{x})}$$

with $C = 2(\sqrt{2} + 1)$. As the set $\cup_n \tau_n \mathbb{N}$ is dense in $(0, \infty)$, since Lemma 12.7 provides the Hölder continuity of x, we can extend x to an Hölder continuous function on $[0, \infty)$. Moreover, by (asymptotic) equi-continuity again, $S_{\tau_k}(\bar{x}, t)$ converge to $x(t)$ locally uniformly in $[0, \infty)$. The proof of (i) and (iii) is then achieved if we show that $x(t)$ does not depend on the initial time step τ_0. This will be achieved by showing that $x(t)$ is the unique EVI solution starting from $\bar{x}$. □

To prove (ii) of Theorem 12.5 we need a third lemma.

Lemma 12.8 *Let (X, d) be a metric space and assume that f_i, $f : X \to [0, \infty]$ are Borel functions such that*

$$\liminf_{i \to \infty} f_i(x_i) \ge f(x) \qquad \text{whenever } x_i \to x. \tag{12.14}$$

Assume that $\nu_i \in \mathscr{M}_+(X)$ weakly converge, in duality with $C_b(X)$, to some $\nu \in \mathscr{M}_+(X)$. Then

$$\liminf_{i\to\infty} \int_X f_i \,\mathrm{d}\nu_i \geq \int_X f \,\mathrm{d}\nu.$$

Proof By Cavalieri's formula, $\int_X f \,\mathrm{d}\nu = \int_0^\infty \nu(\{f > t\})\,\mathrm{d}t$ and a similar formula holds for f_i. Hence the equivalent formulation of the statement is

$$\liminf_{i\to\infty} \int_0^\infty \nu_i(\{f_i > t\})\,\mathrm{d}t \geq \int_0^\infty \nu(\{f > t\})\,\mathrm{d}t$$

which, thanks to Fatou's lemma, follows from

$$\liminf_{i\to\infty} \nu_i(\{f_i > t\}) \geq \nu(\{f > t\}) \qquad \forall t > 0. \tag{12.15}$$

Let us then prove (12.15). First, we notice that (12.14) immediately yields

$$\liminf_{j\to\infty} f_{i(j)}(x_{i(j)}) \geq f(x) \qquad \text{whenever } x_{i(j)} \to x \tag{12.16}$$

for any subsequence $i(j)$. Then, we notice that (12.16) yields

$$\{f \leq t\} \supset \bigcap_j \overline{\bigcup_{i\geq j} \{f_i \leq t\}} \qquad \forall t > 0.$$

Indeed, if x belongs to the set in the right hand side, for any $j \geq 1$ we can find $i(j)$ and $x_{i(j)}$ with $d(x_{i(j)}, x) \leq 1/j$ and $f_{i(j)}(x_{i(j)}) \leq t$, so that (12.16) yields $f(x) \leq t$.

Now, passing to the complementary sets, and denoting by A^{o} the interior of a set A, we get

$$\{f > t\} \subset \bigcup_j \Big(\bigcap_{i\geq j} \{f_i > t\}\Big)^{\mathrm{o}} \qquad \forall t > 0 \tag{12.17}$$

and, from

$$\liminf_{i\to\infty} \nu_i(\{f_i > t\}) \geq \liminf_{i\to\infty} \nu_i\left(\Big(\bigcap_{i\geq j} \{f_i > t\}\Big)^{\mathrm{o}}\right) \geq \nu\left(\Big(\bigcap_{i\geq j} \{f_i > t\}\Big)^{\mathrm{o}}\right) \quad \forall j$$

we can let $j \to \infty$ to obtain (12.15) from (12.17). □

Remark 12.9 Note that (12.14) cannot be replaced by the pointwise condition $\liminf_{i\to\infty} f_i \geq f$, as the following simple example shows: $f_i(t) = 1 - \chi_{\{1/i\}}$, $f \equiv 1$, $\nu_i = \delta_{1/i}$, $\nu = \delta_0$, with $i \geq 1$, $t \in \mathbb{R}$.

Proof of Theorem 12.5(ii) Let us drop for simplicity the starting point $\bar{x}$ in the notation $S_\tau(\bar{x}, t)$. The piecewise constant function $\frac{1}{2}|S_\tau(\cdot) - y|^2$ has distributional derivative

$$D\frac{1}{2}|S_\tau(\cdot) - y|^2 = \sum_{k=0}^{\infty} \left(\frac{1}{2}|S_\tau((k+1)\tau) - y|^2 - \frac{1}{2}|S_\tau(k\tau) - y|^2 \right) \delta_{k\tau}.$$

Therefore, starting from EVI^τ (see (12.6)), i.e.

$$\frac{1}{2\tau}|S_\tau((k+1)\tau) - y|^2 - \frac{1}{2\tau}|S_\tau(k\tau) - y|^2 \le f(y) - f(S_\tau((k+1)\tau)),$$

multiplying by $\tau\delta_{k\tau}$ and summing over $k \in \mathbb{N}$ yields

$$D\frac{1}{2}|S_\tau(\cdot) - y|^2 \le \sum_{k=1}^{\infty} (f(y) - f(S_\tau((k+1)\tau)))\, \tau\delta_{k\tau}.$$

As $S_\tau(k\tau) = S_\tau(\tau\lfloor t/\tau \rfloor)$ for every $t \in ((k-1)\tau, k\tau]$, integration against any positive test function $\varphi \in C_c^1([0,\infty))$ gives

$$-\frac{1}{2}\int_0^\infty \varphi'(t)|S_\tau(t) - y|^2\,\mathrm{d}t \le \int_0^\infty \varphi(t)\,(f(y) - f(S_\tau(\tau\lfloor t/\tau \rfloor)))\,\mathrm{d}\mu_\tau(t), \tag{12.18}$$

where $\mu_\tau := \sum_0^\infty \tau\delta_{k\tau}$. The left hand side in (12.18) converges to $-\frac{1}{2}\int_0^\infty \varphi'(t)|x(t) - y|^2\,\mathrm{d}t$ as $\tau \to 0^+$. Since μ_τ weakly converge in duality with $C_c([0,\infty))$ to $\mathscr{L}^1$ we have

$$\begin{aligned}
-\frac{1}{2}\int_0^\infty \varphi'(t)|x(t) - y|^2\,\mathrm{d}t &\le \limsup_{\tau\to 0} \int_0^\infty \varphi(t)\,(f(y) - f(S_\tau(\tau\lfloor t/\tau \rfloor)))\,\mathrm{d}\mu_\tau(t) \\
&\le \int_0^\infty \varphi(t) f(y)\,\mathrm{d}t - \liminf_{\tau\to 0^+} \int_0^\infty \varphi(t) f(S_\tau(\tau\lfloor t/\tau \rfloor))\,\mathrm{d}\mu_\tau(t) \\
&\le \int_0^\infty \varphi(t) f(y)\,\mathrm{d}t - \int_0^\infty \varphi(t) f(x(t))\,\mathrm{d}t\,.
\end{aligned}$$

Above we used Lemma 12.8 for the last inequality. Note that (12.14) is provided by the lower semicontinuity of f and Lemma 12.7. Since φ is arbitrary, the distributional formulation of EVI is proved. □

Lecture 13: Examples of Gradient Flows in PDEs



In this lecture we provide several examples of gradient flows, more specifically we will interpret several parabolic PDE's as gradient flows of suitable energies (also called "entropies" in the literature) in suitable Hilbert spaces.

In this lecture we provide several examples of gradient flows, more specifically we will interpret several parabolic PDE's as gradient flows of suitable energies (also called "entropies" in the literature) in suitable Hilbert spaces. As crucially pointed out in [94, 95], a remark however is in order: if we are given a C^1-differentiable manifold M and a (say) C^1 curve $\gamma : [0, 1] \to M$ as well as a C^1 function $f : M \to \mathbb{R}$, then the objects

$$\gamma'(t), \qquad d_{\gamma(t)} f$$

make sense, but the first one belongs to the tangent space $T_{\gamma(t)}M$ to M at $\gamma(t)$, while the second one belongs to the dual space $T^*_{\gamma(t)}M$, the cotangent space at $\gamma(t)$. Hence, with $x = \gamma(t)$, a metric tensor $\mathbf{g}_x : T_x M \times T_x M \to \mathbb{R}$ is needed to move from the differential $d_x f$ to the gradient $\nabla f(x)$ via the relation

$$\mathbf{g}_x(\nabla f(x), v) = d_x f(v) \qquad \forall x \in M,\ \forall v \in T_x M. \tag{13.1}$$

Therefore, in order to speak about gradient flows, we need not only the energy f but also the metric $\mathbf{g}$, or at least its integrated version, namely the distance d. Finally, let us remark that the interpretation of a PDE as gradient flow is a kind of ill-posed inverse problem, since many triples $(M, f, \mathbf{g})$ might lead to the same PDE. We will see that this happens even for the heat equation (and this will be an important example).

Besides Example 11.6, we are now going to introduce several new examples, always with $M = H$, H Hilbert space, and suitable choices of f.

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1 p-Laplace Equation, Heat Equation in Domains, Fokker-Planck Equation

In order to treat our first new example without artificial assumptions, it is better to decouple the integrability of the function and the integrability of the gradient, setting

$$L^{p,q}(\mathbb{R}^n) := \left\{u \in L^p(\mathbb{R}^n) : \ \nabla u \in L^q(\mathbb{R}^n;\mathbb{R}^n)\right\} \qquad 1 \le p,q \le \infty,$$

where of course the gradient is understood in the sense of distributions (recall that $L^p \subset L^1_{\rm loc}$).

Example 13.1 (The p-Laplace Equation) We define $H = L^2(\mathbb{R}^n)$ and, for $p > 1$, $\mathbf{D}_p : H \to [0,\infty]$ by

$$\mathbf{D}_p(u) := \begin{cases} \frac{1}{p}\int_{\mathbb{R}^n} |\nabla u|^p \,\mathrm{d}x & \text{if } u \in L^{2,p}(\mathbb{R}^n); \\ \\ +\infty & \text{otherwise.} \end{cases}$$

Then, computations very similar to the ones of Example 11.6 show that the gradient flow of this convex and lower semicontinuous functional coincides with the PDE

$$\frac{\mathrm{d}}{\mathrm{d}t}u = \mathrm{div}\big(|\nabla u|^{p-2}\nabla u\big)$$

where, as usual, the time derivative is understood in the strong L^2 sense and the divergence is understood in the distributional sense. This is stronger than the weak formulation

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathbb{R}^n} u(t,x)\phi(x)\,\mathrm{d}x = -\int_{\mathbb{R}^n} |\nabla u(t,x)|^{p-2}\langle \nabla u(t,x), \nabla\phi(x)\rangle \,\mathrm{d}x \qquad \forall \phi \in C^\infty_c(\mathbb{R}^n)$$

that we shall adopt in the sequel for equations, like the continuity equation, not enjoying the regularizing effect due to the gradient flow structure (or to the diffusion).

Example 13.2 (The Heat Equation in a Domain Ω of $\mathbb{R}^n$) Let $\Omega \subset \mathbb{R}^n$ be an open set, not necessarily bounded. As in Example 11.6, we can now consider the functional $\mathbf{D} : L^2(\Omega) \to [0,\infty]$ defined by

$$\mathbf{D}(u) := \begin{cases} \frac{1}{2}\int_{\Omega} |\nabla u|^2 \,\mathrm{d}x & \text{if } u \in H^1(\Omega); \\ \\ +\infty & \text{otherwise.} \end{cases}$$

The gradient flow of $\mathbf{D}$ in the Hilbert space $H := L^2(\Omega)$ still corresponds to the heat equation (11.6), but with the addition of the homogeneous Neumann boundary condition

$$\frac{\partial u}{\partial n}(t,\cdot) = 0 \quad \text{on } \partial\Omega,$$

understood as always in the weak sense. More precisely, in this case the characterization of the subdifferential of $\mathbf{D}$ easily gives

$$\xi \in \partial\mathbf{D}(u) \qquad \Longleftrightarrow \qquad \int_\Omega \xi\phi \, \mathrm{d}x = \int_\Omega \langle \nabla u, \nabla\phi\rangle \, \mathrm{d}x \quad \forall \phi \in C^\infty(\overline{\Omega}) \cap H^1(\Omega). \tag{13.2}$$

Since for $u \in C^1(\overline{\Omega}) \cap H^1(\Omega)$ and $\partial\Omega$ regular the validity of the formula in the right hand side of (13.2) implies that $\xi = -\Delta u$ inside Ω and that the normal derivative of u is null on $\partial\Omega$, even when $u \in H^1(\Omega)$ we may interpret (13.2) as the validity of these facts in the weak sense, more precisely in the sense of distributions.

Example 13.3 (The Fokker-Planck Equation in $\mathbb{R}^n$) Let $V : \mathbb{R}^n \to \mathbb{R}$ be a locally Lipschitz function with $\int_{\mathbb{R}^n} e^{-V} \, \mathrm{d}x < \infty$. We consider the Fokker-Planck equation (from now on abbreviated as FP)

$$\frac{\mathrm{d}}{\mathrm{d}t} u = \mathrm{div}(\nabla u + u\nabla V) \tag{13.3}$$

and we want to interpret it as a gradient flow. In view of the integrability assumption, it is natural to consider the positive and finite measure $\gamma = e^{-V}\mathscr{L}^n$ and to interpret (at least when u is nonnegative) u as a density of a measure $\mu = u\mathscr{L}^n$, making the change of variable $w = ue^V$. In this way, μ can also be represented as $w\gamma$. Formally, this change of variables transforms the Fokker Planck equation into

$$e^{-V}\frac{\mathrm{d}}{\mathrm{d}t} w = \mathrm{div}(e^{-V}\nabla w) \tag{13.4}$$

which suggests the interpretation of the equation as gradient flow in $H := L^2(\gamma)$ of the functional $\mathbf{D}_\gamma : L^2(\gamma) \to [0,\infty]$ defined by

$$\mathbf{D}_\gamma(w) := \begin{cases} \frac{1}{2}\int_{\mathbb{R}^n} |\nabla w|^2 \, \mathrm{d}\gamma & \text{if } \nabla w \in L^2(\gamma;\mathbb{R}^n); \\ \\ +\infty & \text{otherwise} \end{cases}$$

(since $L^2(\gamma) \subset L^1_{\rm loc}(\mathbb{R}^n)$, one can still define the gradient ∇w in the sense of distributions, so the definition makes sense). Indeed, with computations analogous to the ones of Example 11.6, one easily checks that

$$\xi \in \partial \mathbf{D}_\gamma(w) \qquad \Longleftrightarrow \qquad \int_{\mathbb{R}^n} \xi\phi \, \mathrm{d}\gamma = \int_{\mathbb{R}^n} \langle \nabla w, \nabla \phi\rangle \, \mathrm{d}\gamma \quad \forall \phi \in C^\infty_c(\mathbb{R}^n),$$

which means that $e^{-V}\xi$ is equal to $-\mathrm{div}(e^{-V}\nabla w)$ in the sense of distributions.

A more appealing formulation of (13.4), closer to Riemannian Geometry (see Sect. 2 below), is

$$\frac{\mathrm{d}}{\mathrm{d}t} w = e^V \mathrm{div}(e^{-V}\nabla w) = \Delta_\gamma w$$

where the "γ-Laplacian" (also called *drift* or *Witten* Laplacian in the literature) is given by

$$\Delta_\gamma w = \Delta w - \langle \nabla V, \nabla w\rangle. \tag{13.5}$$

An important choice of V is of course $V(x) = \frac{1}{2}|x|^2 + \frac{n}{2}\ln(2\pi)$, which corresponds to the Gaussian probability measure

$$\gamma = \frac{1}{(2\pi)^{n/2}} e^{-|x|^2/2} \mathscr{L}^n.$$

In this case the γ-Laplacian is the celebrated Ornstein-Uhlenbeck operator $\Delta - \langle x, \nabla\rangle$. Notice also that the integrability condition $\int_{\mathbb{R}^n} e^{-V}\,\mathrm{d}x < \infty$ could be dropped, the only difference is that γ becomes locally finite in $\mathbb{R}^n$ (and, in particular, σ-finite).

The FP equation is important in many applications, because it provides the simplest model combining in an additive way the effects of diffusion and transport. More generally one can replace in (13.3) the vector field ∇V by a general vector field $\mathbf{b}$, but in this more general case the variational interpretation is lost, and one has to use PDE or probabilistic methods to study the equation.

2 The Heat Equation in Riemannian Manifolds

Assume for the moment that on $\mathbb{R}^n$ we replace the Euclidean metric by a Riemannian metric $\mathbf{g}_x$, namely a positive definite bilinear form. Since the volume element is $\sqrt{\det \mathbf{g}_{xij}}$, it is tempting to set $V(x) := -\frac{1}{2}\ln\det \mathbf{g}_{xij}$, so that $\gamma = e^{-V}\mathscr{L}^n$ coincides with the volume measure. However, with this choice, the γ-Laplacian in (13.5) does not coincide with the Laplace-Beltrami operator (the canonical

Laplacian on Riemannian manifolds) $\Delta_{\mathbf{g}}$, given in local coordinates by

$$\Delta_{\mathbf{g}} w(x) := \frac{1}{\sqrt{\det \mathbf{g}_x}} \sum_i \frac{\partial}{\partial x_i} \Big(\sqrt{\det \mathbf{g}_x} \sum_j \big(\mathbf{g}^{-1}_{xij} \frac{\partial w}{\partial x_j}(x) \big) \Big). \tag{13.6}$$

The equality can be restored if we adapt, as it should be, also the distance to the Riemannian structure: since the Riemannian gradient $\nabla^{\mathbf{g}} w(x)$ is given, according to (13.1), by $\mathbf{g}_x^{-1} \nabla w(x)$, the "correct" energy is

$$\mathbf{D}_{\mathbf{g}}(w) := \begin{cases} \frac{1}{2} \int_{\mathbb{R}^n} \mathbf{g}(\nabla^{\mathbf{g}} w, \nabla^{\mathbf{g}} w) \, \mathrm{d}\gamma = \frac{1}{2} \int_{\mathbb{R}^n} \langle \mathbf{g}^{-1} \nabla w, \nabla w \rangle \mathrm{d}\gamma & \text{if } \nabla^{\mathbf{g}} w \in L^2(\gamma; \mathbb{R}^n); \\ \\ +\infty & \text{otherwise} \end{cases} \tag{13.7}$$

whose gradient flow with respect to the $L^2(\gamma)$ distance corresponds to solutions to the PDE

$$\frac{\mathrm{d}}{\mathrm{d}t} w = \Delta_{\mathbf{g}} w.$$

Notice that since $L^2(\gamma) \subset L^1_{\mathrm{loc}}(\mathbb{R}^n)$, in (13.7) the gradient ∇w can still be defined in the sense of distributions, and then the Riemannian gradient $\nabla^{\mathbf{g}} w = \mathbf{g}^{-1} \nabla w$ in the formula makes sense.

More generally, one can adapt the definition of $\mathbf{D}_{\mathbf{g}}$ to any Riemannian manifold M, choosing as γ the volume measure vol_M and using the coordinate-free formula (coming from Leibniz's rule, where X is any smooth section of the tangent bundle)

$$\int_M \phi \mathbf{g}(\nabla^{\mathbf{g}} w, X) \, \mathrm{dvol}_M = -\int_M \big(w \mathbf{g}(\nabla \phi, X) + w \phi \mathrm{div} X \big) \, \mathrm{dvol}_M \qquad \phi \in C_c^\infty(M) \tag{13.8}$$

to define the "distributional" Riemannian gradient $\nabla^{\mathbf{g}} w$ of w, now seen as a measurable section of the tangent bundle. Moreover, we say that $\nabla^{\mathbf{g}} w$ belongs to $L^2(\mathrm{vol}_M)$ if

$$\int_M \mathbf{g}(\nabla^{\mathbf{g}} w, \nabla^{\mathbf{g}} w) \, \mathrm{dvol}_M < \infty.$$

3 Dual Sobolev Space H^{-1} and Heat Flow in H^{-1}

In order to introduce the next example we need to consider the so-called homogeneous Sobolev norms and their dual spaces. In order to avoid technical complications, we shall work in a nice bounded domain. Assume then that $\Omega \subset \mathbb{R}^n$ is a

bounded domain, sufficiently regular to provide, for some $\lambda > 0$, the L^2 Poincaré inequality

$$\int_\Omega |u - \bar{u}|^2 \, \mathrm{d}x \le \frac{1}{\lambda} \int_\Omega |\nabla u|^2 \, \mathrm{d}x \qquad \forall u \in H^1(\Omega) \tag{13.9}$$

with $\bar{u}$ equal to the mean value of u in Ω. Notice that a sufficient condition is that Ω is connected, and that $\partial\Omega$ has a Lipschitz boundary. It turns out that the best constant in (13.9) can be obtained precisely choosing λ equal to the smallest positive eigenvalue of $-\Delta$ with homogeneous Neumann boundary conditions.

Let us now consider the subspace

$$\mathring{H}^1(\Omega) := \left\{ u \in H^1(\Omega) : \int_\Omega u \, \mathrm{d}x = 0 \right\},$$

which can also be considered as a realization of the quotient space $H^1(\Omega)/\!\sim$, with the equivalence relation $f \sim g$ if $f - g$ is (equivalent to) a constant. In view of (13.9), the vector space $\mathring{H}^1(\Omega)$, endowed with the norm

$$\|u\|_{\mathring{H}^1} := \|\nabla u\|_2$$

is a Banach space which embeds continuously in $H^1(\Omega)$. In the next two examples we denote by $-\Delta$ the subdifferential of the energy

$$\mathbf{D}(u) := \begin{cases} \frac{1}{2} \int_\Omega |\nabla u|^2 \, \mathrm{d}x & \text{if } u \in H^1(\Omega); \\ +\infty & \text{if } u \in L^2(\Omega) \setminus H^1(\Omega). \end{cases}$$

It is an $L^2(\Omega)$-valued operator defined on the set $D(\Delta)$ of all $u \in H^1(\Omega)$ where the subdifferential of $\mathbf{D}$ is not empty (and it is a singleton whenever it is not empty, as we have seen for the Dirichlet energy in $\mathbb{R}^n$). Notice that since we are working in a bounded domain and we are using test functions which need not be null at the boundary of Ω, the condition $\Delta u = f \in L^2(\Omega)$ incorporates also the homogeneous Neumann boundary condition, in a weak sense as explained in Example 13.2.

In the sequel we shall use the following two basic properties of Δ:

(i) for all $g \in L^2(\Omega)$ with $\int_\Omega g \, \mathrm{d}x = 0$ there exists $u \in D(\Delta)$ with $\Delta u = g$;
(ii) the set $D(\Delta)$ is dense in H^1 with respect to the $H^1(\Omega)$ norm.

Both these facts can be proved by variational methods, the first one by minimizing

$$v \mapsto \int_\Omega (|\nabla v|^2 + 2gv) \, \mathrm{d}x$$

whose minimizers u are unique up to a constant and solve $\Delta u = g$ (see also the proof of Proposition 13.4 below), the second one by minimizing, for $u \in H^1(\Omega)$

$$v \mapsto \int_\Omega (|\nabla v|^2 + 2\alpha(v-u)^2)\,\mathrm{d}x$$

whose unique minimizers $u_\alpha \in D(\Delta)$ (they actually solve $\Delta u_\alpha = \alpha(u_\alpha - u)$) converge to u strongly in $H^1(\Omega)$ as $\alpha \to +\infty$.

We can now define $\mathring{H}^{-1}(\Omega)$ as the dual of $\mathring{H}^1(\Omega)$, endowed with the dual norm. We collect in the next proposition a few properties of $\mathring{H}^{-1}$ and of the *distributional Laplacian* $\mathbf{\Delta}$ which should not be (totally) confused with Δ, since the former is $\mathring{H}^{-1}$-valued, while the latter is L^2-valued.

Proposition 13.4 (Properties of $\mathring{H}^{-1}$) *The distributional Laplacian $\mathbf{\Delta}$, defined by the formula*

$$\mathbf{\Delta} g(u) := -\int_\Omega \langle \nabla g, \nabla u\rangle\,\mathrm{d}x$$

provides an isometric isomorphism from $\mathring{H}^1(\Omega)$ to $\mathring{H}^{-1}(\Omega)$. In addition, the map

$$L^2(\Omega) \ni g \mapsto L_g(u) = \int_\Omega gu\,\mathrm{d}x$$

provides a continuous embedding of $L^2(\Omega)$ into $\mathring{H}^{-1}(\Omega)$, with a dense image, coinciding with

$$\{\mathbf{\Delta} u :\ u \in D(\Delta)\}.$$

Proof It is clear from the definition that $\|\mathbf{\Delta} u\|_{\mathring{H}^{-1}} = \|\nabla u\|_2 = \|u\|_{\mathring{H}^1}$, hence $\mathbf{\Delta}$ is a linear isometry. Surjectivity can be proved, for instance, by minimizing for all $L \in \mathring{H}^{-1}(\Omega)$, the functional

$$\mathring{H}^1(\Omega) \ni v \mapsto \int_\Omega |\nabla v|^2\,\mathrm{d}x + 2L(v).$$

The unique minimizer u (whose existence is ensured by strict convexity of the energy, and its lower semicontinuity and coercivity with respect to the weak topology), satisfies the Euler-Lagrange equation

$$\int_\Omega \langle \nabla u, \nabla v\rangle\,\mathrm{d}x + L(v) = 0 \qquad \forall v \in \mathring{H}^1(\Omega), \tag{13.10}$$

which means precisely that $\mathbf{\Delta} u = L$. Conversely, it is easily seen that if $u \in D(\Delta)$, then $L_{\Delta u} = \mathbf{\Delta} u$.

Let us now prove the second part of the statement. For $g \in L^2(\Omega)$ and $u \in \mathring{H}^1(\Omega)$, we estimate

$$|L_g(u)| = \left| \int_\Omega gu \, \mathrm{d}x \right| = \left| \int_\Omega g(u - \bar{u}) \, \mathrm{d}x \right| \le \frac{\|g\|_2}{\lambda} \|u\|_{\mathring{H}^1},$$

which proves the continuity of the embedding. Now, if we apply (13.10) to $L = L_g$, we find that the minimizer u belongs to $D(\Delta)$ and that $g = \Delta u$, so that $L_g = \boldsymbol{\Delta} u$.

By polarization, we obtain that the scalar product of $\mathring{H}^{-1}(\Omega)$ is given by

$$\langle L, L' \rangle = \int_\Omega \langle \nabla \boldsymbol{\Delta}^{-1} L, \nabla \boldsymbol{\Delta}^{-1} L' \rangle \, \mathrm{d}x. \tag{13.11}$$

Then, if $L = \boldsymbol{\Delta} u \in \mathring{H}^{-1}(\Omega)$ is orthogonal to all functions $\boldsymbol{\Delta} v$ with $v \in D(\Delta)$, we obtain

$$\int_\Omega \langle \nabla u, \nabla v \rangle \, \mathrm{d}x = 0 \qquad \forall v \in D(\Delta).$$

By the density of $D(\Delta)$ in $\mathring{H}^1(\Omega)$ it follows that $\nabla u = 0$, hence $u = 0$. This proves the density of the embedding of L^2 in $\mathring{H}^{-1}$. □

In view of the comparison with the Wasserstein metric tensor of Lecture 18, it is convenient to write (13.11) in the form

$$\langle L, L' \rangle = \int_\Omega \langle \nabla g, \nabla g' \rangle \, \mathrm{d}x \qquad \text{where } \boldsymbol{\Delta} g = L, \ \boldsymbol{\Delta} g' = L'. \tag{13.12}$$

Example 13.5 (Another Interpretation of the Heat Equation as Gradient Flow) Let $\mathbf{E} : \mathring{H}^{-1}(\Omega) \to [0, \infty]$ be defined by

$$\mathbf{E}(L) := \begin{cases} \frac{1}{2} \int_\Omega u^2 \, \mathrm{d}x & \text{if } L = L_u, \ u \in L^2(\Omega); \\ +\infty & \text{otherwise} \end{cases}$$

and let us check that the gradient flow of the convex, lower semicontinuous and densely defined (i.e. with a dense finiteness domain) functional $\mathbf{E}$ provides another variational interpretation of the heat equation (11.6). According to this new interpretation, both sides are not viewed as elements of $L^2(\Omega)$, but as elements of $\mathring{H}^{-1}(\Omega)$, with homogeneous Neumann boundary conditions on $\partial\Omega$.

Indeed, we need only to prove this characterization of the subdifferential:

$$\partial \mathbf{E}(L_u) \neq \emptyset \qquad \Longleftrightarrow \qquad u \in D(\Delta) \subset H^1(\Omega)$$

with the additional property $\partial \mathbf{E}(L_u) = \{-L_{\Delta u}\}$ whenever the subdifferential is not empty. Having this characterization, one obtains the equation

$$\frac{\mathrm{d}}{\mathrm{d}t} L_{u(t,\cdot)} = L_{\Delta u(t,\cdot)}, \qquad \lim_{t\to 0^+} u(t,\cdot) = \bar{L}$$

where both the time derivative and the initial condition are understood in the Hilbert space $\mathring{H}^{-1}(\Omega)$. This implies in particular the existence of the "weak derivative", testing with L_v, $v \in H^1(\Omega)$:

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_\Omega u(t,\cdot) v \,\mathrm{d}x = -\int_\Omega \langle \nabla u(t,\cdot), \nabla v\rangle \,\mathrm{d}x. \tag{13.13}$$

The latter formulation, weaker than the classical one of Examples 11.6 and 13.2, is very close to the formulation of evolution problems in the space of probability measures (and of the continuity equation) that will appear in the next sections and that allows also to treat measures not absolutely continuous with respect to $\mathscr{L}^n$. Notice also that, when we consider an initial condition $\bar{L}$ in $D(\mathbf{E})$, namely of the form $L_{\bar{u}}$ for some $\bar{u} \in L^2(\Omega)$, then the mean value of $\bar{u}$ is preserved. In particular, if $\bar{u}$ has null mean value, then $u(t,\cdot) \in \mathring{H}^1(\Omega)$ and we can also write (13.13) in the form

$$\frac{\mathrm{d}}{\mathrm{d}t} L_{u(t,\cdot)} = \mathbf{\Delta} u(t,\cdot).$$

In order to prove the claimed characterization of the subdifferential we follow the same strategy of Example 11.6 for the proof of $\Rightarrow$ (the proof of the converse implication is similar). Let $L_u \in D(\mathbf{E})$, $\xi = \mathbf{\Delta}\xi' \in \partial \mathbf{E}(L_u)$ with $\xi' \in \mathring{H}^1(\Omega)$ and let us choose a perturbation $L_v = \mathbf{\Delta} v'$ with $v = \Delta v'$, $v' \in D(\Delta)$. From the inequality

$$\mathbf{E}(L_u + \epsilon L_v) \geq \mathbf{E}(L_u) + \epsilon \langle L_v, \xi\rangle \qquad \forall \epsilon \in \mathbb{R}\setminus\{0\}$$

(notice that the scalar product is the one of $\mathring{H}^{-1}$!) we get

$$O(\epsilon^2) + \epsilon \int_\Omega uv \,\mathrm{d}x \geq \epsilon \int_\Omega \langle \nabla v', \nabla \xi'\rangle \,\mathrm{d}x.$$

Therefore, dividing by ϵ and taking into account the fact that ϵ can be positive and negative we get

$$\int_\Omega uv \,\mathrm{d}x = \int_\Omega \langle \nabla v', \nabla \xi'\rangle \,\mathrm{d}x \qquad \forall v' \in D(\Delta).$$

Writing it in the form

$$\int_\Omega (u + \xi') v \,\mathrm{d}x = 0 \qquad \forall v = \Delta v', \;\; v' \in D(\Delta)$$

and using the fact that $\{\Delta v' : v' \in D(\Delta)\}$ coincides with all $L^2(\Omega)$ functions with 0 mean we obtain $\xi' = -u + c$ for some $c \in \mathbb{R}$, hence $u \in H^1(\Omega)$ and $\xi = \mathbf{\Delta}\xi' = -L_{\Delta u}$.

Example 13.6 (The Porous Medium Equation) More generally, if $m > 0$, let us consider the evolution equation (called porous medium equation)

$$\frac{d}{dt}u = \Delta u^m, \tag{13.14}$$

with the homogeneous Neumann boundary condition

$$\frac{\partial u^m}{\partial n} = 0$$

and let us look for nonnegative solutions, given a nonnegative initial condition $\bar{u} \in L^2(\Omega)$. For simplicity we shall consider only the case $m > 1$, the so-called slow diffusion ($0 < m < 1$ is the so-called fast diffusion). One is led to define the functional $\mathbf{E}_m : \mathring{H}^{-1}(\Omega) \to [0, \infty]$ as follows:

$$\mathbf{E}_m(L) := \begin{cases} \frac{1}{m+1}\int_\Omega |u|^{m+1}\,\mathrm{d}x & \text{if } L = L_u,\ u \in L^{m+1}(\Omega); \\ \\ +\infty & \text{otherwise.} \end{cases}$$

The functional $\mathbf{E}_m$ is convex, lower semicontinuous, with a dense domain and produces, in the space $H = \mathring{H}^{-1}(\Omega)$, solutions to the porous medium equation in the "signed variant" $\frac{\mathrm{d}u}{\mathrm{d}t} = \Delta(|u|^m \operatorname{sign} u)$, because of the following characterization of the subdifferential:

$$\partial\mathbf{E}_m(L_u) \neq \emptyset \qquad \Longleftrightarrow \qquad u \in L^{m+1}(\Omega) \text{ and } |u|^m \operatorname{sign} u \in D(\Delta) \subset H^1(\Omega)$$

with $\partial\mathbf{E}_m(L_u) = \{-\mathbf{\Delta}(|u|^m \operatorname{sign} u)\}$ whenever the subdifferential is not empty.

This characterization of the subdifferential can be obtained arguing exactly as in the previous example, with the only additional care of making perturbations $v = \Delta v'$ with $v' \in D(\Delta)$ and $\Delta v' \in L^{m+1}(\Omega)$.

To obtain solutions of (13.14) we need only to prove that the positive sign of the initial condition $\bar{u}$ is preserved. The *formal* proof is based on the differentiation

$$t \mapsto \int_\Omega \beta(u(t,\cdot))\,\mathrm{d}x \qquad \text{with } \beta(z) := \sqrt{1 + (z^-)^2} - 1$$

and on an integration by parts, which show that the function is nonincreasing and therefore identically null in $[0, \infty)$ (being null at $t = 0$). However, making this argument rigorous requires several regularizations that are beyond the scope of these notes.

Lecture 14: Gradient Flows: The EDE and EDI Formulations



In this lecture we look at the convergence of the implicit Euler scheme with a more variational perspective, providing convergence results valid in general metric spaces (E, d). A basic reference for this part is [12].

1 EDE, EDI Solutions and Upper Gradients

Let us begin by describing the heuristic idea behind De Giorgi's approach to gradient flows, based on maximal energy dissipation rate (see [47, 48]).

Let us consider the general gradient flow

$$x'(t) = -\nabla f(x(t)), \qquad x : (0, \infty) \to \mathbb{R}^n, \tag{14.1}$$

for some differentiable map $f : \mathbb{R}^n \to \mathbb{R}$.

De Giorgi noticed that there is a way to encode this system of equations in a *single* differential inequality. Indeed, if we take any absolutely continuous curve $y(t)$, we have

$$-\frac{\mathrm{d}}{\mathrm{d}t} f(y(t)) = -\langle \nabla f(y(t)), y'(t)\rangle \le |\nabla f(y(t))||y'(t)| \le \frac{1}{2}|\nabla f(y(t))|^2 + \frac{1}{2}|y'(t)|^2, \tag{14.2}$$

for $\mathscr{L}^1$-a.e. $t > 0$. Now, the first inequality is an equality if and only if the vectors $-\nabla f(y(t))$ and $y'(t)$ are parallel, while the second one is an equality if and only if $|\nabla f(y(t))| = |y'(t)|$. Hence, equality in (14.2) encodes the gradient flow system and we can say that $y(t)$ is a solution to (14.1) if and only if

$$-\frac{\mathrm{d}}{\mathrm{d}t} f(y(t)) \ge \frac{1}{2}|\nabla f(y(t))|^2 + \frac{1}{2}|y'(t)|^2. \tag{14.3}$$

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In the abstract setting of metric spaces, $|y'(t)|$ can be interpreted as the metric derivative and $|\nabla f|$ can be interpreted as the descending slope $|\nabla^- f|$, whose role has already been emphasized in the Hilbertian theory of gradient flows. Then, an integral formulation of (14.3) leads to the following definition.

Definition 14.1 (EDI and EDE Solutions) Let (E, d) be a metric space, $f : E \to (-\infty, \infty]$, $x \in AC_{\mathrm{loc}}([0, \infty); E)$ be a lower semicontinuous function with $x(0) = \bar{x} \in \mathrm{Dom}(f)$. We say that $x(t)$ is an EDI (energy dissipation inequality) solution starting at $\bar{x}$ if

$$\int_0^t \frac{1}{2}|x'|^2(s) + \frac{1}{2}|\nabla^- f|^2(x(s))\,\mathrm{d}s + f(x(t)) \le f(\bar{x}) \qquad \forall t \ge 0. \tag{14.4}$$

We say that $x(t)$ is an EDE (energy dissipation equality) solution if equality holds in (14.4).

Remark 14.2 Let us point out that the lower semicontinuity of f yields that the descending slope is a Borel function. Indeed $\left|\nabla^- f\right|$ is the monotone limit of the functions (g_r) as $r \downarrow 0$, where we set

$$g_r(x) := \sup_{y \in B_r(x)\setminus\{x\}} \frac{(f(x) - f(y))^+}{d(x, y)}$$

and the functions g_r are lower semicontinuous due to the lower semicontinuity of f.

We observe that, if E is a Hilbert space and $f : E \to (-\infty, \infty]$ is convex and lower semicontinuous, then EVI implies EDE. Indeed, in this case $f \circ x$ is absolutely continuous in $[0, \infty)$ thanks to Proposition 11.9(ii), that provides the local Lipschitz property of $f \circ x$ and the inequalities $0 \le f \circ x \le f(\bar{x})$.

Definition 14.3 (Upper Gradient) Given a map $f : E \to \mathbb{R}$, we say that a Borel function $g : E \to [0, \infty)$ is an upper gradient of f, and we write $g \in UG(f)$, if

$$|f(\gamma(1)) - f(\gamma(0))| \le \int_\gamma g := \int_0^1 g(\gamma(t))|\gamma'|(t)\,\mathrm{d}t, \tag{14.5}$$

for any $\gamma \in AC([0, 1]; E)$.

Remark 14.4 In order to verify that g is an upper gradient it is sufficient to verify (14.5) only for curves $\gamma \in \mathrm{Lip}([0, 1]; E)$. This is a consequence of the invariance of the right hand side of (14.5) under reparameterization and of the existence of constant speed reparameterizations of absolutely continuous curves provided by Proposition 9.6.

Basically, upper gradients g control the oscillation of f on any absolutely continuous curve: roughly speaking, one may say that g is a pseudo-modulus of the gradient of f.

One of the most popular notions of pseudo-modulus of gradient, in the metric setting, is the notion of *slope* (compare with the descending slope $|\nabla^- f|$ that we introduced in Definition 11.16)

$$|\nabla f|(x) := \limsup_{y \to x} \frac{|f(y) - f(x)|}{d(y, x)}, \tag{14.6}$$

set equal to 0 at isolated points of E. Notice that, in the Hilbert setting and at differentiability points, the metric slope coincides with the modulus of the gradient, so our notation is justified. In the following remark we compare these notions.

Remark 14.5 (Properties of Upper Gradients)

(i) If E is a Banach space and $f \in C^1(E)$, then $|\nabla f|_* \in UG(f)$ (where $|\cdot|_*$ denotes the dual norm) and this is the smallest possible g which can be used in (14.5).

(ii) It is not always true that the metric slope $|\nabla f|$ is an upper gradient, since the validity of this property depends on the regularity of f. It is easy to prove the upper gradient property of $|\nabla f|$ when f is locally Lipschitz (see also (iv) below). On the other hand, if $f : [0, 1] \to [0, 1]$ is the Cantor-Vitali function, then $f' = 0$ $\mathscr{L}^1$-a.e. in $(0, 1)$ and so the metric slope $|f'|$ does not belong to $UG(f)$, since $f(1) = 1$, $f(0) = 0$ and one can choose the curve $\gamma(t) = t$.

(iii) If $g \in UG(f)$ and $\int_\gamma g < \infty$, then $t \mapsto f(\gamma(t))$ is absolutely continuous in $[0, 1]$, since from (14.5) and the invariance of curvilinear integrals under reparameterization it follows that

$$|f(\gamma(t)) - f(\gamma(s))| \le \int_{\gamma|_{[s,t]}} g = \int_s^t g(\gamma(r))|\gamma'|(r)\,\mathrm{d}r \qquad \text{for any } 0 \le s \le t \le 1.$$

(iv) If $f \in \mathrm{Lip}(E)$, then $|\nabla^- f| \in UG(f)$. Indeed, let $\gamma \in AC([0, 1]; E)$, then $f \circ \gamma \in AC([0, 1]; \mathbb{R})$, therefore, by the fundamental theorem of calculus, we obtain

$$|f(\gamma(1)) - f(\gamma(0))| \le \int_0^1 |(f \circ \gamma)'(r)|\,\mathrm{d}r. \tag{14.7}$$

Fix now $r \in (0, 1)$ where the metric derivative of γ and the derivative of $f \circ \gamma$ exist, and suppose that $(f \circ \gamma)'(r) \ge 0$: this means that for any $h > 0$ sufficiently small we have

$$h(f \circ \gamma)'(r) + o(h) = f(\gamma(r)) - f(\gamma(r - h)) \le h|\gamma'|(r)|\nabla^- f|(\gamma(r)) + o(h),$$

which implies $(f\circ\gamma)'(r)\le|\gamma'|(r)|\nabla^- f|(\gamma(r))$. If, instead, $(f\circ\gamma)'(r)\le 0$, still for $h>0$ sufficiently small one has

$$-h(f\circ\gamma)'(r)+o(h)\le f(\gamma(r))-f(\gamma(r+h))\le h|\gamma'|(r)|\nabla^- f|(\gamma(r))+o(h),$$

which implies finally $|(f\circ\gamma)'(r)|\le|\gamma'|(r)|\nabla^- f|(\gamma(r))$ and, in combination with (14.7), the upper gradient property.

Proposition 14.6 *In a metric space (E,d), if $|\nabla^- f|\in UG(f)$ and $x(t)$ satisfies* EDI*, then:*

(i) EDE *holds;*
(ii) $f\circ x\in AC_{\rm loc}([0,\infty))$ *and* $-(f\circ x)'=|x'|^2=|\nabla^- f|^2\circ x$ $\mathscr{L}^1$*-a.e. in* $(0,\infty)$.

Proof For any EDI curve $x(t)$ we have that $f\circ x$ is absolutely continuous, and so, by the fundamental theorem of calculus and the definition of upper gradient, we obtain

$$f(x(0))\le f(x(t))+\int_0^t|x'|(s)|\nabla^- f|(x(s))\,\mathrm{d}s.$$

Applying the Cauchy–Schwarz inequality and taking into account the EDI property (14.4) we get

$$f(x(0))\le f(x(t))+\frac{1}{2}\int_0^t|x'|^2(s)\,\mathrm{d}s+\frac{1}{2}\int_0^t|\nabla^- f|^2(x(s))\,\mathrm{d}s\le f(x(0)).$$

Therefore EDE holds and, since equality holds in the Cauchy–Schwarz inequality, $-(f\circ x)'=|x'|^2=|\nabla^- f|^2\circ x$ $\mathscr{L}^1$-a.e. in $(0,\infty)$.

□

2 Existence of EDE, EDI Solutions

Now we can state a basic existence result of EDI gradient flows. Recall that a function $f:E\to(-\infty,\infty]$ is said to be *coercive* if all sublevel sets $\{f\le t\}$, $t\in\mathbb{R}$, are relatively compact. In particular, sublevel sets of coercive and lower semicontinuous functions are compact.

Theorem 14.7 (De Giorgi) *Let (E,d) be a metric space, let $f:E\to[0,\infty]$ be lower semicontinuous and coercive, and assume that $|\nabla^- f|$ is lower semicontinuous. Then there exists an* EDI *solution starting from any $\bar{x}\in Dom(f)$.*

In general, we cannot hope to have uniqueness of the solution, as it is shown by the following example.

Example 14.8 Let $E = \mathbb{R}^2$, equipped with the ℓ_∞ norm, $\bar{x} = (0,0)$, and $f(x_1, x_2) = x_1$. Then, any curve $\gamma(t) = (-t, a(t))$, with $|a'(t)| \leq 1$, $a(0) = 0$, is an EDE solution. Indeed, $f(\gamma(t)) = -t$, $|\gamma'|(t) = \|(-1, a'(t)\|_\infty = 1$, $|\nabla^- f| = |\nabla f| = 1$, and so we have

$$-t + \int_0^t \frac{1}{2} + \frac{1}{2}\,\mathrm{d}s = 0.$$

It is possible to recover the stronger notion of EDE solution by adding a convexity assumption.

Definition 14.9 (λ-Convexity in Metric Spaces) Given $\lambda \in \mathbb{R}$ and $f : E \to [0, \infty]$, we say that f is λ-convex if for all $x_0, x_1 \in \mathrm{Dom}(f)$ there exists $\gamma \in \mathrm{Geo}(E)$ such that $\gamma(0) = x_0$, $\gamma(1) = x_1$ and $s \mapsto f(\gamma(s))$ is $\lambda d^2(x_0, x_1)$-convex. Equivalenty, this corresponds to the inequality

$$f(\gamma((1-t)a + tb))) \leq (1-t) f(\gamma(a))) + t f(\gamma(b)) - \frac{1}{2}\lambda d^2(\gamma(a), \gamma(b)) t(1-t)$$

for all $a,\ b \in [0,1]$ and all $t \in [0,1]$.

For λ-convex functions in geodesic spaces (compare with (11.19)), thanks to the monotonicity of difference quotients along constant speed geodesics, one still has

$$|\nabla^- f|(u) = \sup_{v \neq u} \left[\frac{f(v) - f(u)}{d(v,u)} - \frac{\lambda}{2} d(v,u) \right]^-. \tag{14.8}$$

The following real analysis lemma will be useful to prove the upper gradient property of the descending slope in the λ-convex case.

Lemma 14.10 *Let $g : [0,1] \to \mathbb{R}$ be a lower semicontinuous function and let $h \in L^1(0,1)$ satisfy*

$$g(s) \leq g(t) + |s - t| h(s) \qquad \forall s,\ t \in [0,1]. \tag{14.9}$$

Then $g \in AC(0,1)$, with $|g'| \leq h$ $\mathscr{L}^1$-a.e. in $(0,1)$.

In the proof of Lemma 14.10 we will use the basic compactness criterion with respect to the weak L^1 topology, we refer to Theorem 1.38 of [10] for a proof.

Theorem 14.11 (Dunford–Pettis) *Let $(\Omega, \mathcal{E}, \theta)$ be a finite measure space and let $\mathcal{F}$ be a bounded subset of $L^1(\Omega, \mathcal{E}, \theta)$. Then the following conditions are equivalent:*

(i) $\mathcal{F}$ is sequentially compact with respect to the weak topology of $L^1(\Omega, \mathcal{E}, \theta)$;
(ii) for any $\epsilon > 0$ there exists $\delta > 0$ such that $\theta(A) < \delta$ implies $\sup_{f \in \mathcal{F}} \int_A |f|\,\mathrm{d}\theta < \epsilon$;
(iii) there exists $\varphi : [0, \infty) \to [0, \infty]$ with more than linear growth at infinity such that $\int_\Omega \varphi(|f|)\,\mathrm{d}\theta \leq 1$ for any $f \in \mathcal{F}$.

Proof of Lemma 14.10 The assumption (14.9) can be symmetrized to get

$$|g(s) - g(t)| \le |s - t|(h(s) + h(t)) \qquad \forall s,\ t \in [0, 1].$$

Fix $[a, b] \subset (0, 1)$. Choosing $s = t + h$, the difference quotients

$$\Delta_h u(t) := h^{-1}(g(t + h) - g(t))$$

are uniformly bounded in $L^1(a, b)$ by $h(t) + h(t + h)$, which strongly converge in $L^1(a, b)$ to $2h(t)$. Therefore, they are equi-integrable, so that Dunford–Pettis Theorem 14.11 provides their relative compactness with respect to the weak topology of $L^1(a, b)$.
Passing to the limit as $h \to 0$ into the discrete integration by parts formula

$$\int_a^b \phi \Delta_h u \, \mathrm{d}t = -\int_a^b \frac{\phi(t - h) - \phi(t)}{-h} u(t) \, \mathrm{d}t, \quad \phi \in C_c^1(a, b),$$

valid for $|h|$ sufficiently small, we obtain that any weak limit is the derivative in the sense of distributions of g. This proves that g belongs to the Sobolev space $W^{1,1}(a, b)$ and that $|g'| \le 2h$. The integrability of h then gives $g \in W^{1,1}(0, 1)$.

Let now $\tilde{g} : [0, 1] \to \mathbb{R}$ be the continuous representative of g, defined by the formula

$$\tilde{g}(t) := \lim_{h \to 0^+} \frac{1}{h} \int_t^{t+h} g(s) \, \mathrm{d}s \qquad \forall t \in [0, 1)$$

(at $t = 1$ one can consider the mean values on $[t - h, t]$). The lower semicontinuity of g guarantees the inequality $\tilde{g} \ge g$ in $[0, 1]$. To prove the converse inequality, fix $t \in [0, 1)$ and let us average (14.9) to get

$$\frac{1}{h} \int_t^{t+h} g(s) \, \mathrm{d}s \le g(t) + \frac{1}{h} \int_t^{t+h} (s - t) g(s) \, \mathrm{d}s \le g(t) + \int_t^{t+h} g(s) \, \mathrm{d}s.$$

As $h \to 0^+$, we get $\tilde{g}(t) \le g(t)$ and the argument at $t = 1$ is similar.

Finally, it is easily seen that (14.9) implies $|g'(s)| \le h(s)$ at any differentiability point $s \in (0, 1)$ of g. □

Theorem 14.12 *Let E be a geodesic metric space, and let $f : E \to (-\infty, \infty]$ be lower semicontinuous, coercive and λ-convex for some $\lambda \in \mathbb{R}$. Then $|\nabla^- f|$ is lower semicontinuous, it belongs to $UG(f)$ and so there exists an* EDE *solution to the gradient flow of f.*

Proof Given $u \neq v$ and (u_n) convergent to u, for n large enough we have $u_n \neq v$ and then (14.8) gives

$$|\nabla^- f|(u_n) \geq \frac{f(u_n) - f(v)}{d(u_n, v)} - \frac{\lambda}{2} d(u_n, v).$$

It follows that

$$\liminf_{n\to\infty} |\nabla^- f|(u_n) \geq \frac{f(u) - f(v)}{d(u, v)} - \frac{\lambda}{2} d(u, v),$$

by the lower semicontinuity of f. Invoking (14.8) again we get the lower semicontinuity of $|\nabla^- f|$.

To prove the upper gradient property we rely on Lemma 14.10 assuming for simplicity that $\lambda \geq 0$. In the general case one further step is needed.
As we already pointed out, to verify (14.5) we can restrict to curves $\gamma \in \mathrm{Lip}((0, 1); E)$ and without loss of generality we can also assume that $|\gamma'| \leq 1$ a.e. in $(0, 1)$. Under these assumptions, setting $g(t) := f(\gamma(t))$ and $h(t) := |\nabla^- f|(\gamma(t))$, it follows from (14.8) that g and h verify (14.9). Therefore g is absolutely continuous with $|g'(t)| \leq |\nabla^- f|(\gamma(t))$ a.e., yielding the upper gradient property.

By Proposition 14.6 and Theorem 14.7, we conclude the existence of an EDE solution. □

3 Proof of Theorem 14.7 via Variational Interpolation

As in the Hilbertian setting, given a time step $\tau > 0$, we construct by recursion a sequence (y_i) by

$$\begin{cases} y_{i+1} \in \arg\min\{f(u) + \frac{1}{2\tau} d^2(u, y_i)\}, \\ y_0 = \bar{x}. \end{cases}$$

We recall that a solution, possibly nonunique, to this minimization problem always exists, since f is coercive.

In the Hilbertian theory we used x_τ, the piecewise affine interpolation, and $\hat{x}_\tau$, the piecewise constant interpolation, let us now present De Giorgi's idea of variational interpolation.

For any $r \in (0, 1]$ we consider

$$x_r \in \arg\min\{f(z) + \frac{1}{2\tau r} d^2(z, \bar{x})\}, \qquad x_0 := \bar{x}$$

so that we can choose $x_1 = y_1$ (notice that minimizers need not to be unique), $x_r \to \bar{x}$ as $r \to 0^+$ and

$$d^2(x_r, \bar{x}) \le 2\tau r\big(f(\bar{x}) - f(x_r)\big). \tag{14.10}$$

This provides a variational interpolation between $\bar{x} = x_0$ and $y_1 = x_1$, possibly discontinuous, but continuous at $r = 0$. The advantage of this interpolation can be appreciated if one looks at the energy dissipation rate. Let us prove, first, that the monotone function

$$\phi(r) := f(x_r) + \frac{1}{2\tau r} d^2(x_r, \bar{x}) = \min_{x \in E} f(x) + \frac{1}{2\tau r} d^2(x, \bar{x}),$$

belongs to $\mathrm{Lip}_{\mathrm{loc}}((0, 1])$. Indeed, we have that $\phi(s) \le f(x_r) + \frac{1}{2\tau s} d^2(x_r, \bar{x})$, for any $r,\ s \in (0, 1]$. Then, it follows that

$$\begin{aligned}\phi(s) - \phi(r) &\le f(x_r) + \frac{1}{2\tau s} d^2(x_r, \bar{x}) - f(x_r) - \frac{1}{2\tau r} d^2(x_r, \bar{x}) \\ &= \frac{r - s}{2\tau r s} d^2(x_r, \bar{x}),\end{aligned}$$

and

$$\begin{aligned}\phi(s) - \phi(r) &\ge f(x_s) + \frac{1}{2\tau s} d^2(x_s, \bar{x}) - f(x_s) - \frac{1}{2\tau r} d^2(x_s, \bar{x}) \\ &= \frac{r - s}{2\tau r s} d^2(x_s, \bar{x}),\end{aligned}$$

that, together, imply the local Lipschitz property.

Therefore, we have

$$\phi(1) - \phi(0_+) = \int_0^1 \phi'(r)\, \mathrm{d}r\,.$$

The lower semicontinuity of f and the definition of ϕ give

$$\phi(0_+) = f(\bar{x})\,, \quad \phi(1) = f(y_1) + \frac{1}{2\tau} d^2(y_1, \bar{x})\,.$$

In order to evaluate $\phi'(r)$, let us fix a differentiability point $r \in (0, 1)$. Then for any h such that $r + h \in (0, 1)$ we get

$$\begin{aligned}h\phi'(r) + o(h) = \phi(r + h) - \phi(r) &\le f(x_r) + \frac{1}{2\tau(r + h)} d^2(x_r, \bar{x}) - f(x_r) - \frac{1}{2\tau r} d^2(x_r, \bar{x}) \\ &= -\frac{h}{2\tau r^2} d^2(x_r, \bar{x}) + o(h).\end{aligned}$$

This proves that

$$\phi'(r) = -\frac{1}{2\tau r^2} d^2(x_r, \bar{x}).$$

Since $x_1 = y_1$, in this way we obtained an energy dissipation identity for the single step minimization:

$$f(y_1) + \frac{1}{2\tau} d^2(y_1, \bar{x}) + \frac{1}{2\tau} \int_0^1 \frac{d^2(x_r, \bar{x})}{r^2} \, \mathrm{d}r = f(\bar{x}). \tag{14.11}$$

We shall need now the following technical result, which allows us to estimate the discrete slope. It can be considered as a (very) weak formulation of the discrete Euler equation $(y - x)/\tau \in -\partial f(y)$ of the Hilbertian setting.

Lemma 14.13 (Slope Estimate) *For any $r \in (0, 1]$, we have*

$$|\nabla^- f|(x_r) \le \frac{1}{\tau r} d(x_r, \bar{x}).$$

Proof For any $z \in E$, we have

$$f(x_r) + \frac{1}{2\tau r} d^2(x_r, \bar{x}) \le f(z) + \frac{1}{2\tau r} d^2(z, \bar{x}).$$

Using the triangle inequality, we obtain

$$\begin{aligned} \frac{f(x_r) - f(z)}{d(x_r, z)} &\le \frac{1}{2\tau r d(x_r, z)} (d(z, \bar{x}) + d(x_r, \bar{x}))(d(z, \bar{x}) - d(x_r, \bar{x})) \\ &\le \frac{1}{2\tau r} (d(z, \bar{x}) + d(x_r, \bar{x})) \, . \end{aligned}$$

Passing to the limsup as $z \to x_r$ we conclude. □

From (14.11) and Lemma 14.13 it follows that

$$f(y_1) + \frac{1}{2\tau} d^2(y_1, \bar{x}) + \frac{\tau}{2} \int_0^1 |\nabla^- f|^2(x_r) \, \mathrm{d}r \le f(\bar{x}). \tag{14.12}$$

For $t \in [0, \tau]$, we define

$$\tilde{x}_\tau(t) = \begin{cases} \bar{x} & \text{if } t = 0, \\ x_{t/\tau} & \text{if } t \in (0, \tau], \end{cases}$$

and then we iterate this procedure on the intervals $[\tau, 2\tau]$, $[2\tau, 3\tau]$, . . ., in order to interpolate. We can now rewrite (14.12) using $\tilde{x}_\tau$:

$$\frac{\tau}{2}\frac{d^2(\tilde{x}_\tau(\tau), \tilde{x}_\tau(0))}{\tau^2} + \frac{1}{2}\int_0^\tau |\nabla^- f|^2(\tilde{x}_\tau(s))\,\mathrm{d}s \le f(\tilde{x}_\tau(0)) - f(\tilde{x}_\tau(\tau)).$$

Now we sum up, getting a telescopic sum in the right hand side:

$$\frac{\tau}{2}\sum_{j=0}^{n-1}\frac{d^2(\tilde{x}_\tau((j+1)\tau), \tilde{x}_\tau(j\tau))}{\tau^2} + \frac{1}{2}\int_0^{n\tau} |\nabla^- f|^2(\tilde{x}_\tau(s))\,\mathrm{d}s$$
$$\le \sum_{j=0}^{n-1}\big(f(\tilde{x}_\tau(j\tau)) - f(\tilde{x}_\tau((j+1)\tau))\big),$$

so that

$$\frac{\tau}{2}\sum_{j=0}^{n-1}\frac{d^2(\tilde{x}_\tau((j+1)\tau), \tilde{x}_\tau(j\tau))}{\tau^2} + \frac{1}{2}\int_0^{n\tau} |\nabla^- f|^2(\tilde{x}_\tau(s))\,\mathrm{d}s \le f(\tilde{x}_\tau(0)) - f(\tilde{x}_\tau(n\tau)). \tag{14.13}$$

Now we pass to the limit as $\tau \to 0$, using the fact that the set $K = \{f \le f(\bar{x})\}$ is compact, and that $\tilde{x}_\tau(t) \in K$. For $t \ge 0$ fixed, the family $\{\tilde{x}_\tau(t)\}$ is sequentially compact as $\tau \to 0^+$. Hence by a diagonal argument there exists an infinitesimal sequence $(\tau_i) \subset (0, \infty)$ satisfying

$$\lim_{i\to\infty} \tilde{x}_{\tau_i}(t) = x(t) \qquad \forall\, t \in \mathbb{Q}^+.$$

Combining the property (derived from (14.10))

$$d^2(\tilde{x}_\tau(t), \tilde{x}_\tau(\tau\lfloor\frac{t}{\tau}\rfloor)) \le 2\tau\big(\frac{t}{\tau} - \lfloor\frac{t}{\tau}\rfloor\big) f(\bar{x}) \le 2\tau f(\bar{x})$$

with the discrete $C^{0,1/2}$ regularity of the piecewise constant interpolation guaranteed by Lemma 12.7, we obtain that $x(t)$ has a $C^{0,1/2}$ extension to $[0, \infty)$, and that

$$\lim_{i\to\infty} \tilde{x}_{\tau_i}(t) = x(t) \qquad \forall\, t \ge 0.$$

For simplicity of notation, from now on we shall write only $\tau \to 0$, instead of $\tau_i \to 0$.

We set

$$v_\tau := \sum_{j=0}^{\infty} d(\tilde{x}_\tau((j+1)\tau), \tilde{x}_\tau(j\tau))\delta_{j\tau}, \qquad \mu_\tau := \sum_{j=0}^{\infty} \tau\delta_{j\tau}.$$

Then, thanks to the triangle inequality, we have

$$\nu_\tau([s,t]) \geq d(\tilde{x}_\tau(s), \tilde{x}_\tau(t)) \qquad \forall\, s,\ t \in \mathbb{N}\tau, \tag{14.14}$$

and

$$\int_0^\infty \left|\frac{\nu_\tau}{\mu_\tau}\right|^2 \mathrm{d}\mu_\tau = \tau \sum_{j=0}^{\infty} \frac{d^2(y_{j+1}, y_j)}{\tau^2} \leq 2 \sum_{j=0}^{\infty} (f(y_j) - f(y_{j+1})) \leq 2 f(\bar{x}). \tag{14.15}$$

Since μ_τ weakly converge to $\mathscr{L}^1$ in the duality with $C_c([0,\infty))$, the joint lower semicontinuity Lemma 10.1 provides (still possibly up to the extraction of a subsequence, recall that we are using a simplified notation) the weak convergence of ν_τ to a measure ν of the form $h\mathscr{L}^1$ with

$$\int_0^t h^2 \,\mathrm{d}s \leq \liminf_{\tau\to 0^+} \int_0^t \left|\frac{\nu_\tau}{\mu_\tau}\right|^2 \mathrm{d}\mu_\tau \leq 2 f(\bar{x}) \qquad \forall t \geq 0. \tag{14.16}$$

For $t > 0$, we set $N_\tau(t) := \lfloor t/\tau \rfloor$, so that $\tau N_\tau(t) \leq t$ and $\tau N_\tau(t) \to t$ as $\tau \to 0$. Passing to the limit as $\tau \to 0$ in (14.14), we obtain

$$\begin{aligned} d(x(s), x(t)) &= \lim_{\tau\to 0^+} d(\tilde{x}_\tau(\tau \lfloor \frac{s}{\tau} \rfloor), \tilde{x}_\tau(\tau \lfloor \frac{t}{\tau} \rfloor)) \\ &\leq \limsup_{\tau\to 0^+} \nu\big([\tau \lfloor \frac{s}{\tau} \rfloor, \tau \lfloor \frac{t}{\tau} \rfloor)]\big) \leq \nu([s,t]) \\ &= \int_s^t h(r)\,\mathrm{d}r \qquad \text{whenever } 0 \leq s \leq t. \end{aligned}$$

This shows that $x \in AC^2([0,\infty); E)$, since $h \in L^2(0,\infty)$, and that $|x'| \leq h$.

By selecting $n = N_\tau(t)$ in (14.13) we get

$$f(\tilde{x}_\tau(\tau N_\tau(t))) + \frac{1}{2}\int_0^{\tau N_\tau(t)} \left|\frac{\nu_\tau}{\mu_\tau}\right|^2 \mathrm{d}\mu_\tau + \frac{1}{2}\int_0^{\tau N_\tau(t)} |\nabla^- f|^2(\tilde{x}_\tau(s))\,\mathrm{d}s \leq f(\bar{x}).$$

Finally, passing to the limit as $\tau \to 0^+$, from the lower semicontinuity of f and $|\nabla^- f|$ and (14.16), we obtain

$$f(x(t)) + \frac{1}{2}\int_0^t h^2(s)\,\mathrm{d}s + \frac{1}{2}\int_0^t |\nabla^- f|^2(x(s))\,\mathrm{d}s \leq f(\bar{x}),$$

that implies, thanks to the inequality $h \geq |x'|$,

$$f(x(t)) + \frac{1}{2}\int_0^t |x'|^2(s)\,\mathrm{d}s + \frac{1}{2}\int_0^t |\nabla^- f|^2(x(s))\,\mathrm{d}s \leq f(\bar{x}).$$

This proves that $x(t)$ is an EDI solution to the gradient flow of f.

We prove now general inequalities related to the convexity of f. Among the applications that these inequalities have there is the exponential convergence to the equilibrium, as $t \to \infty$, of EDE solutions to gradient flows.

Proposition 14.14 *Let E be a geodesic metric space and let $f : E \to (-\infty, \infty]$ be λ-convex, for some $\lambda > 0$. Then, if $\bar{x} \in Dom(f)$ is the unique strict minimizer of f, we have:*

(i) the energy-energy dissipation inequality

$$f(x) - f(\bar{x}) \leq \frac{1}{2\lambda}|\nabla^- f|^2(x) \quad \forall x \in Dom(f); \tag{14.17}$$

(ii) the energy inequality

$$f(x) - f(\bar{x}) \geq \frac{\lambda}{2}d^2(x, \bar{x}) \quad \forall x \in E. \tag{14.18}$$

In addition, by the energy inequality it follows that EDE *solutions $x(t)$ to the gradient flow of f satisfy*

$$f(x(t)) - f(\bar{x}) \leq (f(x(0)) - f(\bar{x}))e^{-2\lambda t}, \quad d^2(x(t), \bar{x}) \leq \frac{2}{\lambda}(f(x(0)) - f(\bar{x}))e^{-2\lambda t}. \tag{14.19}$$

Proof We start by proving (14.17) in a Hilbert space. By λ-convexity and the Young inequality, we have

$$\begin{aligned} f(\bar{x}) - f(x) &\geq \langle \nabla f(x), \bar{x} - x\rangle + \frac{\lambda}{2}|\bar{x} - x|^2 \\ &\geq -|\nabla f(x)||\bar{x} - x| + \frac{\lambda}{2}|\bar{x} - x|^2 \\ &\geq -\frac{1}{2\lambda}|\nabla f(x)|^2 - \frac{\lambda}{2}|\bar{x} - x|^2 + \frac{\lambda}{2}|\bar{x} - x|^2. \end{aligned}$$

In the metric setting, it suffices to apply the Hilbertian inequality to the $\lambda d^2(x, \bar{x})$-convex function $g(t) = f(\gamma(t))$, where $\gamma \in \mathrm{Geo}(E)$ joins $\bar{x}$ to x, taking the inequality

$$|\nabla^- g|(1) \leq |\nabla^- f|(x)d(x, \bar{x})$$

into account.

In order to prove (14.18), we consider once more $\gamma \in \mathrm{Geo}(E)$ from $\bar{x}$ to x. Then, by λ-convexity and minimality of f at $\gamma(0) = \bar{x}$, we have

$$0 \le \frac{f(\gamma(t)) - f(\gamma(0))}{t} \le f(\gamma(1)) - f(\gamma(0)) - \frac{\lambda}{2}(1-t)d^2(\gamma(1), \gamma(0)) \qquad \forall t \in (0, 1].$$

As $t \to 0^+$ we obtain

$$f(x) - f(\bar{x}) - \frac{\lambda}{2} d^2(x, \bar{x}) \ge 0.$$

Finally, we show (14.19). Since $x(t)$ is an EDE solution, we have

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}(f(x(t)) - f(\bar{x})) &= -\frac{1}{2}(|x'|^2(t) + |\nabla^- f|^2(x(t))) \\ &= -|\nabla^- f|^2(x(t)) \le -2\lambda(f(x(t)) - f(\bar{x})), \end{aligned}$$

by (14.17), for $\mathscr{L}^1$-a.e. $t > 0$. By Gronwall lemma, we obtain

$$f(x(t)) - f(\bar{x}) \le (f(x(0)) - f(\bar{x}))e^{-2\lambda t}.$$

Thus, by (14.18) we can conclude. □

Lecture 15: Semicontinuity and Convexity of Energies in the Wasserstein Space

We are now interested in the semicontinuity and convexity properties of the following three types of energy functionals defined on measures: internal energy, potential energy and interaction energy.

1 Semicontinuity of Internal Energies

Definition 15.1 (Relative Entropy) Let (X, τ) be a Polish space and choose a reference measure $\mu \in \mathscr{P}(X)$. We define the relative entropy functional Ent_μ with respect to μ by

$$\mathrm{Ent}_\mu(\nu) := \begin{cases} \int_X \varrho \ln \varrho \, \mathrm{d}\mu & \text{if } \nu = \varrho\mu \in \mathscr{P}(X) \\ +\infty & \text{otherwise.} \end{cases}$$

Let us point out that $\mathrm{Ent}_\mu : \mathscr{P}(X) \to [0, \infty]$. Indeed the energy density $h(s) := s \ln s$ is convex in $[0, \infty)$ and by Jensen's inequality we get

$$\mathrm{Ent}_\mu(\varrho\mu) = \int_X \varrho \ln \varrho \, \mathrm{d}\mu \geq \left(\int_X \varrho \, \mathrm{d}\mu \right) \ln \left(\int_X \varrho \, \mathrm{d}\mu \right) = 0.$$

Let us remark also that the negative part of h will be a source of problems when trying to change the probability reference measure μ into a σ-finite measure. But, even in the case when $\mu \in \mathscr{P}(X)$, it might be better to replace the integrand h with a nonnegative one. To this aim we define $e(s) := s \ln s + (1 - s)$. Observe that e is nonnegative, strictly convex and has a unique minimum point at $s = 1$. Furthermore

$$\mathrm{Ent}_\mu(\nu) = \int_X \varrho \ln \varrho \, \mathrm{d}\mu = \int_X e(\varrho) \, \mathrm{d}\mu, \tag{15.1}$$

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since $\int_X (1-\varrho)\,\mathrm{d}\mu = 0$. Basically we added a so-called "null-Lagrangian" to our functional.

Remark 15.2 Due to the strict convexity of the integrand, $\mathrm{Ent}_\mu(\nu) = 0$ if and only if $\nu = \varrho\mu \ll \mu$ and $\varrho \equiv 1$, otherwise it is strictly positive. Thanks to this property the relative entropy can be considered as an "asymmetric distance" between probability measures.

Take now any $f : [0,\infty) \to [0,\infty]$ proper, lower semicontinuous, convex, and with more than linear growth, namely $f(z)/z \to \infty$ as $z \to \infty$. We can generalize (15.1), defining

$$\mathcal{H}_\mu(\nu) := \begin{cases} \int_X f(\varrho)\,\mathrm{d}\mu & \text{if } \nu = \varrho\mu \\ +\infty & \text{otherwise.} \end{cases}$$

Sometimes we will make explicit the dependence on the energy density f, adopting the notation $\mathcal{H}^f_\mu$ instead of $\mathcal{H}_\mu$.

We define $f^* : \mathbb{R} \to \mathbb{R}$ by $f^*(s^*) := \sup_{s\geq 0}\{ss^* - f(s)\}$ and we observe that, extending the function f to $(-\infty, 0)$ with the value $+\infty$ (this extension preserves convexity and lower semicontinuity), f^* coincides with the convex conjugate function introduced in Definition 3.3. The function f^* is obviously nondecreasing, convex and lower semicontinuous. Furthermore, thanks to the more than linear growth assumption on f, f^* is real valued and in particular locally Lipschitz.

Proposition 15.3 *If $f : [0,\infty) \to [0,\infty]$ is proper, lower semicontinuous, convex, and with more than linear growth, one has*

$$\mathcal{H}^f_\mu(\nu) = \sup\left\{ \int_X S^*\,\mathrm{d}\nu - \int_X f^*(S^*)\,\mathrm{d}\mu : S^* \in C_b(X) \right\}. \tag{15.2}$$

Proof We define $\mathcal{H}'_\mu(\nu)$ to be the expression on the right-hand side of (15.2). We begin by proving that $\mathcal{H}_\mu(\nu) \geq \mathcal{H}'_\mu(\nu)$. We can assume without loss of generality that $\nu \ll \mu$ and $\nu = \varrho\mu$. It follows from the definition of f^* that $ss^* - f^*(s^*) \leq f(s)$ for any $s, s^* \in \mathbb{R}$. Choosing $s = \varrho(x)$ and $s^* = S^*(x)$ the desired conclusion is obtained by integration with respect to μ.

To prove the converse inequality we can assume that $\mathcal{H}'_\mu(\nu) < \infty$. Under this assumption we need to show that $\nu \ll \mu$ and that $\mathcal{H}_\mu(\nu) \leq \mathcal{H}'_\mu(\nu)$. For any $B \in \mathscr{B}(X)$ such that $\mu(B) = 0$ we can find a sequence (K_n) of compact sets and a sequence (A_n) of open sets such that $K_n \subset B \subset A_n$ and $(\mu+\nu)(A_n \setminus K_n) \to 0$. For any fixed $s^* > 0$ we can find a sequence $(\xi_n) \subset C_b(X)$ such that $\xi_n = s^*$ on K_n, $\xi_n = 0$ on $X \setminus A_n$ and $0 \leq \xi_n \leq s^*$. It is clear that, as $n \to \infty$, ξ_n converge to $s^*\chi_B$ in $L^1(\mu+\nu)$ and in particular $\xi_n \to 0$ in $L^1(\mu)$, since $\mu(B) = 0$. The functions ξ_n are admissible in the definition of $\mathcal{H}'_\mu(\nu)$ so that

$$\int_X \xi_n\,\mathrm{d}\nu - \int_X f^*(\xi_n)\,\mathrm{d}\mu \leq \mathcal{H}'_\mu(\nu) < \infty \tag{15.3}$$

and, passing to the limit as n goes to infinity in (15.3), we obtain $s^*\nu(B) - f^*(0) \le \mathcal{H}'_\mu(\nu)$. Thus we conclude that $\nu(B) \le (f^*(0)+\mathcal{H}'_\mu(\nu))/s^*$ and since s^* is arbitrary it follows that $\nu(B) = 0$. From now on we assume that $\nu = \varrho\mu$ with ϱ Borel. Under this assumption we get the simpler expressions for $\mathcal{H}'_\mu$, namely

$$\mathcal{H}'_\mu(\varrho\mu) = \sup\left\{\int_X (\varrho S^* - f^*(S^*))\,\mathrm{d}\mu : S^* \in C_b(X)\right\},$$

and

$$\mathcal{H}_\mu(\varrho\mu) = \int_X \sup_{s^*\in\mathbb{Q}} \{\varrho s^* - f^*(s^*)\}\,\mathrm{d}\mu, \tag{15.4}$$

where the representation in (15.4) is a consequence of the fact that $(f^*)^* = f$. Now we introduce an enumeration of rational numbers $\mathbb{Q} = \{q_n\}_{n\in\mathbb{N}}$ and the family of approximating functionals $\mathcal{H}^k_\mu$ defined by

$$\mathcal{H}^k_\mu(\varrho\mu) := \int_X \sup_{s^*\in\{q_0,\dots,q_k\}} (\varrho(x)s^* - f^*(s^*))\,\mathrm{d}\mu(x).$$

Observe that $\mathcal{H}^k_\mu(\nu) \uparrow \mathcal{H}_\mu(\nu)$ as $k \to \infty$ by Beppo-Levi's theorem (notice that the μ-integrable function $\varrho q_1 - f^*(q_1)$ provides the lower bound needed for the application of the theorem), so for our purposes it suffices to show that $\mathcal{H}' \ge \mathcal{H}^k$ for any fixed $k \in \mathbb{N}$. To this aim we observe that

$$\mathcal{H}^k_\mu(\nu) = \sup\left\{\int_X (\varrho S^* - f^*(S^*))\,\mathrm{d}\mu : S^* \text{ is a simple function with values in } \{q_0,\dots,q_k\}\right\}.$$

Indeed, if we call

$$A_j := \left\{x \in X : q_j\varrho(x) - f^*(q_j) \ge q_i\varrho(x) - f^*(q_i) \ \ \forall\, i \in \{0,\dots,k\}\right\}$$

and

$$A'_0 := A_0, \qquad A'_{j+1} := A_{j+1} \setminus \bigcup_{i=0}^{j} A_i,$$

then all the sets A'_j are Borel and we can define S^* to be equal to q_i on A'_i. Whence the desired conclusion, as any step function can be approximated by a sequence of uniformly bounded continuous functions. □

The following proposition extends Lemma 10.1, where the case of power functions f was considered.

Theorem 15.4 *Assume that $f : [0, \infty) \to [0, \infty]$ is proper, lower semicontinuous, convex, and with more than linear growth. Then the functional $\mathscr{P}(X) \times \mathscr{P}(X) \ni (\mu, \nu) \mapsto \mathcal{H}^f_\mu(\nu)$ is lower semicontinuous.*

Proof The desired conclusion follows from Proposition 15.3 after observing that in (15.2) we are considering the supremum of a family of functionals which are jointly continuous on $\mathscr{P}(X) \times \mathscr{P}(X)$. □

In the Euclidean space $\mathbb{R}^n$ we are interested in considering $\mathscr{L}^n$ as reference measure. In order to get a good definition of the entropy functional with respect to $\mathscr{L}^n$ it is however necessary to impose some restriction on the domain of definition. In particular we will see that the finiteness of the second order moments of $\nu = \varrho \mathscr{L}^n$ ensures that $(\varrho \ln \varrho)^- \in L^1(\mathbb{R}^n)$.

Definition 15.5 We define $\mathrm{Ent} : \mathscr{P}_2(\mathbb{R}^n) \to (-\infty, \infty]$ by

$$\mathrm{Ent}(\nu) := \begin{cases} \int_X \varrho \ln \varrho \, \mathrm{d}x & \text{if } \nu = \varrho \mathscr{L}^n \\ +\infty & \text{otherwise.} \end{cases}$$

The following result relates the entropy with respect to $\mathscr{L}^n$ to the entropy with respect to $\gamma = e^{-V} \mathscr{L}^n$. It applies in particular to the case when γ is the Gaussian measure, with $V(x) = \frac{1}{2}|x|^2 + \frac{n}{2} \ln(2\pi)$.

Proposition 15.6 (Change of Reference Measure) *The entropy functional is well defined and*

$$\mathrm{Ent}(\nu) = \mathrm{Ent}_\gamma(\nu) - \int_X V \, \mathrm{d}\nu \qquad \forall \nu \in \mathscr{P}_2(\mathbb{R}^n) \tag{15.5}$$

whenever $\gamma = e^{-V} \mathscr{L}^n \in \mathscr{P}(\mathbb{R}^n)$ and V has at most quadratic growth at infinity.

Proof We begin by observing that any $\nu \in \mathscr{P}(\mathbb{R}^n)$ is absolutely continuous with respect to $\mathscr{L}^n$ if and only if it is absolutely continuous with respect to γ. More precisely, writing $\nu = \varrho \mathscr{L}^n$, one has $\nu = w\gamma$ with $w = \varrho e^V$. Then, we compute

$$\varrho \ln \varrho = (we^{-V}) \ln(we^{-V}) = w \ln we^{-V} - V\varrho. \tag{15.6}$$

Since $V\varrho \in L^1(\mathbb{R}^n)$, thanks to the quadratic growth assumption on V, and $w \ln we^{-V} \geq -e^{-1-V}$, (15.6) proves that the negative part of $\varrho \ln \varrho$ belongs to $L^1(\mathbb{R}^n)$. By integration of both sides with respect to $\mathscr{L}^n$ we recover (15.5). □

Corollary 15.7 *The functional* Ent *is lower semicontinuous on $\mathscr{P}_2(\mathbb{R}^n)$.*

Proof It follows from Proposition 15.6 that Ent can be written as the sum of the relative entropy functional with respect to the Gaussian measure, which is lower semicontinuous with respect to the weak convergence in $\mathscr{P}(\mathbb{R}^n)$ thanks to Theorem 15.4, the constant $\frac{n}{2} \ln(2\pi)$ and the functional $\int_X |x|^2 \, \mathrm{d}\nu(x) = W_2^2(\nu, \delta_0)$,

obviously continuous (see also Theorem 8.8 for the characterization of convergence in $\mathscr{P}_2(\mathbb{R}^n)$). □

Notice that Ent fails to be lower semicontinuous with respect to the duality with $C_b(\mathbb{R}^n)$: it suffices to consider a sequence $(\nu_k) \subset \mathscr{P}_2(\mathbb{R}^n)$ convergent in this sense to ν with $\mathrm{Ent}_\gamma(\nu_k)$ uniformly bounded and $W_2(\nu_k, \delta_0) \to \infty$ (for instance, in the case $n = 1$, $\nu_k = \frac{1}{2}\big[(1 - k^{-1})\chi_{(-1,1)} + k^{-2}\chi_{(-k,k)}\big]$), so that $\mathrm{Ent}(\nu_k) \to -\infty$.

Theorem 15.8 *Consider an energy density $U : [0, \infty) \to [0, \infty]$ convex, lower semicontinuous, with more than linear growth at infinity and such that $U(0) = 0$. Define $\mathbb{U} : \mathscr{P}(\mathbb{R}^n) \to [0, \infty]$ by*

$$\mathbb{U}(\mu) := \begin{cases} \int U(\varrho)\,\mathrm{d}x & \text{if } \mu = \varrho\mathscr{L}^n \\ +\infty & \text{otherwise.} \end{cases}$$

Then $\mathbb{U}$ is lower semicontinuous in $\mathscr{P}(\mathbb{R}^n)$ with respect to the topology induced by duality with $C_c(\mathbb{R}^n)$.

Proof For any $R > 0$ we define

$$\mathbb{U}_R(\mu) := \begin{cases} \int_{B_R} U(\varrho)\,\mathrm{d}x & \text{if } \mu = \varrho\mathscr{L}^n \text{ in } B_R \\ +\infty & \text{otherwise.} \end{cases}$$

Since $\mathbb{U}$ is the monotone limit of $\mathbb{U}_R$ as $R \to \infty$, it will be sufficient to prove the lower semicontinuity of $\mathbb{U}_R$.

Let us fix a sequence (ν_k) convergent to $\nu \in \mathscr{P}(\mathbb{R}^n)$ in the duality with $C_c(\mathbb{R}^n)$ and such that $\mathbb{U}_R(\nu_k) \le C$ for any $k \in \mathbb{N}$. We claim that the densities ϱ_k of ν_k with respect to $\mathscr{L}^n$ form a weakly convergent sequence in $L^1(B_R)$. To this aim, we apply Theorem 14.11 to the finite measure space $(B_R, \mathscr{B}(B_R), \mathscr{L}^n \llcorner B_R)$ and the family $\mathcal{F} = \{\varrho_k\}_{k\in\mathbb{N}}$. Observe that $\mathcal{F}$ is bounded in $L^1(B_R, \mathscr{B}(B_R), \mathscr{L}^n \llcorner B_R)$ (the dependence on the measure space will be implicit in the rest of the proof) and satisfies condition (iii) in Theorem 14.11 thanks to the more than linear growth condition on the energy density U. Thus, $\mathcal{F}$ admits limit points in the weak topology of L^1. Let ϱ^R be any of these limit points. For any test function $\psi \in C_c(B_R)$ one has

$$\int_{\mathbb{R}^n} \psi\,\mathrm{d}\nu = \lim_{k\to\infty} \int_{\mathbb{R}^n} \psi\,\mathrm{d}\nu_k = \lim_{k\to\infty} \int_{\mathbb{R}^n} \psi\varrho_k\,\mathrm{d}x = \int_{\mathbb{R}^n} \psi\varrho^R\,\mathrm{d}x.$$

Since ψ is arbitrary, it follows that $\nu \llcorner_{B_R} \ll \mathscr{L}^n$ with density ϱ^R. By compactness, we obtain that the whole sequence (ϱ_k) is weakly convergent in $L^1(B_R)$ to ϱ^R. Since R is arbitrary, it follows that $\nu \ll \mathscr{L}^n$ and that its density ϱ is equal to ϱ^R on B_R.

To conclude, it suffices to prove that $\mathbb{U}_R$ is lower semicontinuous with respect to the weak topology of $L^1(B_R)$. This property follows by the lower semicontinuity with respect to the strong topology of $L^1(B_R)$ (ensured by Fatou's lemma and the

lower semicontinuity of U) and by the convexity of $\mathbb{U}_R$ (ensured by the convexity of U). The combination of these two properties yields that all sublevel sets $\{\mathbb{U}_R \le t\}$ are L^1-closed and convex, hence closed with respect to the weak topology. □

The lower semicontinuity results presented so far cover many applications, but not all of them. For instance, one might consider energy densities U having a linear growth at infinity. Also, as we did in Proposition 15.6 and Corollary 15.7 for the relative entropy functional, one might investigate what happens if the nonnegativity assumption on U is dropped, replacing convergence in the duality with $C_c(\mathbb{R}^n)$ with a stronger notion of convergence. Let us present, without proof and in specific cases, two answers to these general questions. We refer to [68] and [30] for a more detailed account about integral functionals defined on measures.

Theorem 15.9 *Let (X,τ) be a Polish space. Assume that $U:[0,\infty)\to[0,\infty]$ is convex and lower semicontinuous and that $\mu\in\mathscr{M}_+(X)$ is finite on bounded sets. Defining $U_\infty := \lim_{t\to+\infty} U(t)/t$, the internal energy*

$$\mathbb{U}(\nu) := \int_X U(\varrho)\,\mathrm{d}\mu + U_\infty \nu^s(X) \qquad\qquad \nu\in\mathscr{P}(X),$$

(where $\nu = \varrho\mu + \nu^s$ is the Radon-Nikodym decomposition of ν), is lower semicontinuous with respect to the weak convergence in duality with continuous functions with bounded support.

Notice that the previous proposition includes Theorem 15.8 as a particular case, since for superlinear energy densities U one has $U_\infty = +\infty$. In the particular case of the energy density $U(t) := \sqrt{1+t^2}$, with $X=\mathbb{R}^n$ and $\mu=\mathscr{L}^n$, we find

$$\mathbb{U}(\nu) = \int_{\mathbb{R}^n} \sqrt{1+\varrho^2}\,\mathrm{d}x + \nu^s(\mathbb{R}^n),$$

a functional strictly related to the area functional in the space of functions of bounded variation (where ϱ plays the role of the modulus of the gradient and ν plays the role of the total variation of the distributional derivative).

Definition 15.10 (Rényi Entropy) The internal energy associated to the energy density $U(s) = -s^\alpha$ where $\frac{n}{n+2} < \alpha < 1$ is defined on $\mathscr{P}_2(\mathbb{R}^n)$ via the expression

$$\mathbb{U}(\mu) := -\int_{\mathbb{R}^n} \varrho^\alpha\,\mathrm{d}x,$$

where $\mu = \varrho\mathscr{L}^n + \mu^s$ is the Lebesgue decomposition of μ, and it is called Rényi entropy.

We will motivate in the next section the assumption $\frac{n}{n+2} < \alpha < 1$. Let us only state here without proof a lower semicontinuity result for the Rényi entropies, referring to Appendix E in [86] for a proof.

Theorem 15.11 *Let $\mathbb{U}$ be the internal energy associated to the energy density $U(s) = -s^\alpha$ for $\frac{n}{n+2} < \alpha < 1$ as in Definition 15.10. Then $\mathbb{U}$ is lower semicontinuous in $\mathscr{P}_2(\mathbb{R}^n)$.*

2 Convexity of Internal Energies

We move now our attention to the study of geodesic convexity of internal energies on $\mathscr{P}_2(\mathbb{R}^n)$. Assume that we are given an energy density $U : [0, \infty) \to (-\infty, \infty]$ convex, lower semicontinuous and with $U(0) = 0$ (this is a natural assumption motivated by the idea that there should be no energy contribution by regions without mass). We will work under the additional assumption that $U^-(s) \le Cs^\alpha$ for some $C \ge 0$ as s goes to 0, where $\alpha \in (\frac{n}{n+2}, 1]$. Thanks to convexity, U^- has at most linear growth at infinity, hence the asymptotic control on U^- near 0 can be turned into a global one, namely $U^-(s) \le C_1 s^\alpha + C_2 s$ for any $s \in [0, \infty)$. These assumptions imply that

$$\begin{aligned}\int_{\mathbb{R}^n} U^-(\varrho)\,\mathrm{d}x &\le C_2 + C_1 \int_{\mathbb{R}^n} \varrho^\alpha\,\mathrm{d}x \\ &\le C_2 + C_1 \int_{\mathbb{R}^n} \varrho^\alpha \frac{(1+|x|)^{2\alpha}}{(1+|x|)^{2\alpha}}\,\mathrm{d}x \\ &\le C_2 + C_1 \left(\int_{\mathbb{R}^n} \varrho(1+|x|)^2\,\mathrm{d}x\right)^\alpha \left(\int_{\mathbb{R}^n} \frac{1}{(1+|x|)^{\frac{2\alpha}{1-\alpha}}}\,\mathrm{d}x\right)^{1-\alpha},\end{aligned} \tag{15.7}$$

where ϱ is the density with respect to $\mathscr{L}^n$ of an absolutely continuous measure in $\mathscr{P}_2(\mathbb{R}^n)$ and the last factor in (15.7) is finite because of the moment assumption on μ and the constraint $\alpha > \frac{n}{n+2}$ (which implies in turn that $\frac{2\alpha}{1-\alpha} > n$). Thanks to this estimate the following definition of internal energy makes sense

$$\mathbb{U}(\mu) := \begin{cases} \int_{\mathbb{R}^n} U(\varrho)\,\mathrm{d}x & \text{if } \mu = \varrho\mathscr{L}^n \\ +\infty & \text{otherwise.} \end{cases} \tag{15.8}$$

Definition 15.12 We say that the energy density U satisfies the condition $(MC)_n$ if the function $(0, \infty) \ni s \mapsto s^n U(s^{-n})$ is convex and nonincreasing.

In the previous definition (MC) stands for R. McCann, who was the first to notice in [89] that this condition is the right one in order to get good convexity properties of internal energies along Wasserstein geodesics.

Example 15.13 Let us consider a few examples of energy densities satisfying $(MC)_n$.

(i) The energy densities of the form $U(s) = s^\alpha$ satisfy the condition $(MC)_n$ if $\alpha \geq 1$ since $s^n U(s^{-n}) = s^{n(1-\alpha)}$ is convex and nonincreasing if $\alpha \geq 1$.
(ii) The energy density of the relative entropy functional satisfies condition $(MC)_n$ too since $\frac{s^\alpha - s}{\alpha - 1} \to s \ln s$ as $\alpha \to 1^+$.
(iii) The energy densities of the form $U(s) = -s^\alpha$ satisfy $(MC)_n$ if $1 - \frac{1}{n} \leq \alpha \leq 1$ since $s^n U(s^{-n}) = -s^{n(1-\alpha)}$ is convex and nonincreasing if $1 - \frac{1}{n} \leq \alpha \leq 1$.

Observe that, in the case when $n \geq 2$, the threshold $\alpha \geq 1 - \frac{1}{n}$ (that we impose in order to satisfy $(MC)_n$) is more restrictive than the threshold $\alpha > \frac{n}{n+2}$ that we imposed to obtain a meaningful definition for the energy.

The following two results will be the key tools in the proof of our main theorem about the convexity of internal energies on $\mathscr{P}_2(\mathbb{R}^n)$.

Proposition 15.14 (Density of Interpolated Measures) *Let $\mu_0 = \varrho \mathscr{L}^n \in \mathscr{P}_2(\mathbb{R}^n)$ and $\mu_1 \in \mathscr{P}_2(\mathbb{R}^n)$. Denote by T the unique optimal transport map with quadratic cost from μ_0 to μ_1 given by Theorem 5.2 (recall that $T = \nabla \phi$, where $\phi : \mathbb{R}^n \to (-\infty, \infty]$ is convex and lower semicontinuous and Ω, the interior of the finiteness domain of ϕ, has full μ_0-measure), and let $(\mu_t)_{t\in[0,1]}$ be the constant speed geodesic between μ_0 and μ_1.*

Then, for any $t \in [0, 1)$ one has that T_t is injective, μ_t is concentrated on $T_t(D_0)$ and $\mu_t = \varrho_t \mathscr{L}^n$ with

$$\varrho_t = \frac{\varrho}{\det \nabla T_t} \circ (T_t)^{-1} \qquad \mathscr{L}^n\text{-a.e. on } T_t(D_0), \tag{15.9}$$

where

$$T_t := (1-t)\mathrm{id} + tT = \nabla \phi_t, \qquad \phi_t := (1-t)\frac{|\cdot|^2}{2} + t\phi$$

and D_0 is the set of full μ_0-measure given, as in Theorem 7.3, by

$$D_0 := \big\{ x \in \Omega : \phi \text{ is differentiable at } x,\ T = \nabla \phi \text{ is differentiable at } x \big\},$$

Proof It follows from Corollary 10.10 that the unique constant speed geodesic $(\mu_t)_{t\in[0,1]}$ connecting μ_0 to μ_1 is given by the formula $(T_t)_\# \mu_0$ for any $t \in [0, 1]$. Therefore, to prove the absolute continuity of μ_t, it is enough to have the Lipschitz continuity of $(T_t)^{-1}$, which we will gain by exploiting that T_t is a strictly monotone operator.

Going into the details, since T is a monotone operator (being the gradient of a convex function) we obtain that $T_t \geq (1-t)\mathrm{id}$ (i.e. $\langle T_t(x) - T_t(y), x-y\rangle \geq (1-t)|x-y|^2$ for any x and y). In particular T_t is invertible for $t < 1$ and $(T_t)^{-1}$ is Lipschitz with $\mathrm{Lip}\big((T_t)^{-1}\big) \leq 1/(1-t)$. Take now any $A \in \mathscr{B}(\mathbb{R}^n)$ such that $\mathscr{L}^n(A) = 0$. Thanks to the Lipschitz continuity of $(T_t)^{-1}$ we obtain $\mathscr{L}^n\big((T_t)^{-1}(A)\big) = 0$ and consequently $\mu_0\big((T_t)^{-1}(A)\big) = 0$, since $\mu_0 \ll \mathscr{L}^n$. It follows that $\mu_t(A) = 0$ and this proves the absolute continuity of μ_t.

Notice that the absolute continuity of μ_t would also follow from (ii) of Theorem 7.3, since in this case the "singular" set Σ of (7.6) corresponding to $T_t = \nabla\phi_t$ is empty. Analogously, the validity of (15.9) follows by (iii) of Theorem 7.3 applied to T_t. □

Lemma 15.15 *The function $A \mapsto (\det(A))^{\frac{1}{n}}$ is concave on the set of nonnegative semi-definite symmetric $n \times n$ matrices.*

Proof By homogeneity, it is sufficient to prove the concavity inequality

$$(\det(A+B))^{\frac{1}{n}} \geq (\det(A))^{\frac{1}{n}} + (\det(B))^{\frac{1}{n}}. \tag{15.10}$$

Also, it suffices to prove (15.10) when the matrices A and B are both positive definite, the inequality in the general case can be recovered by an approximation argument.

Call now C the square root of A, which is a symmetric positive definite matrix and define $D := C^{-1}BC^{-1}$ (observe that, by the definition of square root, $C^{-1}AC^{-1} = \mathrm{id}$). It follows that

$$\det(A)\det(\mathrm{id}+D) = \det(A+B) \qquad \text{and} \qquad \det(A)\det(D) = \det(B),$$

and we are left with the proof of the concavity inequality in the case where $A = \mathrm{id}$ and $B = D$. We can also assume D to be diagonal with positive eigenvalues $\lambda_1, \dots, \lambda_n$, since any positive definite symmetric matrix can be diagonalized and conjugation of both sides in (15.10) leaves the inequality unchanged. All in all, we need to prove that

$$\left(\prod_{i=1}^n (1+\lambda_i)\right)^{\frac{1}{n}} \geq 1 + \left(\prod_{i=1}^n \lambda_i\right)^{\frac{1}{n}},$$

which immediately follows by adding the inequalities

$$\frac{1}{n}\sum_{i=1}^n \frac{\lambda_i}{1+\lambda_i} \geq \left(\prod_{i=1}^n \frac{\lambda_i}{1+\lambda_i}\right)^{1/n}, \qquad \frac{1}{n}\sum_{i=1}^n \frac{1}{1+\lambda_i} \geq \left(\prod_{i=1}^n \frac{1}{1+\lambda_i}\right)^{1/n},$$

both derived from the classical arithmetic/geometric mean inequality. □

Theorem 15.16 *Let $\mathbb{U}$ be the internal energy defined in (15.8) and suppose that U satisfies $(MC)_n$. Then $\mathbb{U}$ is convex along geodesics in $\mathscr{P}_2(\mathbb{R}^n)$.*

Proof Let $(\mu_t)_{t\in[0,1]}$ be any constant speed geodesic in $\mathscr{P}_2(\mathbb{R}^n)$. Observe that we can assume that μ_0 and μ_1 are absolutely continuous with respect to $\mathscr{L}^n$, otherwise there is nothing left to prove. Under these assumptions, with the same notation of Proposition 15.14, $\mu_t = (T_t)_\#\mu_0 = \varrho_t\mathscr{L}^n$ for any $t \in [0,1)$, with the explicit formula for the interpolated density

$$\varrho_t = \frac{\varrho}{\det\nabla T_t}\circ(T_t)^{-1}$$

on $T_t(D_0)$, where μ_t is concentrated. Recalling that $U(0) = 0$, an application of Corollary 7.2 yields

$$\mathbb{U}(\mu_t) = \int_{T_t(D_0)} U\left(\frac{\varrho}{\det\nabla T_t}\circ(T_t)^{-1}\right)\mathrm{d}y = \int_{D_0} U\left(\frac{\varrho}{\det\nabla T_t}\right)\det\nabla T_t\,\mathrm{d}x$$

for all $t \in [0,1)$. The absolute continuity of μ_1 guarantees that that singular set Σ is $\mathscr{L}^n$-negligible, hence we can also write

$$\mathbb{U}(\mu_t) = \int_{D_0\setminus\Sigma} U\left(\frac{\varrho}{\det\nabla T_t}\right)\det\nabla T_t\,\mathrm{d}x \qquad \forall t \in [0,1). \tag{15.11}$$

Analogously, for $t = 1$, Corollary 7.2 and (7.7) guarantee

$$\mathbb{U}(\mu_1) = \int_{T(D_0\setminus\Sigma)} U\left(\frac{\varrho}{\det\nabla T}\circ(T|_{D_0\setminus\Sigma})^{-1}\right)\mathrm{d}y = \int_{D_0\setminus\Sigma} U\left(\frac{\varrho}{\det\nabla T}\right)\det\nabla T\,\mathrm{d}x,$$

so that (15.11) holds also for $t = 1$. We fix now a point $x \in D_0 \setminus \Sigma$ and we study the convexity properties of the function

$$f(t) := U\left(\frac{\varrho(x)}{\det\nabla T_t(x)}\right)\det\nabla T_t(x).$$

The function f can be written as $b\circ a$, where

$$a(t) := (\det\nabla T_t(x))^{\frac{1}{n}}, \qquad b(z) := U\left(\frac{\varrho(x)}{z^n}\right)z^n$$

and we observe that a is concave thanks to Lemma 15.15, while b is convex and nonincreasing because the energy density U satisfies $(MC)_n$. It follows that $b\circ a$ is convex, indeed

$$b(a(\lambda_1 t_1 + \lambda_2 t_2)) \le b(\lambda_1 a(t_1) + \lambda_2 a(t_2)) \le \lambda_1 b(a(t_1)) + \lambda_2 b(a(t_2))$$

for any $\lambda_1, \lambda_2 \in [0, 1]$ with $\lambda_1 + \lambda_2 = 1$. The desired conclusion about geodesic convexity easily follows from the convexity of the integrand in (15.11). □

3 Potential Energy Functional

For a given Borel potential $V : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ whose negative part V^- has at most quadratic growth at infinity, we would like to study the functional $\mathbb{V} : \mathscr{P}_2(\mathbb{R}^n) \to \mathbb{R} \cup \{+\infty\}$ defined by

$$\mathbb{V}(\mu) := \int_{\mathbb{R}^n} V \, \mathrm{d}\mu.$$

Notice that the growth assumption on V^- ensures that $\mathbb{V}$ is well posed and that it takes its values in $\mathbb{R} \cup \{+\infty\}$. As in the case of the internal energy functional $\mathbb{U}$, convexity and lower semicontinuity of V give rise to convexity and lower semicontinuity of $\mathbb{V}$, at least if V is nonnegative.

Recall that, according to Theorem 8.8, convergence in $\mathscr{P}_2(\mathbb{R}^n)$ is equivalent to weak convergence in duality with $C_b(\mathbb{R}^n)$ plus convergence of quadratic moments. More precisely, the following result holds.

Lemma 15.17 *If $\mu_k \to \mu$ in $\mathscr{P}_2(\mathbb{R}^n)$, then $|\cdot|^2\mu_k$ weakly converge to $|\cdot|^2\mu$ in the duality with respect to $C_b(\mathbb{R}^n)$.*

Proof Set $\sigma_k = z_k^{-1}|\cdot|^2\mu_k \in \mathscr{P}(\mathbb{R}^n)$ and $\sigma = z^{-1}|\cdot|^2\mu \in \mathscr{P}(\mathbb{R}^n)$, with z_k, z suitable normalization constants. Note that $z_k \to z$ as $k \to \infty$, hence Remark 2.7 gives $\liminf_k \sigma_k(A) \geq \sigma(A)$ for any open set $A \subset \mathbb{R}^n$. From the convergence criterion for probability measures stated in Lemma 8.13 it follows that σ_k weakly converge in duality with $C_b(\mathbb{R}^n)$ to σ, and the result follows. □

Theorem 15.18 (Properties of the Potential Energy Functional) *Assume that V is Borel, with V^- having at most quadratic growth at infinity. Then:*

(i) *$\mathbb{V}$ is λ-convex in $(\mathscr{P}_2(\mathbb{R}^n), W_2)$ if and only if V is λ-convex;*
(ii) *$\mathbb{V}$ is continuous (resp. lower semicontinuous) in duality with $C_b(\mathbb{R}^n)$ if $V \in C_b(\mathbb{R}^n)$ (resp. V is lower semicontinuous and bounded from below);*
(iii) *$\mathbb{V}$ is continuous (resp. lower semicontinuous) in $(\mathscr{P}_2(\mathbb{R}^n), W_2)$ if V is continuous and has at most quadratic growth (resp. V is lower semicontinuous and V^- is continuous).*

Proof

(i) Assume that $\mathbb{V}$ is λ-convex. Observe that for any $x, y \in \mathbb{R}^n$, the Wasserstein geodesic between δ_x and δ_y is $\delta_{(1-t)x+ty}$, and that $\mathbb{V}(\delta_x) = V(x)$: the λ-convexity of V follows easily.

Assume now that V is λ-convex. As we know that any $\mu_t \in \mathrm{Geo}(\mathscr{P}_2(\mathbb{R}^n))$ can be written as $\mu_t = \big((1-t)x + ty\big)_{\#}\Sigma$ for some $\Sigma \in \Gamma_o(\mu_0, \mu_1)$, we get

$$\begin{aligned}\mathbb{V}(\mu_t) &= \int_{\mathbb{R}^n} V \,\mathrm{d}\mu_t = \int_{\mathbb{R}^n\times\mathbb{R}^n} V((1-t)x+ty)\,\mathrm{d}\Sigma(x,y)\\ &\le (1-t)\int_{\mathbb{R}^n\times\mathbb{R}^n} V(x)\,\mathrm{d}\Sigma(x,y) + t\int_{\mathbb{R}^n\times\mathbb{R}^n} V(y)\,\mathrm{d}\Sigma(x,y)\\ &\quad - \frac{\lambda}{2}t(1-t)\int_{\mathbb{R}^n\times\mathbb{R}^n}|x-y|^2\,\mathrm{d}\Sigma(x,y)\\ &= (1-t)\mathbb{V}(\mu_0) + t\mathbb{V}(\mu_1) - \frac{\lambda}{2}t(1-t)W_2^2(\mu_0,\mu_1).\end{aligned}$$

(ii) The continuity statement is immediate. For the lower semicontinuity, we argue as in the proof of Theorem 2.6, representing V as $\sup_i V_i$, with $V_i \in C_b(\mathbb{R}^n)$ and $V_i \le V_{i+1}$.

(iii) Let $\mu_k \to \mu$ in $\mathscr{P}_2(\mathbb{R}^n)$. As V is continuous, it can approximated by a sequence of functions $V_i \in C_b(\mathbb{R}^n)$ which coincide with V on B_i. Moreover such functions V_i can be taken with uniform quadratic growth, i.e. in such a way that there exists $C \ge 0$ independent of i such that $|V_i(x)| \le C(1+|x|^2)$. Let us prove that

$$\limsup_{i\to\infty}\limsup_{k\to\infty}\int_{\mathbb{R}^n}|V_i - V|\,\mathrm{d}\mu_k = 0. \tag{15.12}$$

Observe that

$$\int_{\mathbb{R}^n}|V_i - V|\,\mathrm{d}\mu_k = \int_{\mathbb{R}^n\setminus B_i}|V_i - V|\,\mathrm{d}\mu_k \le 2C\int_{\mathbb{R}^n\setminus B_i}(1+|x|^2)\,\mathrm{d}\mu_k.$$

Taking Lemma 15.17 and the weak convergence of μ_k to μ into account, the limsup of the right-hand side can be estimated from above with $\int_{\mathbb{R}^n\setminus B_i}(1+|x|^2)\,\mathrm{d}\mu$, which in turn converges to 0 as $i\to\infty$. This proves (15.12). Defining $\mathbb{V}_i(\nu) = \displaystyle\int_{\mathbb{R}^n} V_i\,\mathrm{d}\nu$, one has

$$|\mathbb{V}(\mu_k) - \mathbb{V}(\mu)| \le |\mathbb{V}(\mu_k) - \mathbb{V}_i(\mu_k)| + |\mathbb{V}_i(\mu_k) - \mathbb{V}_i(\mu)| + |\mathbb{V}_i(\mu) - \mathbb{V}(\mu)|.$$

The middle term in the right-hand side obviously goes to 0 as $k\to\infty$, implying

$$\limsup_{k\to\infty}|\mathbb{V}(\mu_k) - \mathbb{V}(\mu)| \le \limsup_{k\to\infty}\int_{\mathbb{R}^n}|V - V_i|\,\mathrm{d}\mu_k + |\mathbb{V}_i(\mu) - \mathbb{V}(\mu)|.$$

Thanks to (15.12) and the dominated convergence theorem, sending i to ∞ we obtain the result.

Finally, the lower semicontinuity of $\mathbb{V}$ when V is lower semicontinuous and V^- is continuous follows by the decomposition $V = V^+ - V^-$, applying statement (ii).

□

4 Interaction Energy

For a given Borel interaction energy $W : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ uniformly bounded from below, set

$$\mathbb{W}(\mu) = \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} W(x, y)\, \mathrm{d}\mu(x)\, \mathrm{d}\mu(y), \qquad \mu \in \mathscr{P}(\mathbb{R}^n).$$

Theorem 15.19 (Properties of the Interaction Energy Functional) *If W is lower semicontinuous and bounded from below, then $\mathbb{W}$ is lower semicontinuous with respect to convergence in duality with $C_b(\mathbb{R}^n)$.*

If W is λ-convex, then $\mathbb{W}$ is 2λ-convex in $(\mathscr{P}_2(\mathbb{R}^n), W_2)$.

Proof If W is lower semicontinuous and bounded from below, the lower semicontinuity of $\mathbb{W}$ follows immediately from Theorem 15.18(ii), since $\mu_k \to \mu$ weakly in the duality with $C_b(\mathbb{R}^n)$ easily implies the convergence of the product measures $\mu_k \times \mu_k$ to $\mu \times \mu$ in the duality with $C_b(\mathbb{R}^{2n})$.

In order to prove 2λ-convexity, let $\mu_t \in \mathrm{Geo}(\mathscr{P}_2(\mathbb{R}^n))$ and let $\Sigma \in \Gamma_{\mathrm{o}}(\mu_0, \mu_1)$ be such that $\big((1-t)x + ty)\big)_{\#}\Sigma = \mu_t$ for every $t \in [0, 1]$. Then

$$\begin{aligned}
\mathbb{W}(\mu_t) &= \int_{\mathbb{R}^n \times \mathbb{R}^n} W(x, y)\, \mathrm{d}\mu_t(x)\, \mathrm{d}\mu_t(y) \\
&= \int_{\mathbb{R}^n \times \mathbb{R}^n} W((1-t)x_0 + tx_1, (1-t)y_0 + ty_1)\, \mathrm{d}\Sigma(x_0, x_1)\, \mathrm{d}\Sigma(y_0, y_1) \\
&\le \int_{\mathbb{R}^n \times \mathbb{R}^n} [(1-t)W(x_0, y_0) + tW(x_1, y_1)]\, \mathrm{d}\Sigma(x_0, x_1)\, \mathrm{d}\Sigma(y_0, y_1) \\
&\quad - \frac{\lambda}{2} t(1-t) \int_{\mathbb{R}^n \times \mathbb{R}^n} |(x_0, y_0) - (x_1, y_1)|^2\, \mathrm{d}\Sigma(x_0, x_1)\, \mathrm{d}\Sigma(y_0, y_1) \\
&= (1-t)\mathbb{W}(\mu_0) + t\mathbb{W}(\mu_1) - \lambda t(1-t) W_2^2(\mu_0, \mu_1).
\end{aligned}$$

□

5 Functional Inequalities via Optimal Transport

One of the remarkable features of the optimal transport theory is that it allows to derive new proofs of well-known functional inequalities, in the unifying perspective provided by the intrinsic convexity properties on $\mathscr{P}_2(\mathbb{R}^n)$ or even by the gradient flow structure. Here we will present three fundamental inequalities:

- the Talagrand transport inequality,
- the Brunn-Minkowski (and isoperimetric) inequality,
- the Log-Sobolev inequality.

Let us start with the transport inequality. This inequality provides an estimate of the Wasserstein distance between the standard Gaussian measure $\gamma = (2\pi)^{-n/2} e^{-|\cdot|^2/2} \mathscr{L}^n$ and any other measure μ in terms of the entropy of μ relative to γ. Among other things, it makes more rigorous the use of the relative entropy as an asymmetric distance between probability measures.

Proposition 15.20 (Talagrand Inequality) *For any $\varrho\gamma \in \mathscr{P}_2(\mathbb{R}^n)$ one has*

$$W_2^2(\varrho\gamma, \gamma) \leq 2\mathrm{Ent}_\gamma(\varrho\gamma). \tag{15.13}$$

Note that this inequality is dimension free. Talagrand noted in [113] that it is actually tensorizable, meaning that if $W_2^2(\varrho\gamma, \gamma) \leq C\mathrm{Ent}_\gamma(\varrho\gamma)$ holds with the same constant C on two metric measure spaces $(X_1, \mathrm{d}_1, \gamma_1)$ and $(X_2, \mathrm{d}_2, \gamma_2)$, then it holds with the same constant C on the "Hilbertian" product metric measure space

$$(X_1 \times X_2, (\mathrm{d}_1^2 + \mathrm{d}_2^2)^{1/2}, \gamma_1 \times \gamma_2).$$

From this observation, Talagrand deduced the general case $n \geq 1$ from the one-dimensional case. The latter can be achieved by a direct computation based on the monotone rearrangement. Let us give the proof via optimal transport.

Proof of Proposition 15.20 Recall that, according to the energy inequality of Proposition 14.14, on geodesic metric spaces one has

$$\frac{\lambda}{2} d^2(x, x_{\min}) \leq F(x) - F(x_{\min})$$

for λ-convex functionals F attaining their minimum value at $x_{\min}$, provided $\lambda \geq 0$. We can apply this property to the 1-convex functional Ent_γ in $\mathscr{P}_2(\mathbb{R}^n)$, whose minimal value is 0, with $\lambda = 1$. Note that the 1-convexity of Ent_γ is a direct consequence of Proposition 15.6 as Ent and $\mu \mapsto \frac{1}{2}\int_{\mathbb{R}^n} |x|^2 \,\mathrm{d}\mu(x)$ are respectively convex and 1-convex. □

Let us discuss now the Brunn-Minkowski inequality.

Proposition 15.21 (Brunn-Minkowski Inequality) *For all compact sets A, $B \subset \mathbb{R}^n$ one has*

$$\big(\mathscr{L}^n((1-t)A+tB)\big)^{1/n} \geq (1-t)\big(\mathscr{L}^n(A)\big)^{1/n} + t\big(\mathscr{L}^n(B)\big)^{1/n} \qquad \forall t \in [0,1]. \tag{15.14}$$

One of the classical proofs of the Brunn-Minkowski is based on the approximation of A and B by finite unions of n-cubes and on the application of the inequality between geometric and arithmetic mean, see Theorem 3.2.41 of [60] for instance. As shown in the proof below, the optimal transport theory provides an even stronger inequality, where the left hand side in (15.14) is replaced by the Rényi n-dimensional entropy of a measure concentrated on $(1-t)A + tB$. The proof uses the convexity of Rényi entropy, which once more relies, as we have seen, on the inequality between geometric and arithmetic mean.

Proof of Proposition 15.21 In this proof we denote for simplicity by $|C|$ the Lebesgue measure of a Borel set C. Let us point out that whenever A and B are compact, $(1-t)A + tB$ is compact and therefore Borel for any $t \in [0,1]$. Recall that the Rényi n-dimensional entropy $\mathcal{E}_n$ is defined by

$$\mathcal{E}_n(\mu) = -\int_{\mathbb{R}^n} \varrho^{1-\frac{1}{n}}\, \mathrm{d}\mathscr{L}^n$$

if $\mu \in \mathscr{P}_2(\mathbb{R}^n)$ has density ϱ with respect to $\mathscr{L}^n$. Thus, for a bounded Borel set C one can rewrite

$$|C|^{1/n} = \int_C |C|^{1/n-1}\, \mathrm{d}\mathscr{L}^n = -\mathcal{E}_n(\mu_C),$$

where $\mu_C = \frac{\chi_C}{|C|}\mathscr{L}^n$. If $\mu_t \in \mathrm{Geo}(\mathscr{P}_2(\mathbb{R}^n))$ is the Wasserstein geodesic from μ_A and μ_B, the convexity of $\mathcal{E}_n$ guaranteed by Theorem 15.16 and by the fact that $-z^{1-1/n}$ satisfies McCann's condition of Definition 15.12 give

$$-\mathcal{E}_n(\mu_t) \geq -(1-t)\mathcal{E}_n(\mu_A) - t\mathcal{E}_n(\mu_B) \qquad \forall t \in (0,1). \tag{15.15}$$

Note that $-\mathcal{E}_n(\mu) \leq |C|^{1/n}$ whenever a probability measure $\mu = \varrho\mathscr{L}^n$ is supported in C. This is a simple consequence of Jensen's inequality

$$-\mathcal{E}_n(\mu) = \int_C \varrho^{1-1/n}\, \mathrm{d}x = |C| \int_{\mathbb{R}^n} \varrho^{1-1/n}\, \mathrm{d}\mu_C \leq |C| \left(\int_{\mathbb{R}^n} \varrho\, \mathrm{d}\mu_C \right)^{1-1/n} = |C|^{1/n}.$$

Since μ_t is concentrated on $C = (1-t)A + tB$, we obtain (15.14) from (15.15). □

A classical consequence of the Brunn-Minkowski inequality is the isoperimetric inequality, stated using the Minkowski content

$$\mathcal{M}(E) = \liminf_{r\to 0^+} \frac{\mathscr{L}^n(E + B_r) - \mathscr{L}^n(E)}{r}$$

as a measurement of the surface area. For sets E with a sufficiently regular boundary (for instance bounded open sets with a Lipschitz boundary, see §2.13 of [10]) this notion coincides with other notions of surface area, coming from the theory of BV functions or based on the Hausdorff $(n-1)$-dimensional measure $\mathscr{H}^{n-1}$. In particular, if $E = B_1$ is the unit ball, then $\mathscr{L}^n(E) = \omega_n$ and $\mathcal{M}(E) = n\omega_n$.

Theorem 15.22 (Isoperimetric Inequality) *Let E be Borel a subset of $\mathbb{R}^n$ such that $\mathscr{L}^n(E) = \mathscr{L}^n(B_1)$. Then*

$$\mathcal{M}(E) \geq \mathcal{M}(B_1).$$

Proof By the Brunn-Minkowski inequality, $\left(\mathscr{L}^n(E+B_r)\right)^{1/n} \geq \omega_n^{1/n} + (\omega_n r^n)^{1/n}$, then

$$\liminf_{r\to 0^+} \frac{\mathscr{L}^n(E + B_r) - \mathscr{L}^n(E)}{r} \geq \liminf_{r\to 0^+} \frac{\left(\omega_n^{1/n} + \omega_n^{1/n} r\right)^n - \omega_n}{r} = n\omega_n.$$

□

Let us come now to the Log-Sobolev inequality. Even though the proof we give is by no means the simplest one available (because the technical characterization in Theorem 15.25 of the slope of Ent_γ is needed), it shows the deep relation between the Log-Sobolev inequality and the general energy-energy dissipation typical of uniformly convex functions. We refer to [83] for a survey about the different approaches to the study of Log-Sobolev inequalities.

Proposition 15.23 (Log-Sobolev Inequality) *If $f \in W^{1,2}_{\mathrm{loc}}(\mathbb{R}^n)$ satisfies $\int_{\mathbb{R}^n} f^2 \, \mathrm{d}\gamma = 1$, one has*

$$\int_{\mathbb{R}^n} f^2 \ln f^2 \, \mathrm{d}\gamma \leq 2 \int_{\mathbb{R}^n} |\nabla f|^2 \, \mathrm{d}\gamma. \tag{15.16}$$

Remark 15.24 Let us compare the Log-Sobolev inequality with the classical Sobolev one:

$$\|f\|_{p^*} \leq C(n,p)\|\nabla f\|_p, \qquad f \in W^{1,p}(\mathbb{R}^n) \tag{15.17}$$

for $1 < p < n$, $p^* = np/(n-p)$. In (15.17) the function f gains a higher integrability exponent, which is not the case with (15.16). However, (15.16) is dimension-free (notice also that $p^* \to p$ as $n \to \infty$) and, still, provides a higher integrability of the function.

Proof of Proposition 15.23 Recall that, according to Proposition 14.14, for any λ-convex functional F on a geodesic space E the following energy-energy dissipation inequality holds:

$$F(x) - F(x_{\min}) \le \frac{1}{2\lambda} |\nabla^- F|^2(x), \quad \forall x \in E,$$

provided $\lambda > 0$ and $x_{\min}$ is the minimizer of F. As in the proof of Proposition 15.20 we can apply this property to the 1-convex relative entropy Ent_γ with $E = \mathscr{P}_2(\mathbb{R}^n)$, $x_{\min} = \gamma$ and $x = f^2\gamma$:

$$\int_{\mathbb{R}^n} f^2 \ln f^2 \, \mathrm{d}\gamma \le \frac{1}{2} |\nabla^- \mathrm{Ent}_\gamma|^2(f^2\gamma).$$

The result follows from Theorem 15.25 below, with $\varrho = f^2$, since our assumptions on f imply that $\varrho \in W^{1,1}_{\mathrm{loc}}(\mathbb{R}^n)$ and that (using the chain rule and the fact that $\nabla f = 0$ $\mathscr{L}^n$-a.e. on $\{f = 0\}$)

$$4 \int_{\mathbb{R}^n} |\nabla f|^2 \, \mathrm{d}\gamma = \int_{\{\varrho > 0\}} \frac{|\nabla \varrho|^2}{\varrho} \, \mathrm{d}\gamma < \infty.$$

□

According to the differential calculus that we are going to develop in the next sections, Theorem 15.25 below can be interpreted as follows: the "Wasserstein gradient" $\nabla^W \mathrm{Ent}_\gamma$ of Ent_γ at $\mu = \varrho\gamma$ is $\nabla \ln \varrho = \varrho^{-1} \nabla \varrho$, understood as an element of the space $L^2(\mathbb{R}^n; \varrho\gamma)$, so that its norm is precisely the slope in (15.18).

Theorem 15.25 (Slope of Ent_γ) *Let $\varrho\gamma \in D(\mathrm{Ent}_\gamma)$. Then Ent_γ has finite descending slope at $\varrho\gamma$ if and only if $\varrho \in W^{1,1}_{\mathrm{loc}}(\mathbb{R}^n)$ and $|\nabla \varrho|^2/\varrho \in L^1(\mathbb{R}^n, \gamma)$, where the quotient $|\nabla \varrho|^2/\varrho$ is conventionally set to be 0 on $\{\varrho = 0\}$. In this case one has the identity*

$$|\nabla^- \mathrm{Ent}_\gamma|^2(\varrho\gamma) = \int_{\mathbb{R}^n} \frac{|\nabla \varrho|^2}{\varrho} \, \mathrm{d}\gamma. \tag{15.18}$$

Proof Let us prove the inequality $\le$ in (15.18), under the assumption that ϱ is Lipschitz and continuously differentiable, with $\inf \varrho > 0$. The general case when $\varrho \in W^{1,1}_{\mathrm{loc}}(\mathbb{R}^n)$ and $|\nabla \varrho|^2/\varrho \in L^1(\mathbb{R}^n, \gamma)$ can be recovered by approximation, using the lower semicontinuity of the slope guaranteed by Theorem 14.12.

Fix $\eta\gamma \in D(\mathrm{Ent}_\gamma)$. Since $x \mapsto x\ln x$ is convex in $[0,\infty)$, the convexity inequality $f(x)-f(y)\le f'(x)(x-y)$ implies

$$\begin{aligned}\mathrm{Ent}_\gamma(\varrho\gamma)-\mathrm{Ent}_\gamma(\eta\gamma)&=\int_{\mathbb{R}^n}(\varrho\ln\varrho-\eta\ln\eta)\,\mathrm{d}\gamma\\&\le\int_{\mathbb{R}^n}(1+\ln\varrho)(\varrho-\eta)\,\mathrm{d}\gamma=\int_{\mathbb{R}^n}\ln\varrho(\varrho-\eta)\,\mathrm{d}\gamma=(*).\end{aligned}$$

To estimate the difference $(\varrho-\eta)$, the idea introduced by Villani in [118] is to use an optimal transport $\Sigma\in\Gamma_{\mathrm{o}}(\varrho\gamma,\eta\gamma)$ that, according to Theorem 5.2, is induced by a map T via $\Sigma=(\mathrm{id}\times T)_\#(\varrho\gamma)$. Then

$$\begin{aligned}(*)&=\int_{\mathbb{R}^n\times\mathbb{R}^n}(\ln\varrho(x)-\ln\varrho(y))\,\mathrm{d}\Sigma(x,y)=\int_{\mathbb{R}^n}(\ln\varrho(x)-\ln\varrho(T(x)))\varrho(x)\,\mathrm{d}\gamma(x)\\&=\int_{\mathbb{R}^n}\left(\int_0^1\langle\nabla\ln\varrho(T_t(x)),x-T(x)\rangle\,\mathrm{d}t\right)\varrho(x)\,\mathrm{d}\gamma(x),\end{aligned}$$

where $T_t:=\mathrm{id}+t(T-\mathrm{id})$ is the standard affine interpolation from id to T. Fubini, Cauchy-Schwarz and Hölder theorems can be applied to get

$$\mathrm{Ent}_\gamma(\varrho\gamma)-\mathrm{Ent}_\gamma(\eta\gamma)\le\|T-\mathrm{id}\|_{L^2(\varrho\gamma)}\int_0^1\left(\int_{\mathbb{R}^n}|\nabla\ln\varrho|^2(T_t(x))\varrho(x)\,\mathrm{d}\gamma(x)\right)^{1/2}\mathrm{d}t.$$

Observing that $\|T-\mathrm{id}\|_{L^2(\varrho\gamma)}=W_2(\varrho\gamma,\eta\gamma)$ we get

$$\frac{\mathrm{Ent}_\gamma(\varrho\gamma)-\mathrm{Ent}_\gamma(\eta\gamma)}{W_2(\varrho\gamma,\eta\gamma)}\le\int_0^1\left(\int_{\mathbb{R}^n}|\nabla\ln\varrho|^2(y)\varrho_t(y)\,\mathrm{d}\gamma(y)\right)^{1/2}\mathrm{d}t,\qquad(15.19)$$

where ϱ_t is the density of $\mu_t=(T_t)_\#(\varrho\gamma)$, the constant speed Wasserstein geodesic between $\varrho\gamma$ and $\eta\gamma$. Now, by convexity of Ent_γ along μ_t one has

$$\frac{\mathrm{Ent}_\gamma(\varrho\gamma)-\mathrm{Ent}_\gamma(\eta\gamma)}{W_2(\varrho\gamma,\eta\gamma)}\le\frac{\mathrm{Ent}_\gamma(\varrho\gamma)-\mathrm{Ent}_\gamma(\varrho_{t_0}\gamma)}{W_2(\varrho\gamma,\varrho_{t_0}\gamma)}$$

for all $t_0\in(0,1)$. By applying (15.19) with ϱ_{t_0} in place of η, a simple rescaling argument gives that the right-hand side can be estimated from above with

$$\frac{1}{t_0}\int_0^{t_0}\left(\int_{\mathbb{R}^n}|\nabla\ln\varrho|^2\varrho_s\,\mathrm{d}\gamma\right)^{1/2}\mathrm{d}s,$$

whence

$$\frac{\mathrm{Ent}_\gamma(\varrho\gamma) - \mathrm{Ent}_\gamma(\eta\gamma)}{W_2(\varrho\gamma, \eta\gamma)} \le \frac{1}{t_0}\int_0^{t_0} \left(\int_{\mathbb{R}^n} |\nabla \ln \varrho|^2 \varrho_s \, \mathrm{d}\gamma\right)^{1/2} \mathrm{d}s. \tag{15.20}$$

By our assumption on ϱ, the function $s \mapsto \int_{\mathbb{R}^n} |\nabla \ln \varrho|^2 \varrho_s \, \mathrm{d}\gamma$ is continuous at 0. This leads to the result by letting $t_0 \to 0$ in (15.20).

Denoting $S = |\nabla^- \mathrm{Ent}_\gamma|(\varrho\gamma)$ and assuming with no loss of generality that $S < \infty$, let us now prove the implication from finiteness of slope to Sobolev regularity of ϱ, as well as the inequality $\ge$ in (15.18). For $\psi \in C_c^\infty(\mathbb{R}^n, \mathbb{R}^n)$ and $\epsilon > 0$ sufficiently small the map $T_\epsilon = I + \epsilon\psi$ is smooth, with a smooth inverse and with positive Jacobian determinant. Even though T_ϵ is not optimal (unless ψ is a gradient) between $\varrho\gamma$ and $\varrho_\epsilon\gamma := (T_\epsilon)_\#(\varrho\gamma)$, we can still estimate

$$W_2(\varrho_\epsilon\gamma, \varrho\gamma) \le \epsilon\left(\int_{\mathbb{R}^n} |\psi|^2 \varrho \, \mathrm{d}\gamma\right)^{1/2},$$

so that

$$\mathrm{Ent}_\gamma(\varrho\gamma) - \mathrm{Ent}_\gamma(\varrho_\epsilon\gamma) \le S\epsilon\left(\int_{\mathbb{R}^n} |\psi|^2 \varrho \, \mathrm{d}\gamma\right)^{1/2} + o(\epsilon). \tag{15.21}$$

Let us find the first order Taylor expansion with respect to ϵ of the left hand side of (15.21). To this aim, recall that, setting $V(x) = |x|^2/2$, the formula for the density (with respect to $\mathscr{L}^n$) of the push-forward measure proved in Proposition 15.14 gives

$$\varrho_\epsilon e^{-V} = \frac{\varrho e^{-V}}{\det \nabla T_\epsilon} \circ T_\epsilon^{-1},$$

so that

$$\begin{aligned}
\mathrm{Ent}_\gamma(\varrho_\epsilon\gamma) - \mathrm{Ent}_\gamma(\varrho\gamma) &= \int_{\mathbb{R}^n} \varrho_\epsilon \ln \varrho_\epsilon \, \mathrm{d}\gamma - \int_{\mathbb{R}^n} \varrho \ln \varrho \, \mathrm{d}\gamma \\
&= \int_{\mathbb{R}^n} (\ln \varrho_\epsilon \circ T_\epsilon - \ln \varrho) \, \mathrm{d}(\varrho\gamma) \\
&= \int_{\mathbb{R}^n} \left(\ln(e^V \circ T_\epsilon) + \ln(e^{-V}) - \ln \det \nabla T_\epsilon\right) \mathrm{d}(\varrho\gamma).
\end{aligned}$$

Using the expansions

$$e^V \circ T_\epsilon = e^V(1 + \epsilon\langle x, \psi\rangle) + o(\epsilon), \qquad \det \nabla T_\epsilon = 1 + \epsilon \mathrm{div}\, \psi + o(\epsilon)$$

we find

$$\mathrm{Ent}_\gamma(\varrho_\epsilon\gamma) - \mathrm{Ent}_\gamma(\varrho\gamma) = \epsilon \int_{\mathbb{R}^n} (\langle x, \psi\rangle - \mathrm{div}\ \psi)\, \mathrm{d}(\varrho\gamma) + o(\epsilon),$$

so that (15.21) and the same inequalities applied to $-\psi$ yield

$$\left| \int_{\mathbb{R}^n} (\langle x, \psi\rangle - \mathrm{div}\ \psi)\varrho\, \mathrm{d}\gamma \right| \le S\|\psi\|_{L^2(\varrho\gamma)}. \tag{15.22}$$

We now conclude by applying L^2 duality. Indeed, if we already knew that $\varrho \in W^{1,1}_{\mathrm{loc}}(\mathbb{R}^n)$ we could integrate by parts, getting the cancellation of the term $\langle x, \psi\rangle$ and

$$\left| \int_{\{\varrho>0\}} \langle \frac{\nabla\varrho}{\varrho}, \psi\rangle\, \mathrm{d}(\varrho\gamma) \right| = \left| \int_{\mathbb{R}^n} \langle \nabla\varrho, \psi\rangle\, \mathrm{d}\gamma \right| \le S\|\psi\|_{L^2(\varrho\gamma)} \qquad \forall \psi \in C^\infty_c(\mathbb{R}^n;\mathbb{R}^n). \tag{15.23}$$

Since $C^\infty_c(\mathbb{R}^n;\mathbb{R}^n)$ is dense in $L^2(\varrho\gamma;\mathbb{R}^n)$, this provides the desired estimate on $\varrho^{-1}\nabla\varrho$ in $L^2(\varrho\gamma)$. Notice that the first equality in (15.23) is justified by the property

$$\nabla\varrho = 0 \quad \mathscr{L}^n\text{-a.e. in } \{\varrho = 0\},$$

valid for all Sobolev functions, while as we said the subsequent inequality comes from the integration by parts in (15.22). However, we can use (15.22) even to obtain the local Sobolev regularity of ϱ as follows: using the identity

$$\mathrm{div}\,\psi - \langle x, \psi\rangle = e^V \mathrm{div}\,(\psi e^{-V})$$

(which relates the γ-divergence to the Euclidean divergence) and setting $\phi = \psi e^{-V}$ we can write (15.22) as follows

$$\left| \int_{\mathbb{R}^n} \varrho \mathrm{div}\,\phi\, \mathrm{d}x \right| \le S\left(\int_{\mathbb{R}^n} |\phi|^2 e^V\, \mathrm{d}x \right)^{1/2} \qquad \forall \phi \in C^\infty_c(\mathbb{R}^n;\mathbb{R}^n). \tag{15.24}$$

Since e^V is locally bounded, (15.24) easily implies that $\varrho \in W^{1,2}_{\mathrm{loc}}(\mathbb{R}^n)$. □

In analogous way one can prove a similar result for the Entropy with respect to $\mathscr{L}^n$. We will also see that the Wasserstein gradient of the functional $\mathbb{V}(\mu) = \int_{\mathbb{R}^n} V\, \mathrm{d}\mu$ is ∇V so that, with the notation $u = \varrho e^{-V}$, the additivity of the logarithmic derivative of products and the formula $\mathrm{Ent} = \mathrm{Ent}_\gamma - \mathbb{V}$ give

$$\nabla^W \mathrm{Ent}(u\mathscr{L}^n) = \nabla^W \mathrm{Ent}_\gamma(\varrho\gamma) - \nabla^W \mathbb{V}(\varrho\gamma) = \frac{\nabla\varrho}{\varrho} - \nabla V = \frac{\nabla u}{u}.$$

This provides an interpretation of the following formula for the slope of Ent, whose rigorous proof can be obtained with methods analogous to those of the previous theorem.

Theorem 15.26 (Slope of Ent**)** *Let* $\mu = u\mathscr{L}^n \in D(\mathrm{Ent})$. *Then* Ent *has finite descending slope at* μ *if and only if* $u \in W^{1,1}_{\mathrm{loc}}(\mathbb{R}^n)$ *and* $|\nabla u|^2/u \in L^1(\mathbb{R}^n)$, *where the quotient* $|\nabla u|^2/u$ *is conventionally set to* 0 *on* $\{u = 0\}$. *In this case one has the identity*

$$|\nabla^- \mathrm{Ent}|^2(u\mathscr{L}^n) = \int_{\mathbb{R}^n} \frac{|\nabla u|^2}{u}\, \mathrm{d}\mathscr{L}^n. \tag{15.25}$$

Lecture 16: The Continuity Equation and the Hopf-Lax Semigroup



We are now going to explore the connections between the classical ODE Cauchy problem, the continuity equation and the transport equation. These results will be applied to give a differential description (through the continuity equation) of the geodesics in the Wasserstein space.

1 Continuity Equation and Transport Equation

Let $T \in (0, \infty]$ and let $v : (0, T) \times \mathbb{R}^n \to \mathbb{R}^n$ be a Borel and possibly time-dependent vector field. Often we will write v_t to identify the vector field at time t. We want to study the following three differential equations:

Ordinary Differential Equation (ODE) Given an initial point $x \in \mathbb{R}^n$, we are interested in a locally absolutely continuous curve $\gamma : [0, T) \to \mathbb{R}^n$ such that

$$\begin{cases} \gamma'(t) = v_t(\gamma(t)), \\ \gamma(0) = x. \end{cases} \tag{16.1}$$

The Cauchy problem (16.1) is understood in the integral sense, namely

$$\gamma(t) = x + \int_0^t v_r(\gamma(r))\, \mathrm{d}r \quad \text{for all} \quad t \in [0, T).$$

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Continuity Equation (CE) Given a measure $\bar{\mu} \in \mathscr{M}(\mathbb{R}^n)$ we want to find a weakly continuous curve in the space of measure $\mu : [0, \infty) \to \mathscr{M}(\mathbb{R}^n)$ such that

$$\begin{cases} \mu_0 = \bar{\mu}, \\ \frac{\mathrm{d}}{\mathrm{d}t}\mu + \operatorname{div}(v \cdot \mu) = 0, \end{cases} \tag{16.2}$$

where the second identity has to be interpreted in the distributional sense.

Transport Equation (TE) Given a function $\psi : [0, \infty) \times \mathbb{R}^n \to \mathbb{R}$ and a fixed time $T > 0$ we are looking for a function $f : [0, T] \times \mathbb{R}^n \to \mathbb{R}$ such that

$$\begin{cases} f(T, \cdot) = 0, \\ \frac{\mathrm{d}}{\mathrm{d}t} f + v \cdot \nabla f = \psi. \end{cases} \tag{16.3}$$

From here on we will refer to the stated equations with their abbreviations. The key results that hold under proper technical assumptions are that a solution to (ODE) automatically induces a solution both for (CE) and (TE) and that as soon as (TE) admits a solution, we have uniqueness for (CE).

Remark 16.1 We are studying a transport equation with a Cauchy condition at the final time as it turns out to be more convenient to prove uniqueness result for the continuity equation. All results would hold almost unchanged if we were instead focusing our attention on the more natural problem $f(0, \cdot) = 0$.

Let us start with a classical result of existence and uniqueness for (ODE). It is a slightly more general version of the Cauchy–Lipschitz theorem, since we are not assuming continuity with respect to t of the velocity field and, consequently, existence and uniqueness hold in the class of absolutely continuous functions.

Theorem 16.2 (Cauchy–Lipschitz) *Let $v : (0, T) \times \mathbb{R}^n \to \mathbb{R}^n$ be a Borel vector field such that:*

(i) $v(t, \cdot)$ is a Lipschitz function for any $t \in (0, T)$ and the Lipschitz constant belongs to $L^1_{\mathrm{loc}}([0, T))$;
(ii) v has uniform linear growth, namely, $|v(t, x)| \le C(1 + |x|)$ for any $(t, x) \in (0, T) \times \mathbb{R}^n$, for some constant C.

Under these assumptions, for any initial point $x \in \mathbb{R}^n$, (ODE) admits a unique solution $\gamma_x \in AC_{\mathrm{loc}}([0, T))$. Furthermore, denoting by $X(t, x) = \gamma_x(t)$ the associated flow, we have the following two growth estimates:

(a) $|X(t, x)| \le \hat{C}_t \cdot (1 + |x|)$ where $\hat{C} : [0, \infty) \to [0, \infty)$ is a function depending only on C;
(b) $|X(t, x) - X(t, y)| \le \exp(\int_0^t \mathrm{Lip}(v_s)\, \mathrm{d}s)|x - y|$.

Before proving that a solution to (ODE) induces naturally a solution to (CE), we need to investigate how a solution to (CE) is done. The following proposition

gives us neat equivalence between solving (CE) in a distributional sense and a more manageable statement.

Proposition 16.3 *Let $\mu_t : (0,\infty) \to \mathscr{P}(\mathbb{R}^n)$ be a weakly continuous curve of probability measures and let v be a vector field such that $|v_t| \in L^1(\mu_t)$ and $\|v_t\|_{L^1(\mu_t)} \in L^1_{\mathrm{loc}}([0,\infty))$. The following facts are equivalent:*

(i) the curve μ_t solves $\frac{\mathrm{d}}{\mathrm{d}t}\mu_t + \operatorname{div}(v\cdot\mu) = 0$ in the distributional sense;
(ii) for any $\varphi \in C_c^\infty(\mathbb{R}^n)$ it holds that $t \mapsto \int_{\mathbb{R}^n} \varphi \,\mathrm{d}\mu_t$ is in $AC_{\mathrm{loc}}([0,\infty))$ and its derivative is $\int_{\mathbb{R}^n} \nabla\varphi \cdot v_t \,\mathrm{d}\mu_t$.

Proof The local finiteness of μ_t and the fact that $\|v_t\|_{L^1(\mu_t)} \in L^1_{\mathrm{loc}}([0,\infty))$ ensure that both μ and $v\cdot\mu$ are distributions on $(0,\infty)\times\mathbb{R}^n$. The first item above is true if and only if for any test function $\varrho \in C_c^\infty((0,\infty)\times\mathbb{R}^n)$ it holds

$$0 = \langle \frac{\mathrm{d}}{\mathrm{d}t}\mu_t, \varrho\rangle + \langle \operatorname{div}(v\cdot\mu), \varrho\rangle = -\langle \mu, \frac{\mathrm{d}}{\mathrm{d}t}\varrho + v\cdot\nabla\varrho\rangle. \tag{16.4}$$

Recalling that the family

$$D := \left\{ \sum_{i=1}^N \alpha_i(t)\varphi_i(x) :\ \alpha_i \in C_c^\infty((0,\infty)),\ \varphi_i \in C_c^\infty(\mathbb{R}^n),\ i\in\{1,\dots,N\},\ N\in\mathbb{N} \right\}$$

is dense in $C_c^\infty((0,\infty)\times\mathbb{R}^n)$, it is enough to check (16.4) only for $\varrho(t,x) = \alpha(t)\varphi(x)$ where $\alpha \in C_c^\infty((0,\infty))$ and $\varphi \in C_c^\infty(\mathbb{R}^n)$. Substituting in (16.4) and explicitly computing the value of the expression as an integral we get

$$\begin{aligned} 0 = \langle \mu, \alpha'(t)\varphi + \alpha(t) v\cdot\nabla\varphi\rangle &= \int_0^\infty \int_{\mathbb{R}^n} \left(\alpha'(t)\varphi + \alpha(t) v\cdot\nabla\varphi\right) \mathrm{d}\mu_t(x)\,\mathrm{d}t \\ &= \int_0^\infty \left(\alpha'(t)\int_{\mathbb{R}^n} \varphi \,\mathrm{d}\mu_t + \alpha(t)\int_{\mathbb{R}^n} v\cdot\nabla\varphi \,\mathrm{d}\mu_t \right) \mathrm{d}t. \end{aligned} \tag{16.5}$$

As easy consequences of the bounds on v, we find that $\int_{\mathbb{R}^n}\varphi\,\mathrm{d}\mu_t \in L^\infty([0,\infty)) \subset L^1_{\mathrm{loc}}([0,\infty))$ and $\int_{\mathbb{R}^n} v\cdot\nabla\varphi\,\mathrm{d}\mu_t \in L^1_{\mathrm{loc}}([0,\infty))$. Therefore, as (16.5) holds for any test function α, we have proven the equivalence between i) and ii). □

Now we want to transform a solution to (ODE) into a solution to (CE). The solution is built as the push-forward of the initial measure through the flow associated to the (ODE).

Proposition 16.4 *Let $(v_t)_{t>0}$ be a time dependent vector field satifying the assumptions of Theorem 16.2 and let X be the associated flow. For any $\bar\mu \in \mathscr{P}_1(\mathbb{R}^n)$ the curve of probabilities $\mu_t := (X_t)_\#\bar\mu$ is contained in $\mathscr{P}_1(\mathbb{R}^n)$ and it solves (CE) with the initial measure $\bar\mu$.*

Proof First of all let us check that $\mu_t \in \mathscr{P}_1(\mathbb{R}^n)$:

$$\int_{\mathbb{R}^n} |x| \, \mathrm{d}\mu_t(x) = \int_{\mathbb{R}^n} |X(t,x)| \, \mathrm{d}\bar{\mu}(x) \leq \hat{C}_t \int_{\mathbb{R}^n} (1+|x|) \, \mathrm{d}\bar{\mu} < \infty,$$

where we have applied the growth estimate (a) in Theorem 16.2.

Fixed any function $\varphi \in C_c^\infty$, we want to consider $\int \varphi \, \mathrm{d}\mu_t$ as a function of t. Recalling Corollary 16.3, if we were able to prove that such a map is $AC_{\mathrm{loc}}([0,\infty))$ and that the derivative is $\int_{\mathbb{R}^n} \nabla\varphi \cdot v_t \, \mathrm{d}\mu_t$ we would have finished the proof. In order to prove this claim, we do explicitly compute the difference for two different times $s < t$ and prove that it is equal to the integral of the claimed derivative:

$$\begin{aligned}
\int_{\mathbb{R}^n} \varphi \, \mathrm{d}\mu_t - \int_{\mathbb{R}^n} \varphi \, \mathrm{d}\mu_s &= \int_{\mathbb{R}^n} (\varphi(X_t) - \varphi(X_s)) \, \mathrm{d}\bar{\mu} = \int_{\mathbb{R}^n} \int_s^t \frac{\mathrm{d}}{\mathrm{d}r} \varphi(X_r) \, \mathrm{d}r \, \mathrm{d}\bar{\mu} \\
&= \int_{\mathbb{R}^n} \int_s^t \nabla\varphi(X_r) \cdot v_r \, \mathrm{d}r \, \mathrm{d}\bar{\mu} = \int_s^t \int_{\mathbb{R}^n} \nabla\varphi(X_r) \cdot v_r(X_r) \, \mathrm{d}\bar{\mu} \, \mathrm{d}r \\
&= \int_s^t \int_{\mathbb{R}^n} \nabla\varphi \cdot v_r \, \mathrm{d}\mu_r \, \mathrm{d}r.
\end{aligned}$$

Here the use of Fubini–Tonelli theorem is justified by the fact that v has linear growth and $\bar{\mu} \in \mathscr{P}_1(\mathbb{R}^n)$, hence $\nabla\varphi(X_r) \cdot v_r$ is integrable. □

We want now to build a solution to (TE) starting from a solution to (ODE) as we have just done for (CE). This time the key insight is that, at least on a formal level, if γ is a solution to (ODE) and f is a solution to (TE) then $\frac{\mathrm{d}}{\mathrm{d}t}(f(t,\gamma(t))) = \psi(t)$. Integrating along characteristic curves of the (ODE) then leads to a representation formula for the solution of (TE). However this approach relies on the existence of a solution, which we will prove directly in Proposition 16.5.

In the proof we will need the concept of enriched flow map. Given a vector field v satisfying the assumptions of Theorem 16.2, we will denote by $X : [0,\infty) \times [0,\infty) \times \mathbb{R}^n \to \mathbb{R}^n$ the map such that $X(t,s,y)$ is the solution at time t starting from y at time s. Such a map exists and it is unique by Theorem 16.2. Let us recall a bunch of elementary properties of the enriched flow that hold under the same assumptions of Theorem 16.2:

(i) For any $s \geq 0$ and $y \in \mathbb{R}^n$ we have $X(s,s,y) = y$.
(ii) For any $s \geq 0$ and $y \in \mathbb{R}^n$ it holds $\frac{\partial X}{\partial t}(t,s,y) = v_t(X(t,s,y))$ for almost every $t > 0$. Moreover, for any $t \geq 0$ and $y \in \mathbb{R}^n$ it holds $\frac{\partial X}{\partial s}(t,s,y) = -v_s(X(t,s,y))$ for almost every $s > 0$.
(iii) As an easy consequence of the uniqueness of the trajectories we have

$$X(t,s,X(s,s',y)) = X(t,s',y).$$

(iv) The growth estimate (b) in Theorem 16.2 can be generalized to

$$|X(t,s,x) - X(t,s,y)| \le \exp\left(\int_s^t \mathrm{Lip}(v_r)\,\mathrm{d}r\right)|x-y|. \tag{16.6}$$

Therefore, fixing s, t, the map $x \mapsto X(s,t,x)$ has the Lipschitz property and, thanks to Rademacher theorem, it is differentiable almost everywhere. Furthermore we get that $\mathrm{d}_x X$ is locally bounded in $[0,\infty)\times[0,\infty)$.

These formulas will be used in the proof of the following proposition.

Proposition 16.5 *Let $\psi \in C_c^\infty([0,T]\times\mathbb{R}^n)$ be a compactly supported smooth function, and let $(v_t)_{t\in(0,T)}$ be a time dependent vector field satisfying all the assumptions of Theorem 16.2. Let us denote by $X : [0,\infty)\times[0,\infty)\times\mathbb{R}^n \to \mathbb{R}^n$ the enriched flow associated to v. Then, the function $f : [0,T]\times\mathbb{R}^n \to \mathbb{R}$ defined as*

$$f(s,y) := -\int_s^T \psi(t, X(t,s,y))\,\mathrm{d}t \tag{16.7}$$

satisfies (TE) $\mathscr{L}^{1+n}$-almost everywhere in $(0,T)\times\mathbb{R}^n$. In particular, f is a C^1 classical solution to (TE) if v is continuous. Furthermore, there exists a constant $A \ge 0$ (that depends on ψ, T, v) such that

$$\left|\frac{\mathrm{d}}{\mathrm{d}s} f(s,y)\right| \le A(1+|v_s(y)|). \tag{16.8}$$

Proof It is obvious that $f(T,\cdot) = 0$. It is also easy to prove that f is a locally Lipschitz function in $(0,T)\times\mathbb{R}^n$, thanks to the local Lipschitz estimates on the extended flow map and the regularity of ψ.

To prove that f satisfies almost everywhere the transport equation we reverse the heuristic argument presented before, of integration along characteristics. For $r \in [0,T]$ fixed, using the semigroup property we get

$$f(s, X(s,r,z)) = -\int_s^T \psi(t, X(t,r,z))\,\mathrm{d}t$$

and therefore differentiation of both sides with respect to s gives

$$\frac{\mathrm{d}}{\mathrm{d}s} f(s, X(s,r,z)) + \nabla f(s, X(s,r,z))\cdot\frac{\mathrm{d}}{\mathrm{d}s} X(s,r,z) = \psi(s, X(s,r,z)). \tag{16.9}$$

The computation above can be rigorously justified due to the following observation. For any $r \in [0,T]$ it holds that $X(s,r,\cdot)_\sharp\mathscr{L}^n \ll \mathscr{L}^n$. Therefore for $\mathscr{L}^n$-a.e. $z \in \mathbb{R}^n$ we have that $X(s,r,z)$ is a differentiability point of $f(s,\cdot)$ and a differentiability point of $X(\cdot,r,z)$ with $\frac{\partial X}{\partial s}(s,r,z) = v_s(X(s,r,z))$ for almost every $s \in (0,T)$,

where Rademacher's theorem and property ii) of the enriched flow play a role in the first and second conclusion, respectively.
Changing back variables with $y = X(s, r, z)$ in (16.9) we obtain the claimed property.

Given the representation formula it is straightforward to check (16.8) relying on the properties (ii) and (iv) of the enriched flow above and on the assumptions about ψ. □

We have proven that as soon as we can find a solution to (ODE) we have solutions for (CE) and (TE) too. What about uniqueness? We are going to see a result exactly in this direction, based on the general principle that in appropriate function spaces, existence for (TE) implies uniqueness for (CE). We apply this principle in the simplified situation when v satisfies the assumptions of Theorem 16.2.

Proposition 16.6 *If $(v_t)_{t>0}$ is a time dependent vector field satisfying all assumptions of Theorem 16.2, then for any $\bar{\mu} \in \mathscr{P}_1(\mathbb{R}^n)$ there is a unique solution $\mu_t \in \mathscr{P}_1(\mathbb{R}^n)$ to (CE) with $\int_{\mathbb{R}^n} |x| \, \mathrm{d}\mu_t \in L^1_{\mathrm{loc}}([0, \infty))$.*

Proof Let us consider two solutions μ_t, μ'_t and set $\sigma_t = \mu_t - \mu'_t$. Then $\sigma_0 = 0$ and, since the continuity equation is linear, σ_t still solves (CE). Let us choose $T > 0$ and an arbitrary compactly supported function $\psi : [0, T] \times \mathbb{R}^n \to \mathbb{R}$ with $0 \le \psi \le 1$. We would like to prove that $\int_0^T \int_{\mathbb{R}^n} \psi \, \mathrm{d}\sigma_t \, \mathrm{d}t = 0$ (that would yield the conclusion by arbitrariness of ψ) plugging into the continuity equation the solution of the transport equation drifted by v_t and with source ψ. This strategy requires some smoothing and cut-off arguments that we describe below.

Let us introduce a family of smooth cut-off functions $\chi_R \in C^\infty_c(\mathbb{R}^n)$ for $R > 0$, with $0 \le \chi_R \le 1$,

$$\chi_R \equiv 1 \text{ on } B(0, R), \quad \chi_R \equiv 0 \text{ on } \mathbb{R}^n \setminus B(0, 2R) \quad \text{and} \quad |\nabla \chi_R| \le 2/R.$$

Next we consider a vector field w_t such that $w_t = v_t$ on $B(0, R) \times (0, T)$, $w_t = 0$ if $t \notin (0, T)$ and

$$\sup_{\mathbb{R}^n} |w_t| + \mathrm{Lip}(w_t, \mathbb{R}^n) \le \sup_{B(0,2R)} |v_t| + \mathrm{Lip}(v_t, B(0, 2R)), \quad \text{for any } t \in [0, T]. \tag{16.10}$$

Then, for any $\epsilon > 0$, we let w^ϵ_t be obtained from w_t by double mollification in space and time (through convolution against a smooth kernel, as we already did in other occasions). Notice that, by (16.10),

$$\sup_{0<\epsilon<1} \int_0^T \left[\sup_{\mathbb{R}^n} \left| w^\epsilon_t \right| + \mathrm{Lip}(w^\epsilon_t, \mathbb{R}^n) \right] \mathrm{d}t < \infty.$$

Arguing as in the proof of Proposition 16.5 by the method of characteristics and taking into account the smoothness of w_t^ϵ, we find a smooth function $f_t^\epsilon \in C^\infty([0,T)\times\mathbb{R}^n)$ such that

$$\begin{cases} f_T^\epsilon(\cdot) = 0, \\ \frac{\mathrm{d}}{\mathrm{d}t} f_t^\epsilon + w_t^\epsilon \cdot \nabla f_t^\epsilon = \psi \quad \text{on } (0,T)\times\mathbb{R}^n. \end{cases} \tag{16.11}$$

Through the representation formula via characteristics of f_t^ϵ and thanks to the bounds on w_t^ϵ it is easy to infer that $0 \geq f_t^\epsilon \geq -T$ and $\left|\nabla f_t^\epsilon\right|$ is uniformly bounded with respect to space-time and regularization parameter.

Plugging $\chi_R f_t^\epsilon$ as a test function in the continuity equation and taking into account the assumption $\mu_0 = 0$ together with (16.11), we infer that

$$\begin{aligned} 0 &= \int_0^T \int_{\mathbb{R}^n} \left[\chi_R \frac{\partial f_t^\epsilon}{\partial t} + v_t \cdot \left(\chi_R \nabla f_t^\epsilon + f_t^\epsilon \nabla \chi_R \right) \right] \mathrm{d}\sigma_t \, \mathrm{d}t \\ &= \int_0^T \int_{\mathbb{R}^n} \chi_R \left[\psi + \left(v_t - w_t^\epsilon \right) \cdot \nabla f_t^\epsilon \right] \mathrm{d}\sigma_t \, \mathrm{d}t + \int_0^T \int_{\mathbb{R}^n} f_t^\epsilon \nabla \chi_R \cdot v_t \, \mathrm{d}\sigma_t \, \mathrm{d}t. \end{aligned}$$

Letting first $\epsilon \downarrow 0$ and then $R \to \infty$, taking into account the uniform bounds on f_t^ϵ that we pointed out above, as well as the linear growth of v and the property $\int_{\mathbb{R}^n} |x| \, \mathrm{d}|\sigma_t| \in L^1(0,T)$, we infer that

$$\int_0^T \int_{\mathbb{R}^n} \psi \, \mathrm{d}\sigma_t \, \mathrm{d}t = 0.$$

Since ψ is arbitrary we get that $\sigma_t = 0$ for almost every $t > 0$ and, by weak continuity, for every $t \geq 0$. □

2 Continuity Equation of Geodesics in the Wasserstein Space

Our goal is to prove that geodesics of $\mathscr{P}_2(\mathbb{R}^n)$ satisfy a continuity equation, finding out the associated gradient vector field. The answer will be very simple, and it will easily provide an inequality in the Benamou–Brenier formula (see Theorem 17.2) of the next lecture.

Let us fix $\mu_0, \mu_1 \in W_2(\mathbb{R}^n)$ with $\mu_0 \ll \mathscr{L}^n$. We know that there exists a unique (up to μ_0-negligible sets) optimal map $T : \mathbb{R}^n \to \mathbb{R}^n$ such that $\mu_1 = T_\# \mu_0$. Furthermore, the constant speed geodesic between μ_0 and μ_1 is given by $\mu_t := (T_t)_\# \mu_0$, where $T_t = (1-t)\mathrm{id} + tT$.

As we want to find a continuity equation for μ_t, thanks to Proposition 16.3, we have to fix $\psi \in C_c^\infty(\mathbb{R}^n)$ and differentiate the function $t \mapsto \int_{\mathbb{R}^n} \psi \, \mathrm{d}\mu_t$. We get:

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{\mathbb{R}^n} \psi \, \mathrm{d}\mu_t = \frac{\mathrm{d}}{\mathrm{d}t} \int_{\mathbb{R}^n} \psi(T_t) \, \mathrm{d}\mu_0 = \int_{\mathbb{R}^n} \langle \nabla \psi(T_t), T - \mathrm{id} \rangle \, \mathrm{d}\mu_0$$
$$= \int_{\mathbb{R}^n} \langle \nabla \psi, (T - \mathrm{id}) \circ T_t^{-1} \rangle \, \mathrm{d}\mu_t$$

and therefore

$$v_t := (T - \mathrm{id}) \circ T_t^{-1} = \frac{\mathrm{id} - T_t^{-1}}{t} \tag{16.12}$$

is the vector field we were looking for.

We would like to understand a little bit better who is v_t and how it is linked to the Kantorovich potentials (ϕ_t, ψ_t) from μ_0 to μ_t. We already know that T_t can be written as $\mathrm{id} - \nabla \phi_t$ and that its inverse T_t^{-1} can be written as $\mathrm{id} - \nabla \psi_t$. Just looking at (16.12), we get for free that $v_t = t^{-1} \nabla \psi_t$.

In addition, comparing the two formulas for T_t, namely $T_t = \mathrm{id} - \nabla \phi_t$ and $T_t = (1-t)\mathrm{id} + tT = \mathrm{id} - t\nabla \phi$, we might guess that a good choice for ϕ_t is $t\phi$. Assuming that, we obtain

$$\psi_t(y) = \inf_{x \in \mathbb{R}^n} \left\{ -\phi_t(x) + \frac{1}{2}|x - y|^2 \right\} = t \inf_{x \in \mathbb{R}^n} \left\{ -\phi(x) + \frac{1}{2t}|x - y|^2 \right\}.$$

We are going to prove in the next section that these formulas for the Kantorovich potentials (ϕ_t, ψ_t) are indeed correct even in the general setting of metric spaces. The proof does not involve indeed any linear structure and uses only the triangle inequality and the elementary inequality

$$\frac{a^2}{s} + \frac{b^2}{t} \geq \frac{(a+b)^2}{s+t} \qquad a, \, b \in \mathbb{R}, \; s, \, t > 0, \tag{16.13}$$

whose case of equality occurs only when $a^2 t^2 = b^2 s^2$.

3 Hopf-Lax Semigroup

Definition 16.7 (Hopf-Lax Semigroup) Let (X, d) be a metric space and $f \in C_b(X)$. For $t > 0$, we define $Q_t f \in C_b(X)$ by

$$Q_t f(y) = \inf_{x \in X} \left\{ f(x) + \frac{1}{2t} d^2(x, y) \right\},$$

setting also $Q_0 f = f$. The family of operators $(Q_t)_{t \geq 0}$ is called the Hopf-Lax semigroup.

Remark 16.8 We will see in Theorem 16.10 that Q_t is indeed a semigroup in geodesic (or even length) metric spaces, namely $Q_{s+t} = Q_s \circ Q_t$.

In general, thanks to (16.13) it is not difficult to verify that $Q_{s+t} \le Q_s \circ Q_t$. To see that some further assumption is needed in order for the opposite inequality to hold we can consider the following example. We let $X = \{0, 1\}$ equipped with the discrete distance and $f : X \to \{0, 1\}$ be defined by $f(0) = 0$ and $f(1) = 1$. Then it is easy to verify that, for any s sufficiently large, $Q_{s+t} f(1) = 1/(2(s + t))$ while $Q_s(Q_t f)(1) = 1/(2s)$, therefore the semigroup property fails.

We can now prove the formulas for the Kantorovich potentials along geodesics claimed in the previous section.

Theorem 16.9 (Interpolation of Kantorovich Potentials) *Let (X, d) be a geodesic metric space and let $\mu_0, \mu_1 \in \mathscr{P}_2(X)$. Let $(\phi, Q_1(-\phi))$ be a pair of Kantorovich potentials from μ_0 to μ_1. Then, for all constant speed geodesics μ_t from μ_0 to μ_1 and for all $t \in [0, 1]$, one has that the pair*

$$(t\phi, Q_1(-t\phi)) = (t\phi, t\, Q_t(-\phi))$$

provides Kantorovich potentials from μ_0 to μ_t.

Proof The case $t = 0$ is obvious. It is clear that $(t\phi, Q_1(-t\phi))$ is an admissible pair of Kantorovich potentials, namely $t\phi(x) + Q_1(-t\phi)(y) \le d^2(x, y)/2$ for all $(x, y) \in X \times X$. To prove optimality, we need to show that equality holds Σ-a.e. for at least one optimal plan Σ from μ_0 to μ_t. Recalling the dynamic formulation of the optimal transport problem and the related description of geodesics, we can take

$$\Sigma = (e_0, e_t)_\# \eta$$

for some optimal geodesic plan $\eta \in \mathscr{P}(\mathrm{Geo}(X))$ whose marginals provide the geodesic μ_t, and prove that

$$t\phi(\gamma(0)) + Q_1(-t\phi)(\gamma(t)) = \frac{1}{2} d^2(\gamma(0), \gamma(t)) \qquad \text{for } \eta\text{-a.e. } \gamma. \tag{16.14}$$

Notice now that the optimality of $(\phi, Q_1(-\phi))$ gives that

$$\phi(\gamma(0)) + Q_1(-\phi)(\gamma(1)) = \frac{1}{2} d^2(\gamma(0), \gamma(1)) \qquad \text{for } \eta\text{-a.e. } \gamma.$$

Then, with $x = \gamma(0)$, $y = \gamma(1)$ and $z = \gamma(t)$, the proof of the equality (16.14) reduces to this *pointwise* implication

$$\phi(x) + Q_1(-\phi)(y) = \frac{1}{2} d^2(x, y) \qquad \Longrightarrow \qquad t\phi(x) + Q_1(-t\phi)(z) = \frac{1}{2} d^2(x, z)$$

whenever z is a t-intermediate proint from x to y, namely $d(x,z) = td(x,y)$, $d(z,y) = (1-t)d(x,y)$. We need only to prove the inequality $\geq$, in the equivalent form

$$\phi(x) + Q_t(-\phi)(z) \geq \frac{1}{2t} d^2(x,z).$$

We proceed as follows:

$$\begin{aligned}
\phi(x) + Q_t(-\phi)(z) &= \inf_{w \in X} \phi(x) - \phi(w) + \frac{1}{2t} d^2(w,z) \\
&= \inf_{w \in X} \frac{1}{2} d^2(x,y) - Q_1(-\phi)(y) - \phi(w) + \frac{1}{2t} d^2(w,z) \\
&\geq \inf_{w \in X} \frac{1}{2} d^2(x,y) - \frac{1}{2} d^2(w,y) + \frac{1}{2t} d^2(w,z).
\end{aligned}$$

So, we are left with the proof of

$$\frac{1}{2} d^2(x,y) + \inf_{w \in X} -\frac{1}{2} d^2(y,w) + \frac{1}{2t} d^2(w,z) \geq \frac{1}{2t} d^2(x,z)$$

which, by the t-intermediate properties of z, corresponds to

$$\frac{d^2(y,z)}{2(1-t)} + \frac{d^2(z,w)}{2t} \geq \frac{1}{2} d^2(y,w) \qquad \forall w \in X,$$

which in turn is a direct consequence of (16.13). □

Below we collect some relevant properties of the Hopf-Lax semigroup. Before doing so, we introduce the notion of *asymptotic Lipschitz constant* of a Lipschitz function $\psi : X \to \mathbb{R}$ as

$$\left|D^* \psi\right|(x) := \lim_{r \to 0} \sup_{\substack{y,z \in B_r(x) \\ y \neq z}} \frac{|\psi(z) - \psi(y)|}{|z - y|}. \tag{16.15}$$

We remark that the asymptotic Lipschitz constant $|D^*\psi|$ of any Lipschitz function ψ is upper semicontinuous, leaving the verification of this property to the reader. Moreover we also point out that

$$|\nabla \psi|(x) \leq \left|D^* \psi\right|(x), \tag{16.16}$$

for any $x \in X$, as one can easily argue from the definitions.

Theorem 16.10 (Properties of the Hopf-Lax Semigroup) *Let (X, d) be a metric space and $f \in C_b(X)$. Then the function $(t, x) \mapsto Q_t f(x)$ satisfies the following properties:*

(i) $\inf f \le Q_t f \le f \le \sup f$;
(ii) $Q_t f \uparrow f$ *pointwise as* $t \downarrow 0$ *and the convergence is uniform if* f *is uniformly continuous;*
(iii) $Q_t f$ *is Lipschitz in* $(\epsilon, \infty) \times X$ *for all* $\epsilon > 0$;
(iv) if X *is a length space (i.e. the distance is given by the infimum of the length of curves), then for all* $x \in X$, *for* $\mathscr{L}^1$*-a.e.* $t > 0$ *one has*

$$\frac{\mathrm{d}}{\mathrm{d}t} Q_t f(x) + \frac{1}{2} |\nabla Q_t f(x)|^2 = 0; \tag{16.17}$$

(v) let $\varphi : X \to \mathbb{R}$ *be Lipschitz. Then* $Q_t \varphi$ *is Lipschitz on* $[0, \infty) \times X$ *and it verifies*

$$\frac{\mathrm{d}^+}{\mathrm{d}t} Q_t \varphi(x) + \frac{1}{2} \left| D^* Q_t \varphi \right|^2 (x) \le 0 \qquad \text{for any } (t, x) \in [0, \infty) \times X, \tag{16.18}$$

where we denoted by

$$\frac{\mathrm{d}^+}{\mathrm{d}t} Q_t \varphi(x) := \limsup_{h \downarrow 0} \frac{Q_{t+h} \varphi(x) - Q_t \varphi(x)}{h} \tag{16.19}$$

the upper Dini derivative;
(vi) if X *is a length space then* Q_t *is a semigroup, i.e.* $Q_t(Q_s f) = Q_{t+s} f$.

Proof

(i) It is a trivial consequence of the definition.
(ii) Let us denote by $E(y, t)$ the closed ball $\overline{B}(y, \sqrt{2tR_f})$ where $R_f = \sup f - \inf f$. We want to show that the value of $Q_t f(y)$ does not change if the infimum is taken only over $E(y, t)$. That is proven by the following chain of inequalities stating that the infimum out of $E(y, t)$ is too large:

$$\inf_{x \in E(y,t)^c} \frac{1}{2t} d^2(x, y) + f(x) \ge \inf_{x \in E(y,t)^c} R_f + f(x) \ge \sup f \ge Q_t f(y).$$

Exploiting what we have just gained we have

$$f(y) \ge Q_t f(y) = \inf_{x \in E(y,t)} \frac{1}{2t} d^2(x, y) + f(x) \ge \inf_{x \in E(y,t)} f(x)$$

and this easily implies our conclusion.

(iii) First of all, thanks to what we obtained throughout the proof of (ii) we know that for any $y_1, y_2 \in X$ and $t_1, t_2 > 0$

$$Q_{t_1} f(y_1) - Q_{t_2} f(y_2) \geq \inf_{x \in E(y_1, t_1)} \frac{d^2(x, y_1)}{2t_1} - \frac{d^2(x, y_2)}{2t_2} \tag{16.20}$$

where we have implicitly applied some easy inequalities involving the infimum on a set. It is sufficient to prove the local Lipschitz property separately with respect to t and x:

Lipschitz with respect to t: Applying (16.20) with $y_1 = y_2 = y$ we have

$$Q_{t_1} f(y) - Q_{t_2} f(y) \geq -\left|\frac{1}{2t_1} - \frac{1}{2t_2}\right| \sup_{x \in E(y, t_1)} d^2(x, y) \geq -\left|\frac{1}{t_1} - \frac{1}{t_2}\right| t_1 R_f.$$

Hence, swapping t_1 and t_2, we obtain

$$\left|Q_{t_1} f(y) - Q_{t_2} f(y)\right| \leq \left|\frac{1}{t_1} - \frac{1}{t_2}\right| \max(t_1, t_2) R_f = \frac{R}{\min(t_1, t_2)} |t_1 - t_2|.$$

Lipschitz with respect to x: Applying (16.20) with $t_1 = t_2 = t$ we get

$$\begin{aligned}
Q_t f(y_1) - Q_t f(y_2) &\geq -\frac{1}{2t} \inf_{x \in E(y_1, t)} d^2(x, y_1) - d^2(x, y_2) \\
&= -\frac{1}{2t} \inf_{x \in E(y_1, t)} (d(x, y_1) - d(x, y_2))\,(d(x, y_1) + d(x, y_2)) \\
&\geq -\frac{1}{2t} \sup_{x \in E(y_1, t)} d(y_1, y_2)\,(2d(x, y_1) + d(y_1, y_2)) \\
&\geq -\frac{1}{2t} \left(2t R_f \cdot d(y_1, y_2) + d^2(y_1, y_2)\right) \\
&= -R_f d(y_1, y_2) - \frac{1}{2t} d^2(y_1, y_2).
\end{aligned}$$

Swapping y_1 and y_2, we obtain

$$|Q_t f(y_1) - Q_t f(y_2)| \leq R_f d(y_1, y_2) + \frac{1}{2t} d^2(y_1, y_2). \tag{16.21}$$

To gain a global Lipschitz property from (16.21) it is sufficient to recall that the function $Q_t f$ is globally bounded.

(iv) For simplicity we shall prove the statement only in the Euclidean space $\mathbb{R}^n$ (because it is needed in the next lecture) and in a weaker form, namely for $\mathscr{L}^n$-a.e. x.

First of all, notice that thanks to the local Lipschitz property and to Rademacher Theorem 5.1, for $\mathscr{L}^n$-a.e. x one has that the property

$$y \mapsto Q_t f(y) \text{ is differentiable at } x$$

is satisfied for $\mathscr{L}^1$-a.e. $t > 0$. On the other hand, we already proved in the theory of gradient flows (see Sect. 3 in Lecture 14) that for all $x \in X$ one has

$$\frac{\mathrm{d}}{\mathrm{d}t} Q_t f(x) = -\frac{d^2(x, J_t(x))}{2t^2} \quad \text{for } \mathscr{L}^1\text{-a.e. } t > 0,$$

where $J_t(x)$ is any minimizer in the definition of $Q_t f(x)$ (whose existence is ensured by coercivity, since $\mathbb{R}^n$ is a proper metric space). Therefore, to prove the statement in the weaker form it is sufficient to check that

$$|\nabla Q_t f(x)|^2 = \frac{d^2(x, J_t(x))}{t^2} \tag{16.22}$$

at any differentiability point x of $Q_t f$.

Still in the metric theory of gradient flows, we already proved the inequality $\leq$. Let us refine the argument to get the equality $\nabla Q_f(x) = (x - J_t(x))/t$ (in particular, $J_t(x)$ is unique at any differentiability point), which implies (16.22) recalling that the metric slope coincides with the modulus of the gradient at any differentiability point. We have indeed

$$\begin{aligned} Q_t f(y) - Q_t f(x) &\leq f(J_t(x)) + \frac{1}{2t}|y - J_t(x)|^2 - \left(f(J_t(x)) + \frac{1}{2t}|x - J_t(x)|^2\right) \\ &= \langle \frac{x - J_t(x)}{t}, y - x\rangle + \frac{|y - x|^2}{2t} \end{aligned}$$

which implies the claimed equality.

(v) We leave the verification of the global Lipschitz continuity to the reader and prove that the Hopf-Lax semigroup satisfies (16.18) under the additional assumption that (X, d) is locally compact and $\varphi \in \mathrm{Lip}_b(X)$, that guarantees existence of minimizers. Under these simplified assumptions, for any $(x, t) \in X \times (0, \infty)$ we let x_t be such that

$$Q_t\varphi(x) = \varphi(x_t) + \frac{1}{2t} d^2(x, x_t).$$

Moreover, for any (x, t) we set $D^+(x, t)$ to be the maximum distance between x and x_t where x_t is as above. The general case can be handled using minimizing sequences instead of minimizers.

Then, for any $t, h > 0$ we can estimate

$$Q_{t+h}\varphi(x) - Q_t\varphi(x) \le \left(\varphi(x_t) + \frac{1}{2(t+h)}d^2(x, x_t)\right) - \left(\varphi(x_t) + \frac{1}{2t}d^2(x, x_t)\right)$$
$$\le -\frac{h}{2t(t+h)}d^2(x, x_t).$$

Dividing both sides by h and letting $h \downarrow 0$ we obtain

$$\frac{\mathrm{d}^+}{\mathrm{d}t} Q_t\varphi(x) \le -\frac{1}{2t^2}d^2(x, x_t).$$

Therefore to conclude it is sufficient to verify that

$$\left|D^* Q_t\varphi\right|(x) \le \frac{D^+(x, t)}{t}. \tag{16.23}$$

In order to do so we first observe that $(x, t) \mapsto D^+(x, t)$ is an upper semicontinuous function. This property can be verified with a completely elementary argument that we skip, referring to Proposition 3.2 in [13] for a detailed proof.

Then we can estimate, letting y_t be such that $Q_t\varphi(y) = \varphi(y) + d^2(y, y_t)/2$,

$$\lim_{r\to 0} \sup_{y,z\in B_r(x)} \frac{Q_t\varphi(z) - Q_t\varphi(y)}{d(y, z)} \le \lim_{r\to 0} \sup_{y,z\in B_r(x)} \frac{1}{2td(y, z)} \left(d^2(z, y_t) - d^2(y, y_t)\right)$$
$$\le \lim_{r\to 0} \sup_{y,z\in B_r(x)} \frac{1}{2t} \left(d(z, y_t) + d(y, y_t)\right)$$
$$\le \frac{1}{t} \lim_{r\to 0} \sup_{y\in B_r(x)} D^+(y, t) \le \frac{D^+(x, t)}{t},$$

where the last inequality follows from the upper semicontinuity of D^+. Combining this estimate with what we obtained before we get (16.18).

(vi) We already observed that the inequality $Q_{s+t} \le Q_s \circ Q_t$ holds for general metric spaces, let us sketch the proof of the converse inequality under the assumption that (X, d) is a length space. To this aim it is sufficient to prove that, for every $x, z \in X$ and for every $s, t > 0$ it holds

$$\inf_{y\in X} \left\{\frac{d^2(x, y)}{2s} + \frac{d^2(y, z)}{2t}\right\} \le \frac{d^2(x, z)}{2(t+s)}. \tag{16.24}$$

In order to prove (16.24) we observe that, since (X, d) is a length space, for every $\epsilon > 0$ there exists a curve $\gamma_\epsilon : [0, 1] \to X$ such that $\gamma_\epsilon(0) = x$, $\gamma_\epsilon(1) = z$ and $\ell(\gamma_\epsilon) < d(x, z) + \epsilon$. Choosing $\tau_\epsilon \in (0, 1)$ such that $d(\gamma_\epsilon(\tau_\epsilon), z)s =$

$d(x, \gamma_\epsilon(\tau_\epsilon))t$ we have that the equality case in (16.13) occurs and therefore, setting $y_\epsilon := \gamma_\epsilon(\tau_\epsilon)$,

$$\frac{d^2(z, y_\epsilon)}{2t} + \frac{d^2(x, y_\epsilon)}{2s} = \frac{(d(x, y_\epsilon) + d(z, y_\epsilon))^2}{2(t+s)} \leq \frac{(d(x, z) + \epsilon)^2}{2(t+s)}.$$

Passing to the infimum as $\epsilon \downarrow 0$ we get (16.24).

□

Lecture 17: The Benamou–Brenier Formula



We are going to show the tight connection between absolutely continuous curves in $(\mathscr{P}_2(\mathbb{R}^n), W_2)$ and solutions to the continuity equation. The first ingredient of this deep link is the Benamou–Brenier formula [20].

1 Benamou–Brenier Formula

Definition 17.1 (Quadratic Action) Given a probability measure $\mu \in \mathscr{P}_2(\mathbb{R}^n)$ and a Borel vector field $v : \mathbb{R}^n \to \mathbb{R}^n$ we define the quadratic action $\mathcal{A}(v, \mu)$ as

$$\mathcal{A}(v, \mu) := \int_{\mathbb{R}^n} |v|^2 \, \mathrm{d}\mu.$$

Theorem 17.2 (Benamou–Brenier Formula) *For all $\mu_0, \mu_1 \in \mathscr{P}_2(\mathbb{R}^n)$ one has:*

$$W_2^2(\mu_0, \mu_1) = \min \left\{ \int_0^1 \mathcal{A}(v_t, \mu_t) \, \mathrm{d}t : \frac{\mathrm{d}}{\mathrm{d}t}\mu_t + \mathrm{div}(v_t \mu_t) = 0 \ \text{ in } (0,1) \times \mathbb{R}^n \right\}, \tag{17.1}$$

where the minimization is among all curves $\mu_t : [0, 1] \to \mathscr{P}_2(\mathbb{R}^n)$ continuous w.r.t. the weak topology.

This formula should be seen as a first hint to a "Riemannian" structure of $\mathscr{P}_2(\mathbb{R}^n)$ which is discussed in Sect. 1 of Lecture 18. Indeed, we are saying that the squared distance is exactly the infimum taken over all connecting curves of a quantity that resembles the classical action.

We give separate proofs of the two inequalities that imply the desired identity. Furthermore we are going to give two different proofs of the *less than* inequality.

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The next lemma is of technical nature but, in the context of general measurable spaces and measurable maps, it succeeds in being quite interesting on its own.

Lemma 17.3 (Absolute Continuity of Push-Forward Measures) *Let X, Y be measurable spaces, $e : X \to Y$ a measurable function, $\mu \in \mathscr{M}(X)$ a finite nonnegative measure and $v \in L^p(\mu; \mathbb{R}^k)$ with $p \in (1, \infty)$. Then $e_\#(v\mu) \ll e_\#\mu$ with density $w \in L^p(e_\#\mu; \mathbb{R}^k)$ satisfying*

$$\|w\|_{L^p(e_\#\mu)} \leq \|v\|_{L^p(\mu)}.$$

Proof We prove this result by exploiting the basic duality between L^p and $L^{p'}$. Given a bounded measurable function $\phi : Y \to \mathbb{R}^k$, applying Hölder's inequality we obtain

$$\left| \int_Y \phi \, \mathrm{d}e_\#(v\mu) \right| = \left| \int_X \phi(e) \cdot v \, \mathrm{d}\mu \right| \leq \|v\|_{L^p(\mu)} \|\phi\|_{L^{p'}(e_\#\mu)}$$

and therefore the linear map $\phi \mapsto \langle e_\#(v\mu), \phi \rangle$ is continuous with respect to the $L^{p'}$ norm. By the density of the space of bounded measurable functions in $L^{p'}(e_\#\mu; \mathbb{R}^k)$ the map is representable by a function $w \in L^p(e_\#\mu; \mathbb{R}^k)$ such that $\|w\|_{L^p(e_\#\mu)} \leq \|v\|_{L^p(\mu)}$. It is now easy to check that w is exactly the density of $e_\#(v\mu)$ with respect to $e_\#\mu$. □

In the case of Polish spaces, the theory of disintegration provides a more constructive approach to the problem above.

Remark 17.4 Using the disintegration $\mu = \mu_y \otimes e_\#\mu$, with μ_y probability measures in X concentrated on $e^{-1}(y)$, it is not hard to check that

$$w(y) = \int_X v(x) \, \mathrm{d}\mu_y(x) \qquad \text{for } e_\#\mu\text{-a.e. } y \in Y.$$

Using this more explicit formula, from Jensen's inequality we obtain the pointwise estimate

$$|w(y)|^p \leq \int_X |v(x)|^p \, \mathrm{d}\mu_y(x)$$

that, by integration, provides the L^p estimate on w.

Proof of $\geq$ ***in Theorem 17.2*** It is of course enough to prove the existence of a weakly continuous curve μ_t that solves the continuity equation with respect to a velocity field v_t such that

$$W_2^2(\mu_0, \mu_1) \geq \int_0^1 \mathcal{A}(v_t, \mu_t) \, \mathrm{d}t. \tag{17.2}$$

We are going to explicitly construct both the curve and the velocity field.

Let us fix $\Sigma \in \Gamma_o(\mu_0, \mu_1)$. As curve we consider the geodesic induced by the optimal plan through the formula $\mu_t = (e_t)_\# \Sigma$, where $e_t : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ is the map $e_t(x, y) := (1-t)x + ty$. Thanks to Lemma 17.3, there exists $v_t \in L^2(\mu_t; \mathbb{R}^n)$ such that $(e_t)_\#((y-x)\Sigma) = v_t \mu_t$ and $\|v_t\|_{L^2(\mu_t)} \le \|y - x\|_{L^2(\Sigma)}$. This last inequality directly implies (17.2), since $\|y - x\|_{L^2(\Sigma)} = W_2(\mu_0, \mu_1)$.

Now it remains to check that μ_t is a solution to the continuity equation with velocity field v_t. As shown in Proposition 16.3, it is enough to observe that, for any $\phi \in C_c^\infty(\mathbb{R}^n)$, one has

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathbb{R}^n} \phi \,\mathrm{d}\mu_t &= \frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathbb{R}^n} \phi(e_t)\,\mathrm{d}\Sigma = \int_{\mathbb{R}^n} \langle \nabla\phi(e_t), y - x\rangle \,\mathrm{d}\Sigma \\ &= \int_{\mathbb{R}^n} \nabla\phi \cdot \mathrm{d}(e_t)_\#((y-x)\Sigma) = \int_{\mathbb{R}^n} \langle \nabla\phi, v_t\rangle \,\mathrm{d}\mu_t.\end{aligned}$$

□

In order to prove the inequality $\le$ in (17.1) we will need to mollify our measures, in order to gain regularity. Let us fix some notation and state a preliminary lemma.

Let us denote by $\varrho \in C^\infty(\mathbb{R}^n)$ a convolution kernel whose support is the whole $\mathbb{R}^n$ and such that $\int |x|^2 \varrho(x)\,\mathrm{d}x < \infty$. Then, for any $\epsilon > 0$, let us define the family of rescaled kernels $\varrho_\epsilon(x) := \epsilon^{-n}\varrho(\frac{x}{\epsilon})$.

Lemma 17.5 *Given $\mu \in \mathscr{P}_2(\mathbb{R}^n)$, the functions $\mu * \varrho_\epsilon$ are smooth and positive. Moreover, $\mu * \varrho_\epsilon \mathscr{L}^n$ converge to μ in $\mathscr{P}_2(\mathbb{R}^n)$ as $\epsilon \to 0^+$.*

Proof The first part of the statement is a simple consequence of the explicit formula

$$\mu * \varrho_\epsilon(x) = \int_{\mathbb{R}^n} \varrho_\epsilon(x - y)\,\mathrm{d}\mu(y) \tag{17.3}$$

and of the global positivity of ϱ.

To prove the W_2-convergence we observe that $\Sigma_\epsilon := \varrho_\epsilon(x - \cdot)\mathscr{L}^n \otimes \mu$ is an admissible plan from μ to $\mu * \varrho_\epsilon$. Therefore it holds

$$\begin{aligned}W_2^2(\mu, \mu * \varrho_\epsilon) &\le \int_{\mathbb{R}^n \times \mathbb{R}^n} |x - y|^2 \,\mathrm{d}\Sigma_\epsilon(x, y) \\ &= \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} |x - y|^2 \varrho_\epsilon(x - y)\,\mathrm{d}y\,\mathrm{d}\mu(x) = \epsilon^2 \int_{\mathbb{R}^n} |z|^2 \,\mathrm{d}\varrho(z),\end{aligned}$$

yielding that $W_2(\mu, \mu * \varrho_\epsilon) \to 0$ as $\epsilon \to 0^+$. □

Proof of $\le$ in Theorem 17.2 by Explicit Transport Construction We have to prove that, for every (v_t, μ_t) solving the continuity equation, it holds

$$W_2^2(\mu_0, \mu_1) \le \int_0^1 \mathcal{A}(v_t, \mu_t)\,\mathrm{d}t. \tag{17.4}$$

If $(v_t)_{0\le t\le 1}$ were smooth and with no more than linear growth, we could apply Proposition 16.4 and obtain that $\mu_t = (X_t)_\# \mu_0$, where X_t is the flow map of v_t. Thus, defining $\Sigma = (\mathrm{id} \times X_1)_\# \mu_0$, we would have that Σ is admissible and

$$\begin{aligned}\int_{\mathbb{R}^n} |x-y|^2 \,\mathrm{d}\Sigma(x,y) &= \int_{\mathbb{R}^n} |X_0(x) - X_1(x)|^2 \,\mathrm{d}\mu_0(x) = \int_{\mathbb{R}^n} \left| \int_0^1 \frac{\mathrm{d}}{\mathrm{d}t} X_t(x)\,\mathrm{d}t \right|^2 \mathrm{d}\mu_0(x)\\ &= \int_{\mathbb{R}^n} \left| \int_0^1 v_t(X_t(x))\,\mathrm{d}t \right|^2 \mathrm{d}\mu_0(x) \le \int_{\mathbb{R}^n} \int_0^1 |v_t(X_t(x))|^2 \,\mathrm{d}t\,\mathrm{d}\mu_0(x)\\ &= \int_0^1 \int_{\mathbb{R}^n} |v_t(X_t(x))|^2 \,\mathrm{d}\mu_0(x)\,\mathrm{d}t = \int_0^1 \int_{\mathbb{R}^n} |v_t|^2 \,\mathrm{d}\mu_t\,\mathrm{d}t\\ &= \int_0^1 \mathcal{A}(v_t,\mu_t)\,\mathrm{d}t,\end{aligned}$$

that proves (17.4) in the smooth setting.

We want to adapt this reasoning to the non-smooth case.

Recalling Lemma 17.5, we are allowed to define $\mu_t^\epsilon := \mu_t * \varrho_\epsilon$ and $v_t^\epsilon := \frac{(v_t\mu_t)*\varrho_\epsilon}{\mu_t^\epsilon}$. Through an easy computation we can check that the pair $(\mu_t^\epsilon \mathscr{L}^n, v_t^\epsilon)$ still solves the continuity equation and then, applying the smooth case of (17.4), we obtain

$$W_2^2(\mu_0^\epsilon, \mu_1^\epsilon) \le \int_0^1 \mathcal{A}(v_t^\epsilon, \mu_t^\epsilon)\,\mathrm{d}t.$$

We did not pay attention to the hypotheses on the growth needed to have existence of the flow X_ϵ associated to v_ϵ until time 1, this is a more delicate technical point that we omit for simplicity (see Proposition 8.1.8 of [12] for details).

Thanks to Lemma 17.5 we know that the left hand side converges to $W_2^2(\mu_0,\mu_1)$ as $\epsilon \to 0^+$. Moreover we are going to prove that for any $\epsilon > 0$

$$\mathcal{A}(v_t^\epsilon, \mu_t^\epsilon) \le \mathcal{A}(v_t, \mu_t) \qquad \forall t \in [0,1] \tag{17.5}$$

and this, of course, is enough to conclude the proof. Recalling (17.3), what we have to show is

$$\int_{\mathbb{R}^n} \frac{\left|\int \varrho_\epsilon(x-y) v_t(y)\,\mathrm{d}\mu_t(y)\right|^2}{\int \varrho_\epsilon(x-y)\,\mathrm{d}\mu_t(y)}\,\mathrm{d}x \le \int_{\mathbb{R}^n} |v_t(y)|^2 \,\mathrm{d}\mu_t(y). \tag{17.6}$$

Denoting by ν_x the probability measure $\frac{\varrho_\epsilon(x-\cdot)\,\mathrm{d}\mu_t}{\int \varrho_\epsilon(x-y)\,\mathrm{d}\mu_t(y)}$, (17.6) can be obtained integrating over $x \in \mathbb{R}^n$, with respect to the measure $\int \varrho_\epsilon(\cdot - y)\,\mathrm{d}\mu_t(y)\mathscr{L}^n$, the inequality

$$\left|\int_{\mathbb{R}^n} v_t(y)\,\mathrm{d}\nu_x(y)\right|^2 \le \int_{\mathbb{R}^n} |v_t(y)|^2\,\mathrm{d}\nu_x(y),$$

which holds thanks to Cauchy–Schwarz inequality. □

Now we are going to provide another proof of the inequality $\le$ in (17.1), based on a remarkable idea of K. Kuwada [79] which exploits the duality result established in Lecture 2.

We shall need two intermediate results. The first one improves upon some of the statements we proved so far. The second one is a technical lemma.

Lemma 17.6 *Let (v_t, μ_t) be a solution of the continuity equation on $[0,1]$ such that $\int_0^1 \|v_t\|_{L^1(\mu_t)}\,\mathrm{d}t < \infty$. Then for any bounded Lipschitz function $\varphi : \mathbb{R}^n \to \mathbb{R}$ and for any $0 \le s \le t \le 1$ it holds*

$$\left|\int_{\mathbb{R}^n} \varphi\,\mathrm{d}\mu_t - \int_{\mathbb{R}^n} \varphi\,\mathrm{d}\mu_s\right| \le \int_s^t \int_{\mathbb{R}^n} |D^*\varphi||v_t|\,\mathrm{d}\mu_r\,\mathrm{d}r, \tag{17.7}$$

where we denoted by $|D^*\varphi|$ the asymptotic Lipschitz constant of ϕ, that we introduced in (16.15)*.*

Proof If $\psi : \mathbb{R}^n \to \mathbb{R}$ is smooth then it follows from the distributional formulation of the continuity equation that

$$\left|\int_{\mathbb{R}^n} \psi\,\mathrm{d}\mu_t - \int_{\mathbb{R}^n} \psi\,\mathrm{d}\mu_s\right| = \left|\int_s^t \int_{\mathbb{R}^n} \nabla\psi \cdot v_t\,\mathrm{d}\mu_r\,\mathrm{d}r\right| \le \int_s^t \int_{\mathbb{R}^n} |\nabla\psi||v_r|\,\mathrm{d}\mu_r\,\mathrm{d}r, \tag{17.8}$$

for any $s,\ t \in [0,1]$. The estimate (17.7) can be achieved arguing by regularization. In order to do so we fix a smooth convolution convolution kernel $\varrho : \mathbb{R}^n \to [0,\infty)$ with compact support in $B_1(0)$. Let us also define the family of the rescaled kernels ϱ_ϵ with the usual procedure. Then we observe that, setting $\varphi_\epsilon := \varphi * \varrho_\epsilon$, it holds that φ_ϵ are smooth and they converge uniformly to φ as $\epsilon \to 0$. Moreover, an application of Jensen's inequality yields

$$|\nabla\varphi_\epsilon|(x) = |\nabla(\varphi * \varrho_\epsilon)|(x) \le (|\nabla\varphi| * \varrho_\epsilon)(x) \le \|\nabla\varphi\|_{L^\infty(B_\epsilon(x))},$$

for any $\epsilon > 0$ and for any $x \in \mathbb{R}^n$. Due to the convexity of the Euclidean ball and a standard argument we can also infer that

$$\|\nabla\varphi\|_{L^\infty(B_\epsilon(x))} = \sup_{y,z \in B_\epsilon(x)} \frac{|\varphi(z) - \varphi(y)|}{|z - y|}. \tag{17.9}$$

Combining the previous two estimates we obtain that

$$\limsup_{\epsilon\to 0} |\nabla\varphi_\epsilon|(x) \le \left|D^*\varphi\right|(x), \tag{17.10}$$

for any $x \in \mathbb{R}^n$.

Applying (17.8) with $\psi = \varphi_\epsilon$ for any $\epsilon > 0$ and then passing to the limit we obtain that

$$\left|\int_{\mathbb{R}^n} \varphi \,\mathrm{d}\mu_t - \int_{\mathbb{R}^n} \varphi \,\mathrm{d}\mu_s\right| \le \limsup_{\epsilon\to 0} \int_s^t \int_{\mathbb{R}^n} |\nabla\varphi_\epsilon||v_r| \,\mathrm{d}\mu_r \,\mathrm{d}r \le \int_s^t \int_{\mathbb{R}^n} \left|D^*\varphi\right||v_r| \,\mathrm{d}\mu_r \,\mathrm{d}r,$$

where we applied Fatou lemma and (17.10). □

Lemma 17.7 *Let (v_t, μ_t) be a solution of the continuity equation on $(0,1)\times\mathbb{R}^n$ such that $\int_0^1 \|v_t\|_{L^1(\mu_t)}\,\mathrm{d}t < \infty$. Then there exists a negligible set $\mathcal{N} \subset (0,1)$ such that, for any upper semicontinuous and bounded function $u : \mathbb{R}^n \to \mathbb{R}$ and for any $t \in (0,1)\setminus\mathcal{N}$, it holds*

$$\limsup_{h\to 0} \frac{1}{h}\int_t^{t+h} \int_{\mathbb{R}^n} u|v_s| \,\mathrm{d}\mu_s \,\mathrm{d}s \le \int_{\mathbb{R}^n} u|v_t| \,\mathrm{d}\mu_t. \tag{17.11}$$

Proof It is enough to prove the result for $u \in C_b(\mathbb{R}^n)$, the general case follows from a simple regularization argument, reminiscent of that in the proof of Theorem 2.6. Indeed, given an upper semicontinuous and bounded function u, for any $\epsilon > 0$ we can set

$$u^\epsilon(x) := \sup_{y\in\mathbb{R}^n} \left\{u(y) - \frac{1}{2\epsilon}|x-y|^2\right\},$$

and notice that $u^\epsilon \in C_b(\mathbb{R}^n)$, $u(x) \le u^\epsilon(x) \le \|u\|_{L^\infty}$, and $\lim_{\epsilon\to 0} u^\epsilon(x) = u(x)$ for any $x \in \mathbb{R}^n$. Assuming that (17.11) holds for bounded continuous functions one has

$$\limsup_{h\to 0} \frac{1}{h}\int_t^{t+h} \int_{\mathbb{R}^n} u|v_s| \,\mathrm{d}\mu_s \,\mathrm{d}s \le \limsup_{h\to 0} \frac{1}{h}\int_t^{t+h} \int_{\mathbb{R}^n} u^\epsilon|v_s| \,\mathrm{d}\mu_s \,\mathrm{d}s \le \int_{\mathbb{R}^n} u^\epsilon|v_t| \,\mathrm{d}\mu_t,$$

and letting $\epsilon \to 0$ we get the sought conclusion.

For any $f \in C_b(\mathbb{R}^n)$ we denote by $\mathcal{N}(f) \subset (0,1)$ the set of times $t \in (0,1)$ satisfying

$$\lim_{h\to 0} \frac{1}{h}\int_t^{t+h} \int_{\mathbb{R}^n} f|v_s| \,\mathrm{d}\mu_s \,\mathrm{d}s = \int_{\mathbb{R}^n} f|v_t| \,\mathrm{d}\mu_t.$$

Notice that $\mathscr{L}^1(\mathcal{N}(f)) = 0$ since $(0,1)\setminus\mathcal{N}(f)$ contains the set of Lebesgue points of $t \mapsto \int_{\mathbb{R}^n} f|v_t|\,\mathrm{d}\mu_t$, which belongs to $L^1(0,1)$. Set

$$\mathcal{N} := \bigcup_{\phi,\,\psi\in D\cup\{1\}} \mathcal{N}(\phi\psi),$$

where $D \subset C_c(\mathbb{R}^n)$ is countable, dense with respect to the uniform convergence on compact sets, and 1 denotes the constant function. Notice that $\mathcal{N}$ is $\mathscr{L}^1$-negligible.

Let $u : \mathbb{R}^n \to \mathbb{R}$ be bounded and upper semicontinuous. For any $\epsilon > 0$ there are and $\phi,\ \psi \in D$ with $0 \le \psi \le 1$ and $|u-\phi| \le \epsilon$ on $\operatorname{supp}\psi$. Given $t \in (0,1)\setminus\mathcal{N}$, it holds

$$\begin{aligned}
&\limsup_{h\to 0}\frac{1}{h}\int_t^{t+h}\int_{\mathbb{R}^n} u|v_s|\,\mathrm{d}\mu_s\,\mathrm{d}s\\
&\quad= \limsup_{h\to 0}\frac{1}{h}\int_t^{t+h}\int_{\mathbb{R}^n} \psi u|v_s|\,\mathrm{d}\mu_s\,\mathrm{d}s + \limsup_{h\to 0}\frac{1}{h}\int_t^{t+h}\int_{\mathbb{R}^n} (1-\psi)u|v_s|\,\mathrm{d}\mu_s\,\mathrm{d}s\\
&\quad\le \limsup_{h\to 0}\frac{1}{h}\int_t^{t+h}\int_{\mathbb{R}^n} \psi\phi|v_s|\,\mathrm{d}\mu_s\,\mathrm{d}s + \epsilon\limsup_{h\to 0}\frac{1}{h}\int_t^{t+h}\int_{\mathbb{R}^n} |v_s|\,\mathrm{d}\mu_s\,\mathrm{d}s\\
&\qquad+ \|u\|_{L^\infty}\limsup_{h\to 0}\frac{1}{h}\int_t^{t+h}\int_{\mathbb{R}^n} (1-\psi)|v_s|\,\mathrm{d}\mu_s\,\mathrm{d}s\\
&\quad\le \int_{\mathbb{R}^n}\psi\phi|v_t|\,\mathrm{d}\mu_t + \epsilon\int_{\mathbb{R}^n}|v_t|\,\mathrm{d}\mu_t + \|u\|_{L^\infty}\int_{\mathbb{R}^n}(1-\psi)|v_t|\,\mathrm{d}\mu_t\\
&\quad\le \int_{\mathbb{R}^n}\psi u|v_t|\,\mathrm{d}\mu_t + 2\epsilon\int_{\mathbb{R}^n}|v_t|\,\mathrm{d}\mu_t + \|u\|_{L^\infty}\int_{\mathbb{R}^n}(1-\psi)|v_t|\,\mathrm{d}\mu_t.
\end{aligned}$$

Letting $\epsilon \to 0$ and $\psi \uparrow 1$ we conclude the proof. □

The combination of Lemmas 17.6 and 17.7 gives the following estimate that will be used in the proof of the Benamou–Brenier formula via duality. Let (v_t, μ_t) be a solution of the continuity equation such that $\int_0^1 \|v_t\|_{L^1(\mu_t)}\,\mathrm{d}t < \infty$. Then there exists a negligible set $\mathcal{N} \subset (0,1)$ such that for any Lipschitz function $\psi : \mathbb{R}^n \to \mathbb{R}$ and for any $t \in (0,1)\setminus\mathcal{N}$ it holds

$$\limsup_{h\to 0}\frac{1}{|h|}\left|\int_{\mathbb{R}^n}\psi\,\mathrm{d}\mu_{t+h} - \int_{\mathbb{R}^n}\psi\,\mathrm{d}\mu_t\right| \le \int_{\mathbb{R}^n}\left|D^*\psi\right||v_t|\,\mathrm{d}\mu_t. \tag{17.12}$$

In order to get (17.12) it is sufficient to apply (17.7) between times $t+h$ and t and then Lemma 17.7 to the upper semicontinuous function $u = |D^*\varphi|$.

Proof of* $\leq$ *of Theorem 17.2 by Duality Recalling Theorem 3.1, it is enough to show that for every $\varphi \in \mathrm{Lip}_b(\mathbb{R}^n)$ and (v_t, μ_t) solving the continuity equation one has

$$2\left(-\int_{\mathbb{R}^n} \varphi \, \mathrm{d}\mu_0 + \int_{\mathbb{R}^n} Q_1\varphi \, \mathrm{d}\mu_1\right) \leq \int_0^1 \mathcal{A}(v_t, \mu_t) \, \mathrm{d}t,$$

where Q_t is the Hopf-Lax semigroup that we introduced in Definition 16.7 (notice that the factor 2 is due to the fact that we are using the duality with respect to the cost $\frac{1}{2}|x-y|^2$).

Let us consider the map $\eta(s,t) := \int_{\mathbb{R}^n} Q_s\varphi \, \mathrm{d}\mu_t$ for $s,\ t \in [0,1]$. Observe that, since φ is Lipschitz, the Hopf-Lax semigroup is Lipschitz on $[0,1] \times \mathbb{R}^n$. Denoting by L its Lipschitz constant, we can infer by Lemma 17.6 that

$$\big|\eta(s,t) - \eta(s',t)\big| \leq L\big|s - s'\big|, \qquad \big|\eta(s,t) - \eta(s,t')\big| \leq L\int_t^{t'} \|v_r\|_{L^1(\mu_r)} \, \mathrm{d}r, \tag{17.13}$$

for any $0 \leq s \leq s' \leq 1$ and $0 \leq t \leq t' \leq 1$. Hence we can apply the weak chain rule in Lemma 4.3.4 of [12] to obtain that $t \mapsto \eta(t,t) = \int_{\mathbb{R}^n} Q_t\varphi \, \mathrm{d}\mu_t$ is absolutely continuous and that

$$\frac{\mathrm{d}}{\mathrm{d}t}\eta(t,t) \leq \limsup_{h\to 0^+} \frac{1}{h}\int_{\mathbb{R}^n} Q_t\varphi \, \mathrm{d}(\mu_t - \mu_{t-h}) + \limsup_{h\to 0^+} \frac{1}{h}\int_{\mathbb{R}^n} (Q_{t+h}\varphi - Q_t\varphi) \, \mathrm{d}\mu_t, \tag{17.14}$$

for $\mathscr{L}^1$-a.e. $t \in (0,1)$.

Applying (17.12) with the choice $\psi = Q_t\varphi$ and Fatou's lemma (observe that the functions $(Q_{t+h}\varphi - Q_t\varphi)/h$ are uniformly bounded from above thanks to the Lipschitz property of the semigroup) we get that, for $\mathscr{L}^1$-a.e. $t \in (0,1)$, it holds

$$\frac{\mathrm{d}}{\mathrm{d}t}\eta(t,t) \leq \int_{\mathbb{R}^n} \big|D^*Q_t\varphi\big||v_t| \, \mathrm{d}\mu_t + \int_{\mathbb{R}^n} \frac{\mathrm{d}^+}{\mathrm{d}t} Q_t\varphi \, \mathrm{d}\mu_t.$$

Thanks to the Holder inequality and (16.18) we get that

$$\frac{\mathrm{d}}{\mathrm{d}t}\eta(t,t) \leq \int_{\mathbb{R}^n} \left(\frac{\mathrm{d}^+}{\mathrm{d}t} Q_t\varphi + \frac{1}{2}\big|D^*Q_t\varphi\big|^2 + \frac{1}{2}|v_t|^2\right) \mathrm{d}\mu_t \leq \frac{1}{2}\int_{\mathbb{R}^n} |v_t|^2 \, \mathrm{d}\mu_t = \frac{1}{2}\mathcal{A}(v_t, \mu_t).$$

All in all we can estimate

$$-\int_{\mathbb{R}^n} \varphi \, \mathrm{d}\mu_0 + \int_{\mathbb{R}^n} Q_1\varphi \, \mathrm{d}\mu_1 = \int_0^1 \frac{\mathrm{d}}{\mathrm{d}t}\eta(t,t) \, \mathrm{d}t \leq \int_0^1 \frac{1}{2}\mathcal{A}(v_t, \mu_t) \, \mathrm{d}t,$$

getting the sought conclusion. □

2 Correspondence Between Absolutely Continuous Curves in $\mathscr{P}_2(\mathbb{R}^n)$ and Solutions to the Continuity Equation

A way to look at the Benamou–Brenier formula is that geodesics in $\mathscr{P}_2(\mathbb{R}^n)$ are represented by solutions to the continuity equation. Our goal is to generalize the result to absolutely continuous curves in $\mathscr{P}_2(\mathbb{R}^n)$. Therefore what we want is a two-ways connection between absolutely continuous curves and solutions to (CE):

- We would like to say that any solution μ_t to (CE), for some velocity field v_t such that $\int |v_t|^2 \, \mathrm{d}\mu_t \in L^1$, is an AC^2 curve with respect to W_2 and, moreover, that the metric derivative of μ_t can be estimated from above with the action of v_t relative to μ_t.
- On the other hand, we might also hope that, given an AC^2 curve μ_t w.r.t. W_2, there exists an "optimal" vector field v_t such that, when paired with μ_t, (CE) is satisfied and the action of v_t relative to μ_t equals the metric derivative of the curve.

Both statements hold and we prove them in this section.

Lemma 17.8 *Let $\gamma : [0,1] \to X$ be a curve with values in a metric space (X,d) satisfying*

$$d^2(\gamma(s),\gamma(t)) \le (t-s)\int_s^t g^2 \, \mathrm{d}r \qquad \forall s,\ t \in [0,1] \text{ with } s \le t$$

for some $g \in L^2(0,1)$. Then $\gamma \in AC^2([0,1];X)$ and $|\gamma'| \le g$ holds $\mathscr{L}^1$-a.e. in $(0,1)$.

Proof Since

$$d(\gamma(s),\gamma(t)) \le \frac{1}{2}(t-s)+\frac{1}{2}\int_s^t g^2 \, \mathrm{d}r = \frac{1}{2}\int_s^t (1+g^2)\, \mathrm{d}r \quad \forall s,\ t \in [0,1] \text{ with } s \le t$$

we obtain that γ is absolutely continuous.

At any metric differentiability point $t \in (0,1)$ which is a Lebesgue point of g^2, choosing $s = t+h$, dividing both sides by h^2 and taking the limit as $h \to 0$ provides the estimate on the metric derivative. □

Given Lemma 17.8, the proofs of the two statements anticipated above are not very hard.

Proposition 17.9 *Let $[0,1] \ni t \mapsto \mu_t \in \mathscr{P}_2(\mathbb{R}^n)$ be a solution to the continuity equation induced by the vector field $(v_t)_{0\le t\le 1}$ and assume that $t \mapsto \mathcal{A}(v_t,\mu_t)$ belongs to $L^1(0,1)$. Then $\mu_t \in AC^2([0,1];\mathscr{P}_2(\mathbb{R}^n))$ and $|\mu_t'|^2 \le \mathcal{A}(v_t,\mu_t)$ for $\mathscr{L}^1$-almost every $t \in (0,1)$.*

Proof Applying a rescaled version of the Benamou–Brenier formula (Theorem 17.2) we obtain

$$W_2^2(\mu_s, \mu_t) \le (t-s) \int_s^t \mathcal{A}(v_r, \mu_r)\, \mathrm{d}r \qquad \text{for all } s,\ t \in [0,1] \text{ with } s \le t.$$

The statement then follows by Lemma 17.8. □

Theorem 17.10 *Given $\mu_t \in AC^2([0,1]; \mathscr{P}_2(\mathbb{R}^n))$, there exists a velocity field $(v_t)_{0\le t\le 1}$ such that μ_t solves the associated continuity equation and*

$$\mathcal{A}(v_t, \mu_t) = \left|\mu_t'\right|^2 \qquad \text{for } \mathscr{L}^1\text{-a.e. } t \in (0,1).$$

Proof Thanks to Remark 10.8 there exists a dynamic plan $\eta \in \mathscr{P}(C([0,1], \mathbb{R}^n))$ supported on AC^2 curves that is a lifting for μ_t and satisfies

$$\int_{C([0,1],\mathbb{R}^n)} \mathcal{A}_2(\gamma)\, \mathrm{d}\eta(\gamma) \le \int_0^1 \left|\mu_t'\right|^2 \mathrm{d}t < \infty. \tag{17.15}$$

Therefore, for $\mathscr{L}^1$-almost every $t \in (0,1)$ one has that the map $\gamma \mapsto \gamma'(t)$ belongs to $L^2(\eta)$. As a consequence, Lemma 17.3 ensures that for $\mathscr{L}^1$-almost every $t \in (0,1)$ we can write $(e_t)_\#(\gamma'(t)\eta) = v_t \mu_t$ for a suitable $v_t \in L^2(\mu_t; \mathbb{R}^n)$ with $\|v_t\|_{L^2(\mu_t)} \le \|\gamma'(t)\|_{L^2(\eta)}$. Now we check that, with this choice of v_t, the continuity equation holds. Given $\phi \in C_c^\infty(\mathbb{R}^n)$ we have

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t} \int_{\mathbb{R}^n} \phi\, \mathrm{d}\mu_t &= \frac{\mathrm{d}}{\mathrm{d}t} \int_{C([0,1],\mathbb{R}^n)} \phi(\gamma(t))\, \mathrm{d}\eta(\gamma) = \int_{C([0,1],\mathbb{R}^n)} \langle \nabla\phi(\gamma(t)), \gamma'(t)\rangle\, \mathrm{d}\eta(\gamma) \\
&= \int_{C([0,1],\mathbb{R}^n)} \nabla\phi(\gamma(t)) \cdot \mathrm{d}\left(\gamma'(t)\eta\right)(\gamma) = \int_{\mathbb{R}^n} \nabla\phi \cdot \mathrm{d}(e_t)_\#\left(\gamma'(t)\eta\right) \\
&= \int_{\mathbb{R}^n} \nabla\phi \cdot \mathrm{d}(v_t\mu_t) = \int_{\mathbb{R}^n} \langle \nabla\phi, v_t\rangle\, \mathrm{d}\mu_t
\end{aligned}$$

and this is enough, as proven in Proposition 16.3. Applying Fubini's theorem we have $\int_0^1 \mathcal{A}(v_t, \mu_t)\, \mathrm{d}t \le \int \mathcal{A}_2(\gamma)\, \mathrm{d}\eta(\gamma)$. Therefore recalling (17.15) and the fact that $|\mu_t'|^2 \le \int_0^1 \|\gamma'(t)\|^2_{L^2(\eta)}\, \mathrm{d}t \le \mathcal{A}(v_t, \mu_t)$ thanks to Proposition 17.9, we get also $\mathcal{A}(v_t, \mu_t) = |\dot\mu_t|^2$ for $\mathscr{L}^1$-a.e. $t \in (0,1)$. □

Remark 17.11 As the reader may have noticed, this last proof is nothing more than a straightforward generalization of the proof of the $\ge$ inequality in the Benamou–Brenier formula (17.1).

A nice application of Theorem 17.10 is the following estimate.

Corollary 17.12 *Let $\mu_t \in AC^2([0,1], \mathscr{P}_2(\mathbb{R}^n))$ and $f \in \mathrm{Lip}_b(\mathbb{R}^n) \cap C^1(\mathbb{R}^n)$. Then*

$$\left| \int_{\mathbb{R}^n} f \, \mathrm{d}\mu_1 - \int_{\mathbb{R}^n} f \, \mathrm{d}\mu_0 \right| \leq \int_0^1 \left(\int_{\mathbb{R}^n} |\nabla f|^2 \, \mathrm{d}\mu_t \right)^{1/2} |\mu_t'| \, \mathrm{d}t. \tag{17.16}$$

Proof Thanks to Theorem 17.10 we can find a velocity field v_t such that μ_t solves the associated continuity equation and $|\mu_t'| = \|v_t\|_{L^2(\mu_t)}$ for $\mathscr{L}^1$-almost every $t \in (0,1)$. Therefore we can estimate

$$\begin{aligned}
\left| \int_{\mathbb{R}^n} f \, \mathrm{d}\mu_1 - \int_{\mathbb{R}^n} f \, \mathrm{d}\mu_0 \right| &= \left| \int_0^1 \frac{\mathrm{d}}{\mathrm{d}t} \int_{\mathbb{R}^n} f \, \mathrm{d}\mu_t \, \mathrm{d}t \right| = \left| \int_0^1 \left(\int_{\mathbb{R}^n} \langle \nabla f, v_t \rangle \, \mathrm{d}\mu_t \right) \mathrm{d}t \right| \\
&\leq \int_0^1 \left(\int_{\mathbb{R}^n} |\nabla f|^2 \, \mathrm{d}\mu_t \right)^{\frac{1}{2}} \|v_t\|_{L^2(\mu_t)} \, \mathrm{d}t \\
&= \int_0^1 \left(\int_{\mathbb{R}^n} |\nabla f|^2 \, \mathrm{d}\mu_t \right)^{1/2} |\mu_t'| \, \mathrm{d}t,
\end{aligned}$$

where in the second line we applied Hölder's inequality. □

Remark 17.13 If we assume that $\mu_t \ll \mathscr{L}^n$ for $\mathscr{L}^1$-almost every $t \in (0,1)$, then we can drop the regularity assumption $f \in C^1(\mathbb{R}^n)$ in Corollary 17.12. Indeed, we can observe that, under this additional assumption, the right hand side of (17.16) is well defined for any $f \in \mathrm{Lip}_b(\mathbb{R}^n)$ thanks to Rademacher's theorem and one can prove, by a standard regularization procedure, that the estimate still holds in this case.

Lecture 18: An Introduction to Otto's Calculus



1 Otto's Calculus

In the seminal papers [94, 95], Otto introduced a formal interpretation of the Wasserstein distance as a Riemannian distance in $\mathscr{P}_2(\mathbb{R}^n)$, using this interpretation as a guide to rigorous results on the large time asymptotics of solutions to porous medium equations. To simplify the presentation, let us work in the subspace $\mathscr{P}_2^a(\mathbb{R}^n)$ of absolutely continuous measures μ, identified with their densities ϱ.

According to Otto's calculus, elements s of the tangent space $T_\varrho \mathscr{P}_2^a(\mathbb{R}^n)$ are thought as functions with null mean and gradient tangent vectors $v = \nabla\phi$ are coupled to tangent vectors s by solving the degenerate elliptic PDE

$$-\operatorname{div}\big((\nabla\phi)\varrho\big) = s.$$

Notice that this picture is fully consistent with the structure of the continuity equation, that we extensively discussed in the previous sections. Otto's metric tensor is then

$$\langle s, s'\rangle_\varrho := \int_{\mathbb{R}^n} \langle \nabla\phi, \nabla\phi'\rangle \varrho \, \mathrm{d}x \quad \text{whenever} \quad -\operatorname{div}\big((\nabla\phi)\varrho\big) = s, \ -\operatorname{div}\big((\nabla\phi')\varrho\big) = s'. \tag{18.1}$$

Notice that the vector fields v are *gradient* vector fields and that, when $\varrho \sim 1$, this metric is reminiscent of the flat H^{-1} metric on tangent vectors. The restriction to gradient vector fields can be understood, for instance, on the basis of the fact that optimal transport maps are gradients. In addition, the same holds for the natural velocity field $v_t = t^{-1}\nabla\psi_t$ attached to geodesics (see Sect. 2 in Lecture 16).

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In this perspective, the Benamou–Brenier formula can be precisely understood as the fact that W_2 is the Riemannian distance associated to this metric tensor, as $\int_{\mathbb{R}^n} |\nabla\phi_t|^2 \varrho_t \, \mathrm{d}x$ is precisely the Riemannian action of the curve ϱ_t solving a continuity equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\varrho_t + \mathrm{div}\left((\nabla\phi_t)\varrho_t\right) = 0.$$

More generally, the close link between solutions of the continuity equations and absolutely continuous curves in $\mathscr{P}_2(\mathbb{R}^n)$, proved in Proposition 17.9, shows that this interpretation of $\mathscr{P}_2(\mathbb{R}^n)$ as a kind of infinite-dimensional Riemannian manifold is fully consistent and goes beyond the class of absolutely continuous measures.

2 Formal Interpretation of Some Evolution Equations as Wasserstein Gradient Flows

This section is devoted to discuss at a formal level the possibility of interpreting some evolution equations as gradient flows of energy functionals with respect to the Wasserstein distance, relying on Otto's calculus and computing the "Wasserstein gradient" $\nabla^W \mathcal{E}$. In many cases the metric theory can then be used, among other things, to make this interpretation fully rigorous and in Sect. 3 we will perform this task for the heat equation.

Assume that an energy $\mathcal{E} : \mathscr{P}_2(\mathbb{R}^n) \to (-\infty, \infty]$ is given. We recall that any gradient flow μ_t must be in particular an absolutely continuous curve with metric derivative locally in L^2. At this stage, thanks to Theorem 17.10 we can say that μ_t is a solution of the continuity equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\mu_t + \mathrm{div}(v_t\mu_t) = 0, \tag{18.2}$$

for some velocity field v_t such that $\|v_t\|_{L^2(\mu_t)} \in L^1_{\mathrm{loc}}(0, \infty)$. Thus, if we are able to identify the Wasserstein gradient of $\mathcal{E}$, the condition

$$v_t = -\nabla^W \mathcal{E}(\mu_t) \quad \text{for } \mathscr{L}^1\text{-a.e. } t \in (0, \infty) \tag{18.3}$$

turns the continuity equation (18.2) into

$$\frac{\mathrm{d}}{\mathrm{d}t}\mu_t = \mathrm{div}(\nabla^W \mathcal{E}(\mu_t)\mu_t). \tag{18.4}$$

Let us compute then the Wasserstein gradients of the main examples of energy functionals we introduced so far: internal energy, potential energy and interaction energy.

As we discussed in Lecture 13, having a metric tensor at our disposal, the problem of computing the Wasserstein gradient of $\mathcal{E}$ at any $\varrho \in \mathscr{P}_2^a(\mathbb{R}^n)$ is reduced to the computation of the action of its differential $\mathrm{d}_\varrho \mathcal{E}$ on any $s \in T_\varrho \mathscr{P}_2^a(\mathbb{R}^n)$, and therefore to the computation of

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \mathcal{E}(\varrho_t),$$

where ϱ_t is any absolutely continuous curve passing through ϱ with velocity s at time $t = 0$. To this aim we fix $\varphi \in C_c^\infty(\mathbb{R}^n)$ and we consider the vector field $v = \nabla\varphi$. There are two natural choices for a smooth curve in the space of probabilities with velocity v at time 0, namely

$$\text{either} \quad \varrho_t := (\mathrm{id} + tv)_\# \varrho \quad \text{or} \quad \tilde{\varrho}_t := (X_t)_\# \varrho,$$

where X_t is the flow map associated to v. It is easily seen (recall Lecture 16) that the velocity field for the former is $v_t = v \circ (\mathrm{id} + tv)^{-1}$, while the velocity field for the latter is time-independent, and equal to v.

Let us focus on the case of the internal energy functional $\mathbb{U}$ associated to a density U. As a consequence of the previous remarks, defining $s := \frac{\mathrm{d}}{\mathrm{d}t}|_{t=0}\varrho_t$, we can compute

$$\begin{aligned}
\mathrm{d}_\varrho \mathbb{U}(s) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \mathbb{U}(\varrho_t) &= \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \left(\int_{\mathbb{R}^n} U(\varrho_t)\,\mathrm{d}x \right) \\
&= -\int_{\mathbb{R}^n} U'(\varrho)\,\mathrm{div}(v\varrho)\,\mathrm{d}x \\
&= \int_{\mathbb{R}^n} \langle \nabla U'(\varrho), v\rangle \varrho\,\mathrm{d}x\,.
\end{aligned}$$

This last expression allows us to identify $\nabla^W \mathbb{U}(\varrho)$ with $\nabla U'(\varrho)$, since with this choice we recover

$$\langle \nabla^W \mathbb{U}(\varrho), s\rangle_\varrho = \mathrm{d}_\varrho \mathbb{U}(s),$$

which is the relation that a Riemannian gradient should satisfy.

With analogous computations one can formally compute the Wasserstein gradients also for the potential energy $\mathbb{V}$ (induced by a density V) and for the interaction energy $\mathbb{W}$ (induced by a density W), obtaining

$$\nabla^W \mathbb{V}(\varrho) = \nabla V \qquad \text{and} \qquad \nabla^W \mathbb{W}(\varrho) = \nabla\,(W * \varrho)\,(= (\nabla W) * \varrho)\,.$$

Combining all these ingredients one can consider the general case of an energy

$$\mathcal{E}(\varrho) = \int_{\mathbb{R}^n} (U(\varrho) + V\varrho + (W * \varrho)\,\varrho)\ \mathrm{d}x$$

and the discussion above allows us to formally interpret the gradient flow of $\mathcal{E}$ in $(\mathscr{P}_2(\mathbb{R}^n), W_2)$ as a solution of the following evolution equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\varrho_t = \operatorname{div}\big((\nabla U'(\varrho_t) + \nabla V + \nabla(W * \varrho_t))\varrho_t\big). \tag{18.5}$$

We observe that the right hand side of (18.5) is given by a combination of a diffusion term (induced by the internal energy), a transport term (associated to the potential energy) and an interaction term (induced by the interaction energy).

The rest of this section is devoted to the discussion of a few explicit examples: the heat equation, the Fokker–Planck equation, the porous medium equation and the system of interacting particles.

Example 18.1 (The Heat Equation) If we consider the Wasserstein gradient flow associated to the Shannon-Boltzmann logarithmic entropy Ent (i.e. the internal energy associated to the density $U(\varrho) = \varrho\ln(\varrho)$) we can observe that

$$U'(\varrho) = 1 + \ln(\varrho) \quad \text{and} \quad \nabla U'(\varrho) = \frac{\nabla\varrho}{\varrho}.$$

Thus, the formal computation of the Wasserstein gradient we presented above allows us to interpret (up to the usual identification between densities ϱ and measures μ) the heat flow as gradient flow of Ent, since

$$\frac{\mathrm{d}}{\mathrm{d}t}\varrho_t = \Delta\varrho_t = \operatorname{div}(\nabla\varrho_t) = \operatorname{div}(\frac{\nabla\varrho_t}{\varrho_t}\varrho_t) = \operatorname{div}(\nabla^W \mathrm{Ent}(\varrho_t)\varrho_t).$$

Let us briefly recapitulate the interpretations of the heat equation as a gradient flow we introduced so far:

(i) in Lecture 13 we proved that the heat flow can be interpreted as gradient flow of the Dirichlet energy $\int_{\mathbb{R}^n} |\nabla\varrho|^2$ with respect to the L^2 metric;
(ii) in the same Lecture we proved that it can also be interpreted as gradient flow of the energy $\int_{\mathbb{R}^n} \varrho^2$ with respect to the H^{-1} metric;
(iii) we just proved (formally, for the moment) that the heat flow admits an interpretation as gradient flow of Ent with respect to the Wasserstein metric.

Example 18.2 (The Fokker–Planck Equation) Let us consider the energy $\mathcal{E}$ defined by $\mathcal{E}(\varrho) := \mathrm{Ent}(\varrho) + \int_{\mathbb{R}^n} V\varrho\,\mathrm{d}x$, where V is any smooth potential such that $\int_{\mathbb{R}^n} e^{-V} < \infty$. Then, thanks to the discussion above,

$$\nabla^W \mathcal{E}(\varrho) = \frac{\nabla\varrho}{\varrho} + \nabla V,$$

thus we can say that any gradient flow ϱ_t of $\mathcal{E}$ in $\mathscr{P}$ with respect to the Wasserstein metric is a solution of the Fokker–Planck equation we introduced in Lecture 13 (see Eq. (13.3)). Indeed

$$\frac{\mathrm{d}}{\mathrm{d}t}\varrho_t = \operatorname{div}\left((\frac{\nabla\varrho_t}{\varrho_t} + \nabla V)\varrho_t\right) = \Delta\varrho_t + \operatorname{div}\left((\nabla V)\varrho_t\right).$$

To let the picture be more complete we recall that we already introduced an interpretation of the Fokker–Planck equation as a gradient flow in Lecture 13. We also remark that, according to (15.5) in Proposition 15.6, we can interpret $\mathcal{E}$ as the relative entropy Ent_γ, where $\gamma = e^{-V}\mathscr{L}^n$, hence we can see (FP) as the gradient flow of Ent_γ with respect to the Wasserstein metric (still at a formal level).

At this point of the discussion some questions arise: why are these new interpretations relevant? And can we get anything out of them?

First of all we remark that the interpretations presented above and their many variants provide new contractivity estimates and rates of convergence for many partial differential equations (starting from the seminal paper [95], see also [12] for more details about this topic). We also point out that in the Wasserstein interpretation there is a clear separation between metric and measure (i.e. the entropy functional depends only on the reference measure, while the Wasserstein metric depends only on the underlying distance of the ambient space). In the Hilbertian interpretation presented in Lecture 13 instead, both the distance and the reference measure are involved in the definition of the Dirichlet energy.

To conclude this discussion, let us consider the specific case of the Fokker–Planck equation (FP) when γ is the Gaussian measure: solutions converge to the equilibrium γ as $t \to \infty$. But, while the interpretation of (FP) as an Hilbertian gradient flow in $L^2(\gamma)$ does not provide a rate of convergence (because of the lack of uniform convexity in L^2), in the W_2 picture the exponential rate of convergence holds, and it comes as a consequence the 1-convexity of the relative entropy Ent_γ guaranteed by (15.5). In particular, this provides the estimate

$$W_2(w_t\gamma, \gamma) \le e^{-t} W_2(w_0\gamma, \gamma) \qquad \forall t \ge 0.$$

Example 18.3 (The Porous-Medium Equation) Let us consider for any $m \neq 1$ the internal energy $\mathbb{U}(\varrho) := \int_{\mathbb{R}^n} \frac{1}{m-1}\varrho^m$. With this choice we have that

$$U(\varrho) = \frac{\varrho^m}{m-1}, \quad U'(\varrho) = \frac{m}{m-1}\varrho^{m-1} \quad \text{and} \quad \nabla U'(\varrho) = m\varrho^{m-2}\nabla\varrho = \frac{\nabla\varrho^m}{\varrho},$$

therefore we can say that any W_2 gradient flow of $\mathbb{U}$ is indeed a solution of

$$\frac{\mathrm{d}}{\mathrm{d}t}\varrho_t = \operatorname{div}(\frac{\nabla{\varrho_t}^m}{\varrho_t}\varrho_t) = \Delta\varrho_t^m,$$

that is the porous-medium equation we already discussed in Lecture 13.

Example 18.4 (A System of Interacting Particles) We are given a system of N particles $x_1, \dots, x_N \in \mathbb{R}^n$ with weights $m_1, \dots, m_N$ such that $\sum m_i = 1$ and a (smooth) symmetric potential $W : \mathbb{R}^n \to \mathbb{R}$. Assume for simplicity that $\nabla W(0) = 0$ (even though this assumption is not realistic for many applications) and that the particles evolve according to the following system of first order ordinary differential equations:

$$x_i'(t) = \sum_{j \neq i} m_j \nabla W(x_i(t) - x_j(t)) \tag{18.6}$$

$$= \sum_{j} m_j \nabla W(x_i(t) - x_j(t)) \qquad i = 1, \dots, N. \tag{18.7}$$

If we introduce the time-dependent empirical measures

$$\mu_t := \sum_{i=1}^{N} m_i \delta_{x_i(t)},$$

then (18.6) can be rephrased in the following terms

$$x_i'(t) = ((\nabla W) * \mu_t)(x_i(t)) \qquad i = 1, \dots, N. \tag{18.8}$$

Thanks to the linearity with respect to μ of the continuity equation and the duality results between solutions of (ODE) and (CE) we explored in Lecture 16, we can conclude that the empirical measures μ_t solve the continuity equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\mu_t = \operatorname{div}(((\nabla W) * \mu_t)\, \mu_t)$$

and finally that the system can be interpreted as a gradient flow of the interaction energy $\mathbb{W}$ associated to the potential W.

Moving our attention to the second order problem

$$m_i x_i''(t) = \sum_{i \neq j} m_i m_j \nabla W(x_i(t) - x_j(t)) \qquad i = 1, \dots, N, \tag{18.9}$$

after the introduction of the empirical measure

$$\tilde{\mu}_t = \sum_{i=1}^{N} m_i \delta_{(x_i(t), x_i'(t))}$$

(that is a measure in the phase space with variables (x, p) this time), we can see that also this system evolves according to the continuity equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\tilde{\mu}_t = \mathrm{div}(b_t\tilde{\mu}_t),$$

where now the velocity field b_t is given by

$$b_t(x, p) := (p, \nabla W * \mu_t(x)).$$

In this Hamiltonian framework it is no more possible to get a gradient flow interpretation of (18.9), since the vector field b is not a gradient.

3 Rigorous Interpretation of the Heat Equation as a Wasserstein Gradient Flow

In this section we provide a rigorous interpretation of the heat equation on $\mathbb{R}^n$ as gradient flow of the entropy functional in the Wasserstein space $(\mathscr{P}_2(\mathbb{R}^n), W_2)$. In this connection, it is worth to remember the paper [72] which provides another justification of this fact based on the implicit Euler scheme.

We already know from Theorem 15.16 that Ent is geodesically convex on $\mathscr{P}_2(\mathbb{R}^n)$, thanks to the fact that the energy density $U(\varrho) = \varrho \ln \varrho$ satisfies McCann's condition (MC). Observe also that, as we already pointed out, to give a meaning to the EVI formulation of gradient flows we just need the metric structure of the underlying space: the next results show exactly that the heat flow is an EVI gradient flow of the entropy functional.

Let us recall that, for the heat equation in $\mathbb{R}^n$ with initial datum $\bar{\varrho} \in L^1(\mathbb{R}^n)$, one has the explicit expression for the solution

$$\varrho_t(x) = \int_{\mathbb{R}^n} p_t(x, y)\bar{\varrho}(y)\,\mathrm{d}y, \tag{18.10}$$

where

$$p_t(x, y) := (4\pi t)^{-\frac{n}{2}} \exp(-\frac{1}{4t}|x - y|^2) \tag{18.11}$$

is the so called heat kernel. Assuming that $\bar{\varrho}$ is a probability density with finite quadratic moments, the aim of this section is to check that ϱ_t, seen as a curve $\mu_t = \varrho_t\mathscr{L}^n$ in $\mathscr{P}_2(\mathbb{R}^n)$, is an EVI gradient flow. We will use in part the explicit expression (18.10) of ϱ_t to justify some estimates and computations. First of all, one can use the explicit expression to show that the properties of $\bar{\varrho}$, namely being a probability density with finite quadratic moments, are preserved by the heat flow. Therefore it does make sense to consider the curve $\mu_t = \varrho_t\mathscr{L}^n$ as a curve in $\mathscr{P}_2(\mathbb{R}^n)$.

Theorem 18.5 *Let $\bar\varrho \in L^1(\mathbb{R}^n)$ be nonnegative, such that*

$$\int_{\mathbb{R}^n} \bar\varrho(x)\,\mathrm{d}x = 1, \quad \int_{\mathbb{R}^n} |x|^2 \bar\varrho(x)\,\mathrm{d}x < \infty \quad \textit{and} \quad \mathrm{Ent}(\bar\varrho) < \infty.$$

Then $\mu_t = \varrho_t \mathscr{L}^n$ solves EVI *in* $(\mathscr{P}_2(\mathbb{R}^n), W_2)$.

The proof of Theorem 18.5 will come as a consequence of some intermediate results. First of all, in order to give sense to the EVI formulation of gradient flows, we need to check that μ_t is (locally) absolutely continuous with respect to the Wasserstein metric.

Proposition 18.6 *Under the same assumptions of Theorem 18.5 the curve $t \mapsto \mu_t$ belongs to $AC^2_{\mathrm{loc}}((0,\infty); \mathscr{P}_2(\mathbb{R}^n))$.*

Proof We start from the observation that, as a consequence of the explicit expression of ϱ_t, we can say that $\varrho_t \in C^\infty(\mathbb{R}^n) \cap L^p(\mathbb{R}^n)$ for any $p \in [1,\infty]$, that $\varrho_t > 0$ and that the map $t \mapsto \varrho_t$ is continuously differentiable in $(0,\infty)$, as a $L^2(\mathbb{R}^n)$-valued map, with

$$\frac{\mathrm{d}}{\mathrm{d}t}\varrho_t = \Delta \varrho_t \qquad \forall t \in (0,\infty). \tag{18.12}$$

Thus, if we define

$$v_t := -\frac{\nabla \varrho_t}{\varrho_t}$$

it follows that the continuity equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\mu_t + \mathrm{div}(v_t \mu_t) = 0 \tag{18.13}$$

holds in the sense of distributions.

Therefore, in order to prove the W_2-absolute continuity of the curve μ_t, it will be sufficient to apply Proposition 17.9, after proving that $\|v_t\|^2_{L^2(\mu_t)}$ is locally integrable in $(0,\infty)$. Using the explicit expression

$$v_t(x) = -\frac{\int_{\mathbb{R}^n} \nabla_x p_t(x,y)\bar\varrho(y)\,\mathrm{d}y}{\int_{\mathbb{R}^n} p_t(x,y)\bar\varrho(y)\,\mathrm{d}y}$$

and arguing as in (17.6) one can prove that

$$\int_{\mathbb{R}^n} |v_t|^2(x)\,\mathrm{d}\mu_t(x) \le \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} \frac{|\nabla_x p_t(x,y)|^2}{p_t(x,y)}\bar\varrho(y)\,\mathrm{d}x\,\mathrm{d}y$$

so that $\|v_t\|_{L^2(\mu_t)} \in L^\infty(\epsilon, \infty)$ for any $\epsilon > 0$. However, we prefer to provide another proof of integrability, inspired by Ambrosio et al. [14] (dealing with the theory of heat flow in metric measure spaces) which provides a space-time sharp estimate of $\|v_t\|^2_{L^2(\mu_t)}$ very much in the spirit of the EDI theory of gradient flows.

To this aim, for $\epsilon \in (0, e^{-1})$, in order to take care of the singularity at $r = 0$ of the density $e(r) = r \ln r$, we define the regularized energy densities $e_\epsilon(r)$ as follows:

$$\begin{cases} e_\epsilon(r) = (1 + \ln \epsilon) r & \text{in } [0, \epsilon] \\ e_\epsilon(r) = r \ln r + \epsilon & \text{in } [\epsilon, \infty). \end{cases}$$

Observe that $e(r) \le e_\epsilon(r) \le \frac{1}{2} r^2$, $e_\epsilon(r) \downarrow e(r)$ as ϵ goes to 0 and $e_\epsilon \in C^{1,1}(0, \infty)$. Then extend e_ϵ to $C^{1,1}(\mathbb{R})$ convex densities $\tilde{e}_\epsilon$ writing $\tilde{e}_\epsilon(r) = e_\epsilon(r) - (1 + \ln \epsilon) r$ for $r \ge 0$ and $\tilde{e}_\epsilon(r) = 0$ for $r < 0$. Observe that $\tilde{e}_\epsilon(\bar{\varrho}) \in L^1(\mathbb{R}^n)$ and one can prove that the map

$$t \mapsto \int_{\mathbb{R}^n} \tilde{e}_\epsilon(\varrho_t(x))\, \mathrm{d}x$$

is locally Lipschitz on $(0, \infty)$, with the explicit expression for its derivative provided by (18.12) and an integration by parts:

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{\mathbb{R}^n} \tilde{e}_\epsilon(\varrho_t(x))\, \mathrm{d}x = \int_{\mathbb{R}^n} \tilde{e}'_\epsilon(\varrho_t) \Delta \varrho_t \, \mathrm{d}x = - \int_{\mathbb{R}^n} \tilde{e}''_\epsilon(\varrho_t(x)) |\nabla \varrho_t(x)|^2 \, \mathrm{d}x. \tag{18.14}$$

Recalling that the total mass is preserved by the heat flow, an integration in time of (18.14) yields to

$$\int_{\mathbb{R}^n} e_\epsilon(\varrho_T(x))\, \mathrm{d}x + \int_0^T \left(\int_{\{\varrho_t > \epsilon\}} \frac{|\nabla \varrho_t(x)|^2}{\varrho_t(x)} \, \mathrm{d}x \right) \mathrm{d}t = \int_{\mathbb{R}^n} e_\epsilon(\bar{\varrho}(x))\, \mathrm{d}x. \tag{18.15}$$

Passing to the limit as $\epsilon \to 0$ in (18.15) we conclude that

$$\mathrm{Ent}(\mu_T) + \int_0^T \left(\int_{\{\varrho_t > 0\}} \frac{|\nabla \varrho_t(x)|^2}{\varrho_t(x)} \, \mathrm{d}x \right) \mathrm{d}t = \mathrm{Ent}(\bar{\mu}) \qquad \forall T \ge 0,$$

yielding in particular the stated local integrability in time of $\|v_t\|^2_{L^2(\mu_t)}$, and concluding the proof. □

Remark 18.7 (The Heat Equation as an EDE *Gradient Flow)* Notice that, as stated in Theorem 15.26, computations analogous to those in Theorem 15.25 (dealing with the slightly simpler case of the relative entropy with respect to a Gaussian) show

that $\int_{\{\varrho_t>0\}} \frac{|\nabla \varrho_t(x)|^2}{\varrho_t(x)}\,\mathrm{d}x$ coincides with the slope of the Entropy at μ_t. In addition, the same quantity provides an upper bound for the metric derivative of the curve μ_t, thanks to Proposition 17.9. Therefore (18.15) implies

$$\mathrm{Ent}(\mu_T) + \int_0^T \frac{1}{2}|\mu_t'|^2 + \frac{1}{2}|\nabla^- \mathrm{Ent}|^2(\mu_t)\,\mathrm{d}t \le \mathrm{Ent}(\bar\mu) \qquad \forall T \ge 0,$$

namely the EDI formulation of gradient flows. However, our goal in this section is to achieve the stronger EVI property.

The forthcoming Lemmas 18.8 and 18.10 encode the information about the "Riemannian-like" behaviour of the Wasserstein distance and the geodesic convexity of the entropy functional, respectively.

Lemma 18.8 *If $\nu \in \mathscr{P}_2(\mathbb{R}^n)$ has compact support then*

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t} W_2^2(\mu_t, \nu) = \int_{\mathbb{R}^n} \langle T_{\mu_t}^\nu - \mathrm{id}, \nabla \varrho_t\rangle\,\mathrm{d}x \quad \text{for } \mathscr{L}^1\text{-a.e. } t \in (0,\infty), \tag{18.16}$$

where $T_{\mu_t}^\nu$ is the unique optimal transport map from μ_t to ν (with quadratic cost), whose existence follows from Theorem 5.2.

Proof Thanks to Proposition 18.6 we know that μ_t is locally absolutely continuous on $(0,\infty)$ with respect to the W_2 distance. It follows in particular that the map

$$t \mapsto \frac{1}{2} W_2^2(\mu_t, \nu)$$

is differentiable $\mathscr{L}^1$-a.e. on $(0,\infty)$. From now on we fix such a differentiability point t and we use the fact, already pointed out in the proof of Proposition 18.6, that (18.12) holds in the strong L^2 sense. The differentiability of $s \mapsto W_2^2(\mu_s, \nu)$ at $s = t$ justifies the following expansion:

$$\frac{1}{2} W_2^2(\mu_{t+h}, \nu) - \frac{1}{2} W_2^2(\mu_t, \nu) = \frac{h}{2}\frac{\mathrm{d}}{\mathrm{d}s}\bigg|_{s=t} W_2^2(\mu_s, \nu) + o(h) \quad (h \to 0)\,. \tag{18.17}$$

Thanks to the compactness assumption on ν we can find a Lipschitz Kantorovich potential φ from μ_t to ν. From the optimality of φ we deduce that

$$\frac{1}{2} W_2^2(\mu_t, \nu) = \int_{\mathbb{R}^n} \varphi\,\mathrm{d}\mu_t + \int_{\mathbb{R}^n} Q_1(-\varphi)\,\mathrm{d}\nu,$$

while

$$\frac{1}{2} W_2^2(\mu_{t+h}, \nu) \ge \int_{\mathbb{R}^n} \varphi\,\mathrm{d}\mu_{t+h} + \int_{\mathbb{R}^n} Q_1(-\varphi)\,\mathrm{d}\nu.$$

Hence

$$\begin{aligned}\frac{1}{2}W_2^2(\mu_{t+h},\nu)-\frac{1}{2}W_2^2(\mu_t,\nu) &\geq \int_{\mathbb{R}^n}\varphi\,\mathrm{d}(\mu_{t+h}-\mu_t)\\ &= h\int_{\mathbb{R}^n}\varphi\Delta\varrho_t\,\mathrm{d}x+o(h)\quad(h\to 0)\qquad(18.18)\\ &= -h\int_{\mathbb{R}^n}\langle\nabla\varphi,\nabla\varrho_t\rangle\,\mathrm{d}x+o(h)\quad(h\to 0),\end{aligned}$$

where in the last two passages we used (18.12) and integrated by parts, respectively. Notice that the integration by parts is justified by the boundedness of $\nabla\varphi$ and by the fact that all derivatives of ϱ_t are rapidly decreasing at infinity.

By comparing (18.18) with (18.17) we conclude that

$$\frac{\mathrm{d}}{\mathrm{d}s}\bigg|_{s=t}W_2^2(\mu_s,\nu)=-\int_{\mathbb{R}^n}\langle\nabla\varphi,\nabla\varrho_t\rangle\,\mathrm{d}x$$

and, after recalling the explicit expression $T_{\mu_t}^{\nu}=\mathrm{id}-\nabla\varphi$ of the optimal transport map in terms of the Kantorovich potential φ, we get the desired conclusion. □

Remark 18.9 The heuristic interpretation of Lemma 18.8 is that if you want to differentiate the square of the distance form a fixed point q along a curve σ on a Riemannian manifold (at a differentiability point $p=\sigma(t)$) you just need to couple with the metric tensor the speed of the curve $\dot\sigma(t)$ and the speed $\dot\gamma(0)$ of a geodesic γ such that $\gamma(0)=p$ and $\gamma(1)=q$. In the case of our interest $T_{\mu_t}^{\nu}-\mathrm{id}$ is the speed at time 0 of the geodesic joining μ_t with ν, while the speed of the curve μ_t should equal $\frac{\nabla\varrho_t}{\varrho_t}$ if we expect the heat flow to be a gradient flow of the entropy functional. Thus at the right hand-side of (18.16) we recover exactly their scalar product with respect to the "Riemannian metric" of $\mathscr{P}_2(\mathbb{R}^n)$ at the point μ_t.

Lemma 18.10 *If $\mu,\nu\in\mathscr{P}_2(\mathbb{R}^n)$ are such that ν has compact support and $\mu=\varrho\mathscr{L}^n$ with $\varrho\in C^\infty(\mathbb{R}^n)$, bounded and with a bounded integrable gradient, then*

$$\mathrm{Ent}(\nu)\geq\mathrm{Ent}(\mu)+\int_{\mathbb{R}^n}\langle T_\mu^\nu-\mathrm{id},\nabla\varrho\rangle\,\mathrm{d}x.\qquad(18.19)$$

In particular this holds with $\varrho=\varrho_t$ for all $t>0$.

Proof Consider the unique constant speed W_2 geodesic μ_s joining $\mu_0=\mu$ with $\mu_1=\nu$ and recall that, defining $T:=T_\mu^\nu$ and $T_s:=(1-s)T+s\mathrm{id}$, we have $\mu_s=(T_s)_\#\mu$. Thanks to Theorem 15.16, which yields the geodesic convexity of Ent, we have that

$$\mathrm{Ent}(\nu)-\mathrm{Ent}(\mu)\geq\frac{\mathrm{Ent}(\mu_s)-\mathrm{Ent}(\mu)}{s}\quad\text{for any } s\in(0,1],$$

thus, in order to get the desired conclusion, it suffices to prove that

$$\liminf_{s\to 0^+} \frac{\mathrm{Ent}(\mu_s) - \mathrm{Ent}(\mu)}{s} \geq \int_{\mathbb{R}^n} \langle T - \mathrm{id}, \nabla \varrho \rangle \, \mathrm{d}x.$$

To this aim we recall that Proposition 15.14 provides the explicit expression for the density ϱ_s of the interpolating measure μ_s, namely

$$\varrho_s = \left(\frac{\varrho}{\det \nabla T_s} \right) \circ T_s^{-1},$$

so that an application of the change of variables formula yields

$$\mathrm{Ent}(\mu_s) = \int_{\mathbb{R}^n} \ln \left(\frac{\varrho}{\det \nabla T_s} \right) \varrho \, \mathrm{d}x. \tag{18.20}$$

Thanks to Brenier's theorem and the compactness of the support of ν, T is given by ∇f, where $f : \mathbb{R}^n \to \mathbb{R}$ is a convex Lipschitz function. Moreover, by Alexandrov's Theorem 6.4, f is twice differentiable $\mathscr{L}^n$-a.e. (and then μ-a.e.), and also in the sense of distributions. Denoting by div the pointwise divergence and by Div the distributional divergence operators, as in the proof of the isoperimetric inequality we use the inequality between measures

$$\mathrm{Div}\, T = \mathrm{tr}\, D^2 f \geq \mathrm{tr}\, \nabla^2 f \mathscr{L}^n = \mathrm{div}\, T \mathscr{L}^n .$$

The inequality above comes from Theorem 6.4, since the nonnegativity of $D^2 f$ guarantees also the nonnegativity of the singular part of $D^2 f$ with respect to the Lebesgue measure, whose trace is the difference $\mathrm{tr}\, D^2 f - \mathrm{tr}\, \nabla^2 f \mathscr{L}^n$. We conclude that, in the sense of distributions,

$$\mathrm{Div}(T - \mathrm{id}) \geq \mathrm{div}(T - \mathrm{id}) \mathscr{L}^n . \tag{18.21}$$

Alexandrov's theorem ensures us also that the first order expansion

$$\det \nabla T_s(x) = 1 + s \, \mathrm{div}(T - \mathrm{id})(x) + o(s) \quad (s \to 0) \tag{18.22}$$

holds for μ-a.e. $x \in \mathbb{R}^n$.

Applying Fatou's lemma and taking into account (18.22), (18.20) and (18.21), we get

$$\begin{aligned} \liminf_{s\to 0^+} \frac{\mathrm{Ent}(\mu_s) - \mathrm{Ent}(\mu)}{s} &\geq - \int_{\mathbb{R}^n} \mathrm{div}(T - \mathrm{id}) \varrho \, \mathrm{d}x \\ &\geq \langle \mathrm{Div}(T - \mathrm{id}), \varrho \rangle \\ &= \int_{\mathbb{R}^n} \langle T - \mathrm{id}, \nabla \varrho \rangle \, \mathrm{d}x. \end{aligned}$$

Notice that the first inequality above is justified by boundedness and continuity of ϱ. The second one can be justified by approximating ϱ with the family $\varrho_R = \varrho\psi(\cdot/R)$, where $\psi \in C_c^\infty(\mathbb{R}^n)$ is identically equal to 1 on the unit ball. □

Remark 18.11 As for Lemma 18.8, also Lemma 18.10 admits a Riemannian interpretation. Indeed, for a geodesically convex (smooth) function f on a Riemannian manifold M, one has

$$f(q) \geq f(p) + \langle \nabla f(p), \dot{\gamma}(0) \rangle,$$

where $p,\ q \in M$ and $\gamma : [0, 1] \to M$ is a geodesic joining p to q. As before, at the right hand-side of (18.19), we can identify all the ingredients which appear in the smooth context, since $\nabla^W \mathrm{Ent}(\mu) = \frac{\nabla \varrho}{\varrho}$ and we already observed that $T_\mu^\nu - \mathrm{id}$ is the speed at time 0 of the geodesic joining μ to ν.

Proof of Theorem 18.5 Applying Lemmas 18.8 and 18.10, we obtain that

$$\mathrm{Ent}(\nu) \geq \mathrm{Ent}(\mu_t) + \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t} W_2^2(\mu_t, \nu)$$

for any $\nu \in \mathscr{P}_2(\mathbb{R}^n)$ with compact support. By a simple approximation procedure we can drop the compactness assumption on the support of ν and conclude that μ_t is the EVI gradient flow of Ent starting from $\bar{\mu}$. □

4 More Recent Ideas and Developments

Here and throughout the last lecture we will briefly present some more recent ideas and developments about Optimal Transport.

Contractivity via Action Estimates We first want to illustrate the idea, appeared in [97] and then refined in [44], that one can obtain contractivity estimates on a semigroup S via action estimates (see also [24] for similar ideas in the context of semigroups induced by conservation laws, where contractivity occurs with respect to L^1-like distances).

We are given a length space (X, d) and a semigroup S_t, $t \geq 0$, on X. We recall the definition of quadratic action of an absolutely continuous curve $\gamma : [0, 1] \to X$

$$\mathcal{A}(\gamma) = \frac{1}{2}\int_0^1 |\gamma'(s)|^2 \, \mathrm{d}s.$$

If we assume that the action estimate

$$\mathcal{A}(\gamma^t) \leq e^{-2kt}\mathcal{A}(\gamma) \qquad t \geq 0 \tag{18.23}$$

holds, where γ^t is the deformation of the curve γ induced by S_t, namely $\gamma^t(s) := S_t\gamma(s)$, then one can prove that S is k-contractive, namely

$$d(S_t x, S_t y) \le e^{-kt} d(x, y) \qquad \forall x,\ y \in X \quad \text{and} \quad \forall t \ge 0. \tag{18.24}$$

Indeed, for any absolutely continuous curve γ joining $\gamma(0) = x$ to $\gamma(1) = y$, the curve γ^t provides an admissible curve between $S_t x$ and $S_t y$, thus

$$\frac{1}{2} d^2(S_t x, S_t y) \le \mathcal{A}(\gamma^t) \le e^{-2kt} \mathcal{A}(\gamma).$$

Therefore, since (X, d) is a length space, passing to the infimum among all the admissible curves γ joining x to y, and taking the square roots of both sides, we conclude that (18.24) holds. Notice also that, conversely, contractivity immediately implies the action estimate (18.23), so that the two properties are equivalent.

We illustrate the validity of the method in the particular case of the heat semigroup in $\mathbb{R}^n$, with $(X, d) = (\mathscr{P}_2(\mathbb{R}^n), W_2)$. This example is also propedeutic to the next lecture, where we are going to investigate the deep relations between contractivity of the heat flow in the space of probability measures and many other geometric and analytic concepts.

Let $p_t(x, y)$ be the Euclidean heat kernel in (18.11) and, as in Sect. 3, define the heat flow starting from $f \in L^1(\mathbb{R}^n)$ with the classical formula:

$$P_t f(x) := \int_{\mathbb{R}^n} p_t(x, y) f(y)\, \mathrm{d}y \qquad t > 0. \tag{18.25}$$

We already remarked in that Section that, since sign and total mass are preserved by the evolution, the heat flow can also be seen as an evolution problem in the space of absolutely continuous probability measures with finite quadratic moments, when identified with their densities.

In addition, one has

$$p_t(x, \cdot) \text{ is a Gaussian probability density for all } (t, x) \in (0, \infty) \times X \tag{18.26}$$

and it is easily seen that, in measure theoretic terms, (18.25) can be read as

$$P_t f \mathscr{L}^n = \int_{\mathbb{R}^n} \left(p_t(x, \cdot) \mathscr{L}^n \right) f(x)\, \mathrm{d}x. \tag{18.27}$$

Lemma 18.12 *Let f, g be probability densities of measures in $\mathscr{P}_2(\mathbb{R}^n)$. Then*

$$W_2(P_t f \mathscr{L}^n, P_t g \mathscr{L}^n) \le W_2(f \mathscr{L}^n, g \mathscr{L}^n). \tag{18.28}$$

Remark 18.13 (Alternative Proofs) We notice that one could try to prove this result directly using the identity (18.27) for $P_t g\mathscr{L}^n$. Indeed, if T is the optimal map from $f\mathscr{L}^n$ to $g\mathscr{L}^n$, one has

$$P_t g\mathscr{L}^n = \int_{\mathbb{R}^n} \Big(p_t(y,\cdot)\mathscr{L}^n\Big) g(y)\,\mathrm{d}y = \int_{\mathbb{R}^n} \Big(p_t(T(x),\cdot)\mathscr{L}^n\Big) f(x)\,\mathrm{d}x.$$

This, together with the observation that $W_2(p_t(x,\cdot)\mathscr{L}^n, p_t(T(x),\cdot)\mathscr{L}^n) = |T(x)-x|$ (since the optimal map is the shift by the vector $T(x)-x$) provides the contractivity property arguing as in (19.22).

Furthermore, since the heat flow satisfies the EVI property (see once more Sect. 3) another, more abstract, proof would follow by the general theory of EVI gradient flows in metric spaces. Indeed, in Lemma 11.13 we proved that EVI_K gradient flows on Hilbert spaces are K-contractive and in the last part of the proof we already pointed out that the conclusion holds even for general metric spaces.

Here we give instead a proof where the monotonicity of the action is employed.

Proof of Lemma 18.12 Let ϱ_s be a geodesic in $\mathscr{P}_2(\mathbb{R}^n)$ from $f\mathscr{L}^n$ to $g\mathscr{L}^n$. If the equation

$$\frac{\mathrm{d}}{\mathrm{d}s}\varrho_s + \mathrm{div}(v_s\varrho_s) = 0,$$

is satisfied for some velocity field v_s then, letting $\varrho_s^t = P_t\varrho_s$ and $\mu_s^t = \varrho_s^t\mathscr{L}^n$ (which is a curve from $P_t f\mathscr{L}^n$ to $P_t g\mathscr{L}^n$), we would like to define a velocity field v_s^t in such a way that

$$\frac{\mathrm{d}}{\mathrm{d}s}\varrho_s^t + \mathrm{div}(v_s^t\varrho_s^t) = 0\,. \tag{18.29}$$

Then Proposition 17.9 would give

$$W_2^2(P_t f\mathscr{L}^n, P_t g\mathscr{L}^n) \le \int_0^1 |(\mu_s^t)'|^2\,\mathrm{d}s \le \int_0^1\int_{\mathbb{R}^n} |v_s^t|^2\varrho_s^t\,\mathrm{d}x\,\mathrm{d}s. \tag{18.30}$$

Moreover, as illustrated in Sect. 4, if our vector field v_s^t satisfies the (pointwise) action monotonicity property

$$\int_{\mathbb{R}^n} |v_s^t|^2\varrho_s^t\,\mathrm{d}x \le \int_{\mathbb{R}^n} |v_s|^2\varrho_s\,\mathrm{d}x, \tag{18.31}$$

for all $s\in(0,1)$, by integration of (18.30) we get

$$W_2^2(P_t f\mathscr{L}^n, P_t g\mathscr{L}^n) \le \int_0^1\int_{\mathbb{R}^n} |v_s|^2\varrho_s\,\mathrm{d}x\,\mathrm{d}s.$$

Finally, choosing the optimal velocity field v_s, see Theorem 17.10, we get (18.28).

Thus, we need to determine a velocity field v_s^t with properties (18.29) and (18.31). Understanding the action of the semigroup on vector fields componentwise, we have

$$-\operatorname{div}(v_s^t \varrho_s^t) = \frac{\mathrm{d}}{\mathrm{d}s}\varrho_s^t = P_t \frac{\mathrm{d}}{\mathrm{d}s}\varrho_s = -P_t \operatorname{div}(v_s \varrho_s) = -\operatorname{div}(P_t(v_s \varrho_s)). \tag{18.32}$$

Therefore any admissible velocity field v_t^t should satisfy

$$v_s^t = \frac{P_t(v_s \varrho_s)}{P_t(\varrho_s)} = \frac{P_t(v_s \varrho_s)}{\varrho_s^t} \tag{18.33}$$

up to a vector field G/ϱ_s^t, with G solenoidal. Choosing v_s^t exactly as in (18.33), we can now check that (18.31) holds. Indeed, let $\psi \in C_c(\mathbb{R}^n; \mathbb{R}^n)$ be a test function. Using the fact that P_t is self-adjoint in $L^2(\mathbb{R}^n)$, we have

$$\begin{aligned} \left| \int_{\mathbb{R}^n} \langle v_s^t, \psi \rangle \varrho_s^t \right| = \left| \int_{\mathbb{R}^n} \langle P_t(v_s \varrho_s), \psi \rangle \right| = \left| \int_{\mathbb{R}^n} \langle v_s, P_t \psi \rangle \varrho_s \right| \\ \leq \|v_s\|_{L^2(\varrho_s \mathscr{L}^n)} \|P_t \psi\|_{L^2(\varrho_s \mathscr{L}^n)}. \end{aligned} \tag{18.34}$$

Hence, taking (18.27) into account, by Jensen inequality we obtain

$$\int_{\mathbb{R}^n} (P_t \psi)^2 \varrho_s \, \mathrm{d}x \leq \int_{\mathbb{R}^n} P_t \psi^2 \varrho_s \, \mathrm{d}x = \int_{\mathbb{R}^n} \psi^2 \varrho_s^t \, \mathrm{d}x = \|\psi\|^2_{L^2(\varrho_s^t \mathscr{L}^n)}. \tag{18.35}$$

Thus, by (18.34) and (18.35) and by duality, we obtain

$$\|v_s^t\|_{L^2(\varrho_s^t \mathscr{L}^n)} \leq \|v_s\|_{L^2(\varrho_s \mathscr{L}^n)}.$$

□

When we move from the Euclidean space to Riemannian manifolds (and even more general spaces) we lose the commutation property between divergence and semigroup crucially used in (18.32). Therefore curvature comes into play and a deeper analysis is necessary: this will be the topic of the last lecture.

Convexity from EVI The following result, taken from [44], conveys the idea that any energy admitting an EVI gradient flow is automatically (geodesically-)convex. We state the result in a simplified form, see the original paper for the more general and useful statement dealing with lower semicontinuous functions with values in $(-\infty, \infty]$.

Theorem 18.14 (EVI **Implies Convexity**) *Let (X, d) be a geodesic space and let $F : X \to \mathbb{R}$ be a lower semicontinuous energy functional. Assume there exists an* EVI *gradient flow S_t of F starting from any $x \in X$. Then F is convex along all geodesics.*

Sketch of Proof Fix any $\gamma \in \mathrm{Geo}(X)$ and any intermediate time $s \in (0,1)$. Define, as before, $\gamma^t : [0,1] \to X$ by $\gamma^t(s) := S_t\gamma(s)$. Here, as in Lemma 11.13, we can use the lower semicontinuity of F to write the EVI differential inequality in a pointwise sense, namely

$$F(w) \geq F(S_r u) + \frac{\mathrm{d}}{\mathrm{d}t}^+\Big|_{t=r} d^2(S_t u, w) \qquad \forall r \geq 0,$$

where $\frac{\mathrm{d}}{\mathrm{d}t}^+$ denotes the upper right derivative. Applying EVI with $u = \gamma(s)$, $r = 0$ and with test points $w = \gamma(0)$, $w = \gamma(1)$ we obtain that

$$F(\gamma(0)) \geq \frac{\mathrm{d}}{\mathrm{d}t}^+\Big|_{t=0} \frac{1}{2} d^2(\gamma^t(s), \gamma(0)) + F(\gamma(s)) \tag{18.36}$$

and

$$F(\gamma(1)) \geq \frac{\mathrm{d}}{\mathrm{d}t}^+\Big|_{t=0} \frac{1}{2} d^2(\gamma^t(s), \gamma(1)) + F(\gamma(s)). \tag{18.37}$$

Multiplying (18.36) by $(1-s)$ and (18.37) by s and adding the two expressions we end up with

$$\begin{aligned}&(1-s)F(\gamma(0)) + sF(\gamma(1)) - F(\gamma(s)) \geq \\ &(1-s)\frac{\mathrm{d}}{\mathrm{d}t}^+\Big|_{t=0} \frac{1}{2} d^2(\gamma^t(s), \gamma(0)) + s\frac{\mathrm{d}}{\mathrm{d}t}^+\Big|_{t=0} \frac{1}{2} d^2(\gamma^t(s), \gamma(1)).\end{aligned}$$

The subadditivity of the upper right derivatives gives

$$(1-s)F(\gamma(0)) + sF(\gamma(1)) - F(\gamma(s)) \geq \frac{\mathrm{d}}{\mathrm{d}t}^+\Big|_{t=0} \Big((1-s)d^2(\gamma(0), \gamma^t(s)) + sd^2(\gamma^t(s), \gamma(1))\Big). \tag{18.38}$$

Observe now that the triangle inequality gives (see also (16.13))

$$(1-s)d^2(x,w) + sd^2(w,y) \geq s(1-s)d^2(x,y),$$

with equality when $w = \gamma(s)$ is the s-intermediate point of γ, therefore the right hand side in (18.38) is nonnegative and we obtain the convexity inequality. □

Monotonicity of Action and Energy Implies EVI The action estimate (18.23) has been cleverly modified in [44] into a differential inequality (with different roles of the deformation parameter t and the interpolation parameter s) involving action and energy which turns out to be sufficient for the validity of the EVI property.

In this case, the deformation scheme used to prove contractivity has to be modified as follows: we keep $v \in X$ fixed and, if $\gamma(s)$ is a continuous curve in $[0,1]$

connecting v to u, where u plays the role of the "test" point in the EVI formulation, we define

$$\gamma^t(s) := S_{st}(\gamma(s))$$

instead of $S_t(\gamma(s))$. With this notation, the following result (stated for simplicity only in the case $\mathrm{EVI} = \mathrm{EVI}_0$, i.e. $\lambda = 0$) holds:

Theorem 18.15 (A Differential Criterion for EVI**)** *Assume that a continuous contraction semigroup S in a length space (X, d) and a lower semicontinuous function $F : X \to \mathbb{R}$ satisfy the following assumptions for all $\gamma \in AC^2([0,1]; X)$:*

(i) $s \mapsto F(\gamma_s^t)$ *is absolutely continuous in* $[0,1]$ *for all* $t > 0$*;*

(ii) the maps $r \mapsto \frac{\mathrm{d}^+}{\mathrm{d}r} d^2(S_r(\gamma(s)), \gamma(s))$, $s \in [0,1]$, *are equi-integrable in all intervals* $[0, T]$*;*

(iii) for all $t > 0$ *one has*

$$\frac{1}{2}\frac{\mathrm{d}^+}{\mathrm{d}t}|(\gamma^t)'_+(s)|^2 + \frac{\mathrm{d}}{\mathrm{d}s}F(\gamma_s^t) \le 0 \qquad \text{for } \mathscr{L}^1\text{-a.e. } s \in (0,1), \tag{18.39}$$

where $\mathrm{d}^+/\mathrm{d}t$ *denotes, as usual, the upper right derivative, while* $|(\gamma^t)'_+(s)|$ *denotes the upper, right metric derivative.*

Then $S_t(v)$ satisfies EVI *and therefore it is the gradient flow of F.*

Proof By the semigroup property, it is sufficient to check EVI in the pointwise form (11.13) at $t = 0$. We fix $\gamma \in AC^2((0,1); X)$ connecting v to u. For all $t > 0$, by integration of (18.39) with respect to s we obtain

$$\frac{1}{2}\frac{\mathrm{d}^+}{\mathrm{d}t}\mathcal{A}(\gamma^t) \le F(\gamma_0^t) - F(\gamma_1^t) = F(v) - F(S_t(u)).$$

Notice that we have been able to pull the upper right derivative out of the integral thanks to Fatou's lemma and to the equi-integrability of the family of maps $s \mapsto |(\gamma^t)'(s)|^2$ in $L^1(0,1)$ (in turn, this equi-integrability is guaranteed by the contractivity of the semigroup S, together with (ii)).

Now, an integration with respect to t gives

$$\begin{aligned}\frac{1}{2}d^2(S_t(u), v) - \frac{1}{2}\mathcal{A}(\gamma^0) &\le \mathcal{A}(\gamma^t) - \mathcal{A}(\gamma^0)\\ &\le tF(v) - \int_0^t F(S_r(u))\,\mathrm{d}r.\end{aligned}$$

Since (X, d) is a length space and γ is arbitrary, we can replace $\mathcal{A}(\gamma^0) = \mathcal{A}(\gamma)$ by $d^2(u, v)$ in the inequality. Dividing both sides by t, the lower semicontinuity of F and the continuity of S yield (11.13). □

Lecture 19: Heat Flow, Optimal Transport and Ricci Curvature



In this final lecture we wish to explore the connections between Heat Flow, Optimal Transport and Ricci curvature on a smooth compact Riemannian manifold. In doing so we will extend and generalise some of the properties we have proved for the Euclidean space. Let us recall that, in particular, we have established the convexity of the logarithmic entropy in $(\mathscr{P}_2(\mathbb{R}^n), W_2)$ in Theorem 15.16 and the fact that the heat flow (defined as the L^2 gradient flow of the Dirichlet energy) is an EVI gradient flow of the relative entropy in $(\mathscr{P}_2(\mathbb{R}^n), W_2)$.

The Ricci curvature tensor is a fundamental geometric invariant of Riemannian manifolds. It is usually introduced as contraction of the Riemann curvature tensor, or equivalently as the average of the sectional curvatures of 2-planes passing through a given direction. That is to say, given $x \in M$ and a unit vector $v \in T_x M$ one chooses vectors $\{e_1, \dots, e_{n-1}\}$ in $T_x M$ such that $\{v, e_1, \dots, e_{n-1}\}$ is an orthonormal basis of $T_x M$ and then sets

$$\mathrm{Ric}(v, v) := \sum_{i=1}^{n-1} \kappa(v, e_i),$$

where we denoted by $\kappa(v, e_i)$ the sectional curvature of $(M, \mathbf{g})$ at x on the plane spanned by v, e_i. It can be proved that the definition is independent of the choice of the basis and that Ric is a quadratic form.

While its meaning might be quite obscure from the definition, it is usually advocated that the Ricci curvature governs the distortion of volumes (while the sectional curvature governs the distortion of lengths). Here we wish to provide some insights about this perspective, mainly connected with optimal transportation and in part inspired by [118], referring to [54, 99] for the basic treatment of curvature on Riemannian manifolds. In particular, in Theorem 19.4 we characterize lower bounds on the Ricci curvature in several ways, either in "Lagrangian" terms, via K-convexity of the logarithmic entropy in the Wasserstein space, or in "Eulerian" terms, via Bochner inequalities or gradient contractivity of the heat flow.

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The connections between Ricci curvature, gradient contractivity and more general functional inequalities, even in the context of diffusion semigroups, emerged in the fundamental paper [17], see also [81, 82, 84] and the more recent monograph [19]. The full picture, including the Optimal Transport side, has been understood with the contribution of many authors starting from the late Nineties, we just mention here [18, 42, 44, 72, 79, 94, 96, 97, 111], without the aim of being complete in this list.

The equivalences stated in Theorem 19.4 in a smooth context paved the way to *synthetic* notions of lower bounds on Ricci curvature (which can also be combined with upper bounds on the dimension) which make sense for abstract *metric measure spaces* $(X, d, \mathfrak{m})$ and *energy/metric structures* induced by diffusion semigroups. We refer to [118] and the more recent survey [6] and the literature mentioned therein for more on the Lagrangian side, and to [19] for more on the Eulerian side.

1 Heat Flow on Riemannian Manifolds

Let $(M, \mathbf{g})$ be a smooth n-dimensional compact and connected Riemannian manifold without boundary. Denoting for simplicity by $\mathfrak{m}$ the volume measure vol_M, the Dirichlet energy of a function $f \in L^2(\mathfrak{m})$ is given by

$$\mathbf{D}_{\mathbf{g}}(f) := \frac{1}{2}\int_M |\nabla f|^2_{\mathbf{g}}\, \mathrm{d}\mathfrak{m}$$

whenever ∇f is an $L^2(\mathfrak{m})$ section of the tangent bundle and $\mathbf{D}_{\mathbf{g}}(f) = +\infty$ otherwise.

Recall that, as we have already observed in Sect. 2 of Lecture 13, the Riemannian gradient ∇f of $f \in L^2(\mathfrak{m})$ is defined in the sense of distributions through the integration by parts formula (13.8).

Being $\mathbf{D}_{\mathbf{g}}$ convex, lower semicontinuous and densely defined, we know from Theorem 11.7 that for any $f \in L^2(\mathfrak{m})$ there exists a unique gradient flow $P_t f$ starting from f satisfying the semi-group property

$$P_t(P_s f) = P_{t+s} f \quad \text{in } L^2(\mathfrak{m}), \text{ for any } t,\ s \geq 0, \tag{19.1}$$

and the contraction estimate

$$\|P_t f\|_{L^2} \leq \|f\|_{L^2} \quad \text{for any } t \geq 0.$$

Moreover, since

$$\partial_G \mathbf{D}_{\mathbf{g}}(f) \neq \emptyset \qquad \Longleftrightarrow \qquad \Delta_{\mathbf{g}} f \in L^2(\mathfrak{m}) \quad \text{and} \quad \partial_G \mathbf{D}_{\mathbf{g}}(f) = \{-\Delta_{\mathbf{g}} f\},$$

the gradient flow $P_t f$ solves the heat equation

$$\frac{\mathrm{d}}{\mathrm{d}t} P_t f = \Delta_{\mathbf{g}} P_t f \qquad \text{for } \mathscr{L}^1\text{-a.e. } t \in (0, \infty), \tag{19.2}$$

where the left-hand side is understood as the derivative of the $L^2(\mathfrak{m})$-valued map $t \mapsto P_t f$. Above $\Delta_{\mathbf{g}} f$, for $f \in C^2(M)$ denotes the Laplace-Beltrami operator (see (13.6) for its formula in local coordinates) which is self-adjoint in $L^2(\mathfrak{m})$, i.e.

$$\int_M f\, \Delta_{\mathbf{g}} g \, \mathrm{d}\mathfrak{m} = \int_M g \Delta_{\mathbf{g}} f \, \mathrm{d}\mathfrak{m} \qquad f,\ g \in C^2(M). \tag{19.3}$$

Starting from (19.3) we can even define $\Delta_{\mathbf{g}} f$ in the sense of distributions for any $f \in L^1(\mathfrak{m})$. In this connection, one has also the formula, analogous to the Euclidean one

$$\Delta_{\mathbf{g}} f = \mathrm{div}_{\mathbf{g}} \nabla f$$

where, for a smooth vector field F, its divergence $\mathrm{div}_{\mathbf{g}}$ is characterized by the integration by parts formula

$$\int_M \mathbf{g}(\nabla g, F) \, \mathrm{d}\mathfrak{m} = -\int_M g \, \mathrm{div}_{\mathbf{g}} F \, \mathrm{d}\mathfrak{m}, \qquad \text{for any } \ g \in C^2(M),$$

or by its expression in local coordinates:

$$\mathrm{div}_{\mathbf{g}} F(x) := \frac{1}{\sqrt{\det \mathbf{g}_x}} \sum_i \frac{\partial}{\partial x_i} \big(\sqrt{\det \mathbf{g}_x} F^i(x) \big).$$

In the following proposition we list a few important properties of the Riemannian heat semigroup.

Proposition 19.1 *Let $(M, \mathbf{g})$ be a smooth compact Riemannian manifold without boundary, let P_t be the heat semigroup as above and let $f,\ g \in L^2(\mathfrak{m})$. Then the following hold:*

(i) the heat semigroup is stochastically complete and self adjoint, i.e.

$$\int_M P_t f \, \mathrm{d}\mathfrak{m} = \int_M f \, \mathrm{d}\mathfrak{m} \quad \text{and} \quad \int_M P_t f \, g \, \mathrm{d}\mathfrak{m} = \int_M f \, P_t g \, \mathrm{d}\mathfrak{m} \qquad \text{for any } t \geq 0;$$

(ii) the heat semigroup P_t commutes with $\Delta_{\mathbf{g}}$, i.e.

$$\Delta_{\mathbf{g}} P_t f = P_t(\Delta_{\mathbf{g}} f) \quad \text{for any } t \geq 0 \tag{19.4}$$

whenever f is in the domain of the Laplacian (i.e. $\partial_G \mathbf{D}_{\mathbf{g}}(f) \neq \emptyset$);

(iii) the following regularization estimates hold

$$\mathbf{D}_{\mathbf{g}}(P_t f) \le \frac{1}{2t}\|f\|_{L^2}^2, \quad \left\|\Delta_{\mathbf{g}} P_t f\right\|_{L^2}^2 \le \frac{1}{t^2}\|f\|_{L^2}^2 \qquad \text{for any } t > 0;$$

(iv) if $e : \mathbb{R} \to [0, \infty]$ is convex and lower semicontinuous then

$$\int_M e(P_t f)\,\mathrm{d}\mathfrak{m} \le \int_M e(f)\,\mathrm{d}\mathfrak{m} \quad \text{for any } t \ge 0;$$

(v) if for some constant $C \in \mathbb{R}$, $f \le C$ (resp. $f \ge C$), then $P_t f \le C$ (resp. $P_t f \ge C$) for any $t \ge 0$.

Proof

(i) A simple interpolation argument along with (19.3) gives

$$\begin{aligned}\int_M P_t f\, g\,\mathrm{d}\mathfrak{m} - \int_M f\, P_t g\,\mathrm{d}\mathfrak{m} &= \int_0^t \int_M \frac{\mathrm{d}}{\mathrm{d}s}(P_s f\, P_{t-s} g)\,\mathrm{d}\mathfrak{m}\,\mathrm{d}s \\ &= \int_0^t \int_M (\Delta_{\mathbf{g}} P_s f\, P_{t-s} g - P_s f\, \Delta_{\mathbf{g}} P_{t-s} g)\,\mathrm{d}\mathfrak{m}\,\mathrm{d}s = 0.\end{aligned}$$

In particular

$$\int_M P_t f\,\mathrm{d}\mathfrak{m} = \int_M f\, P_t 1\,\mathrm{d}\mathfrak{m} = \int_M f\,\mathrm{d}\mathfrak{m},$$

where the last equality follows from $\frac{\mathrm{d}}{\mathrm{d}t} P_t 1 = \Delta_{\mathbf{g}} 1 = 0$.

(ii) By (19.1) and (19.2), for any test function $g \in C^\infty(M)$ we have

$$\begin{aligned}\int_M \Delta_{\mathbf{g}} P_t f\, g\,\mathrm{d}\mathfrak{m} &= \lim_{h\to 0^+} \int_M \left(\frac{P_h - \mathrm{Id}}{h} P_t f\right) g\,\mathrm{d}\mathfrak{m} \\ &= \lim_{h\to 0^+} \int_M \left(P_t \frac{P_h - \mathrm{Id}}{h} f\right) g\,\mathrm{d}\mathfrak{m} = \int_M P_t(\Delta_{\mathbf{g}} f)\, g\,\mathrm{d}\mathfrak{m}\end{aligned}$$

for $\mathscr{L}^1$-a.e. $t > 0$. The sought conclusion follows by exploiting the continuity of $t \mapsto P_t(\Delta_{\mathbf{g}} f)$ in $L^2(\mathfrak{m})$.

(iii) The estimates follow from (iv) in Proposition 11.9.

(iv) Let us additionally assume that $e \in C^2(\mathbb{R})$ and $e' \in L^\infty(\mathbb{R})$. Notice that $t \mapsto e(P_t f)$ belongs to $\mathrm{AC}_{\mathrm{loc}}((0,\infty); L^2(\mathfrak{m}))$. We compute

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_M e(P_t f)\,\mathrm{d}\mathfrak{m} = \int_M e'(P_t f)\Delta_{\mathbf{g}} P_t f\,\mathrm{d}\mathfrak{m} = -\int_M e''(P_t f)|\nabla P_t f|_{\mathbf{g}}^2\,\mathrm{d}\mathfrak{m} \le 0.$$

In the last step we have used the identity $\Delta_{\mathbf{g}} = \mathrm{div}_{\mathbf{g}}\nabla$ and integrated by parts. The general case follows from a simple regularization argument.

(v) It is a particular case of (iv), simply by choosing $e(r) = \max\{C - r, 0\}$ or $e(r) = \max\{r - C, 0\}$.

□

We now prove that the heat flow $P_t f(x)$, that we introduced from the functional analytic viewpoint is smooth in space and time (more precisely, up to a modification in a $\mathscr{L}^{1+n}$-negligible set) and therefore solves the PDE in the classical sense. The proof follows by combining elliptic estimates with the Sobolev embedding theorem. We just sketch the argument, referring to [46] for a detailed account.

Proposition 19.2 *Let $(M, \mathbf{g})$ as above. For any $k \in \mathbb{N}$ and $t > 0$ there exists $C = C(\mathbf{g}, k, t) > 0$ such that*

$$\|P_t f\|_{C^k} \leq C\|f\|_{L^2} \quad \textit{for every } f \in L^2(\mathfrak{m}). \tag{19.5}$$

Moreover, for any $f \in L^2(\mathfrak{m})$ the map $(t, x) \mapsto P_t f(x)$ is smooth in $(0, \infty) \times M$.

Proof Let $t > 0$ and $f \in L^2(\mathfrak{m})$ be fixed. Building upon (19.1) and (ii) in Proposition 19.1 we deduce that

$$\Delta_{\mathbf{g}} P_t f = \Delta_{\mathbf{g}} P_{t/2}(P_{t/2} f) = P_{t/2}(\Delta_{\mathbf{g}} P_{t/2} f).$$

Hence applying twice (iii) in Proposition 19.1 we get

$$\left\|\Delta_{\mathbf{g}}^2 P_t f\right\|_{L^2}^2 = \left\|\Delta_{\mathbf{g}} P_{t/2}(\Delta_{\mathbf{g}} P_{t/2} f)\right\|_{L^2}^2 \leq \frac{4}{t^2}\left\|\Delta_{\mathbf{g}} P_{t/2} f\right\|_{L^2}^2 \leq \frac{16}{t^4}\|f\|_{L^2}^2,$$

where $\Delta_{\mathbf{g}}^2 := \Delta_{\mathbf{g}} \circ \Delta_{\mathbf{g}}$.

Iterating we get

$$\left\|\Delta^k P_t f\right\|_{L^2} \leq \left(\frac{2}{t}\right)^{2k} \|f\|_{L^2}^2 \quad \text{for any } k \in \mathbb{N},$$

which implies, via elliptic estimates,

$$\|P_t f\|_{H^{2k}(M)} \leq C(g, t, k)\|f\|_{L^2} \quad \text{for any } k \in \mathbb{N},$$

where $H^{2k}(M)$ is the space of those $f \in L^2(\mathfrak{m})$ whose distributional $2k$-covariant derivative $\nabla^{2k} f$ is an $L^2(\mathfrak{m})$ section, see for instance Section 7.4.5 in [114]. The sought estimate (19.5) follows from the Sobolev embedding theorem on Riemannian manifolds, see for instance Chapter 2 in [16].

The smoothness of $P_t f$ with respect to the time variable in $(0, \infty)$ can be proven by exploiting (19.2). Indeed, observe that for any $g \in L^2(\mathfrak{m})$ we can write

$$\int_M (P_{t+h} f - P_t f) g \, \mathrm{d}\mathfrak{m} = \int_0^h \int_M \Delta_{\mathbf{g}} P_{t+s} f \, g \, \mathrm{d}\mathfrak{m} \, \mathrm{d}s = \int_0^h \int_M P_s(\Delta_{\mathbf{g}} P_t f) \, g \, \mathrm{d}\mathfrak{m} \, \mathrm{d}s$$

for any $t, h > 0$. Hence, a simple approximation argument relying on (19.5) gives the pointwise identity

$$P_{t+h} f(x) - P_t f(x) = \int_0^h P_s(\Delta_{\mathbf{g}} P_t f)(x) \, \mathrm{d}s \quad \text{for any } x \in X. \tag{19.6}$$

From (19.5) and (19.6) we deduce that $t \mapsto P_t f(x)$ is locally Lipschitz in $(0, \infty)$ uniformly in $x \in X$. Of course the same conclusion holds for $s \mapsto P_s(\Delta_{\mathbf{g}} P_t f)(x)$, yielding

$$\partial_t P_t f(x) = \Delta_{\mathbf{g}} P_t f(x) = P_t(\Delta_{\mathbf{g}} f)(x) \quad \text{for any } (t, x) \in (0, \infty) \times X.$$

A simple iteration argument ensures that for any $k \in \mathbb{N}$ and $x \in X$ the map $t \mapsto \Delta_{\mathbf{g}}^k P_t f(x)$ is smooth in $(0, \infty)$ and satisfies

$$\partial_t^m \Delta_{\mathbf{g}}^k P_t f(x) = \Delta_{\mathbf{g}}^{k+m} P_t f(x) \quad \text{for any } m \in \mathbb{N},$$

which, arguing as in the first part of the proof, gives the sought conclusion. □

From now on we will identify the heat flow $P_t f$ with its space-time continuous representative, for any $f \in L^2(\mathfrak{m})$.

For any $\mu \in \mathscr{P}(M)$ we define $P_t^* \mu$ by duality:

$$\int_M f \, \mathrm{d}P_t^* \mu := \int_M P_t f \, \mathrm{d}\mu \qquad \text{for any } f \in C(M).$$

Notice that $P_t^* : \mathscr{P}(M) \to \mathscr{P}(M)$. Indeed, $P_t^* \mu$ is a monotone functional on $C(M)$ in view of (v) in Proposition 19.1 and

$$P_t^* \mu(M) = \int_M 1 \, \mathrm{d}P_t^* \mu = \int_M P_t 1 \, \mathrm{d}\mu = 1,$$

(where we have used the identity $P_t 1 = 1$) therefore Riesz theorem applies.

Proposition 19.3 *Let $(M, \mathbf{g})$ be as above. There exists a smooth function $p : (0, \infty) \times M \times M \to [0, \infty)$ such that*

$$P_t f(x) = \int_M p_t(x, y) f(y) \, \mathrm{d}\mathfrak{m}(y), \quad \text{for any } t > 0, \ x \in M \tag{19.7}$$

and $p_t(x, y) = p_t(y, x)$ for all $(t, x, y) \in (0, \infty) \times M \times M$. Moreover, $P_t^* \delta_x = p_t(x, \cdot)\mathfrak{m}$.

Proof Let us begin by proving that $P_t^*\delta_x \ll \mathfrak{m}$ for any $t > 0$. Given $f \in C(M)$ it holds

$$\left|\int_M f \, \mathrm{d}P_t^*\mu\right| = \left|\int_M P_t f \, \mathrm{d}\mu\right| \leq \|P_t f\|_{C^0} \leq C(\mathbf{g}, t)\|f\|_{L^2}, \tag{19.8}$$

where the last inequality follows from (19.5). In particular $P_t^*\delta_x \ll \mathfrak{m}$ with density, denoted by $p_t(x, \cdot)$, in $L^2(\mathfrak{m})$.

Building upon (19.1) and (i) in Proposition 19.1 we deduce

$$p_{t+s}(x, y) = P_t(p_s(x, \cdot))(y) \quad \text{for any } s,\ t > 0,$$

therefore p is smooth in $(0, \infty) \times M \times M$ as a consequence of Proposition 19.2 and of the identity $p_t(x, y) = p_t(y, x)$. The latter easily follows from (i) in Proposition 19.1.

It remains only to prove (19.7). When $f \in C(M)$ it is immediate:

$$P_t f(x) = \int_M P_t f \, \mathrm{d}\delta_x = \int_M f \, \mathrm{d}P_t^*\delta_x = \int_M f(y) p_t(x, y) \, \mathrm{d}\mathfrak{m}(y).$$

The general case follows from a simple regularization argument. □

2 Heat Flow, Optimal Transport and Ricci Curvature

In the proof of the convexity over $(\mathscr{P}_2(\mathbb{R}^n), W_2)$ of the relative entropies associated to energies verifying the McCann condition a crucial role was played by the form of the optimal transport geodesics, $\mu_t = (T_t)_\sharp \mu_0$ for $T_t = (1 - t)\mathrm{id} + tT$. This expression allowed to infer the concavity of the map

$$t \mapsto (\det \nabla T_t(x))^{\frac{1}{n}}$$

which was a crucial information to gain the geodesic convexity of relative entropies in Theorem 15.16.

In the case when the ambient space is a Riemannian manifold the expression for the optimal map for intermediate times takes the form

$$T_t(x) = \exp(-t\nabla\phi(x)),$$

for some c-concave potential ϕ, as we have seen in Theorem 7.10 (see also Remark 7.14). Trying to formally reproduce the proof of Theorem 15.16 one is led to the study of the quantity

$$\mathcal{J}(t) := \det(\nabla_x \exp(-t\nabla\phi))$$

and, setting $\gamma(t) := \exp(-t\nabla\phi(x))$, it turns that

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}\left(\mathcal{J}^{\frac{1}{n}}(t)\right) + \frac{\mathrm{Ric}(\gamma'(t),\gamma'(t))}{n}\mathcal{J}^{\frac{1}{n}}(t) \le 0. \tag{19.9}$$

The above differential inequality shows the connection between Ricci curvature and the behaviour of the infinitesimal rate of change of the volume along the transport.

An alternative way to interpret Ricci curvature is given by the Bochner identity. If $f : \mathbb{R}^n \to \mathbb{R}$ is sufficiently smooth then it is not difficult to verify that

$$\Delta\frac{|\nabla f|^2}{2} = \|\mathrm{Hess}\, f\|^2 + \langle \nabla f, \nabla\Delta f\rangle, \tag{19.10}$$

where, from now on, we denote by $\|\mathrm{Hess}\, f\|$ the Hilbert-Schmidt norm of the Hessian matrix and by $|\cdot|$ and $\langle\cdot,\cdot\rangle$ the norm and scalar products induced by $\mathbf{g}$. When the ambient space is a Riemannian manifold $(M,\mathbf{g})$ and $f : M \to \mathbb{R}$ is smooth, an extra term appears in (19.10), depending on Ricci curvature:

$$\Delta_{\mathbf{g}}\frac{|\nabla f|^2}{2} = \|\mathrm{Hess}\, f\|^2 + \langle \nabla f, \nabla\Delta_{\mathbf{g}} f\rangle + \mathrm{Ric}(\nabla f, \nabla f). \tag{19.11}$$

Neglecting the nonnegative contribution of the term $\|\mathrm{Hess}\, f\|^2$, (19.11) can be turned into the so-called Bochner inequality

$$\Delta_{\mathbf{g}}\frac{|\nabla f|^2}{2} - \langle \nabla f, \nabla\Delta_{\mathbf{g}} f\rangle \ge \mathrm{Ric}(\nabla f, \nabla f). \tag{19.12}$$

It turns out that (19.9) and (19.12) are just different perspectives on Ricci curvature: the first one is a *Lagrangian* interpretation, based on the behavior of volumes distortion rates along curves, while choosing $f = \phi$ in (19.12) one is led instead to a *Eulerian* interpretation, where the Ricci curvature governs the behavior of the velocity vector field $\nabla\phi$.

Let us also mention another instance of (Ricci) curvature that we are going to address later and that was already mentioned at the end of the proof of Lemma 18.12. On the Euclidean space the heat flow commutes with the divergence and the gradient, that is to say $\nabla P_t f = P_t \nabla f$ and $\mathrm{div}\, P_t v = P_t\, \mathrm{div}\, v$ whenever f and v are a smooth function and vector field respectively and the heat flow for vector fields is defined component by component.

Over Riemannian manifold this commutation, which is based on the fact that on $\mathbb{R}^n$ the heat kernel p_t has the form $p_t(x,y) = p_t(x-y)$, is not true in general (and even the notion of componentwise action does not make sense). As we shall see below, also the error in this commutation is somehow governed by the Ricci curvature.

Now we can state our main result, concerning various connections between Ricci curvature, convexity of the logarithmic entropy in the Wasserstein space, heat flow and gradient contractivity estimates.

Theorem 19.4 *Let $(M, \mathbf{g})$ be a smooth compact and connected Riemannian manifold without boundary and let us denote by $\mathfrak{m}$ its canonical volume measure. Then the following properties hold:*

(i) the logarithmic entropy $\mathrm{Ent}_{\mathfrak{m}}$ is K-convex in $\mathscr{P}(M)$ if and only if P_t^* is an EVI_K gradient flow of $\mathrm{Ent}_{\mathfrak{m}}$ in $(\mathscr{P}(M), W_2)$;

(ii) (Kuwada equivalence) the heat flow P_t verifies the gradient K-contractivity estimate, that is

$$|\nabla P_t f|^2 \le e^{-2Kt} P_t |\nabla f|^2 \qquad \text{for any } f \in \mathrm{Lip}(M), \tag{19.13}$$

if and only if P_t^* is K-contractive in $(\mathscr{P}(M), W_2)$;

(iii) the K-Bochner inequality

$$\Delta_{\mathbf{g}} \frac{|\nabla f|^2}{2} - \langle \nabla f, \nabla \Delta_{\mathbf{g}} f \rangle \ge K |\nabla f|^2 \tag{19.14}$$

holds for any $f \in C^\infty(M)$ if and only if $\mathrm{Ric} \ge K\mathbf{g}$.

Moreover, items (i), (ii) and (iii) above are all equivalent.

The rest of this last lecture is devoted to the (strategy of the) proofs for the various implications in Theorem 19.4. We start from the "horizontal" ones (i), (ii), (iii) and later we address the "vertical" ones, namely the equivalences between (i), (ii), (iii). In the proof we use a lighter notation, namely using Δ for $\Delta_{\mathbf{g}}$.

Proof of (i) in Theorem 19.4 The implication from the EVI_K property of the dual heat flow P_t^* to the K-convexity of $\mathrm{Ent}_{\mathfrak{m}}$ follows from Theorem 18.14 in the more general version of the original paper [44] (in Theorem 18.14 we considered only for simplicity the case $K = 0$).

The converse implication, from K-convexity of the entropy to the EVI_K property of the heat flow, has been proved in [91]. We refer also to [65] relying on the previous [66] (and dealing with the more general setting of Alexandrov spaces with curvature bounded below). Under the assumption that $\mathrm{Ent}_{\mathfrak{m}}$ is convex the strategy of the proof is very similar to the one we presented in Theorem 18.5 in the Euclidean case, even though some new difficulties arise due to the presence of the cut-locus, and relies on Theorem 7.10 in place of Theorem 5.2. The general case of K-convexity can be handled with minor modifications. □

To prove (ii), the so-called Kuwada equivalence, we will rely on two preliminary results. The first one is just an adaptation to the Riemannian setting of Corollary 17.12 (see also Remark 17.13 addressing the case of Lipschitz functions) and we omit its proof, that can be obtained with minor modifications to the Euclidean one. The second one is originally due to Kuwada [79], recall that

the Hopf-Lax semigroup Q_t associated to exponent $p = 2$ was introduced in Definition 16.7.

Lemma 19.5 *If $\mu_t \in AC^2([0,1]; \mathscr{P}(M))$, $\mu_t \ll \mathfrak{m}$, then for any $f \in \mathrm{Lip}(M)$ we have*

$$\left| \int_M f \, \mathrm{d}\mu_1 - \int_M f \, \mathrm{d}\mu_0 \right| \leq \int_0^1 \left(\int_M |\nabla f|^2 \, \mathrm{d}\mu_t \right)^{\frac{1}{2}} |\mu_t'| \, \mathrm{d}t.$$

Lemma 19.6 *If P_t verifies the gradient K-contractivity estimate (19.13), then for any $f \in \mathrm{Lip}(M)$, for any $t \geq 0$ and $x, y \in M$, it holds*

$$P_t Q_1 f(x) - P_t f(y) \leq \frac{1}{2} d^2(x,y) e^{-2Kt}. \tag{19.15}$$

Proof Let $\gamma \in \mathrm{Geo}(M)$ be a geodesic curve such that $\gamma(1) = x$, $\gamma(0) = y$. Then we have

$$P_t Q_1 f(x) - P_t f(y) = P_t Q_1 f(\gamma(1)) - P_t Q_0 f(\gamma(0)).$$

Next we claim that the map $(0,1] \ni s \mapsto P_t Q_s f(\gamma(s))$ is locally Lipschitz. In order to prove this conclusion we rely on Theorem 16.10(iii). Given $\epsilon > 0$ we denote by L_ϵ the Lipschitz constant of $Q_t f$ on $(\epsilon, \infty) \times M$. Then, for any $s \in (\epsilon, 1]$ and $h > 0$ such that $s + h \in (\epsilon, 1]$, we can write

$$\begin{aligned} P_t Q_{s+h} f(\gamma(s+h)) - P_t f(\gamma(s)) = &(P_t Q_{s+h} f(\gamma(s+h)) - P_t Q_{s+h} f(\gamma(s))) \\ &+ (P_t Q_{s+h} f(\gamma(s)) - P_t Q_s f(\gamma(s))) . \end{aligned} \tag{19.16}$$

The first summand can be estimated as

$$\begin{aligned} |P_t Q_{s+h} f(\gamma(s+h)) - P_t Q_{s+h} f(\gamma(s))| \leq & h d(x,y) \sup_M |\nabla P_t Q_{s+h} f| \\ \leq & h d(x,y) e^{-Kt} \sup_M \left(P_t |\nabla Q_{s+h} f|^2 \right)^{\frac{1}{2}} \\ \leq & h d(x,y) e^{-Kt} L_\epsilon, \end{aligned} \tag{19.17}$$

where we used (19.13) to pass from the first line to the second one and Proposition 19.1(v) to pass from the second one to the last one.

To deal with the second summand in (19.16) instead we estimate

$$P_t Q_{s+h} f(\gamma(s)) - P_t Q_s f(\gamma(s)) \leq L_\epsilon h. \tag{19.18}$$

The combination of (19.17) and (19.18) gives the local Lipschitz continuity of $s \mapsto P_t Q_s f(\gamma(s))$. Therefore, taking into account also the pointwise convergence of $Q_t f$ to f as $t \downarrow 0$, one has

$$P_t Q_1 f(x) - P_t f(y) = \int_0^1 \frac{\mathrm{d}}{\mathrm{d}s} P_t Q_s f(\gamma(s))\,\mathrm{d}s. \tag{19.19}$$

Next, at any differentiability point of $s \mapsto P_t Q_s f(\gamma(s))$ such that $\frac{\mathrm{d}}{\mathrm{d}s} Q_s f = -\frac{1}{2}|\nabla Q_s f|^2$ $\mathfrak{m}$-a.e. in M, we estimate

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}s} P_t Q_s f(\gamma(s)) &= \lim_{h\to 0^+} \frac{P_t Q_{s+h} f(\gamma(s+h)) - P_t Q_s f(\gamma(s))}{h} \\ &\le \limsup_{h\to 0^+} \int_M \frac{Q_{s+h} f - Q_s f}{h}\,\mathrm{d}P_t^* \delta_{\gamma(s+h)} \qquad (19.20)\\ &\quad + \limsup_{h\to 0^+} \frac{P_t Q_s f(\gamma(s+h)) - P_t Q_s f(\gamma(s))}{h} \qquad (19.21)\end{aligned}$$

Now we estimate (19.20) as

$$\begin{aligned}&\limsup_{h\to 0^+} \int_M \frac{Q_{s+h} f - Q_s f}{h}\,\mathrm{d}P_t^* \delta_{\gamma(s+h)} \\ &\quad \le \int_M \limsup_{h\to 0} \frac{(Q_{s+h} f - Q_s f)(z)}{h} p_t(z, \gamma(s+h))\,\mathrm{d}\mathfrak{m}(z) \\ &\quad \le -\int_M \frac{1}{2}|\nabla Q_s f|^2(z)\,\mathrm{d}P_t^* \delta_{\gamma(s)}(z) \\ &\quad = -\frac{1}{2} P_t |\nabla Q_s f|^2(\gamma(s)),\end{aligned}$$

where we applied Fatou's lemma relying on the continuity of the heat kernel p_t, on the Lipschitz bounds on the Hopf-Lax semigroup and on the Hamilton-Jacobi equation (16.17).

Dealing with (19.21), we have

$$\begin{aligned}&\limsup_{h\to 0^+} \frac{P_t Q_s f(\gamma(s+h)) - P_t Q_s f(\gamma(s))}{h} \\ &\quad \le \limsup_{h\to 0^+} \frac{d(x,y)}{h} \int_s^{s+h} |\nabla P_t Q_s f|(\gamma(u))\,\mathrm{d}u \\ &\quad \le \limsup_{h\to 0^+} e^{-Kt} \frac{d(x,y)}{h} \int_s^{s+h} \left(P_t |\nabla Q_s f|^2\right)^{\frac{1}{2}} (\gamma(u))\,\mathrm{d}u \\ &\quad \le e^{-Kt} d(x,y) \left(P_t |\nabla Q_s f|^2(\gamma(s))\right)^{\frac{1}{2}}.\end{aligned}$$

All in all, at $\mathscr{L}^1$-a.e. differentiability point of $s \mapsto P_t Q_s f(\gamma(s))$ it holds

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}s} P_t Q_s f(\gamma(s)) &\le -\frac{1}{2} P_t |\nabla Q_s f|^2(\gamma(s)) + e^{-Kt} d(x,y) \left(P_t |\nabla Q_s f|^2(\gamma(s)) \right)^{\frac{1}{2}} \\ &\le \frac{1}{2} e^{-2Kt} d^2(x,y),\end{aligned}$$

that, combined with (19.19), leads to (19.15). □

Proof of (ii) in Theorem 19.4 Let us assume that P_t^* is K-contractive in $(\mathscr{P}(M), W_2)$. Let $x, y \in M$ and $\gamma \in \mathrm{Geo}(M)$ be such that $\gamma(0) = x$, $\gamma(1) = y$ and $|\gamma'| \equiv d(x,y)$. We set $\mu_s := P_t^*(\delta_{\gamma(s)})$, so that K-contractivity implies $|\mu_s'| \le e^{-Kt} d(x,y)$.

Then, for any $f \in \mathrm{Lip}(M)$, we have

$$\begin{aligned}|P_t f(x) - P_t f(y)| &= \left| \int_M f \,\mathrm{d} P_t^* \delta_{\gamma(0)} - \int_M f \,\mathrm{d} P_t^* \delta_{\gamma(1)} \right| \\ &\le e^{-Kt} d(x,y) \int_0^1 \left(\int_M |\nabla f|^2 \,\mathrm{d}\mu_s \right)^{\frac{1}{2}} \mathrm{d}s,\end{aligned}$$

by Lemma 19.5. This implies that

$$\frac{|P_t f(x) - P_t f(y)|^2}{d^2(x,y)} \le e^{-2Kt} \int_0^1 \left(\int_M |\nabla f|^2 \,\mathrm{d}\mu_s \right) \mathrm{d}s.$$

Now we send $y \to x$, so that $\mu_s \to P_t^* \delta_x$ weakly in duality with $C_b(M)$ and then in duality with $L^1(\mathfrak{m})$, since all measures have uniformly bounded densities. Therefore we get

$$|\nabla P_t f(x)|^2 \le e^{-2Kt} \int_M |\nabla f|^2 \,\mathrm{d} P_t^* \delta_x = e^{-2Kt} P_t |\nabla f|^2(x),$$

which is (19.13).

As for the converse implication, from gradient contractivity to Wasserstein contractivity, we employ Lemma 19.6 and the duality formula to obtain that, for any $x, y \in M$,

$$\begin{aligned}\frac{1}{2} W_2^2(P_t^* \delta_x, P_t^* \delta_y) &= \sup_{f \in \mathrm{Lip}(M)} \int_M Q_1 f \,\mathrm{d} P_t^* \delta_x - \int_M f \,\mathrm{d} P_t^* \delta_y \\ &= \sup_{f \in \mathrm{Lip}(M)} P_t Q_1 f(x) - P_t f(y) \\ &\le \frac{1}{2} e^{-2Kt} d^2(x,y).\end{aligned}$$

This proves the contractivity on Dirac delta measures. Then we use the disintegration theory to extend it to any $\mu,\ \nu \in \mathscr{P}(M)$. Indeed, we have

$$P_t^*\mu = \int_M P_t^*\delta_x \,\mathrm{d}\mu(x), \qquad P_t^*\nu = \int_M P_t^*\delta_y \,\mathrm{d}\nu(y).$$

In addition, for any $x,\ y \in M$, there exists a unique $\Sigma_{x,y} \in \Gamma_{\mathrm{o}}(P_t^*\delta_x, P_t^*\delta_y)$, so that

$$\int_{M\times M} d^2(u,v)\,\mathrm{d}\Sigma_{x,y}(u,v) \le e^{-2Kt} d^2(x,y). \tag{19.22}$$

Given $\Sigma \in \Gamma_{\mathrm{o}}(\mu,\nu)$, then we observe that

$$\tilde{\Sigma} := \int_{M\times M} \Sigma_{x,y}\,\mathrm{d}\Sigma(x,y)$$

satisfies $\tilde{\Sigma} \in \Gamma(P_t^*\mu, P_t^*\nu)$, by linearity.[1] Now we integrate (19.22) with respect to Σ and we obtain the sought contractivity estimate

$$W_2^2(P_t^*\mu, P_t^*\nu) \le e^{-2Kt}\int_{M\times M} d^2(x,y)\,\mathrm{d}\Sigma(x,y) = e^{-2Kt} W_2^2(\mu,\nu).$$

□

Proof of (iii) in Theorem 19.4 If we assume that $\mathrm{Ric} \ge K\mathbf{g}$ then (19.11) yields the Bochner inequality.

Conversely, if we assume that (19.14) holds true, then for any smooth function $f : M \to \mathbb{R}$ it holds

$$\|\mathrm{Hess} f\|^2 + \mathrm{Ric}(\nabla f, \nabla f) = \Delta\frac{|\nabla f|^2}{2} - \langle \nabla f, \nabla\Delta f\rangle \ge K|\nabla f|^2, \tag{19.23}$$

where the first equality follows from the Bochner identity. For any $x \in M$ and $v \in T_xM$ we can now find a smooth function $f_{x,v}$ such that $\nabla f_{x,v}(x) = v$ and $\mathrm{Hess} f_{x,v}(x) = 0$ (for instance working in exponential coordinates centered at x). Applying (19.23) with $f = f_{x,v}$ we get the sought estimate $\mathrm{Ric}(v,v) \ge K|v|^2$. □

As we anticipated, it is also possible to establish equivalences between the conditions (i), (ii) and (iii) in Theorem 19.4. Since any of these conditions has two equivalent forms, for the sake of clarity we explicitly state the implications we are going to address.

[1] As in the proof of Theorem 9.13 we are tacitly using the fact that the map $(x,y) \mapsto \Sigma_{x,y}$ is Σ-measurable. In this particular case, continuity and then measurability follows by Theorem 6.8.

Proposition 19.7 (From (i) to (ii)) *Assume that P_t^* is an EVI_K gradient flow of $\mathrm{Ent}_{\mathfrak{m}}$ in $(\mathscr{P}(M), W_2)$, then P_t^* is K-contractive in $(\mathscr{P}(M), W_2)$ i.e. for any $\mu,\ \nu \in \mathscr{P}(M)$ it holds*

$$W_2(P_t^*\mu, P_t^*\nu) \le e^{-Kt} W_2(\mu, \nu) \qquad \text{for any } t \ge 0.$$

Proof The conclusion follows from the general theory of EVI gradient flows in metric spaces. In particular in Lemma 11.13 we proved that EVI_K gradient flows on Hilbert spaces are K-contractive and in the last part of the proof we already pointed out that the conclusion holds even for general metric spaces. □

Proposition 19.8 (Equivalence of (ii) and (iii)) *The heat flow P_t verifies the gradient estimate (19.13) if and only if the Bochner inequality (19.14) holds.*

Proof Assume that $f : M \to \mathbb{R}$ is smooth. Let us observe that

$$\frac{P_t f - f}{t} \to \Delta f \quad \text{as } t \downarrow 0$$

in the C^k norm for any $k \in \mathbb{N}$. This is a consequence of Proposition 19.1(ii), applied iteratively, and of the embedding theorems on manifolds that we already exploited in its proof. Then we can differentiate (19.13) at $t = 0$, using the fact that at $t = 0$ we have equality, thus obtaining

$$-2K|\nabla f|^2 + \Delta|\nabla f|^2 - 2\langle \nabla f, \nabla \Delta f\rangle \ge 0,$$

which is (19.14).

Conversely, if we assume that (19.14) holds, then for any smooth function $f : M \to \mathbb{R}$ and for any $0 < \epsilon < t$ we have

$$\begin{aligned}
e^{-2Kt} P_t|\nabla f|^2 - e^{-2K\epsilon} P_\epsilon |\nabla P_{t-\epsilon} f|^2 &= \int_\epsilon^t \frac{\mathrm{d}}{\mathrm{d}s}\left(e^{-2Ks} P_s |\nabla P_{t-s} f|^2\right) \mathrm{d}s \\
&= \int_\epsilon^t e^{-2Ks} \Delta P_s(|\nabla P_{t-s} f|^2)\, \mathrm{d}s \\
&\quad - \int_\epsilon^t 2e^{-2Ks} P_s \langle \nabla P_{t-s} f, \nabla \Delta P_{t-s} f\rangle \, \mathrm{d}s \\
&\quad - \int_\epsilon^t 2Ke^{-2Ks} P_s |\nabla P_{t-s} f|^2 \, \mathrm{d}s \\
&= \int_\epsilon^t e^{-2Ks} P_s \Delta(|\nabla P_{t-s} f|^2) \\
&\quad - 2P_s \langle \nabla P_{t-s} f, \nabla \Delta P_{t-s} f\rangle \, \mathrm{d}s \\
&\quad - \int_\epsilon^t e^{-2Ks} 2K P_s |\nabla P_{t-s} f|^2 \, \mathrm{d}s.
\end{aligned}$$

Applying (19.14) to $P_{t-s}f$ for any $s \in [\epsilon, t]$ we obtain that the last expression above is nonnegative, and letting $\epsilon \downarrow 0$ we obtain the gradient contractivity estimate for smooth functions, since

$$e^{-2K\epsilon} P_\epsilon |\nabla P_{t-\epsilon} f|^2 \to |\nabla P_t f|^2$$

pointwise as $\epsilon \downarrow 0$ due to the smoothness of the heat flow on $(0, \infty) \times M$. The general case $f \in \mathrm{Lip}(M)$ follows from a simple regularization argument. □

The last result we state, closing the circle, concerns the implication from the lower Ricci curvature bound $\mathrm{Ric} \geq K\mathbf{g}$ and the K-convexity of the relative entropy. It is worth pointing out it explicitly, also for its huge impact over the literature of lower Ricci curvature bounds for non smooth spaces in the last 15 years. A crucial remark in this regard is that, apart from the condition $\mathrm{Ric} \geq K\mathbf{g}$ which requires the smooth Riemannian structure to be given a meaning, all the other equivalent conditions appearing in Theorem 19.4 do not require the smoothness of the ambient space to be formulated and indeed they make sense on any metric measure space.

Proposition 19.9 (From (iii) to (i)) *The logarithmic entropy* $\mathrm{Ent}_{\mathfrak{m}}$ *is* K*-convex over* $(\mathscr{P}(M), W_2)$ *if* $\mathrm{Ric} \geq K\mathbf{g}$.

This result has been obtained by D. Cordero-Erausquin, McCann and M. Schmuckenschläger in [43]. The proof requires a clever adaptation to the Riemannian setting of the strategy we followed to prove convexity of the relative entropy with respect to $\mathscr{L}^n$ on $\mathbb{R}^n$ in Theorem 15.16. As we pointed out in the preliminary discussion of this section, Ricci curvature enters into play through (19.9) in this approach.

As a final remark, from all these equivalences we know that even the converse implication in Proposition 19.9 holds. However, a direct proof of this fact, that played an important role in the development of the theory, has been given by K.-T. Sturm and M. Von Renesse in [111].

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